Configuration Summary

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DHW ANALYSIS - CONFIGURATION SUMMARY
Project: oisst_regression
Generated: 2026-10-21 20:58:37
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-- Data Preparation --

  input_dir.............................. /g/data/ia39/aus-ref-clim-data-nci/oisst/data/yearly/
  file_pattern........................... oisst-avhrr-v02r01_{year}.nc
  year_range............................. (1981, 2024)
  variable............................... sst
  detrend_method......................... global_linear
  standardise_method..................... global
  fill_method............................ diffusion
  fill_diffusion_iterations.............. 500
  fill_fallback_value.................... 0
  downscale.............................. True
  target_lat_size........................ 256
  target_lon_size........................ 512
  temporal_average....................... none
  subset_region.......................... False
  lat_bounds............................. None
  lon_bounds............................. None
  diagnostic_year........................ 2020
  diagnostic_pdf......................... diagnostics/data_preparation_diagnostics_256x512.pdf
  output_dir............................. data/processed_256x512/
  save_intermediates..................... True
  downscaled_file........................ sst_downscaled.nc
  temporally_averaged_file............... sst_temporally_averaged.nc
  detrended_file......................... sst_detrended.nc
  standardised_file...................... sst_standardised.nc
  filled_file............................ sst_filled.nc
  final_file............................. sst_final.nc
  subset_file............................ sst_subset.nc
  trend_params_file...................... trend_params.nc
  standardise_params_file................ std_params.nc
  resume_after_stage..................... 0
  chunks................................. {'time': 366, 'lat': 720, 'lon': 720}
  dask_scheduler......................... threads
  dask_n_workers......................... 8
  dask_threads_per_worker................ 2
  dask_memory_limit...................... 12GB
  dask_num_threads....................... 16

-- Model Training --

  model_name............................. sfno
  input_history.......................... 10
  output_channels........................ 1
  embed_dim.............................. 80
  num_layers............................. 5
  num_modes_lat.......................... 48
  num_modes_lon.......................... 96
  mlp_ratio.............................. 2.5
  activation............................. gelu
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norm_type.............................. instance_norm
dropout_rate........................... 0.05
use_residual........................... True
spectral_bias.......................... True
add_grid_features...................... True
add_seasonal_encoding.................. False
forecast_horizon....................... 90
initial_rollout_steps.................. 1
max_rollout_steps...................... 5
rollout_epoch_start.................... 5
rollout_epoch_end...................... 50
rollout_loss_decay..................... 0.95
train_years............................ (1982, 2015)
val_years.............................. (2016, 2019)
test_years............................. (2020, 2024)
learning_rate.......................... 0.0002
min_learning_rate...................... 1e-05
batch_size............................. 10
num_epochs............................. 120
weight_decay........................... 1e-05
optimizer.............................. adamw
scheduler.............................. cosine
scheduler_patience..................... 10
scheduler_factor....................... 0.5
warmup_epochs.......................... 5
gradient_clip_norm..................... 1
loss_function.......................... huber
early_stopping_patience................ 20
early_stopping_metric.................. val_loss
input_data_path........................ data/processed_256x512/sst_final.nc
checkpoint_dir......................... checkpoints_256x512/
best_model_path........................ checkpoints_256x512/best_model.pt
training_log_path...................... results_256x512/training_log.csv
random_seed............................ 42
num_workers............................ 4
distributed............................ True
dist_backend........................... nccl
find_unused_parameters................. False
sync_batchnorm......................... False
master_port............................ 29500

-- Model Inference --

model_path............................. checkpoints_256x512/best_model.pt
input_data_path........................ data/processed_256x512/sst_final.nc
run_validation......................... True
run_test............................... True
forecast_lead_steps.................... 90
chain_stride_steps..................... 90
output_dir............................. results_256x512/
val_predictions_file................... predictions_val.nc
test_predictions_file.................. predictions_test.nc

-- Concat Chains --
```

```
  predictions_dir..... /scratch/um09/xl5758/DHW-project/results_256x512
  predictions_file_template..... predictions_{split}.nc
  std_params_path..... data/processed_256x512/std_params.nc
  trend_params_path..... data/processed_256x512/trend_params.nc
  output_dir..... /scratch/um09/xl5758/DHW-project/results_256x512
  actual_file_template..... sst_actual_{split}_continuous.nc
  predicted_file_template..... sst_predicted_{split}_chains.nc
  splits..... ['test', 'val']
  lon_correction_threshold..... 0.6

-- Calc DHW --

  mmm_path..... /home/272/xl5758/ct5km_climatology_v3.1.nc
  results_dir..... /scratch/um09/xl5758/DHW-project/results_256x512
  predicted_file_template..... sst_predicted_{split}_chains.nc
  actual_file_template..... sst_actual_{split}_continuous.nc
  dhw_output_dir..... /scratch/um09/xl5758/DHW-project/dhw_output_256x512
  output_file_template..... dhw_products_{split}.nc
  run_test..... True
  run_validation..... True
  window_days..... 84
  hotspot_threshold..... 1

-- DHW Plots --

  dhw_dir..... /scratch/um09/xl5758/DHW-project/dhw_output_256x512
  output_pdf..... dhw_analysis_combined.pdf
  output_csv..... dhw_metrics_combined.csv
  spatial_lead_steps..... [1, 7, 14, 30, 60, 90]
  n_sample_chains..... 6
  plot_init_dates..... ['2016-05-20', '2016-10-15', '2017-02-10', '2017-07-09', '20
  dhw_vmax..... 20
  dhw_diff_vmax..... 8
  tropic_lat..... 30
  dhw_cmap..... ['#CCe6F0', '#441C94', '#6F47BF', '#815CCC', '#A383E6', '#ED
  error_cmap..... ['#053061', '#2166AC', '#92C5DE', '#F7F7F7', '#F4A582', '#D6
  rmse_cmap..... YlOrRd
  bias_cmap..... RdBu_r
  bleaching_thresholds..... [(0, 4, 'No Stress', '#27AE60'), (4, 8, 'Warning', '#F39C12'
  gbr_lat_bounds..... (-25.0, -10.0)
  gbr_lon_bounds..... (142.0, 157.0)
  reef_boxes..... [{'name': 'Great Barrier Reef', 'lat': (-25.0, -10.0), 'lon'
```

DHW Analysis - Summary

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DEGREE HEATING WEEKS (DHW) - COMBINED ANALYSIS
======================================================================

METHODOLOGY  (NOAA Coral Reef Watch v3.1)
  SSTA      = SST - DC              (DC = daily climatology)
  HotSpot   = max(SST - MMM, 0)     (MMM = Maximum Monthly Mean)
  DHW_i     = (1/7) x sum(HotSpot_j) for j=i-83..i, HotSpot_j >= 1.0C

SLIDING WINDOW CONSTRUCTION  (always 84 days ending on valid date)
  Lead L:  (84-L) actual + L predicted SST days
  Lead 1:  83 actual +  1 predicted
  Lead 30: 54 actual + 30 predicted
  Lead 84:  0 actual + 84 predicted
  Lead 90: previous 84 predicted days (days 7-90)

VERIFICATION
  Actual DHW  : 84 observed days ending on the forecast valid date
  Persistence : observed DHW on the day before lead 1, held constant
  Sample      : chains defined at every lead step (same chains for
                model and persistence at all leads)

BLEACHING ALERT THRESHOLDS
  DHW  0-4  : No Stress
  DHW  4-8  : Warning
  DHW  8-12 : Alert Level 1
  DHW 12-16 : Alert Level 2
  DHW 16-20 : Alert Level 3
  DHW 20->= : Alert Level 4-5

DATASETS
  Test          19 chains x 90 leads x 256 lat x 512 lon
  Val           16 chains x 90 leads x 256 lat x 512 lon
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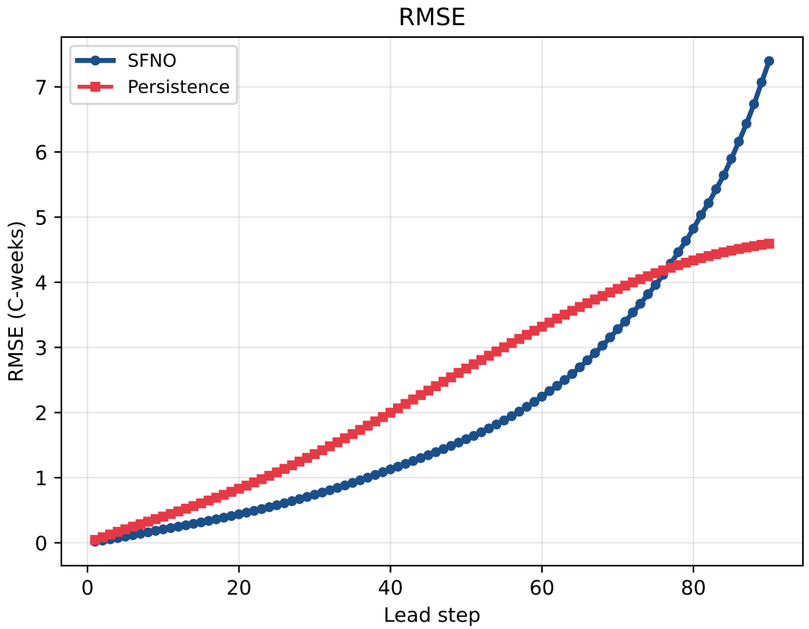
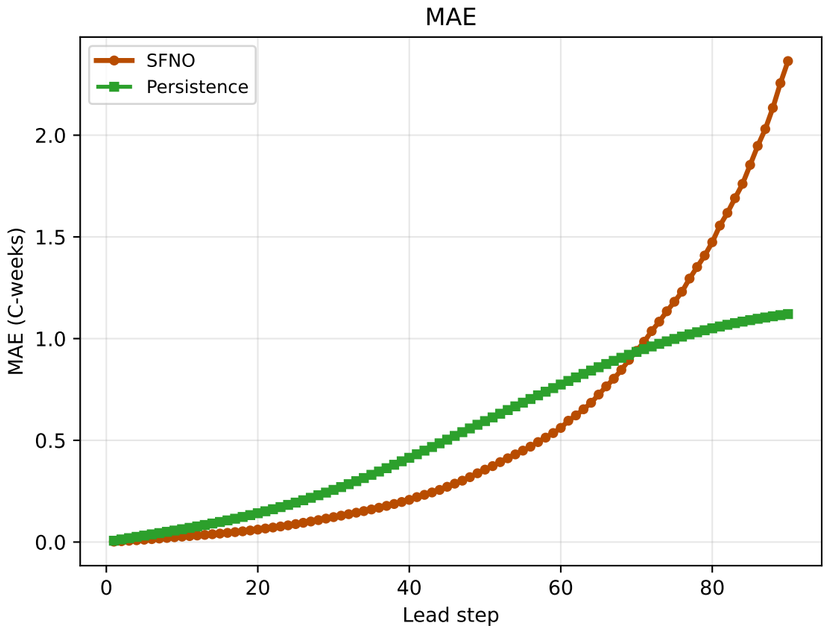
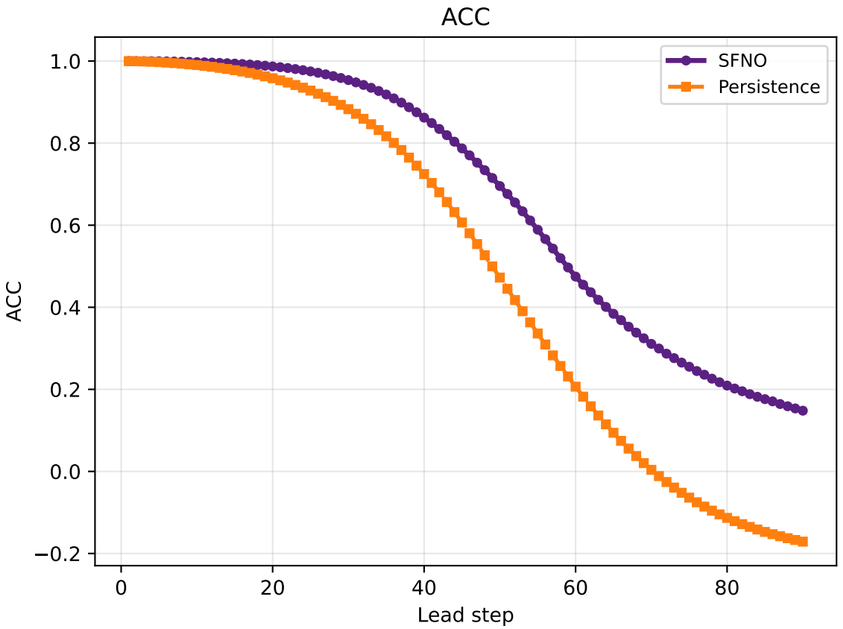
DHW Metrics by Lead Step - Test (18 chains scored at every lead)

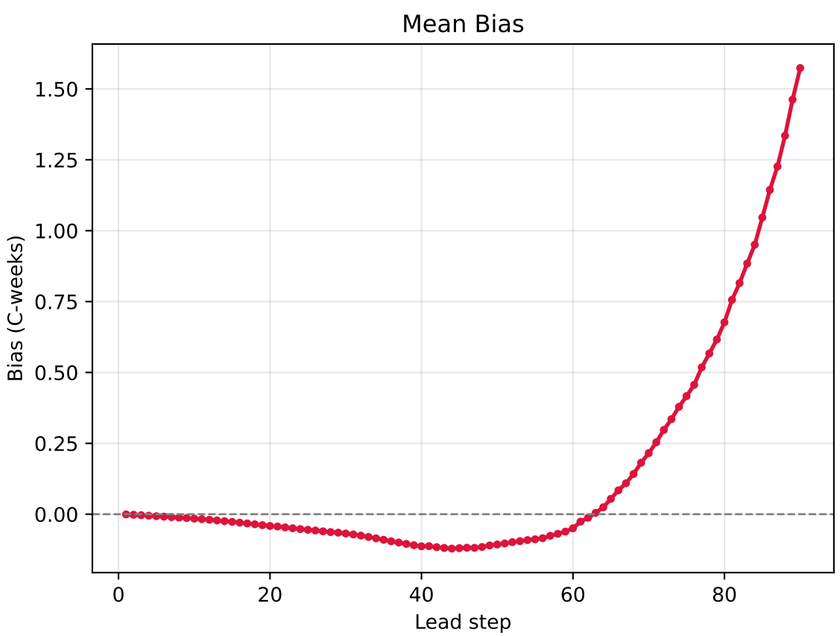
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	Persist MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0183	0.0437	58.2%	0.0021	0.0065	-0.0007	1.0000	0.9999	1.0000
2	0.0353	0.0858	58.9%	0.0041	0.0129	-0.0020	0.9999	0.9996	0.9999
3	0.0539	0.1272	57.6%	0.0064	0.0192	-0.0036	0.9998	0.9990	0.9997
4	0.0740	0.1678	55.9%	0.0089	0.0255	-0.0053	0.9997	0.9983	0.9994
5	0.0951	0.2079	54.2%	0.0116	0.0317	-0.0069	0.9995	0.9974	0.9991
6	0.1170	0.2473	52.7%	0.0145	0.0378	-0.0085	0.9992	0.9964	0.9986
7	0.1389	0.2863	51.5%	0.0174	0.0440	-0.0103	0.9989	0.9951	0.9981
8	0.1611	0.3252	50.5%	0.0203	0.0502	-0.0121	0.9985	0.9937	0.9974
9	0.1840	0.3640	49.4%	0.0236	0.0565	-0.0136	0.9980	0.9921	0.9965
10	0.2067	0.4030	48.7%	0.0267	0.0630	-0.0155	0.9975	0.9902	0.9956
11	0.2263	0.4426	48.9%	0.0295	0.0697	-0.0174	0.9969	0.9882	0.9947
12	0.2475	0.4827	48.7%	0.0323	0.0767	-0.0196	0.9963	0.9860	0.9936
13	0.2691	0.5232	48.6%	0.0353	0.0838	-0.0220	0.9956	0.9835	0.9924
14	0.2914	0.5644	48.4%	0.0384	0.0911	-0.0244	0.9948	0.9808	0.9910
15	0.3139	0.6060	48.2%	0.0416	0.0986	-0.0270	0.9939	0.9778	0.9895
16	0.3371	0.6484	48.0%	0.0450	0.1064	-0.0298	0.9928	0.9746	0.9877
17	0.3611	0.6920	47.8%	0.0488	0.1147	-0.0325	0.9916	0.9710	0.9857
18	0.3860	0.7367	47.6%	0.0527	0.1233	-0.0355	0.9902	0.9671	0.9835
19	0.4117	0.7827	47.4%	0.0569	0.1323	-0.0385	0.9887	0.9628	0.9809
20	0.4380	0.8299	47.2%	0.0612	0.1416	-0.0415	0.9869	0.9582	0.9781
21	0.4633	0.8777	47.2%	0.0666	0.1513	-0.0433	0.9851	0.9532	0.9751
22	0.4903	0.9267	47.1%	0.0714	0.1613	-0.0464	0.9830	0.9477	0.9717
23	0.5181	0.9771	47.0%	0.0766	0.1718	-0.0495	0.9806	0.9417	0.9679
24	0.5471	1.0286	46.8%	0.0821	0.1826	-0.0525	0.9779	0.9352	0.9637
25	0.5769	1.0812	46.6%	0.0882	0.1938	-0.0550	0.9749	0.9281	0.9589
26	0.6078	1.1352	46.5%	0.0946	0.2054	-0.0575	0.9715	0.9204	0.9536
27	0.6395	1.1903	46.3%	0.1010	0.2175	-0.0604	0.9677	0.9121	0.9478
28	0.6718	1.2466	46.1%	0.1078	0.2300	-0.0630	0.9635	0.9030	0.9413
29	0.7048	1.3041	46.0%	0.1153	0.2430	-0.0650	0.9588	0.8933	0.9342
30	0.7385	1.3622	45.8%	0.1225	0.2563	-0.0680	0.9536	0.8828	0.9265
31	0.7719	1.4212	45.7%	0.1297	0.2702	-0.0714	0.9481	0.8715	0.9182
32	0.8070	1.4816	45.5%	0.1370	0.2845	-0.0754	0.9418	0.8593	0.9090
33	0.8432	1.5431	45.4%	0.1443	0.2993	-0.0802	0.9349	0.8461	0.8990
34	0.8811	1.6057	45.1%	0.1521	0.3146	-0.0850	0.9271	0.8319	0.8878
35	0.9201	1.6692	44.9%	0.1602	0.3304	-0.0903	0.9185	0.8167	0.8757
36	0.9598	1.7333	44.6%	0.1686	0.3466	-0.0955	0.9091	0.8006	0.8625
37	1.0003	1.7984	44.4%	0.1777	0.3631	-0.0999	0.8988	0.7832	0.8483
38	1.0422	1.8644	44.1%	0.1871	0.3799	-0.1048	0.8875	0.7647	0.8330
39	1.0850	1.9307	43.8%	0.1969	0.3969	-0.1094	0.8753	0.7452	0.8165
40	1.1282	1.9973	43.5%	0.2074	0.4141	-0.1132	0.8622	0.7247	0.7992
41	1.1688	2.0645	43.4%	0.2208	0.4315	-0.1127	0.8490	0.7030	0.7821
42	1.2122	2.1322	43.1%	0.2320	0.4492	-0.1162	0.8346	0.6802	0.7636
43	1.2561	2.2003	42.9%	0.2437	0.4671	-0.1190	0.8194	0.6565	0.7444
44	1.3009	2.2685	42.7%	0.2562	0.4852	-0.1212	0.8035	0.6319	0.7246
45	1.3457	2.3367	42.4%	0.2716	0.5035	-0.1197	0.7872	0.6065	0.7045
46	1.3917	2.4048	42.1%	0.2870	0.5218	-0.1183	0.7702	0.5805	0.6836
47	1.4394	2.4729	41.8%	0.3016	0.5401	-0.1186	0.7524	0.5539	0.6619
48	1.4883	2.5407	41.4%	0.3189	0.5584	-0.1155	0.7340	0.5269	0.6396
49	1.5387	2.6080	41.0%	0.3380	0.5765	-0.1101	0.7151	0.4997	0.6164
50	1.5908	2.6747	40.5%	0.3559	0.5946	-0.1068	0.6955	0.4724	0.5924
51	1.6413	2.7409	40.1%	0.3733	0.6127	-0.1032	0.6762	0.4450	0.5692
52	1.6973	2.8067	39.5%	0.3929	0.6308	-0.0985	0.6553	0.4177	0.5434
53	1.7555	2.8720	38.9%	0.4117	0.6488	-0.0950	0.6338	0.3904	0.5164
54	1.8165	2.9370	38.2%	0.4308	0.6670	-0.0913	0.6115	0.3632	0.4879
55	1.8786	3.0018	37.4%	0.4493	0.6851	-0.0885	0.5891	0.3361	0.4588
56	1.9444	3.0662	36.6%	0.4688	0.7031	-0.0847	0.5662	0.3093	0.4277
57	2.0145	3.1300	35.6%	0.4913	0.7211	-0.0761	0.5431	0.2828	0.3941
58	2.0882	3.1931	34.6%	0.5132	0.7389	-0.0692	0.5197	0.2569	0.3586
59	2.1634	3.2559	33.6%	0.5357	0.7566	-0.0614	0.4971	0.2314	0.3223
60	2.2446	3.3183	32.4%	0.5610	0.7743	-0.0498	0.4748	0.2065	0.2828
61	2.3294	3.3806	31.1%	0.5963	0.7918	-0.0261	0.4550	0.1823	0.2416
62	2.4098	3.4423	30.0%	0.6225	0.8092	-0.0125	0.4367	0.1588	0.2038
63	2.4992	3.5030	28.7%	0.6526	0.8263	0.0049	0.4182	0.1363	0.1606
64	2.5918	3.5624	27.2%	0.6846	0.8430	0.0243	0.4009	0.1148	0.1155
65	2.6953	3.6209	25.6%	0.7250	0.8593	0.0541	0.3842	0.0942	0.0633
66	2.8018	3.6784	23.8%	0.7652	0.8751	0.0842	0.3687	0.0745	0.0093
67	2.9124	3.7350	22.0%	0.8029	0.8906	0.1091	0.3526	0.0556	-0.0479
68	3.0279	3.7906	20.1%	0.8462	0.9057	0.1419	0.3383	0.0375	-0.1085
69	3.1534	3.8450	18.0%	0.8952	0.9204	0.1816	0.3244	0.0202	-0.1770
70	3.2801	3.8978	15.8%	0.9396	0.9346	0.2153	0.3108	0.0039	-0.2471
71	3.3950	3.9488	14.0%	0.9846	0.9483	0.2536	0.2995	-0.0114	-0.3084
72	3.5362	3.9981	11.6%	1.0369	0.9615	0.2973	0.2869	-0.0258	-0.3907
73	3.6706	4.0462	9.3%	1.0834	0.9742	0.3353	0.2761	-0.0394	-0.4680
74	3.8173	4.0927	6.7%	1.1344	0.9865	0.3788	0.2652	-0.0520	-0.5561
75	3.9585	4.1379	4.3%	1.1811	0.9984	0.4166	0.2551	-0.0640	-0.6405
76	4.1152	4.1815	1.6%	1.2304	1.0099	0.4563	0.2446	-0.0752	-0.7392
77	4.2862	4.2232	-1.5%	1.2951	1.0208	0.5177	0.2357	-0.0857	-0.8525
78	4.4623	4.2628	-4.7%	1.3518	1.0312	0.5668	0.2260	-0.0956	-0.9732
79	4.6344	4.3003	-7.8%	1.4083	1.0411	0.6162	0.2173	-0.1047	-1.0937
80	4.8220	4.3359	-11.2%	1.4738	1.0505	0.6767	0.2092	-0.1132	-1.2315
81	5.0337	4.3699	-15.2%	1.5554	1.0595	0.7562	0.2020	-0.1212	-1.3956
82	5.2148	4.4017	-18.5%	1.6179	1.0680	0.8153	0.1954	-0.1285	-1.5349
83	5.4294	4.4318	-22.5%	1.6903	1.0760	0.8839	0.1882	-0.1352	-1.7116
84	5.6416	4.4602	-26.5%	1.7607	1.0836	0.9505	0.1819	-0.1415	-1.8912
85	5.8961	4.4869	-31.4%	1.8538	1.0907	1.0463	0.1761	-0.1475	-2.1208
86	6.1595	4.5116	-36.5%	1.9472	1.0974	1.1440	0.1708	-0.1529	-2.3690
87	6.4346	4.5346	-41.9%	2.0301	1.1038	1.2264	0.1642	-0.1580	-2.6402
88	6.7353	4.5555	-47.9%	2.1341	1.1097	1.3348	0.1588	-0.1628	-2.9533
89	7.0690	4.5746	-54.5%	2.2558	1.1153	1.4624	0.1535	-0.1671	-3.3195
90	7.3958	4.5922	-61.1%	2.3642	1.1206	1.5737	0.1480	-0.1711	-3.6931

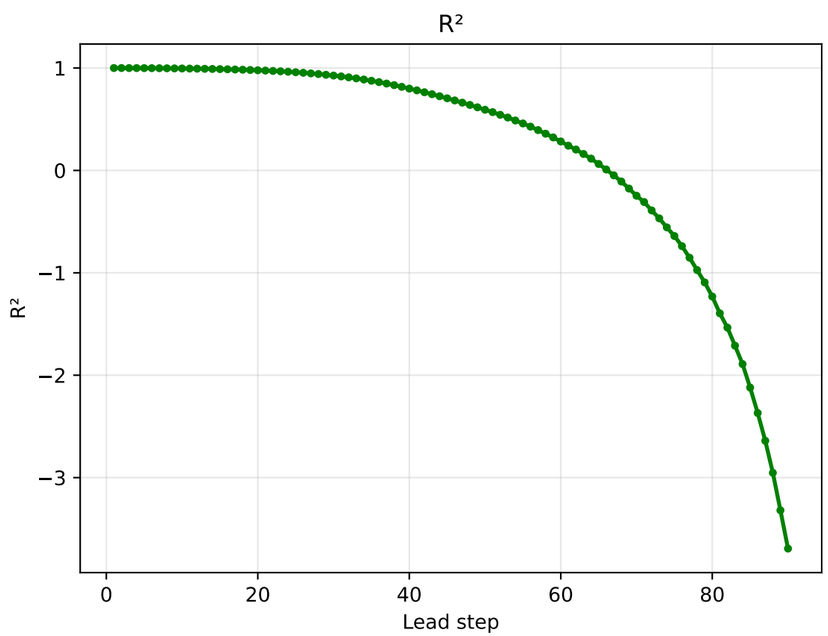
DHW Forecast Skill vs Lead Step - Test

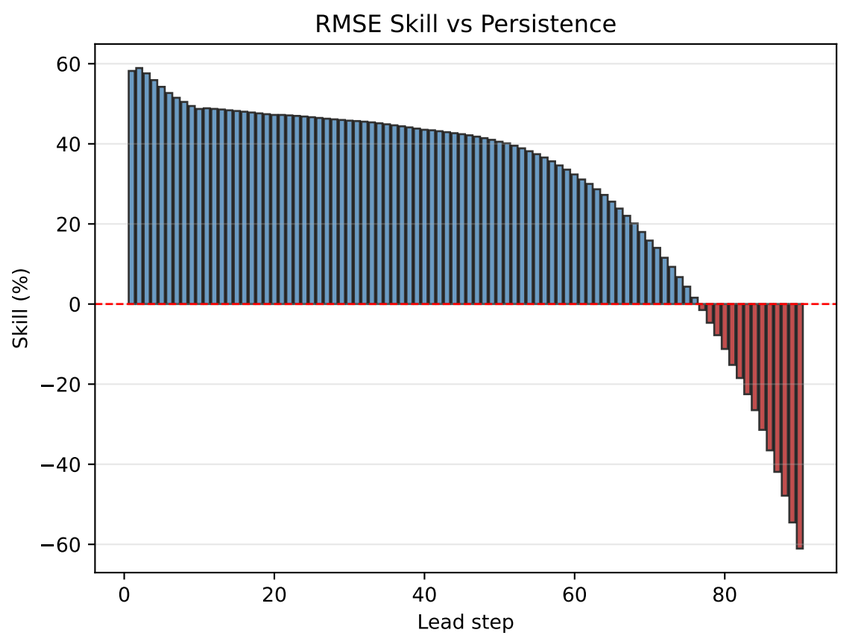


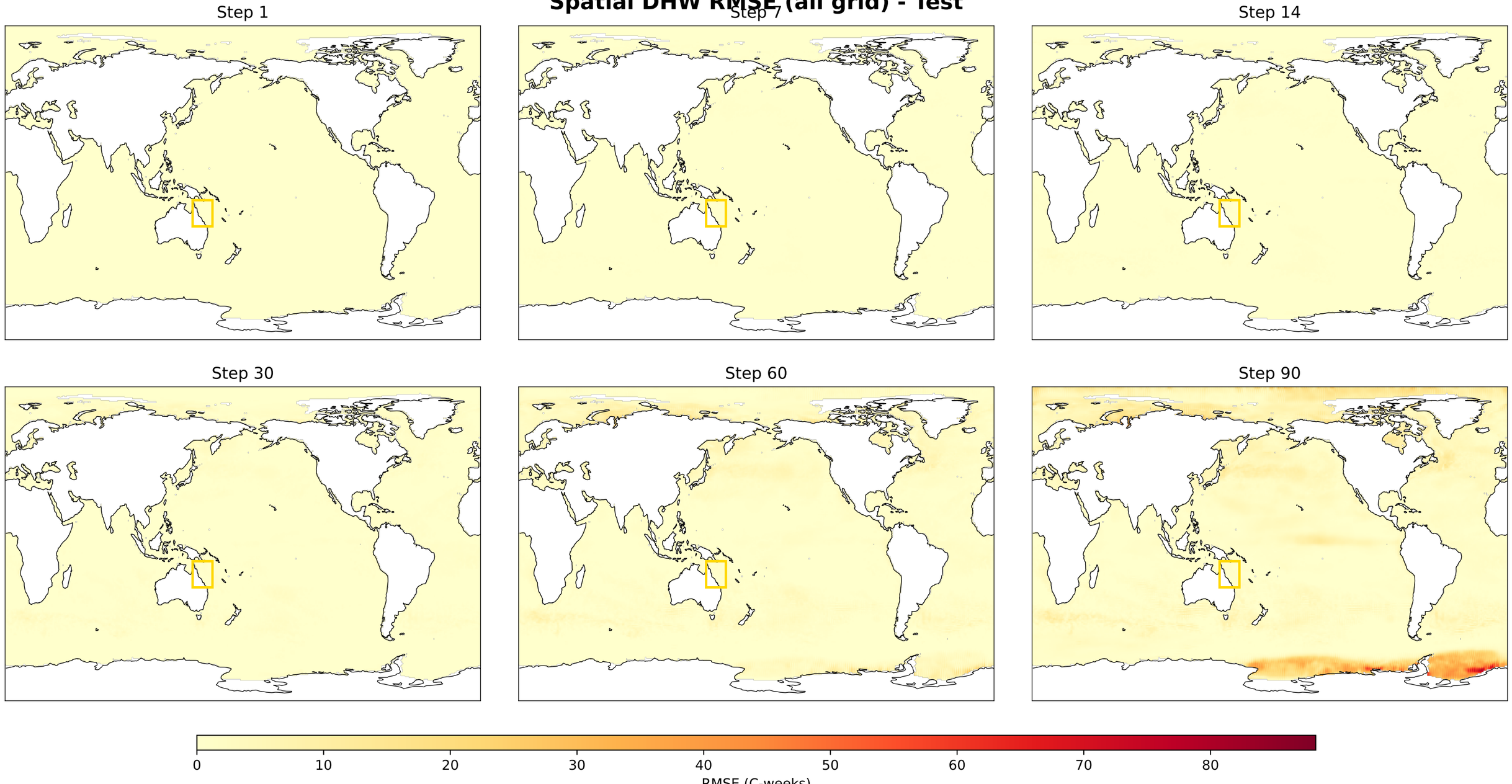


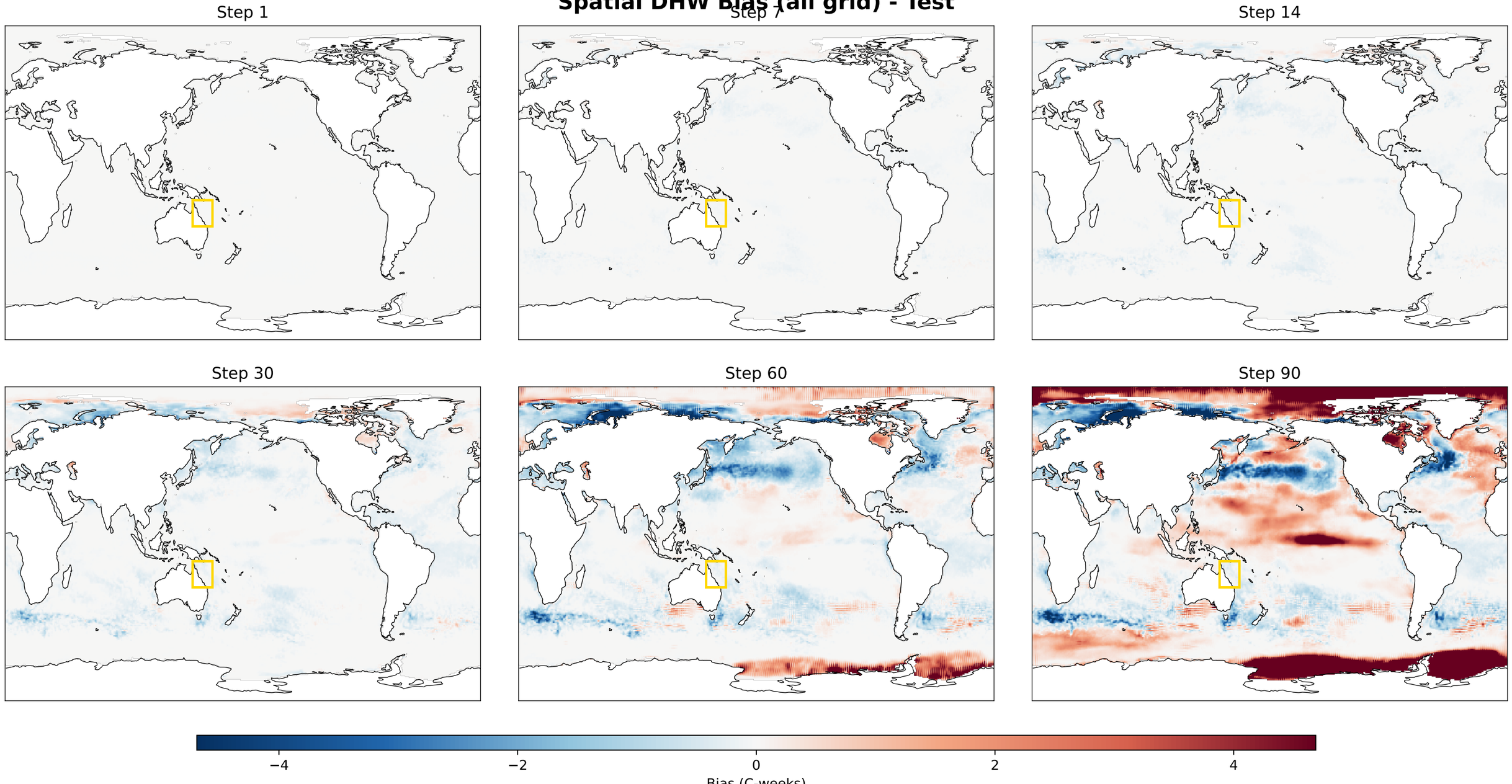


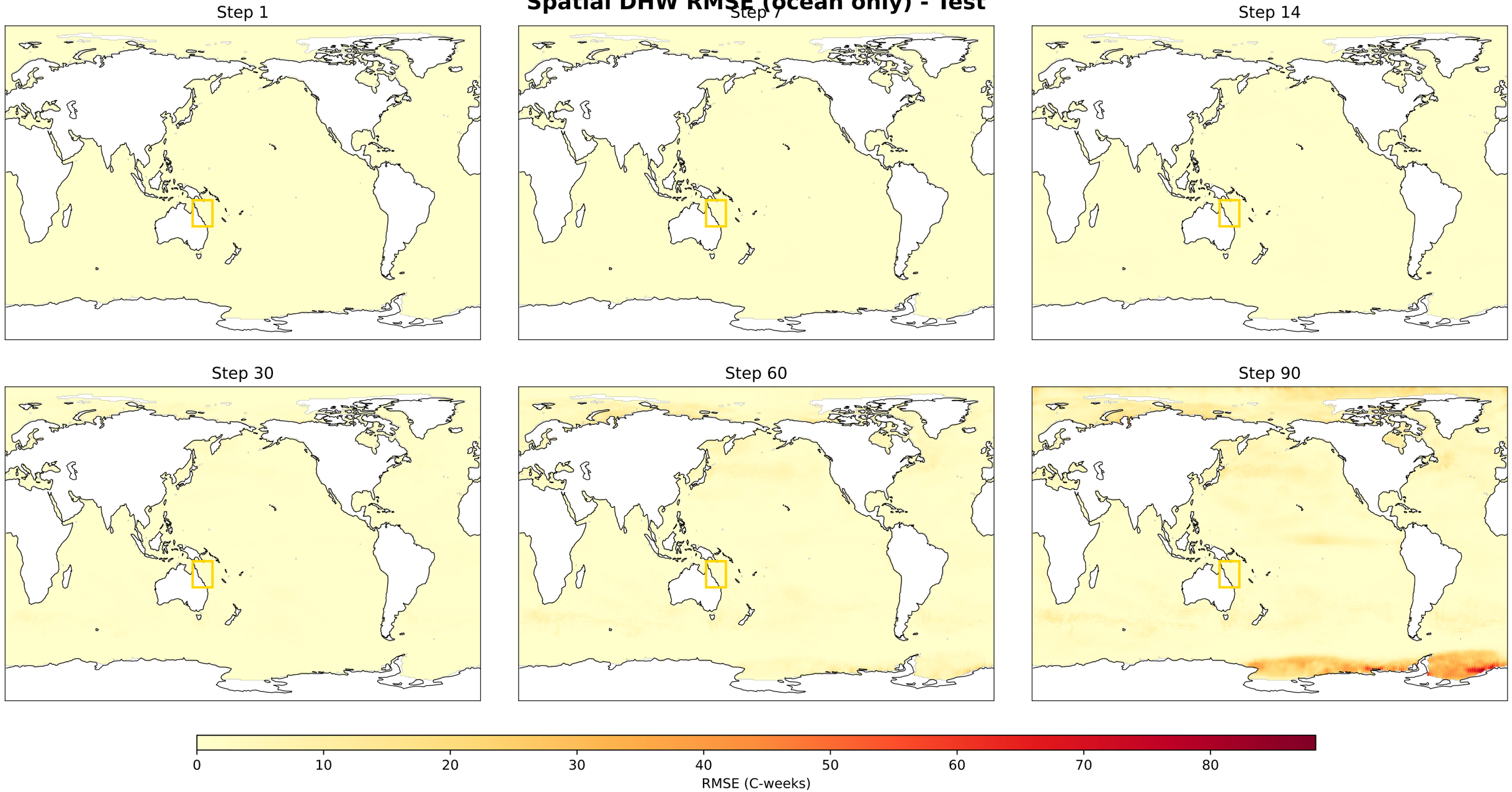


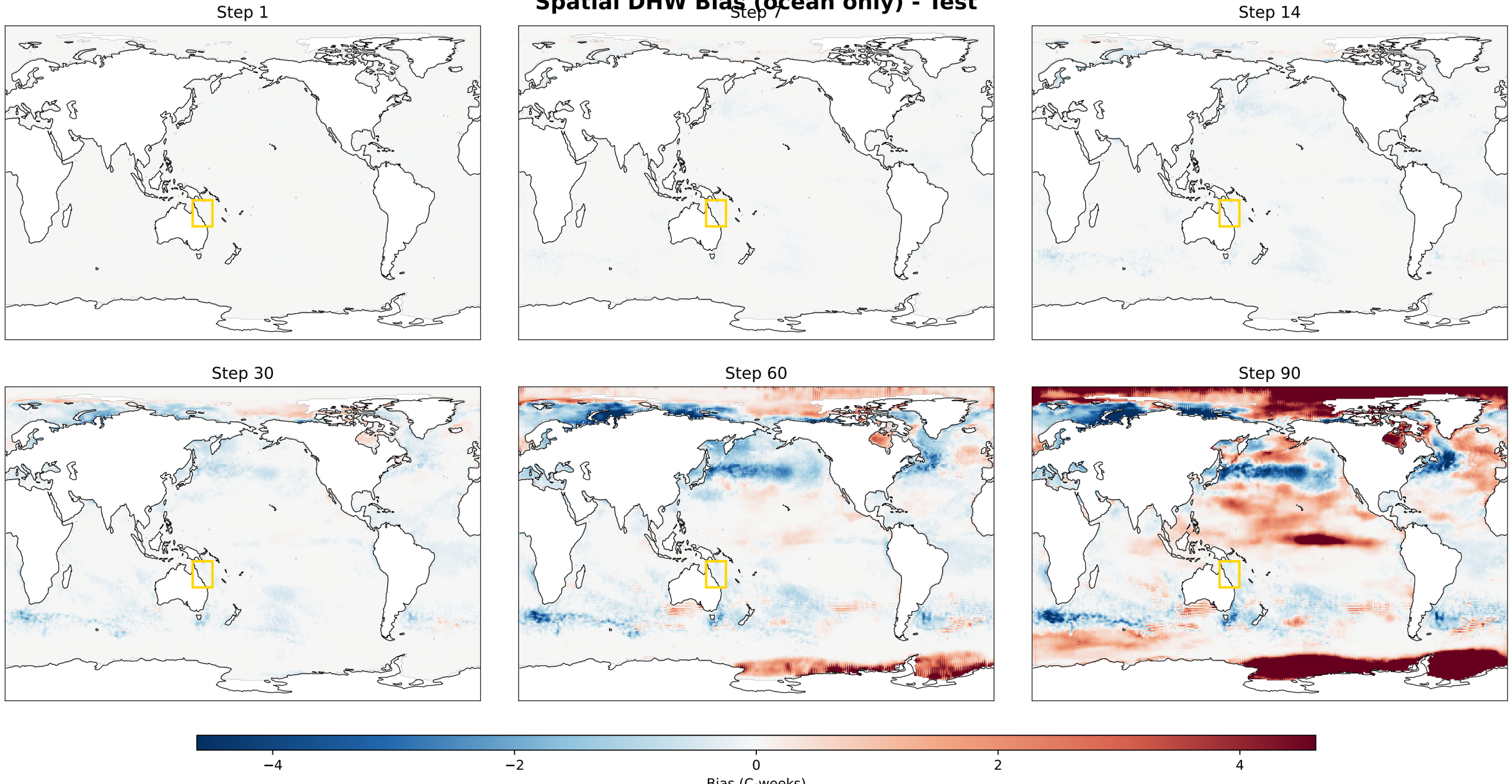





Spatial DHW RMSE (all grid) - Test
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0
10
20
30
40
50
60
70
80
RMSE (C-weeks)

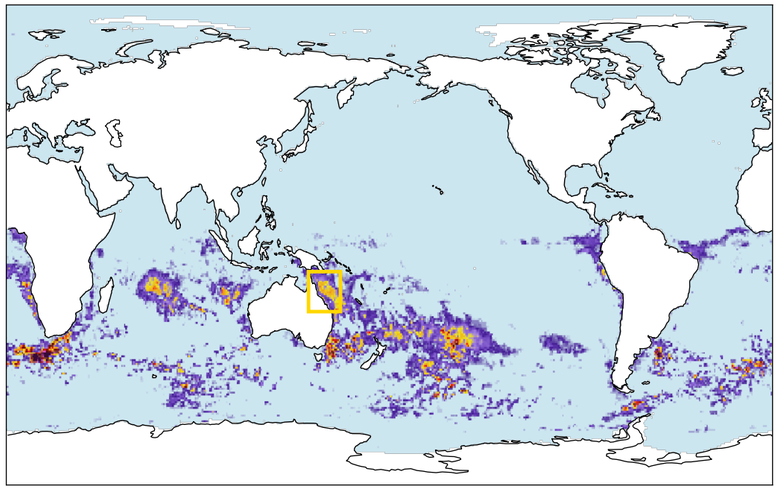

Spatial DHW Bias (all grid) - Test
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−4
−2
0
2
4
Bias (C-weeks)


Spatial DHW RMSE (ocean only) - Test
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0
10
20
30
40
50
60
70
80
RMSE (C-weeks)

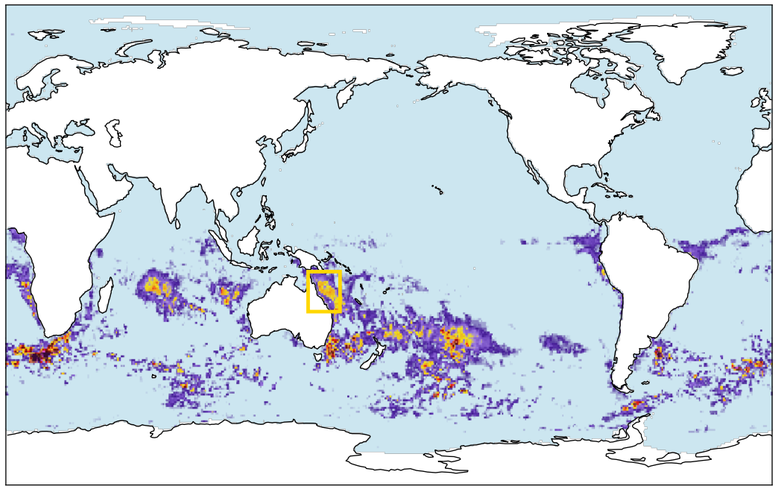

Spatial DHW Bias (ocean only) - Test
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−4
−2
0
2
4
Bias (C-weeks)

DHW Chain init: 2020-04-10 - Test (C-weeks)

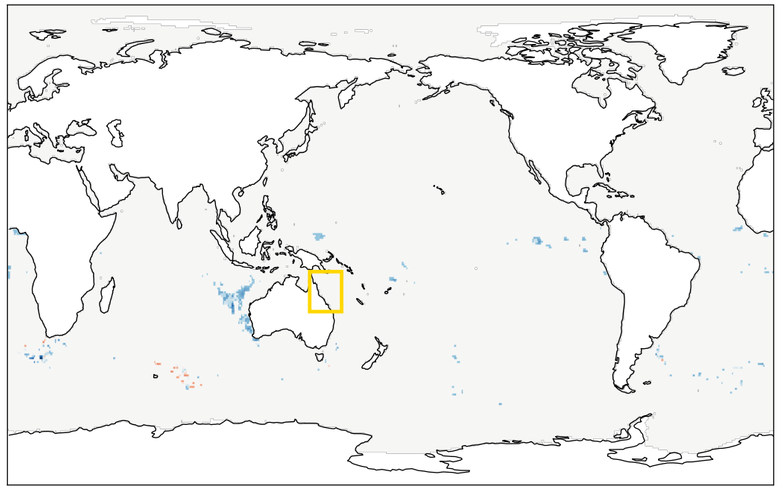
Actual DHW - Step 1



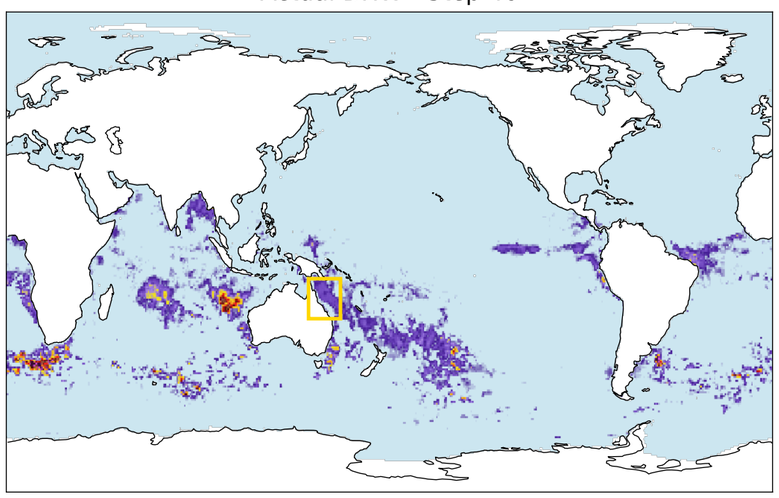
Predicted DHW - Step 1



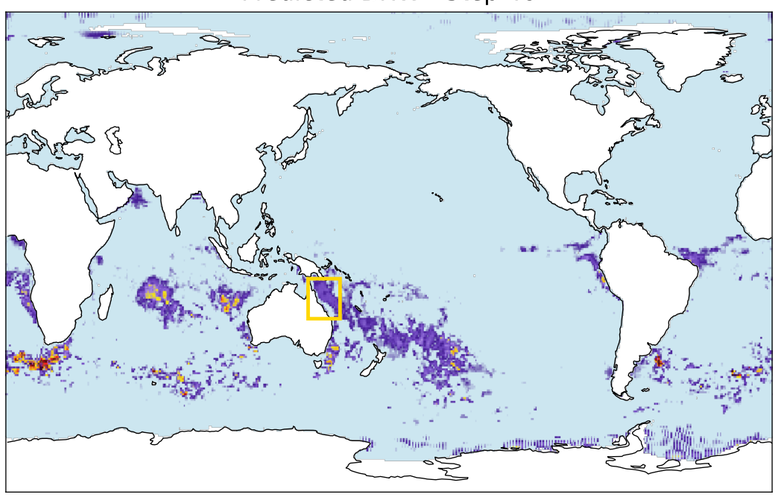
Error (Pred - Actual) - Step 1 [scale ±0.47]



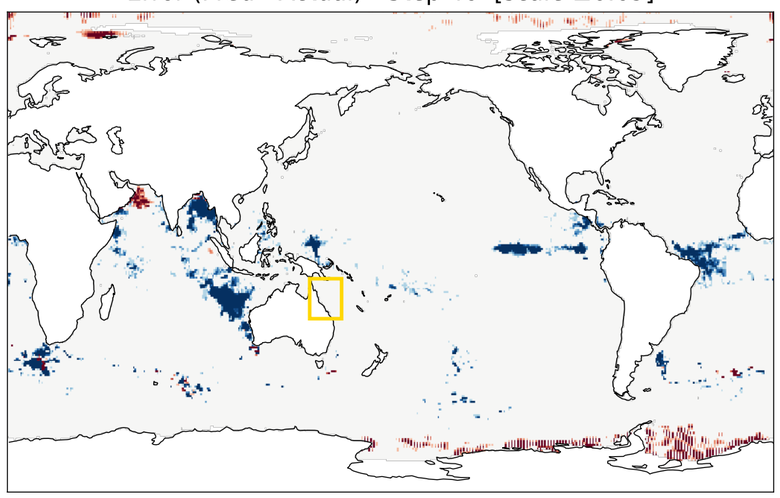
Actual DHW - Step 46



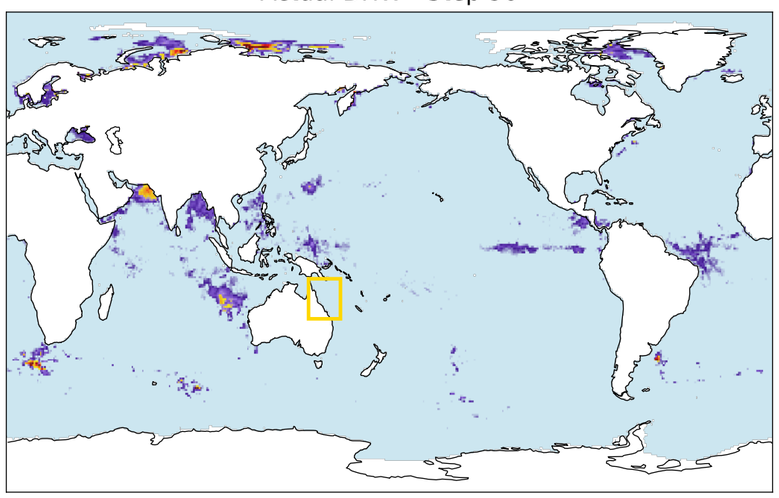
Predicted DHW - Step 46



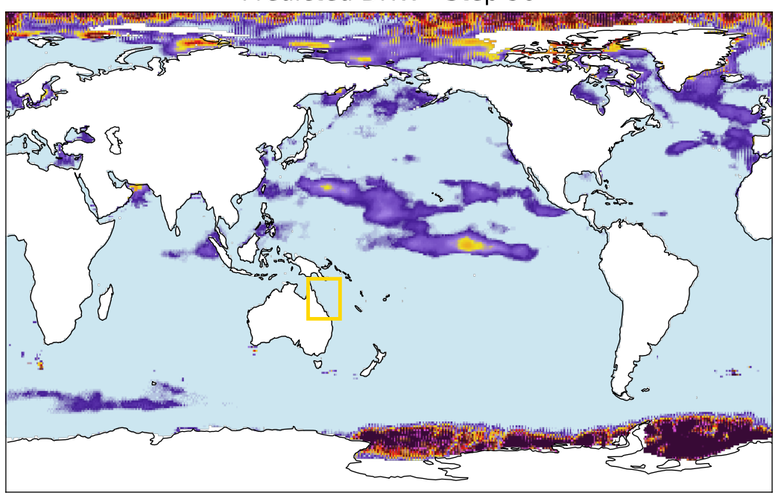
Error (Pred - Actual) - Step 46 [scale ±0.69]



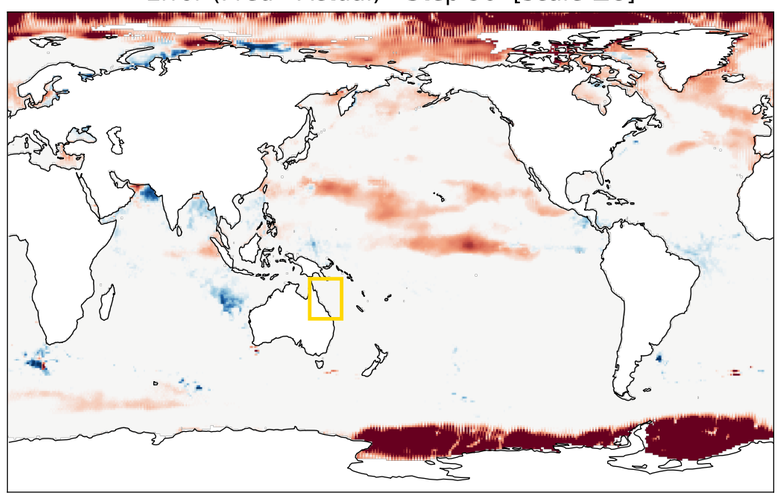
Actual DHW - Step 90

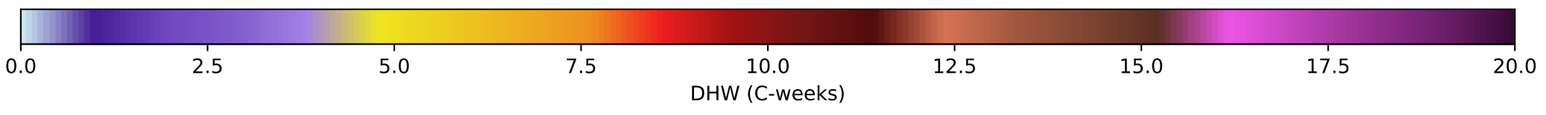


Predicted DHW - Step 90



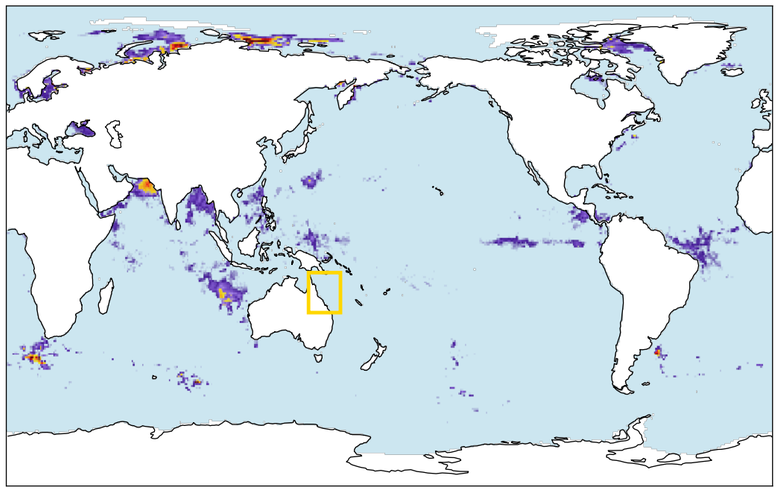
Error (Pred - Actual) - Step 90 [scale ±8]



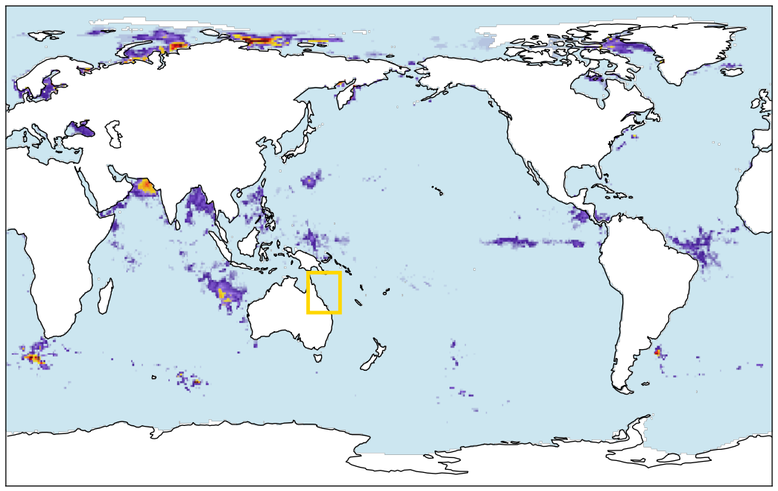


DHW Chain init: 2020-07-09 - Test (C-weeks)

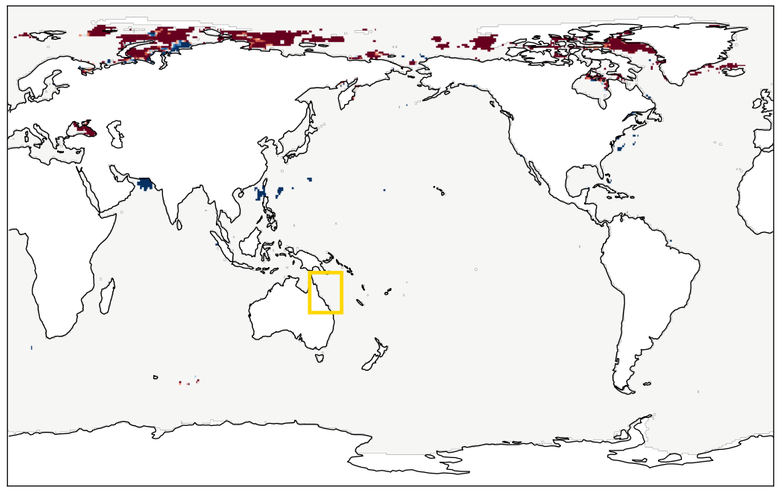
Actual DHW - Step 1



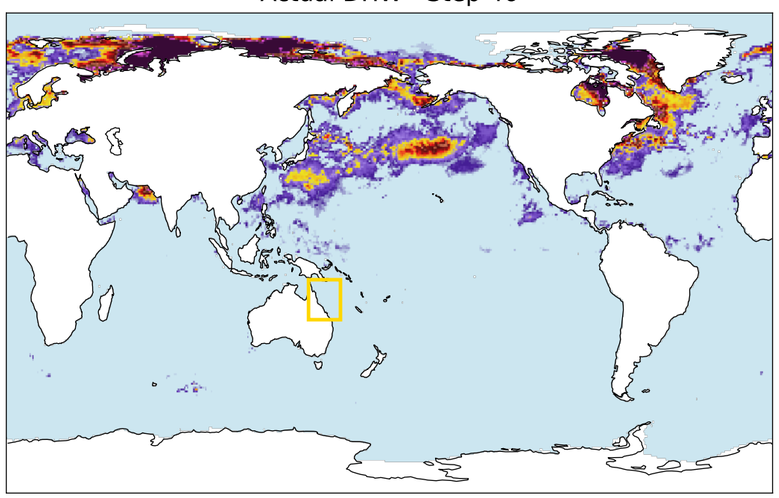
Predicted DHW - Step 1



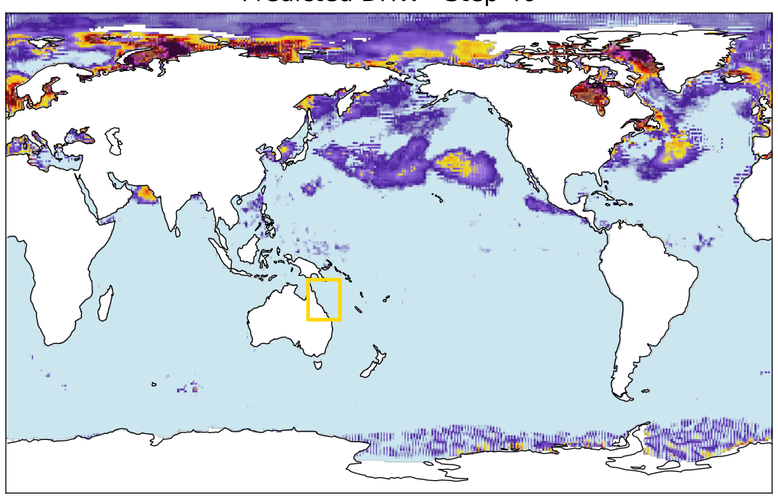
Error (Pred - Actual) - Step 1 [scale ±0.074]



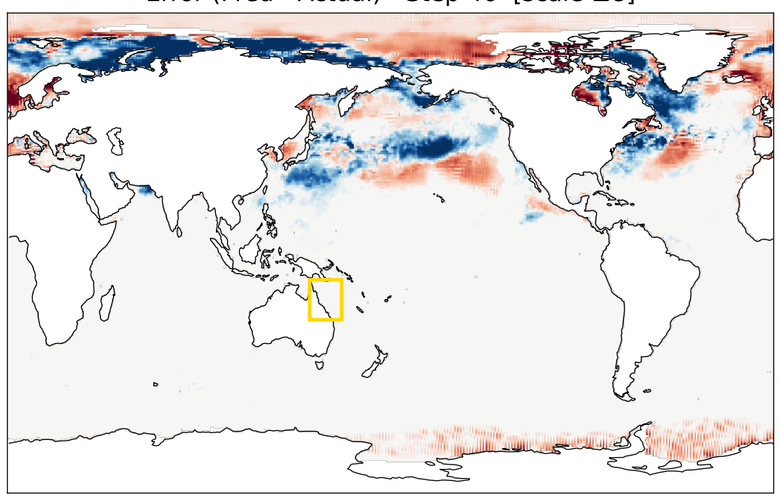
Actual DHW - Step 46



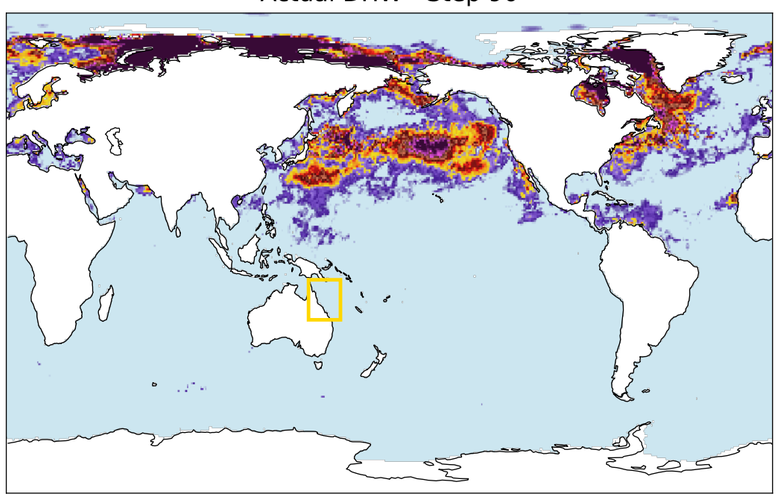
Predicted DHW - Step 46



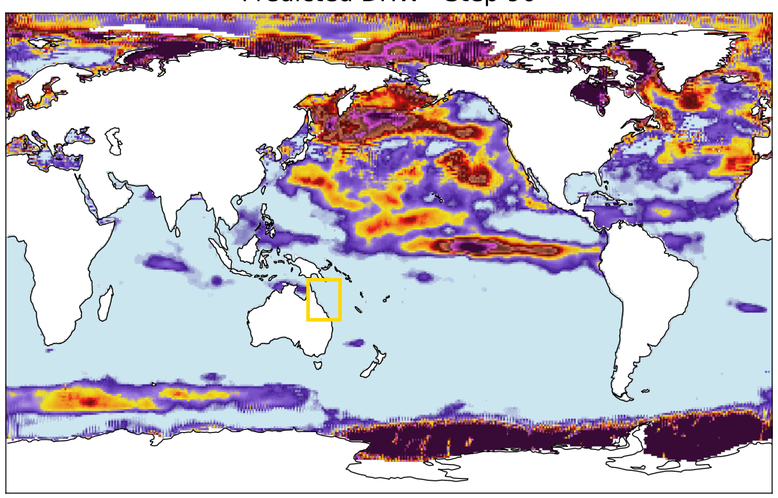
Error (Pred - Actual) - Step 46 [scale ±8]



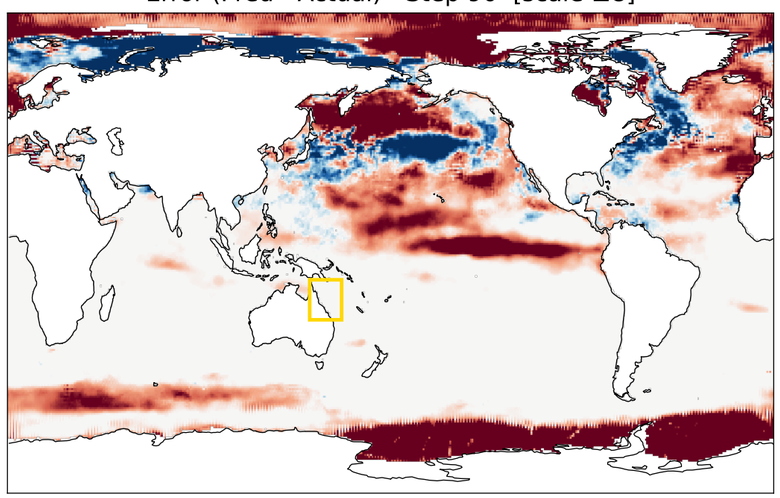
Actual DHW - Step 90

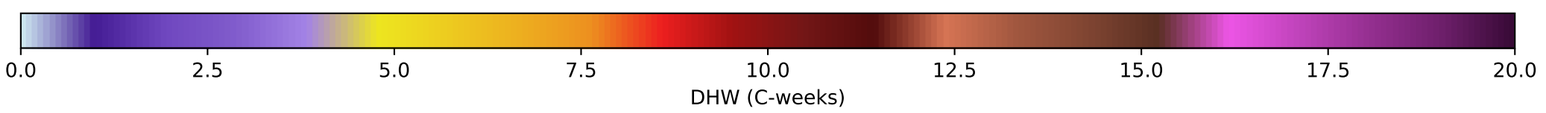


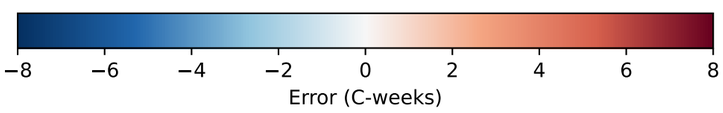
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

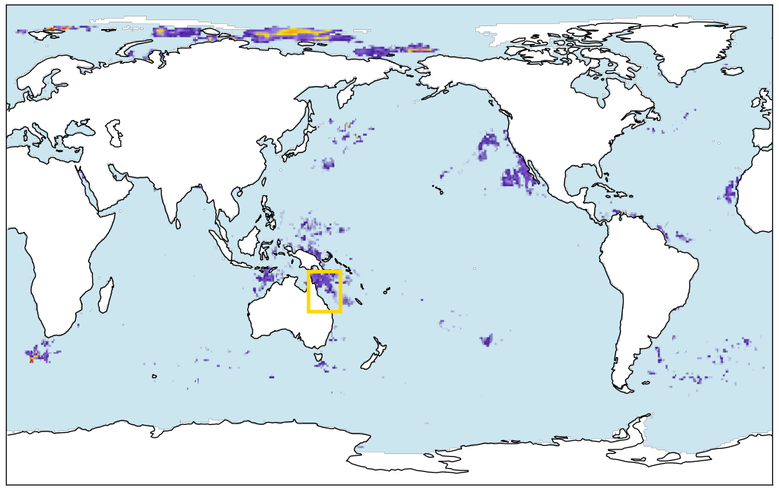




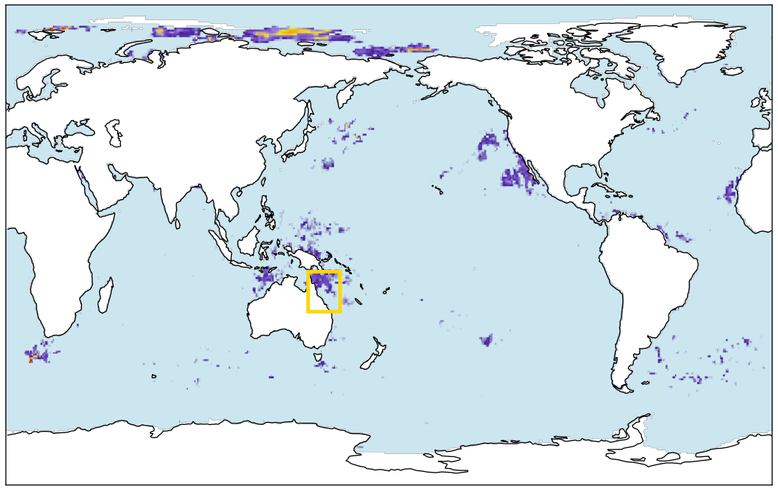


DHW Chain init: 2021-01-06 - Test (C-weeks)

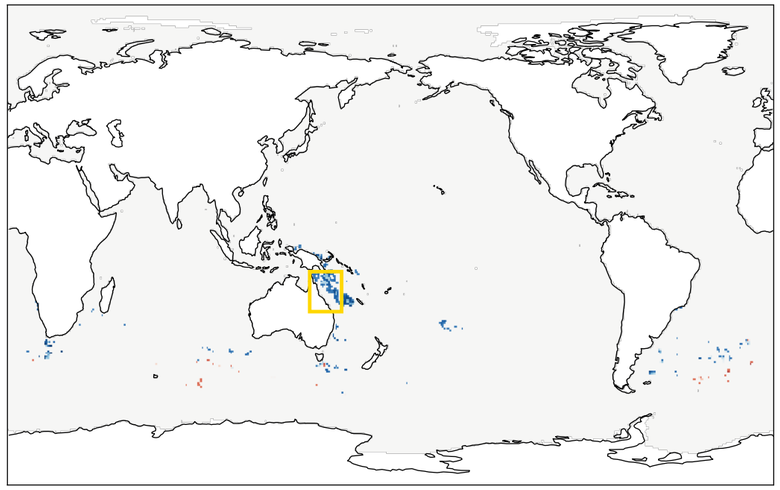
Actual DHW - Step 1



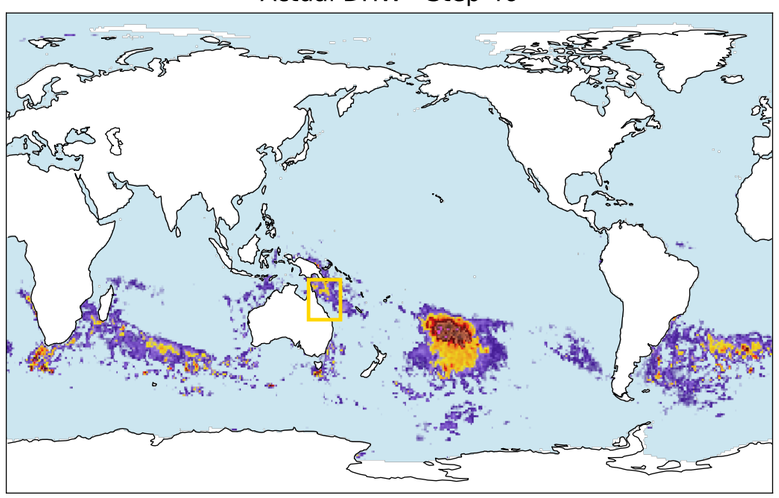
Predicted DHW - Step 1



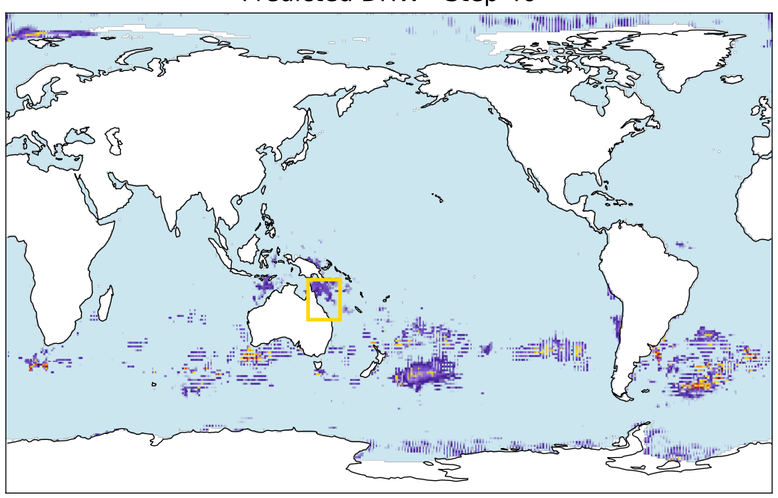
Error (Pred - Actual) - Step 1 [scale ±0.26]



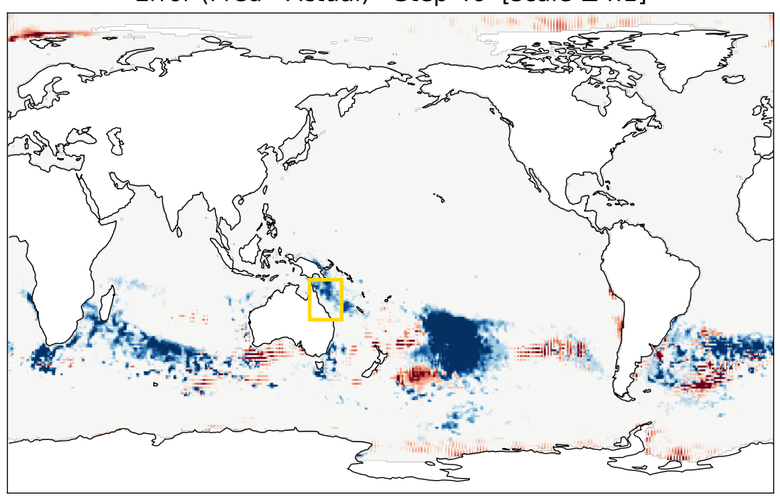
Actual DHW - Step 46



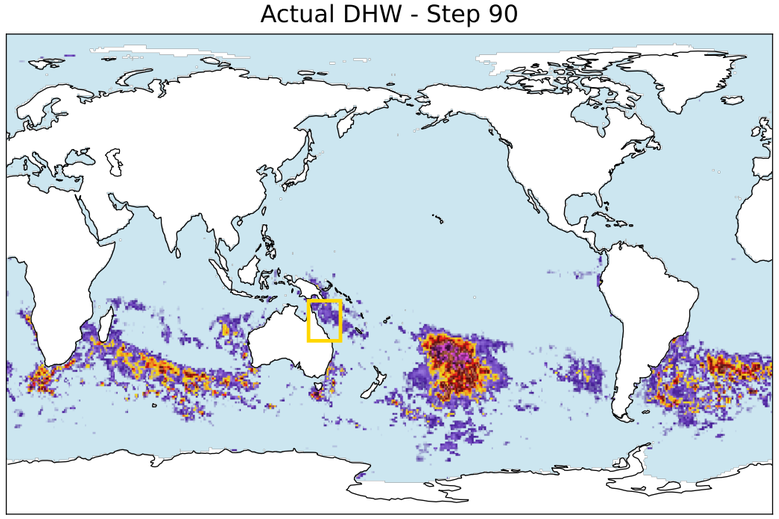
Predicted DHW - Step 46



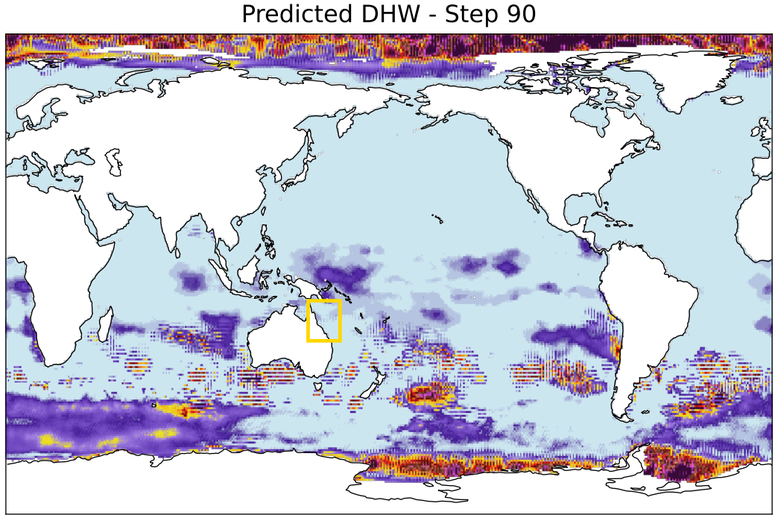
Error (Pred - Actual) - Step 46 [scale ±4.1]



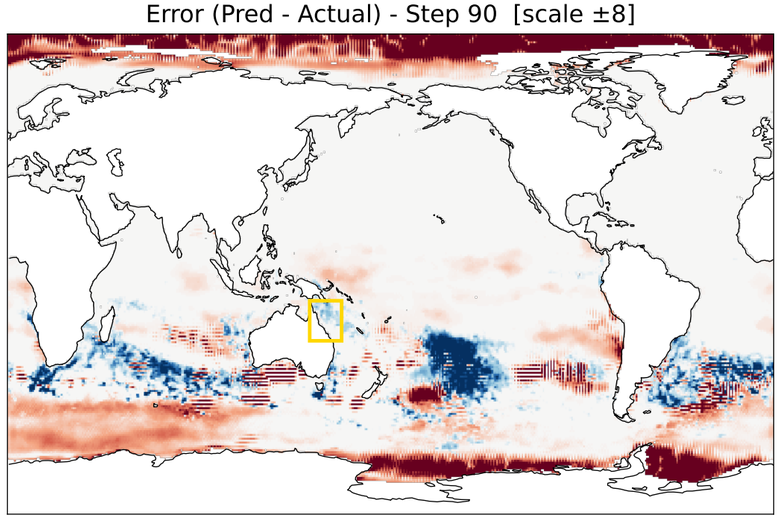
Actual DHW - Step 90

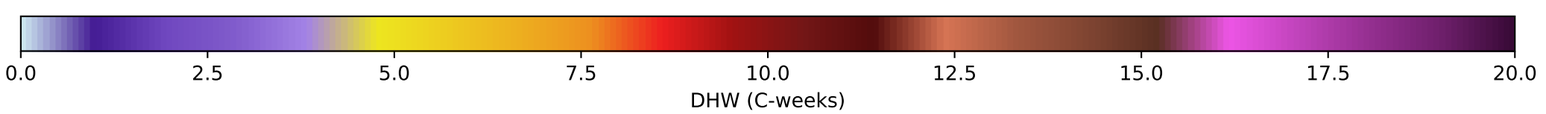


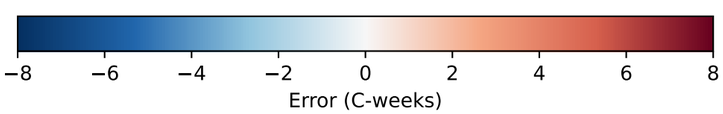
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

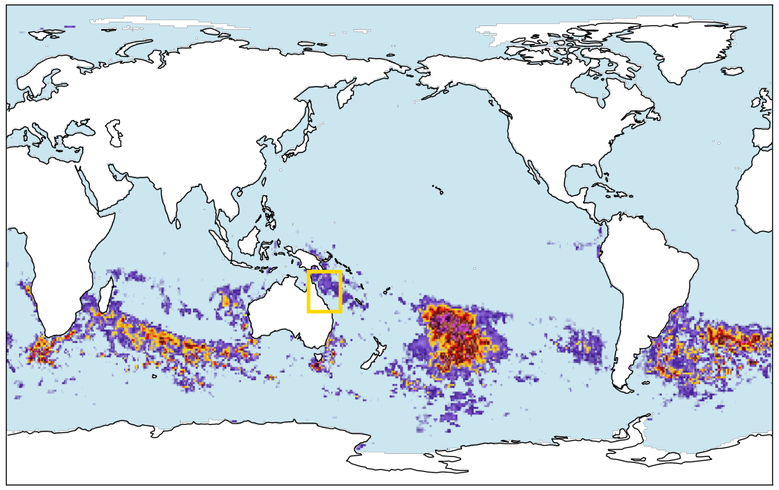




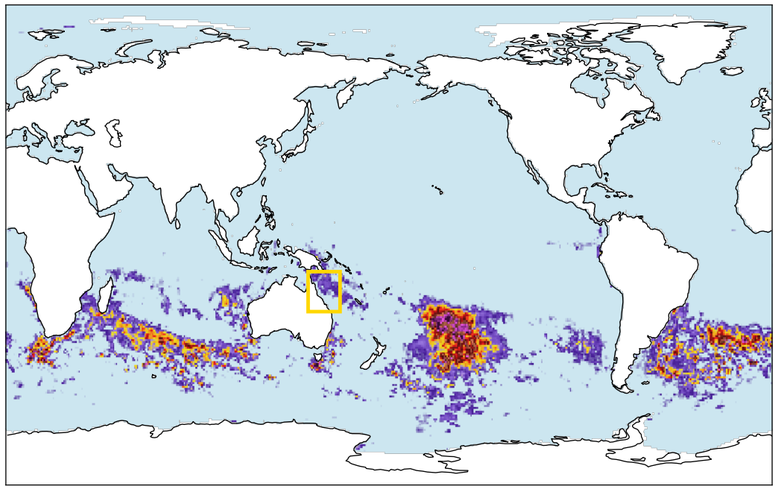


DHW Chain init: 2021-04-06 - Test (C-weeks)

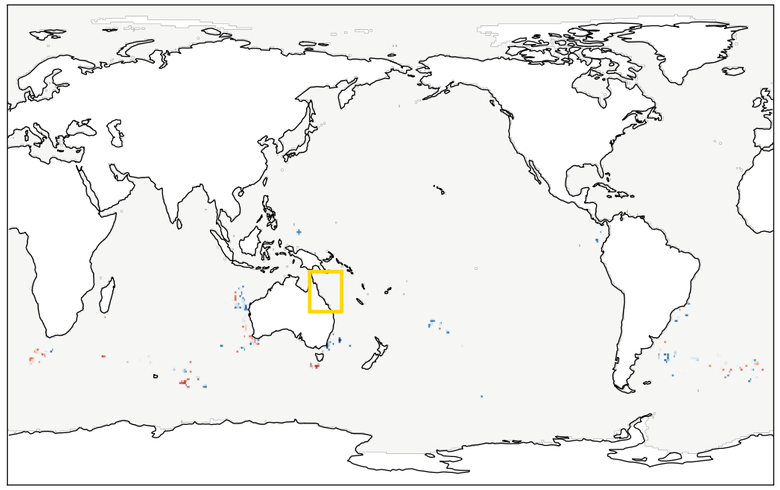
Actual DHW - Step 1



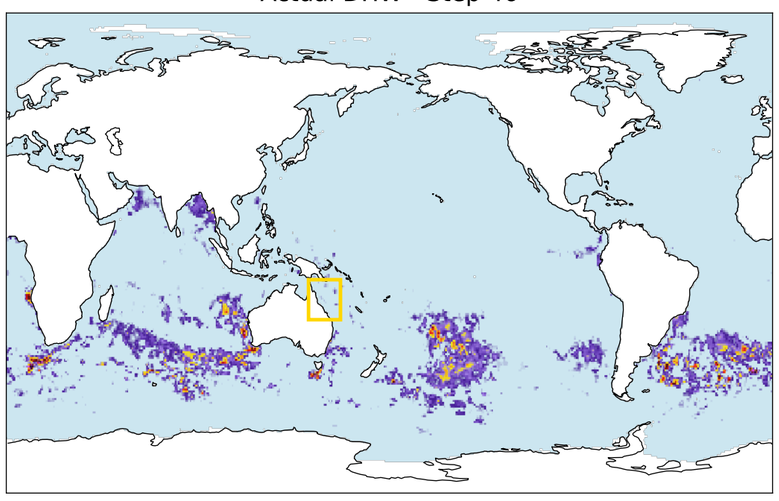
Predicted DHW - Step 1



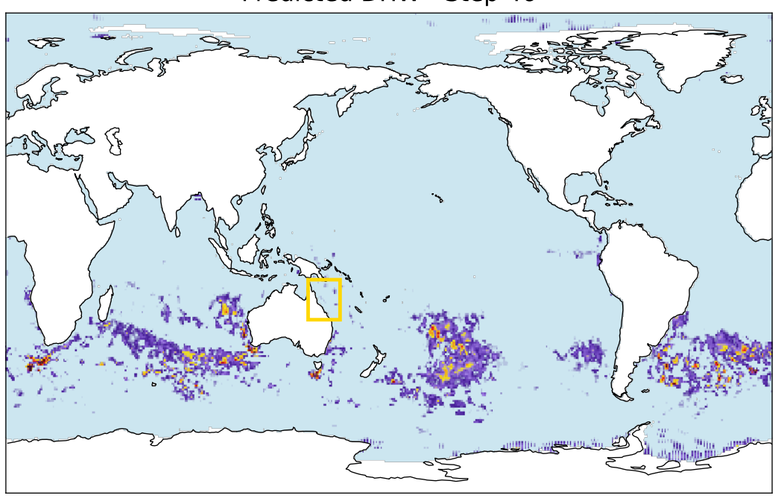
Error (Pred - Actual) - Step 1 [scale ±0.29]



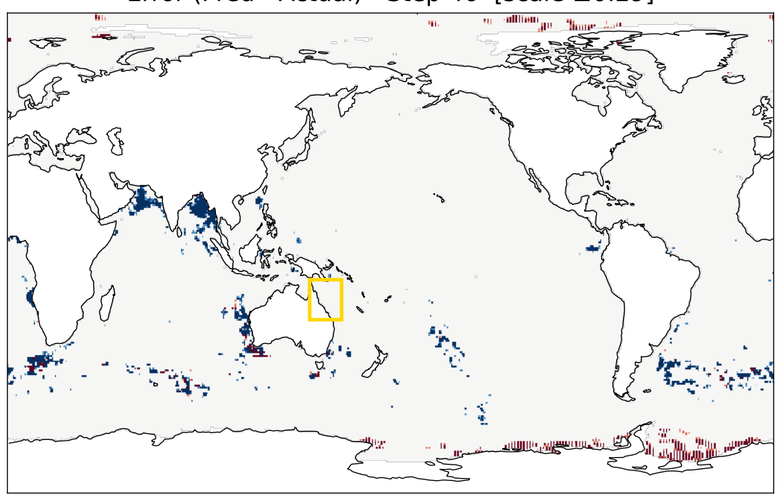
Actual DHW - Step 46



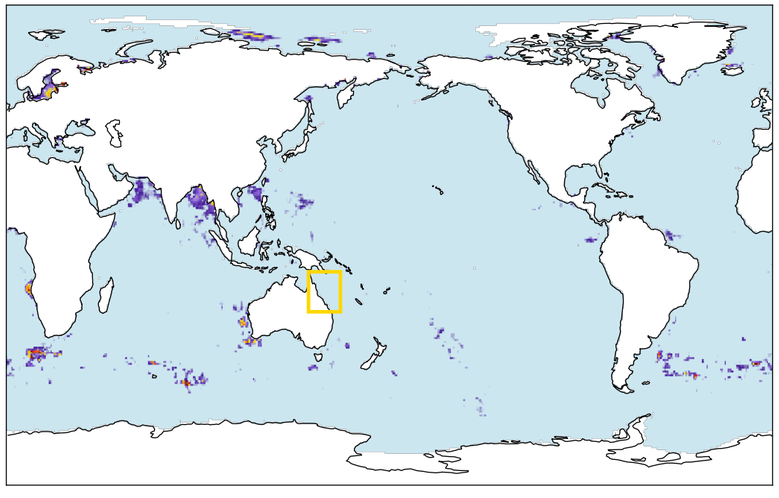
Predicted DHW - Step 46



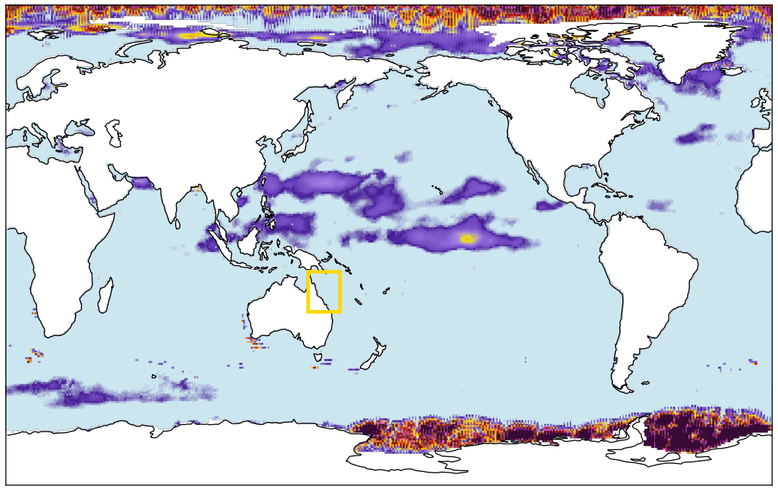
Error (Pred - Actual) - Step 46 [scale ±0.29]



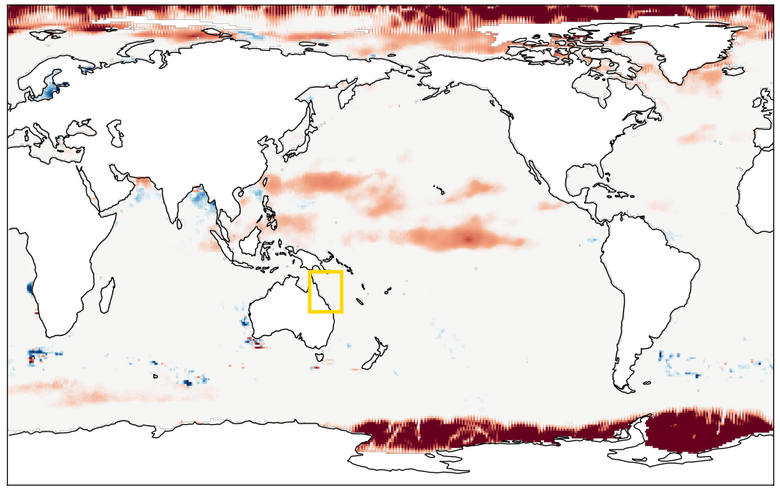
Actual DHW - Step 90

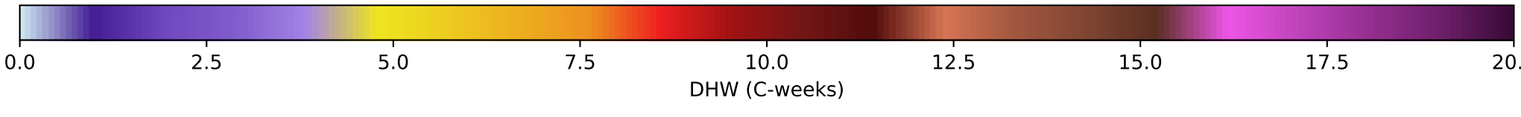


Predicted DHW - Step 90



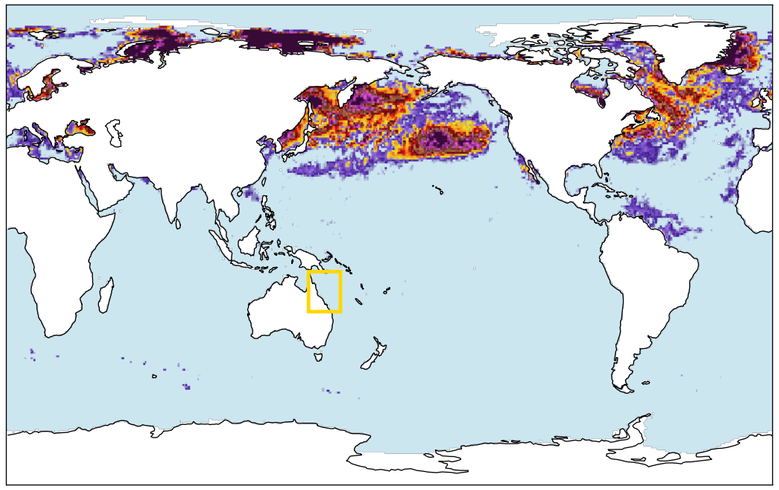
Error (Pred - Actual) - Step 90 [scale ±8]



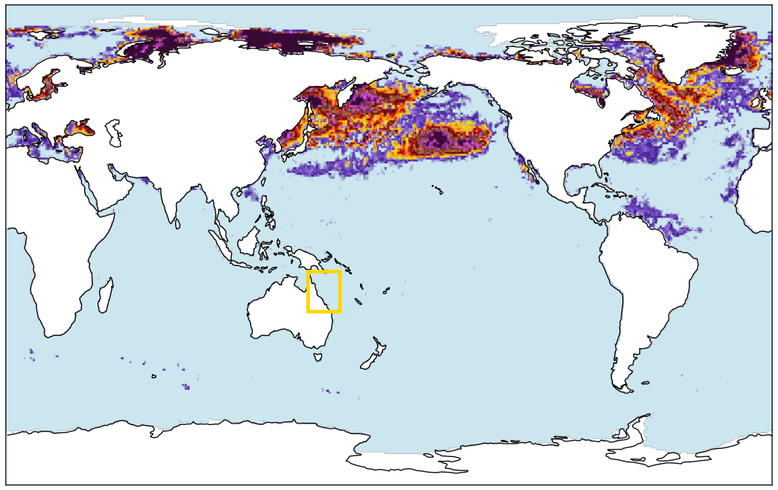


DHW Chain init: 2021-10-03 - Test (C-weeks)

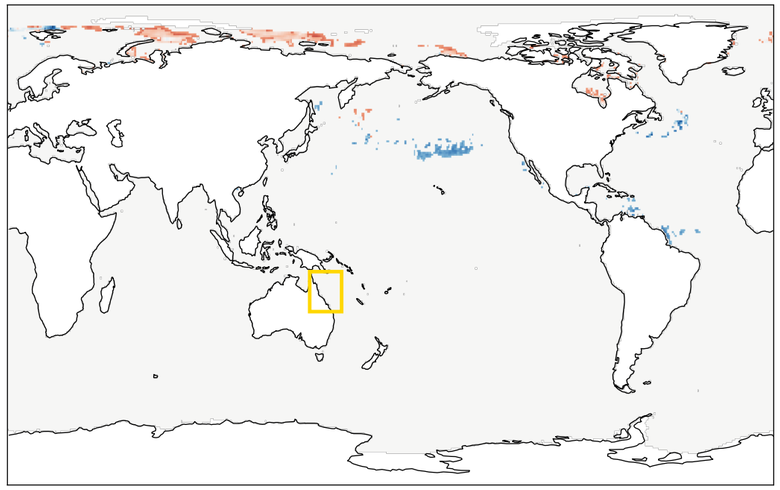
Actual DHW - Step 1



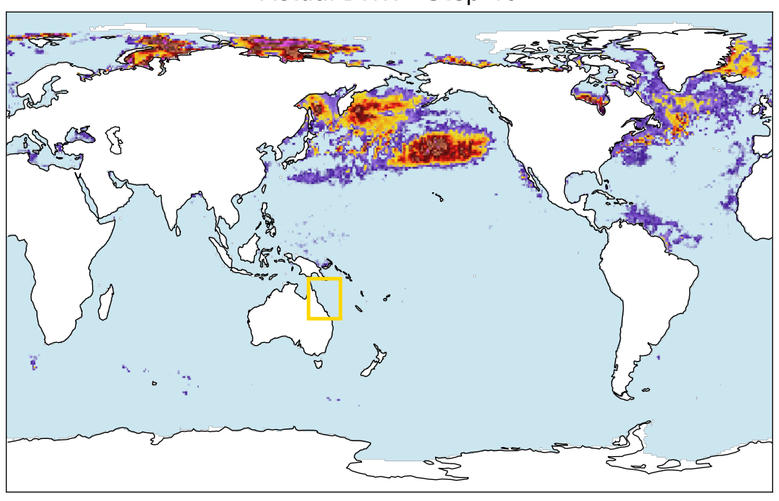
Predicted DHW - Step 1



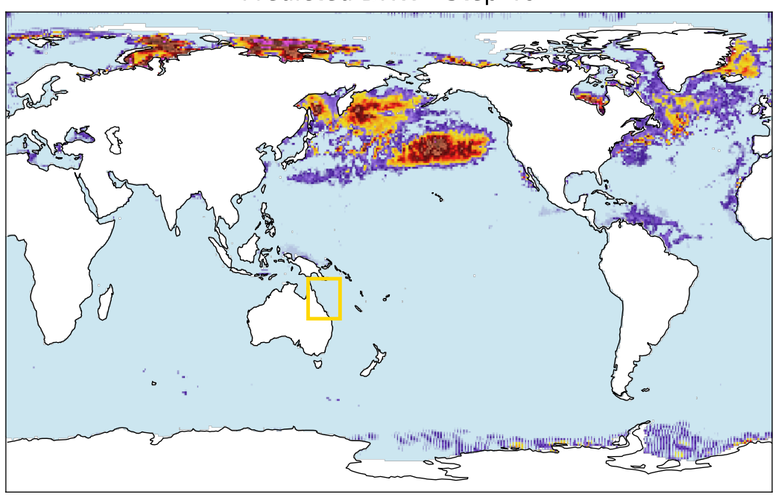
Error (Pred - Actual) - Step 1 [scale ±0.38]



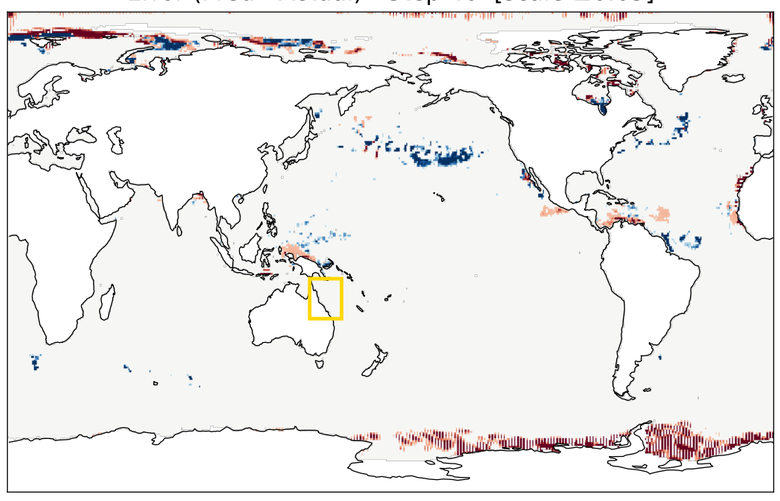
Actual DHW - Step 46



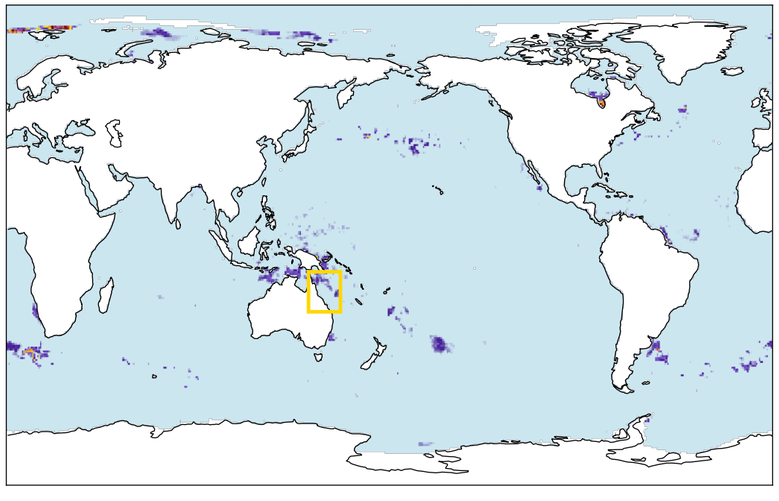
Predicted DHW - Step 46



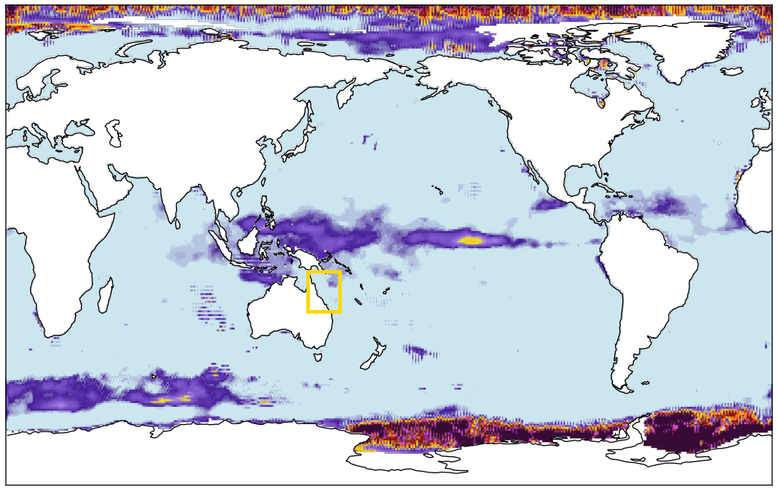
Error (Pred - Actual) - Step 46 [scale ±0.63]



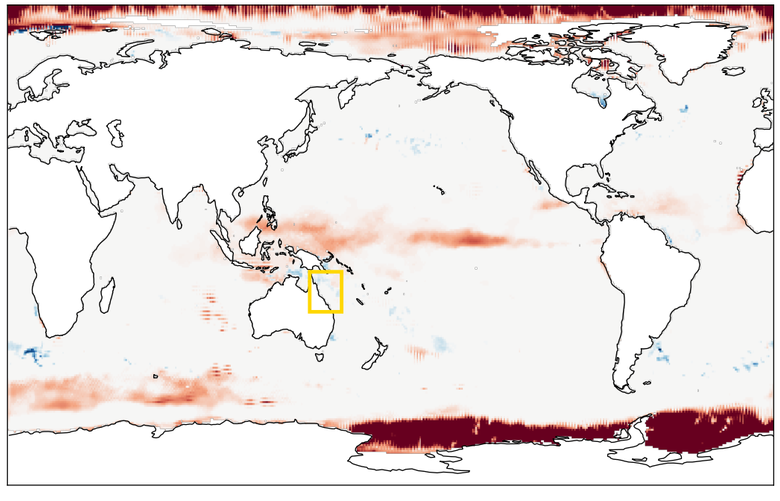
Actual DHW - Step 90

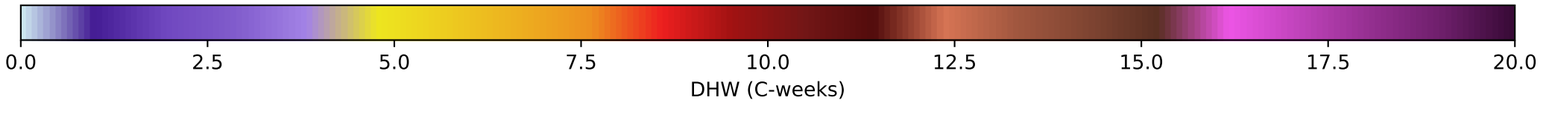


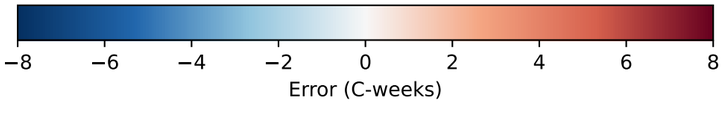
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

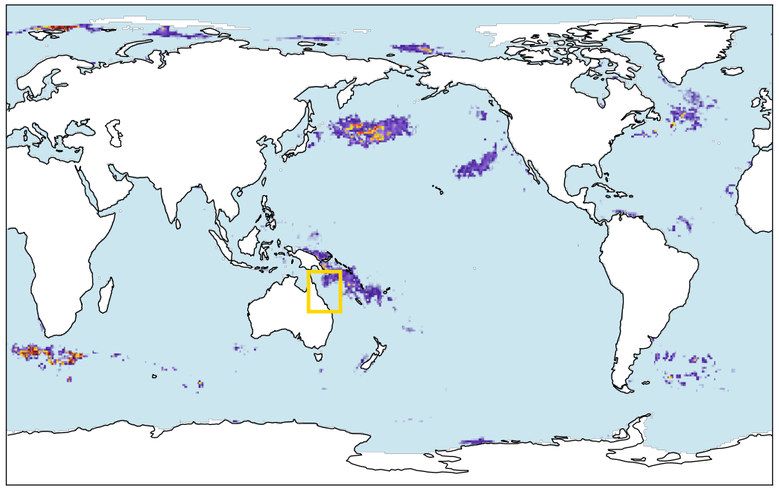




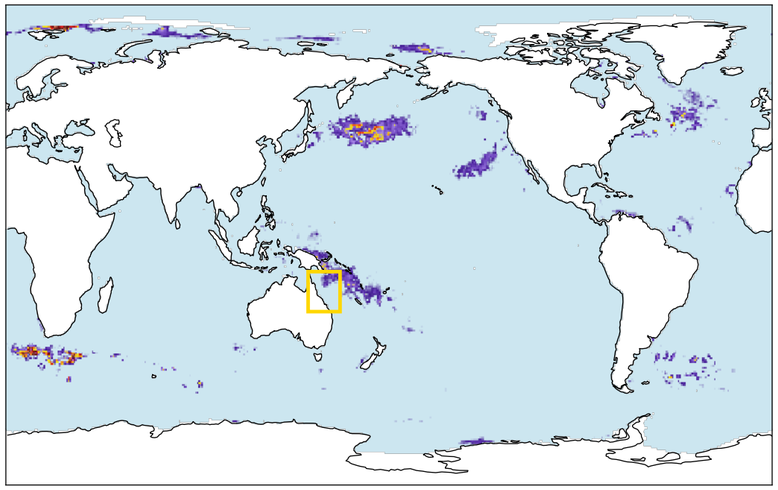


DHW Chain init: 2022-12-27 - Test (C-weeks)

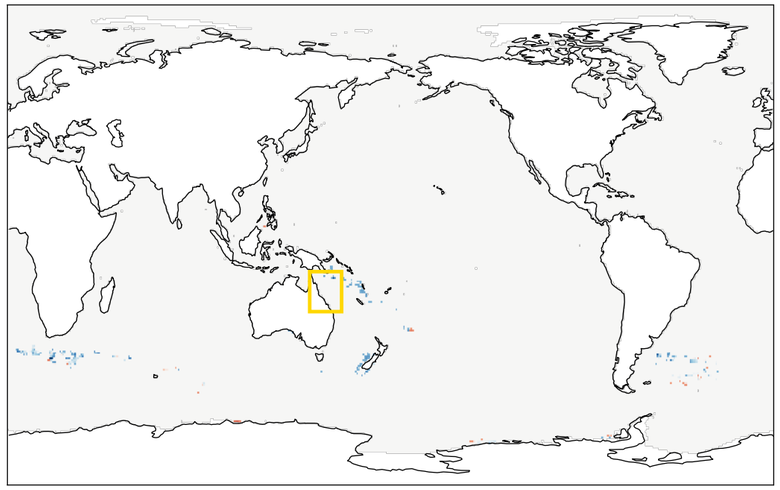
Actual DHW - Step 1



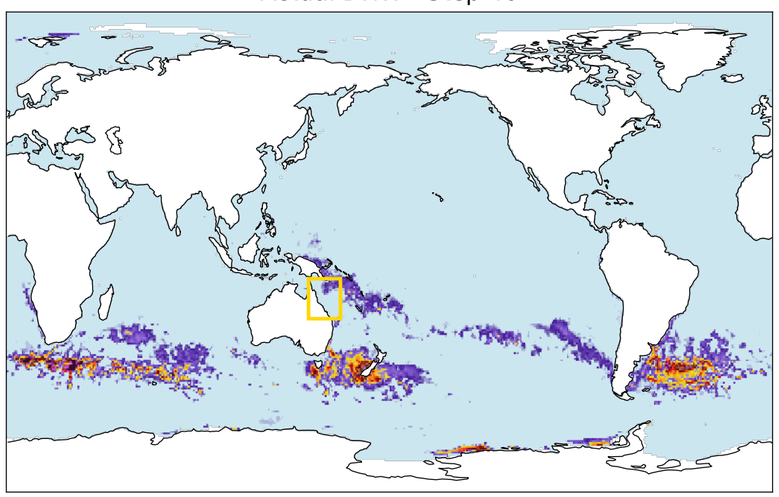
Predicted DHW - Step 1



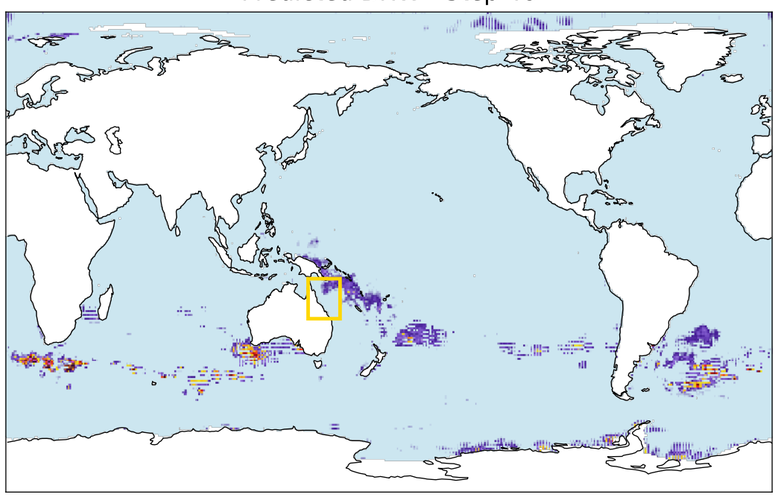
Error (Pred - Actual) - Step 1 [scale ±0.37]



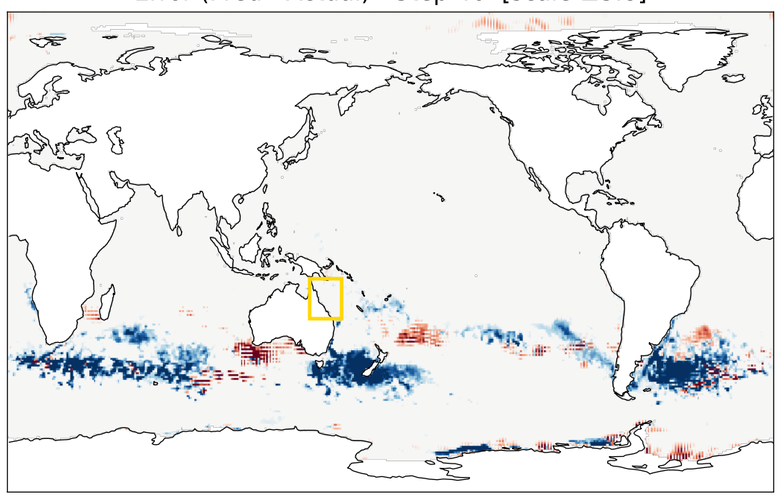
Actual DHW - Step 46



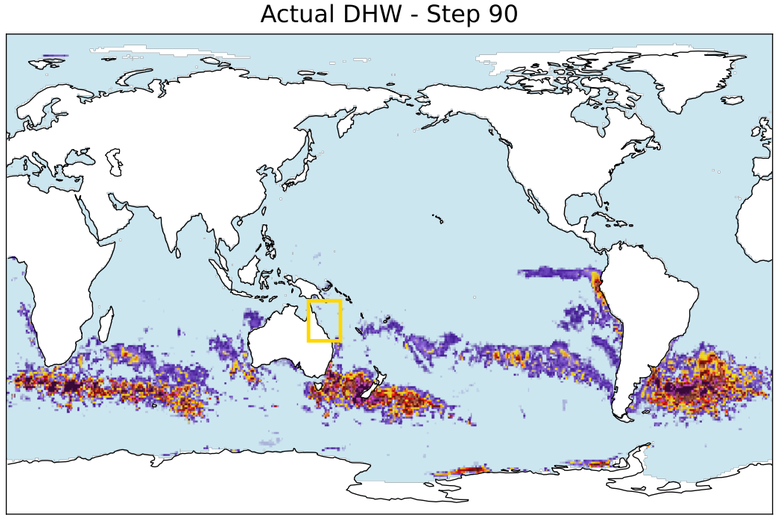
Predicted DHW - Step 46



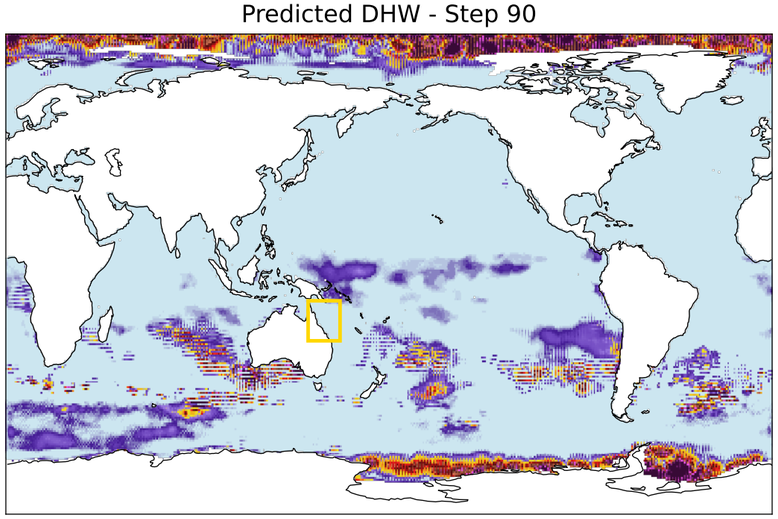
Error (Pred - Actual) - Step 46 [scale ±3.8]



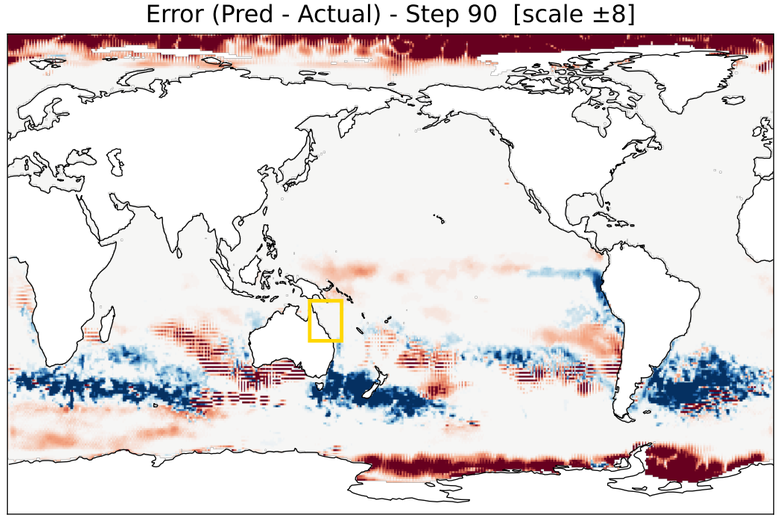
Actual DHW - Step 90

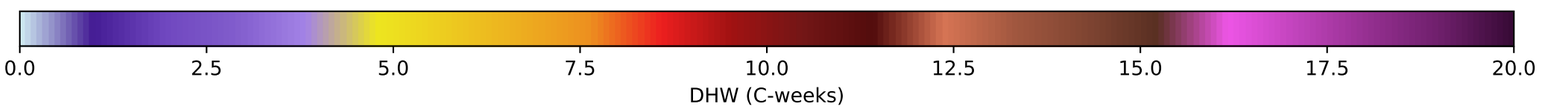


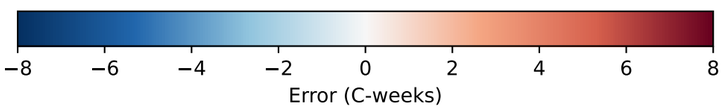
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

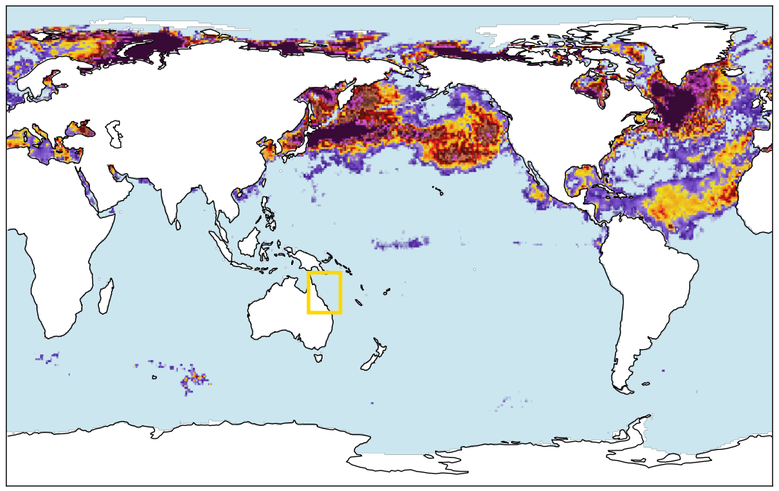




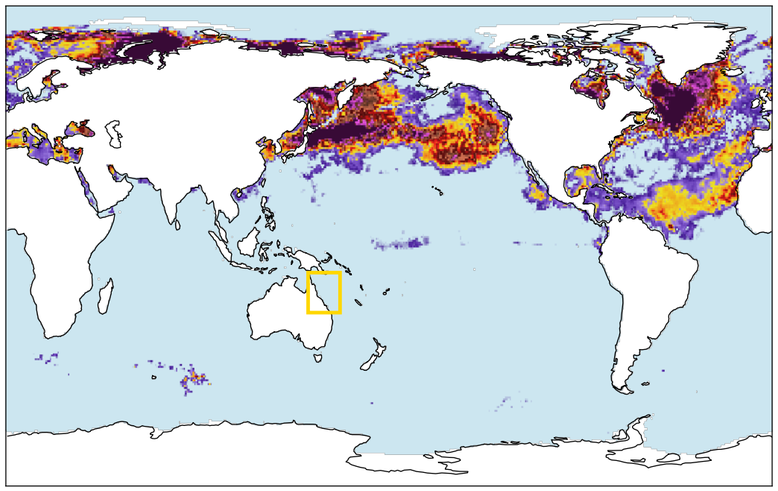


DHW Chain init: 2023-09-23 - Test (C-weeks)

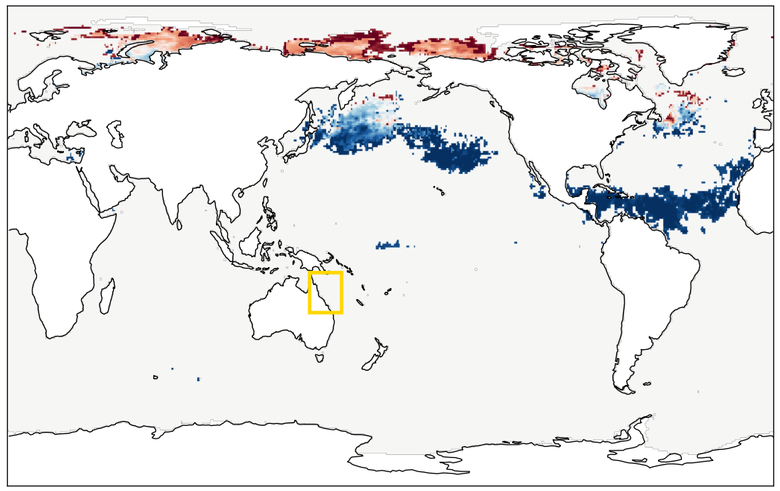
Actual DHW - Step 1



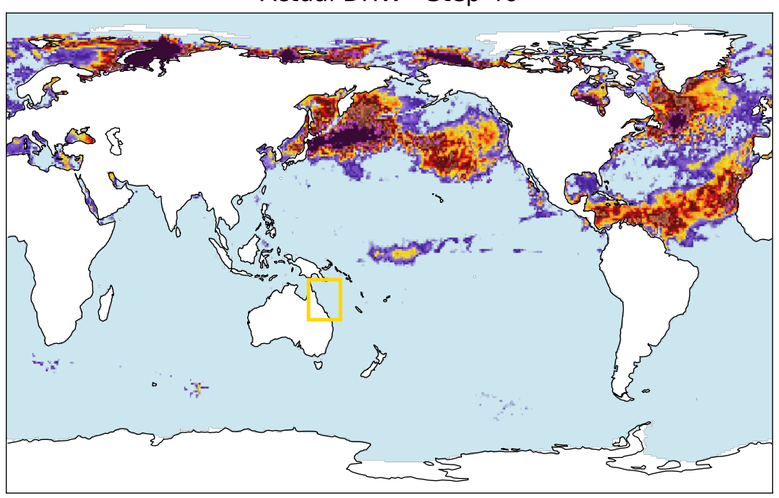
Predicted DHW - Step 1



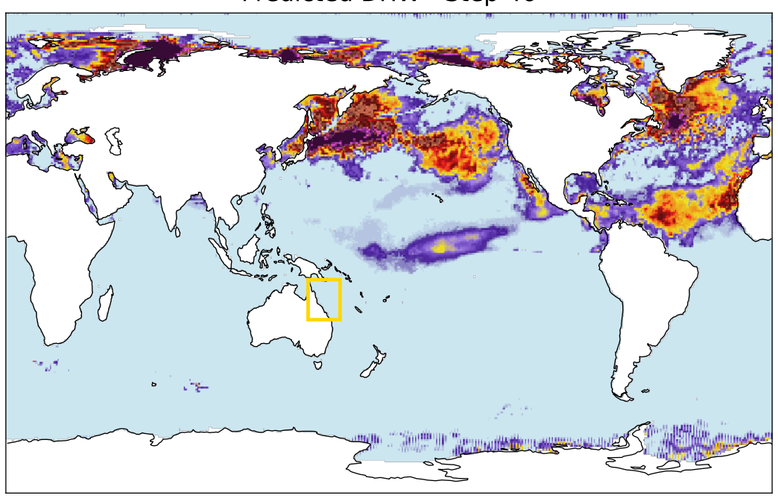
Error (Pred - Actual) - Step 1 [scale ±0.18]



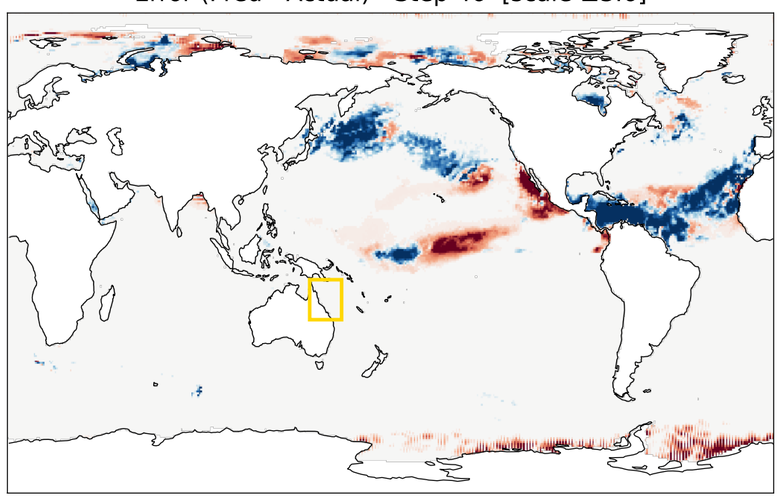
Actual DHW - Step 46



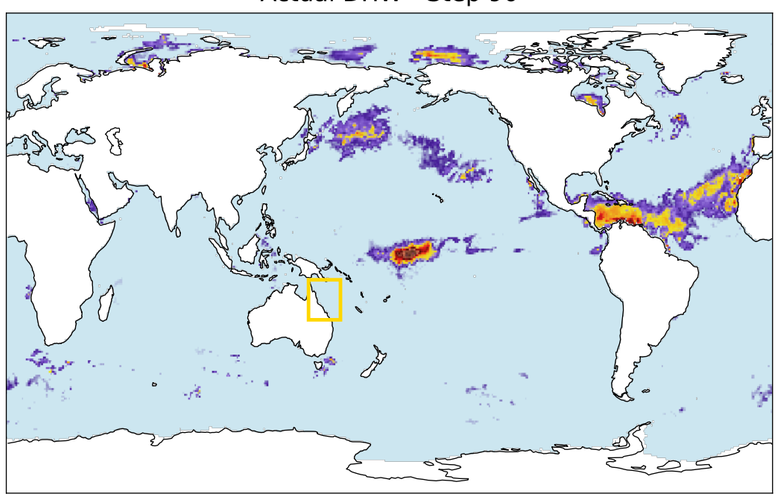
Predicted DHW - Step 46



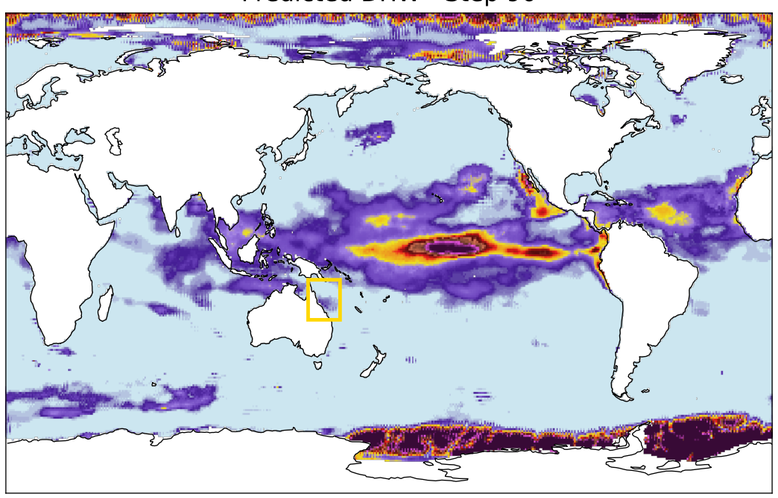
Error (Pred - Actual) - Step 46 [scale ±3.6]



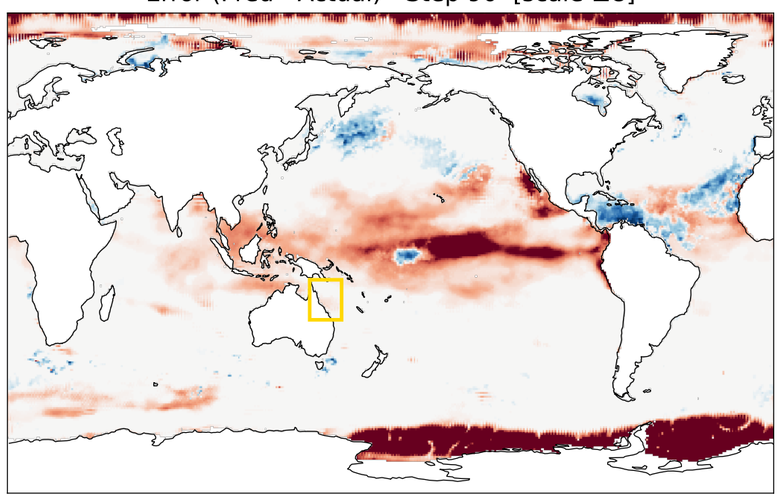
Actual DHW - Step 90

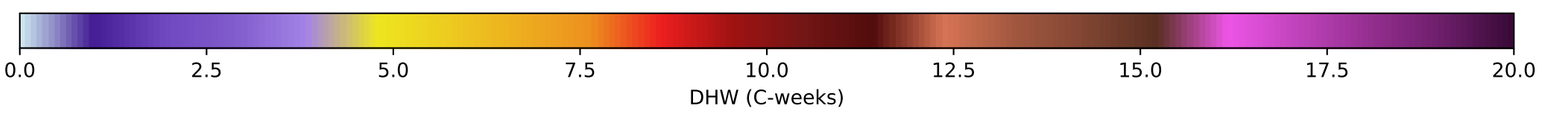


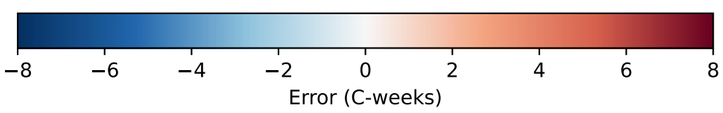
Predicted DHW - Step 90



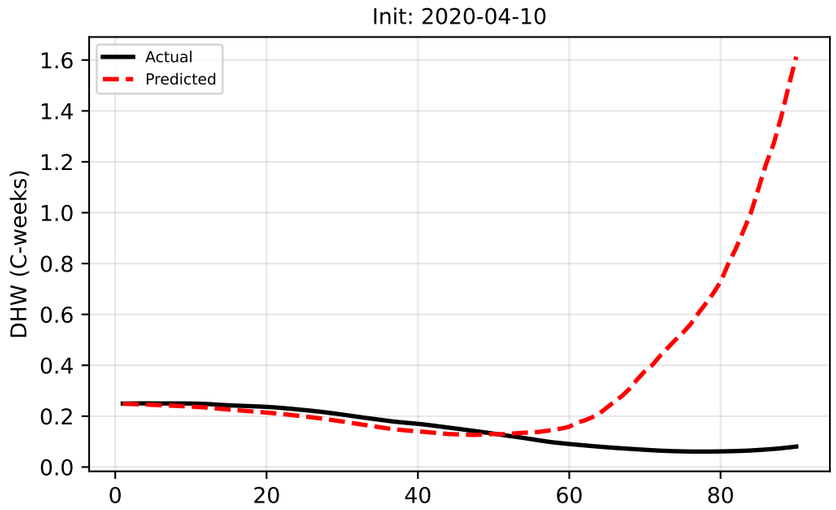
Error (Pred - Actual) - Step 90 [scale ±8]

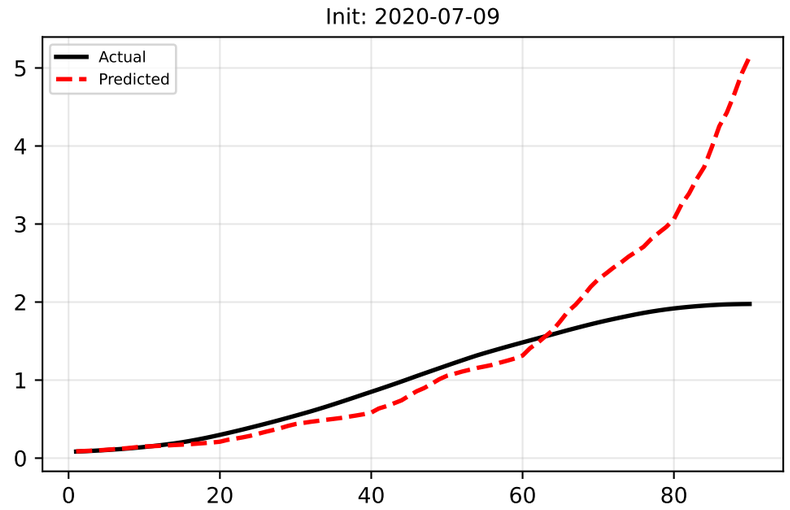


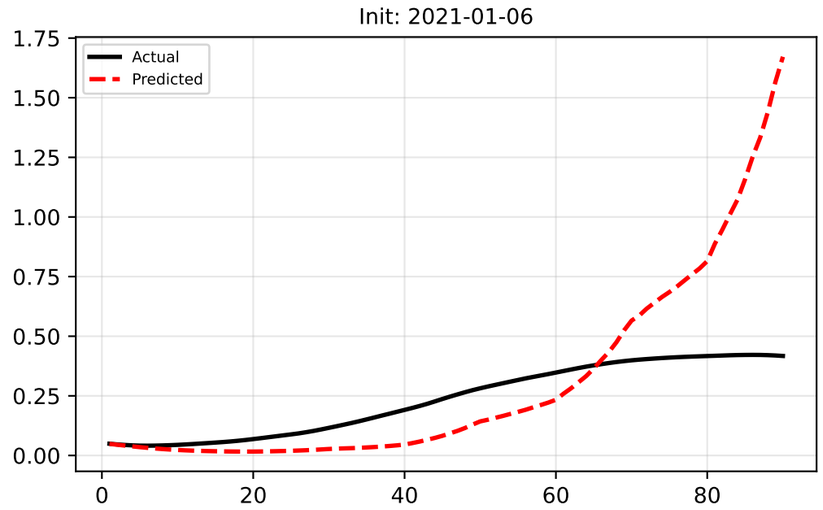


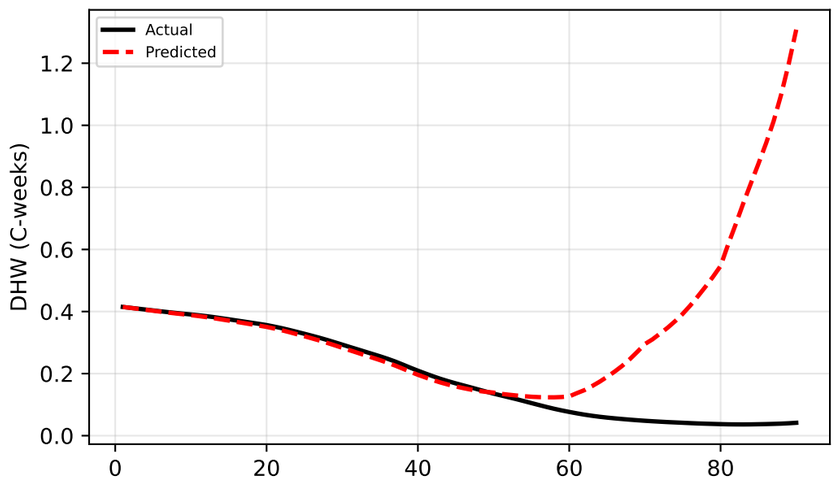


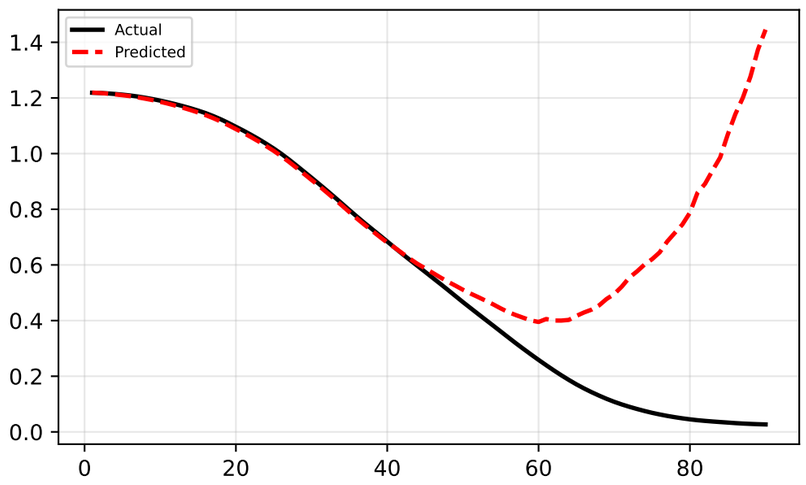
Global-Mean DHW - Test (C-weeks)

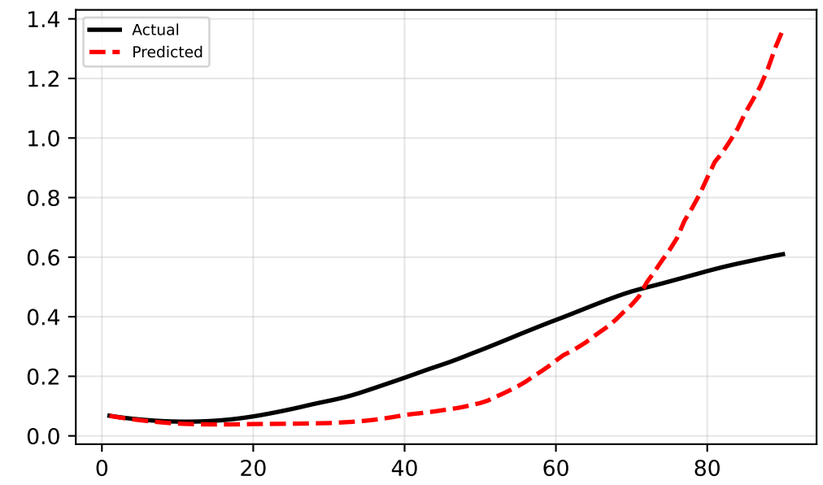


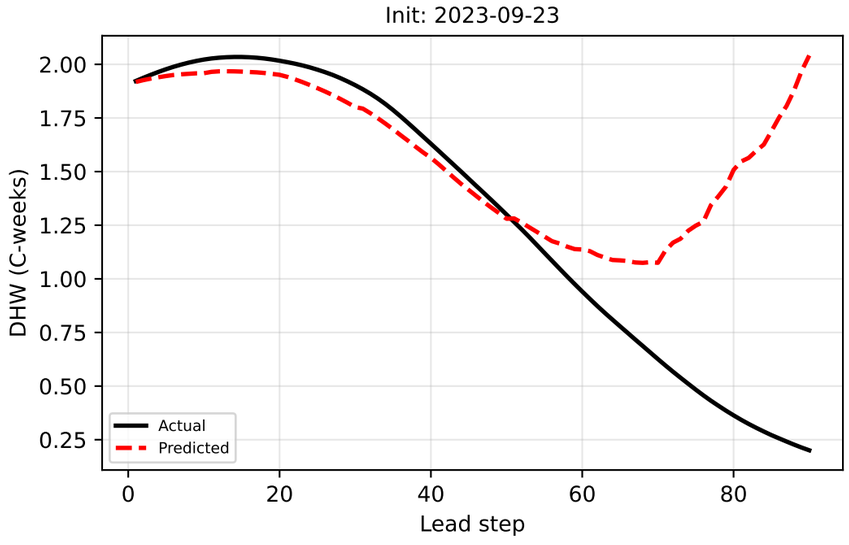




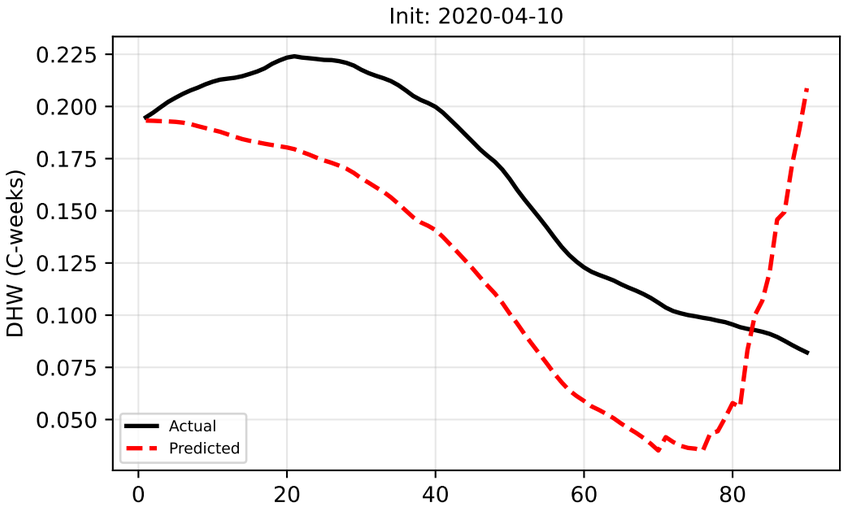


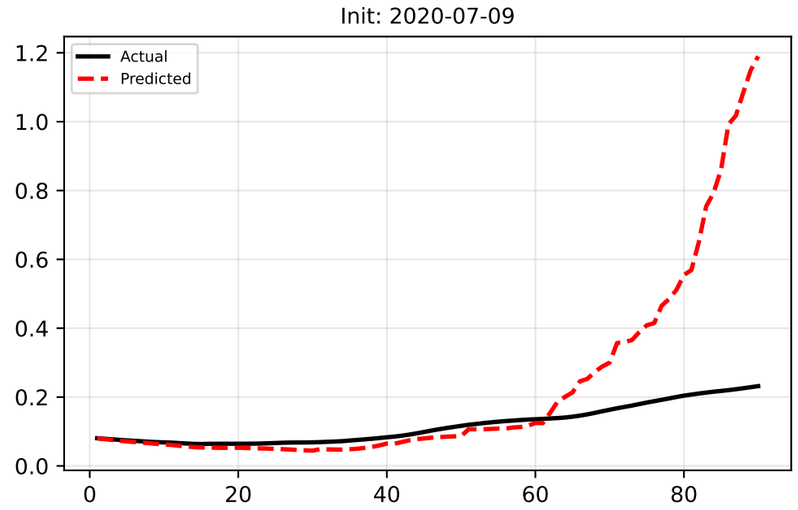


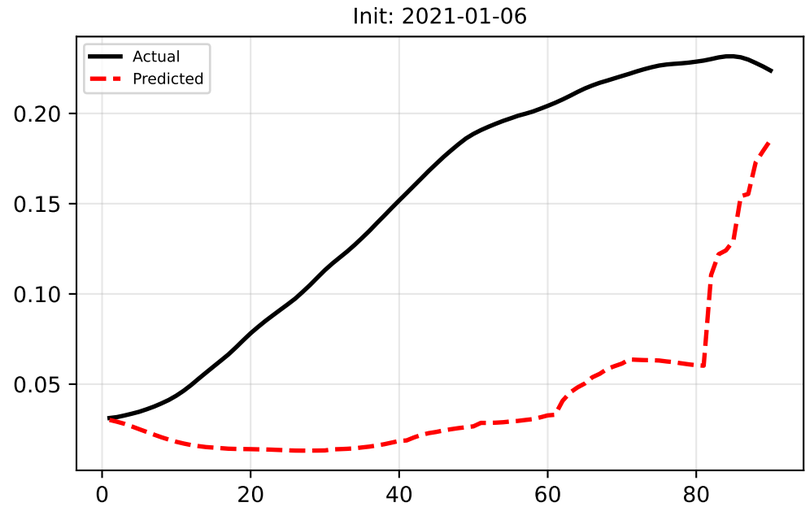


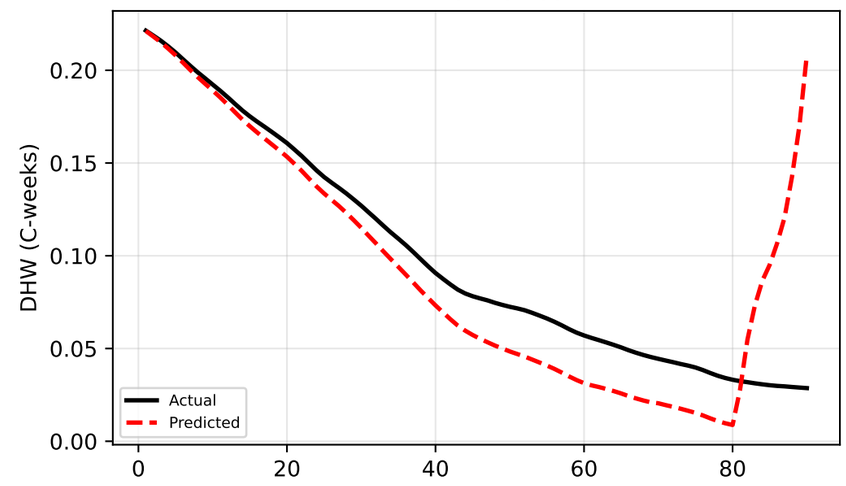


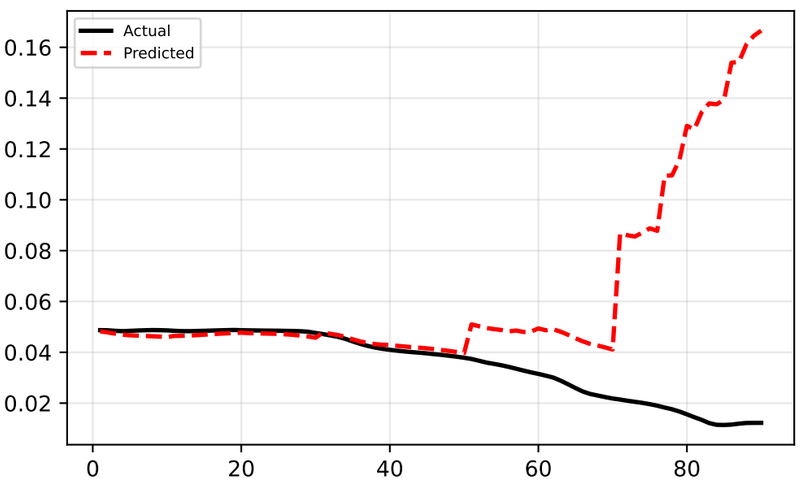
Tropical-Mean DHW (|lat|<=30) - Test (C-weeks)

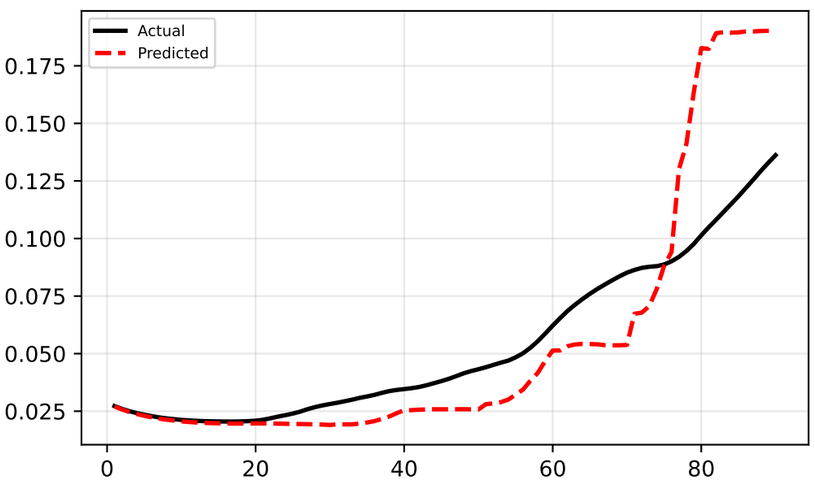


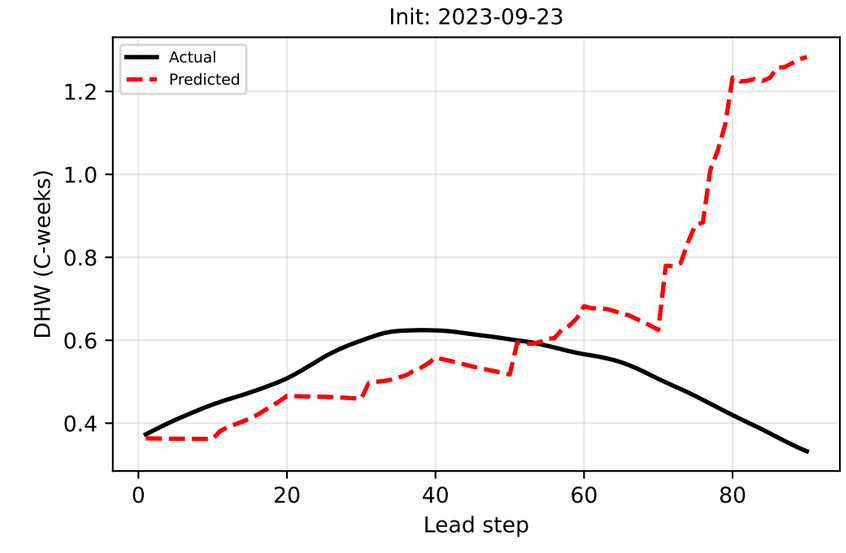


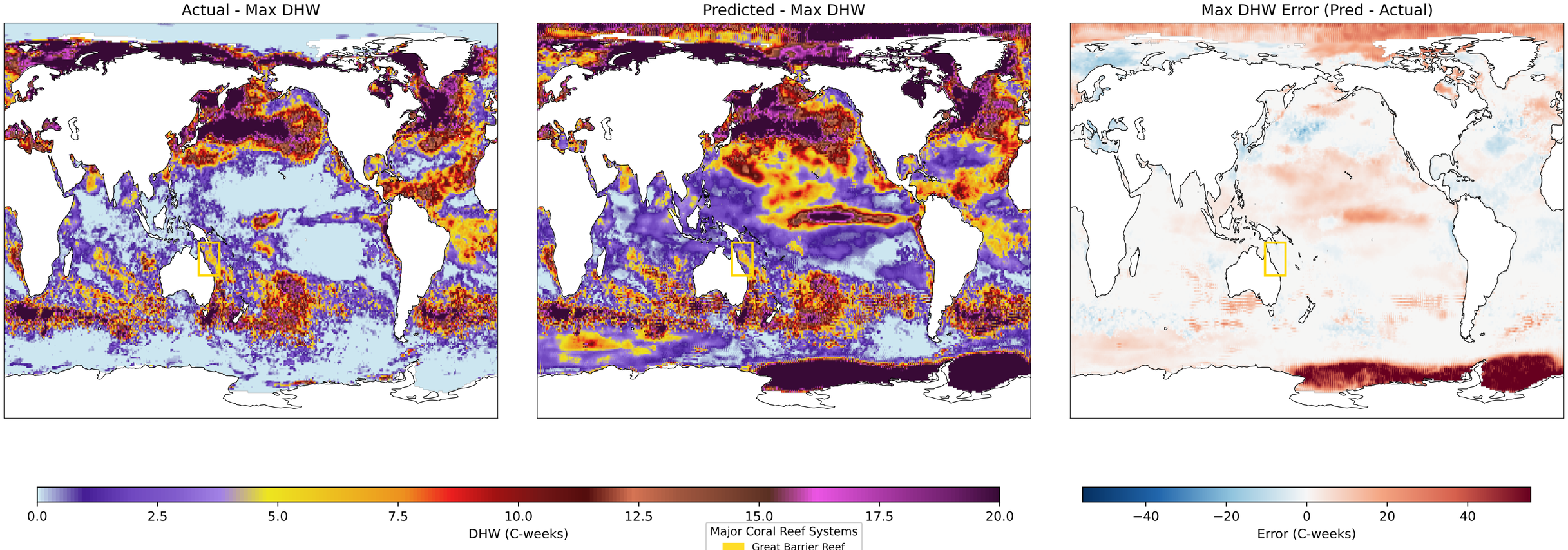


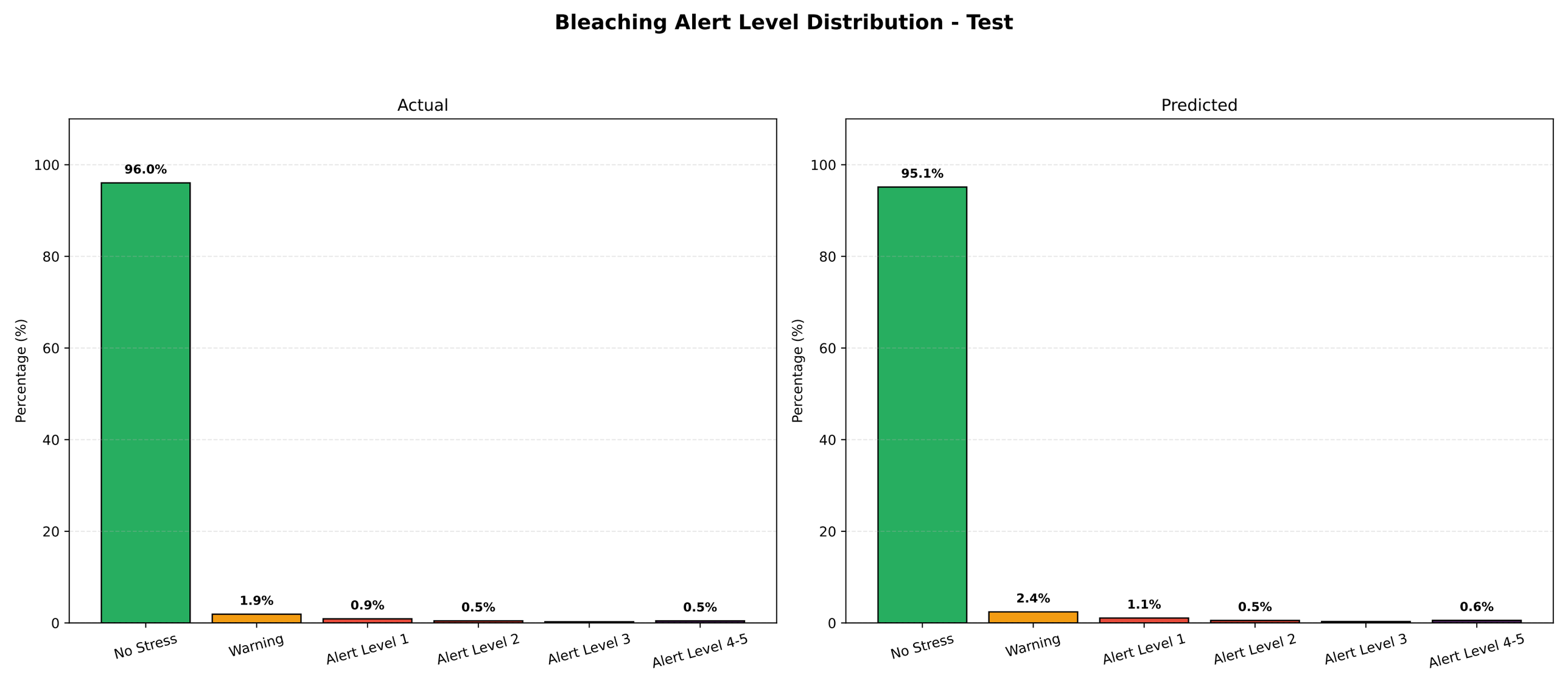






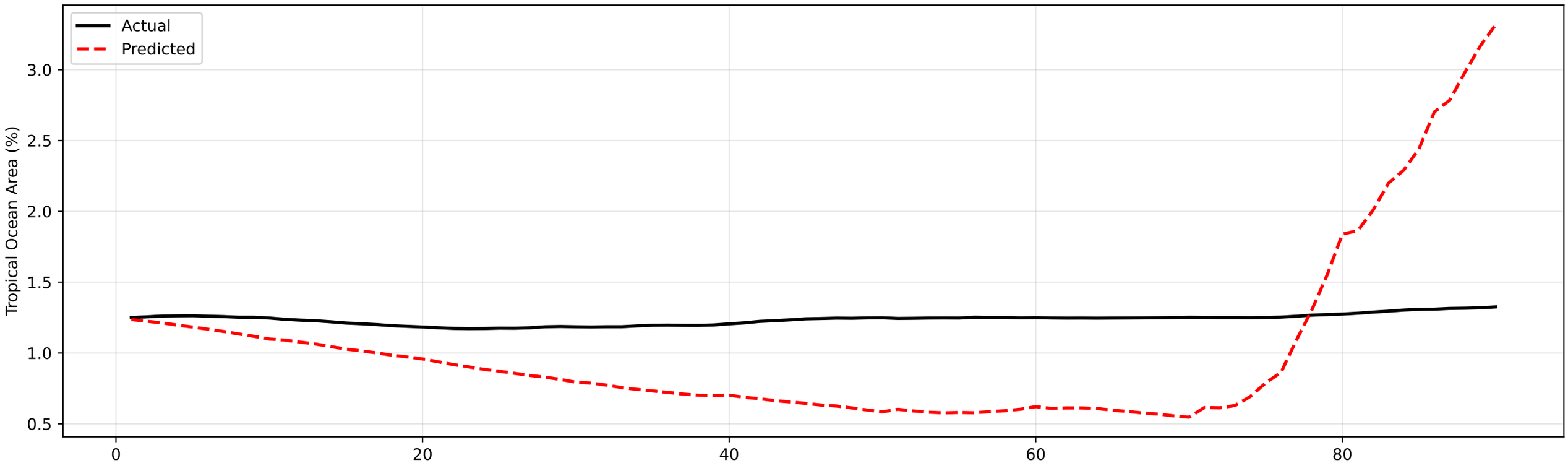



Peak DHW (max over all chains & leads) - Test (C-weeks)
Actual - Max DHW
Predicted - Max DHW
Max DHW Error (Pred - Actual)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
DHW (C-weeks)
Major Coral Reef Systems
Great Barrier Reef
−40
−20
0
20
40
Error (C-weeks)

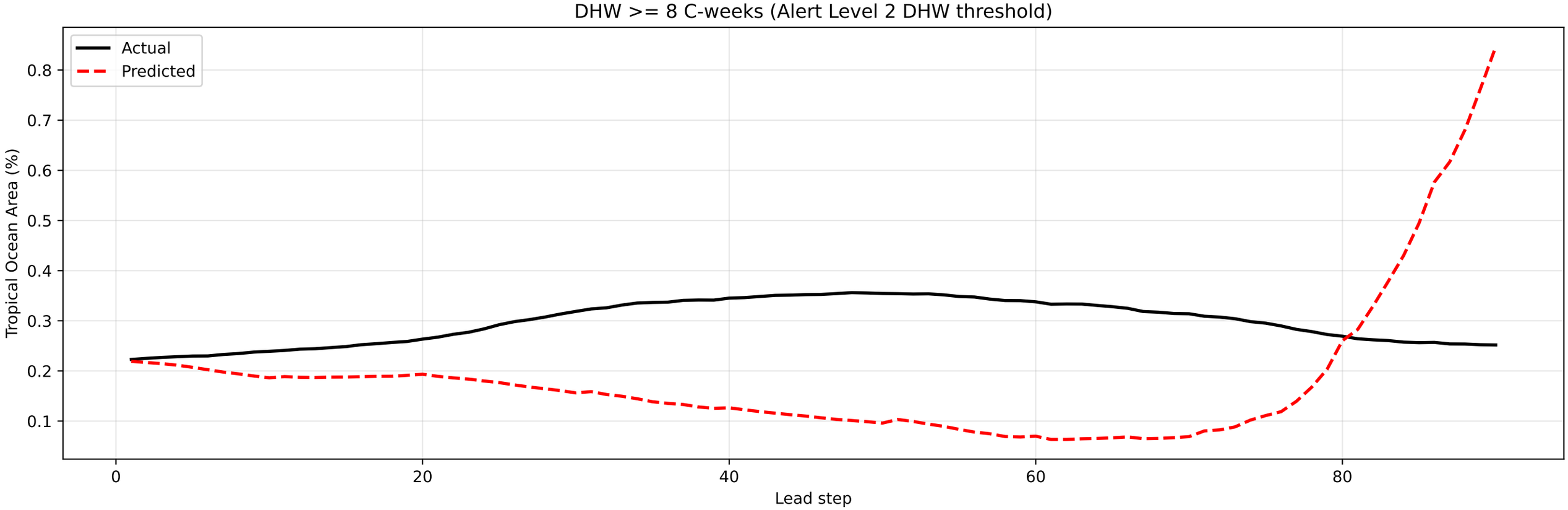

Bleaching Alert Level Distribution - Test
Actual
Predicted
Percentage (%)
0
20
40
60
80
100
96.0%
1.9%
0.9%
0.5%
0.5%
95.1%
2.4%
1.1%
0.5%
0.6%
No Stress
Warning
Alert Level 1
Alert Level 2
Alert Level 3
Alert Level 4-5

Tropical Ocean Area Under Bleaching Stress - Test

DHW >= 4 C-weeks (Alert Level 1 DHW threshold)



DHW >= 8 C-weeks (Alert Level 2 DHW threshold)



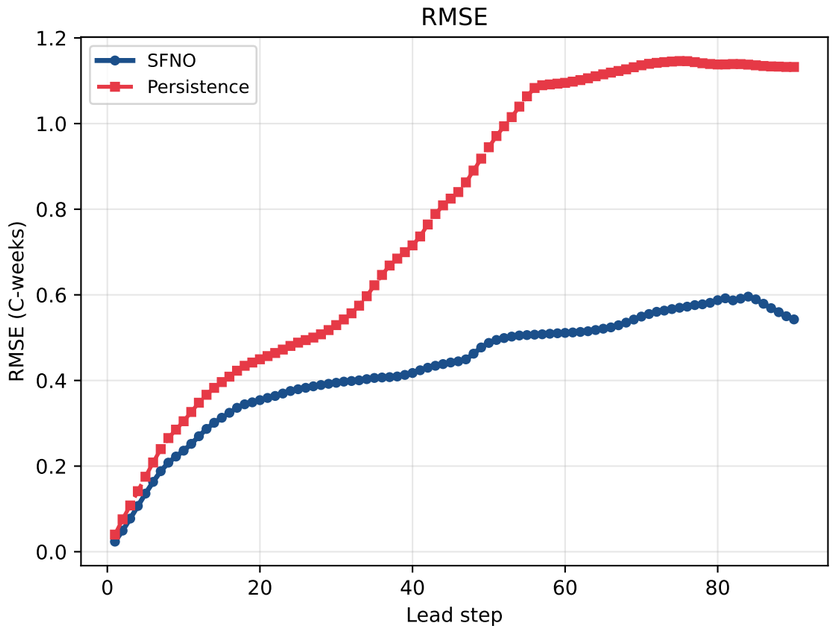
Great Barrier Reef Analysis - Test

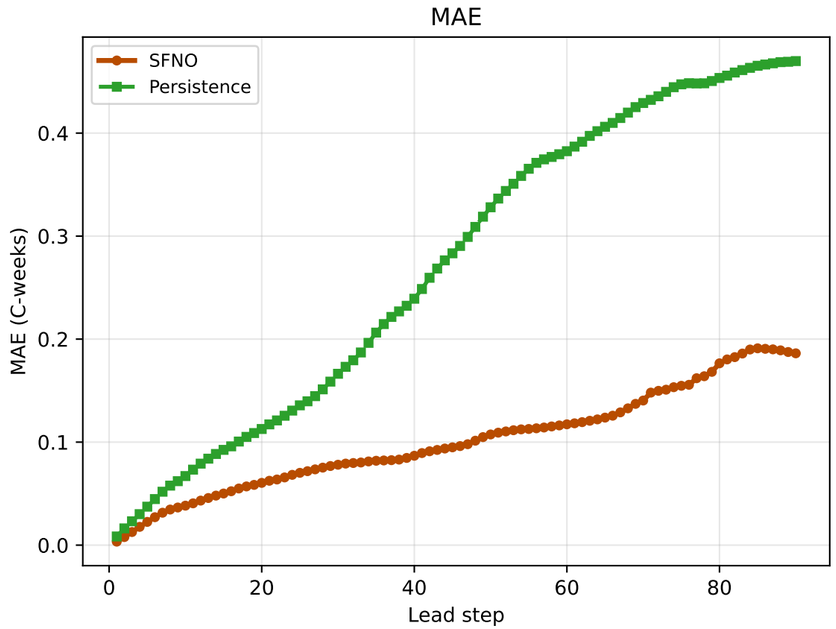
```
                    GREAT BARRIER REEF - TEST
Lat -25.0 deg to -10.0 deg  |  Lon 142.0 deg to 157.0 deg  (0-360)
                    Grid: 21 lat x 21 lon points
```

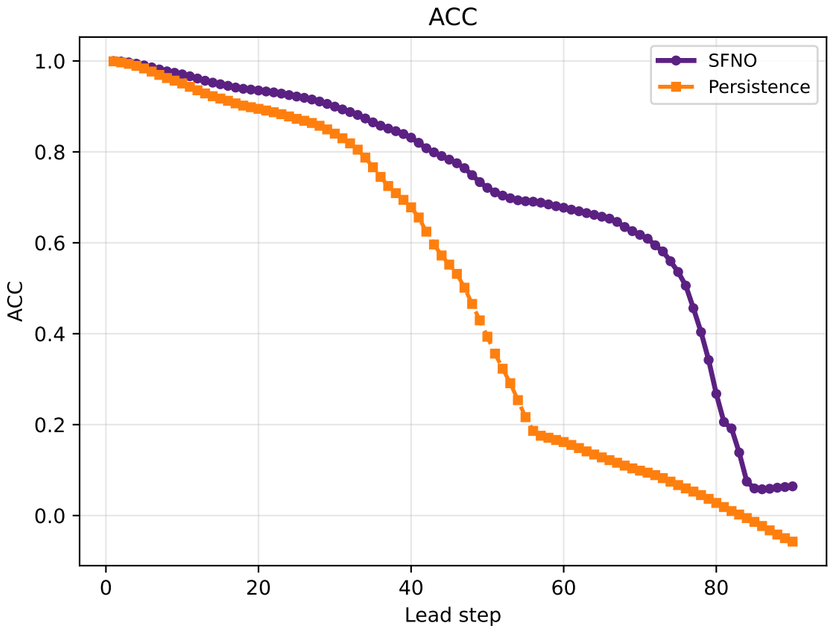
DHW Metrics by Lead Step - Test - GBR (18 chains scored at every lead)

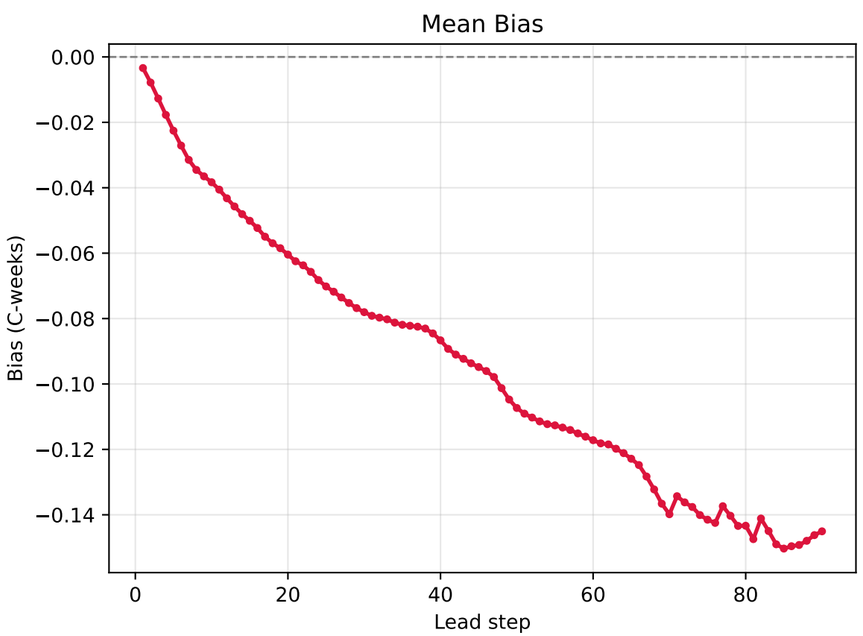
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	Persist MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0236	0.0400	41.1%	0.0034	0.0084	-0.0034	0.9997	0.9991	0.9994
2	0.0493	0.0757	34.9%	0.0078	0.0163	-0.0078	0.9987	0.9969	0.9974
3	0.0774	0.1080	28.3%	0.0127	0.0232	-0.0127	0.9968	0.9937	0.9937
4	0.1068	0.1414	24.5%	0.0177	0.0301	-0.0177	0.9939	0.9892	0.9882
5	0.1358	0.1751	22.5%	0.0226	0.0373	-0.0226	0.9902	0.9835	0.9810
6	0.1631	0.2084	21.7%	0.0271	0.0447	-0.0271	0.9858	0.9767	0.9728
7	0.1882	0.2397	21.5%	0.0315	0.0518	-0.0315	0.9812	0.9693	0.9640
8	0.2081	0.2654	21.6%	0.0345	0.0578	-0.0345	0.9771	0.9624	0.9561
9	0.2223	0.2852	22.0%	0.0365	0.0621	-0.0365	0.9739	0.9567	0.9501
10	0.2360	0.3047	22.6%	0.0383	0.0670	-0.0383	0.9705	0.9505	0.9437
11	0.2519	0.3264	22.8%	0.0406	0.0733	-0.0406	0.9663	0.9431	0.9357
12	0.2697	0.3480	22.5%	0.0432	0.0791	-0.0432	0.9614	0.9354	0.9265
13	0.2868	0.3666	21.8%	0.0457	0.0839	-0.0457	0.9565	0.9286	0.9173
14	0.3012	0.3827	21.3%	0.0481	0.0885	-0.0481	0.9523	0.9225	0.9094
15	0.3129	0.3962	21.0%	0.0501	0.0925	-0.0501	0.9487	0.9173	0.9029
16	0.3245	0.4089	20.6%	0.0523	0.0959	-0.0523	0.9452	0.9123	0.8964
17	0.3361	0.4228	20.5%	0.0550	0.1006	-0.0550	0.9415	0.9067	0.8895
18	0.3443	0.4343	20.7%	0.0570	0.1051	-0.0570	0.9388	0.9018	0.8844
19	0.3489	0.4419	21.0%	0.0585	0.1088	-0.0585	0.9370	0.8983	0.8813
20	0.3541	0.4495	21.2%	0.0604	0.1127	-0.0604	0.9350	0.8947	0.8777
21	0.3592	0.4569	21.4%	0.0625	0.1172	-0.0625	0.9329	0.8909	0.8738
22	0.3638	0.4642	21.6%	0.0637	0.1211	-0.0637	0.9308	0.8872	0.8701
23	0.3695	0.4722	21.8%	0.0657	0.1256	-0.0657	0.9281	0.8829	0.8654
24	0.3754	0.4807	21.9%	0.0682	0.1306	-0.0682	0.9249	0.8779	0.8598
25	0.3795	0.4884	22.3%	0.0702	0.1357	-0.0702	0.9217	0.8727	0.8542
26	0.3829	0.4943	22.5%	0.0718	0.1396	-0.0718	0.9189	0.8684	0.8493
27	0.3863	0.5002	22.8%	0.0736	0.1446	-0.0736	0.9152	0.8636	0.8430
28	0.3894	0.5078	23.3%	0.0752	0.1512	-0.0752	0.9108	0.8574	0.8354
29	0.3922	0.5177	24.2%	0.0768	0.1587	-0.0768	0.9055	0.8494	0.8264
30	0.3946	0.5293	25.4%	0.0780	0.1663	-0.0780	0.8996	0.8401	0.8164
31	0.3973	0.5425	26.8%	0.0792	0.1731	-0.0792	0.8933	0.8298	0.8059
32	0.3987	0.5568	28.4%	0.0797	0.1795	-0.0797	0.8875	0.8187	0.7962
33	0.4003	0.5747	30.3%	0.0802	0.1870	-0.0802	0.8813	0.8049	0.7858
34	0.4031	0.5968	32.5%	0.0812	0.1964	-0.0812	0.8734	0.7873	0.7730
35	0.4059	0.6223	34.8%	0.0819	0.2063	-0.0819	0.8649	0.7661	0.7591
36	0.4070	0.6465	37.1%	0.0822	0.2146	-0.0822	0.8576	0.7450	0.7474
37	0.4078	0.6685	39.0%	0.0825	0.2215	-0.0825	0.8512	0.7248	0.7374
38	0.4093	0.6847	40.2%	0.0830	0.2269	-0.0830	0.8456	0.7092	0.7284
39	0.4129	0.6997	41.0%	0.0846	0.2324	-0.0845	0.8392	0.6944	0.7183
40	0.4174	0.7155	41.7%	0.0868	0.2392	-0.0866	0.8312	0.6780	0.7056
41	0.4238	0.7363	42.4%	0.0894	0.2487	-0.0893	0.8200	0.6556	0.6877
42	0.4297	0.7643	43.8%	0.0911	0.2595	-0.0910	0.8080	0.6245	0.6687
43	0.4343	0.7888	44.9%	0.0924	0.2685	-0.0923	0.7989	0.5963	0.6539
44	0.4383	0.8089	45.8%	0.0938	0.2764	-0.0937	0.7909	0.5720	0.6405
45	0.4419	0.8247	46.4%	0.0949	0.2833	-0.0948	0.7830	0.5520	0.6273
46	0.4447	0.8400	47.1%	0.0961	0.2904	-0.0960	0.7749	0.5316	0.6140
47	0.4493	0.8626	47.9%	0.0980	0.2992	-0.0979	0.7642	0.5012	0.5965
48	0.4625	0.8902	48.0%	0.1014	0.3089	-0.1013	0.7488	0.4654	0.5717
49	0.4771	0.9181	48.0%	0.1049	0.3188	-0.1048	0.7335	0.4293	0.5475
50	0.4877	0.9447	48.4%	0.1074	0.3280	-0.1073	0.7208	0.3933	0.5272
51	0.4946	0.9710	49.1%	0.1092	0.3364	-0.1091	0.7109	0.3562	0.5110
52	0.4990	0.9938	49.8%	0.1104	0.3438	-0.1103	0.7038	0.3229	0.4989
53	0.5025	1.0151	50.5%	0.1115	0.3508	-0.1114	0.6980	0.2911	0.4888
54	0.5051	1.0396	51.4%	0.1124	0.3584	-0.1123	0.6935	0.2538	0.4803
55	0.5061	1.0637	52.4%	0.1127	0.3654	-0.1126	0.6914	0.2164	0.4751
56	0.5070	1.0831	53.2%	0.1134	0.3711	-0.1133	0.6904	0.1862	0.4721
57	0.5079	1.0895	53.4%	0.1142	0.3745	-0.1141	0.6883	0.1755	0.4685
58	0.5091	1.0914	53.3%	0.1152	0.3768	-0.1151	0.6845	0.1712	0.4631
59	0.5102	1.0933	53.3%	0.1162	0.3795	-0.1161	0.6805	0.1663	0.4572
60	0.5111	1.0951	53.3%	0.1173	0.3824	-0.1172	0.6771	0.1618	0.4520
61	0.5121	1.0980	53.4%	0.1183	0.3869	-0.1182	0.6729	0.1553	0.4444
62	0.5133	1.1016	53.4%	0.1194	0.3916	-0.1185	0.6694	0.1484	0.4377
63	0.5150	1.1058	53.4%	0.1207	0.3973	-0.1198	0.6653	0.1411	0.4317
64	0.5178	1.1105	53.4%	0.1221	0.4019	-0.1211	0.6615	0.1343	0.4264
65	0.5209	1.1150	53.3%	0.1238	0.4061	-0.1229	0.6573	0.1279	0.4209
66	0.5240	1.1190	53.2%	0.1256	0.4099	-0.1248	0.6531	0.1219	0.4153
67	0.5289	1.1228	52.9%	0.1289	0.4146	-0.1283	0.6459	0.1162	0.4071
68	0.5351	1.1267	52.5%	0.1328	0.4199	-0.1322	0.6347	0.1098	0.3954
69	0.5423	1.1316	52.1%	0.1371	0.4252	-0.1366	0.6256	0.1037	0.3858
70	0.5492	1.1362	51.7%	0.1404	0.4292	-0.1398	0.6175	0.0987	0.3774
71	0.5552	1.1396	51.3%	0.1481	0.4323	-0.1343	0.6091	0.0943	0.3681
72	0.5603	1.1418	50.9%	0.1498	0.4357	-0.1362	0.5945	0.0888	0.3533
73	0.5632	1.1435	50.8%	0.1509	0.4400	-0.1376	0.5809	0.0823	0.3383
74	0.5667	1.1449	50.5%	0.1533	0.4446	-0.1401	0.5594	0.0746	0.3176
75	0.5698	1.1460	50.3%	0.1547	0.4474	-0.1415	0.5358	0.0670	0.2963
76	0.5724	1.1457	50.0%	0.1556	0.4484	-0.1425	0.5057	0.0598	0.2708
77	0.5760	1.1435	49.6%	0.1620	0.4482	-0.1374	0.4561	0.0527	0.2328
78	0.5779	1.1409	49.4%	0.1640	0.4483	-0.1403	0.4036	0.0450	0.1943
79	0.5813	1.1392	49.0%	0.1682	0.4505	-0.1434	0.3421	0.0368	0.1515
80	0.5875	1.1381	48.4%	0.1765	0.4535	-0.1433	0.2677	0.0277	0.0998
81	0.5919	1.1382	48.0%	0.1803	0.4558	-0.1474	0.2058	0.0187	0.0574
82	0.5871	1.1390	48.5%	0.1825	0.4588	-0.1411	0.1916	0.0098	0.0488
83	0.5911	1.1389	48.1%	0.1860	0.4612	-0.1450	0.1386	0.0022	0.0110
84	0.5958	1.1377	47.6%	0.1899	0.4634	-0.1490	0.0748	-0.0056	-0.0381
85	0.5894	1.1361	48.1%	0.1911	0.4653	-0.1503	0.0596	-0.0139	-0.0527
86	0.5792	1.1345	48.9%	0.1906	0.4666	-0.1496	0.0578	-0.0230	-0.0579
87	0.5689	1.1335	49.8%	0.1901	0.4678	-0.1492	0.0589	-0.0326	-0.0598
88	0.5595	1.1331	50.6%	0.1891	0.4689	-0.1480	0.0612	-0.0418	-0.0604
89	0.5501	1.1323	51.4%	0.1875	0.4692	-0.1462	0.0627	-0.0498	-0.0605
90	0.5426	1.1322	52.1%	0.1862	0.4699	-0.1451	0.0642	-0.0574	-0.0605

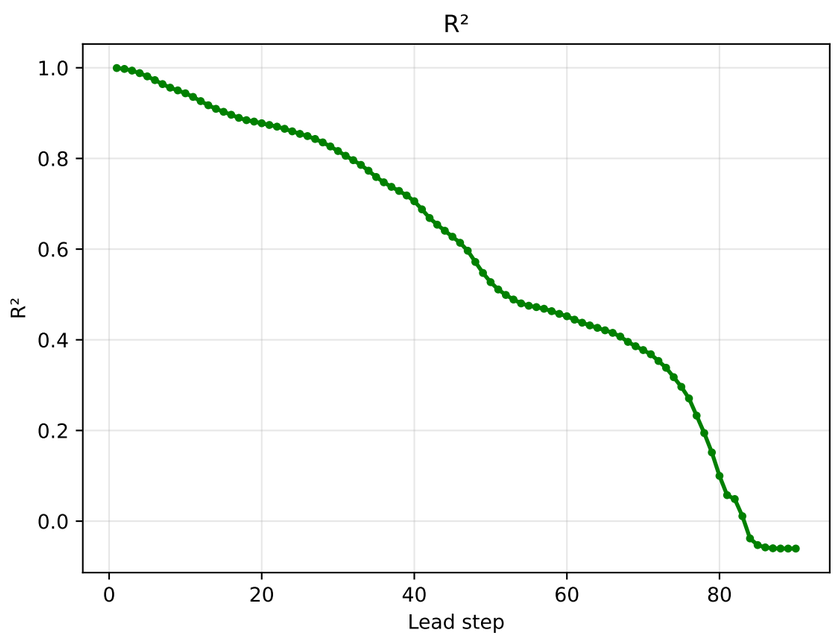
DHW Forecast Skill vs Lead Step - Test - GBR

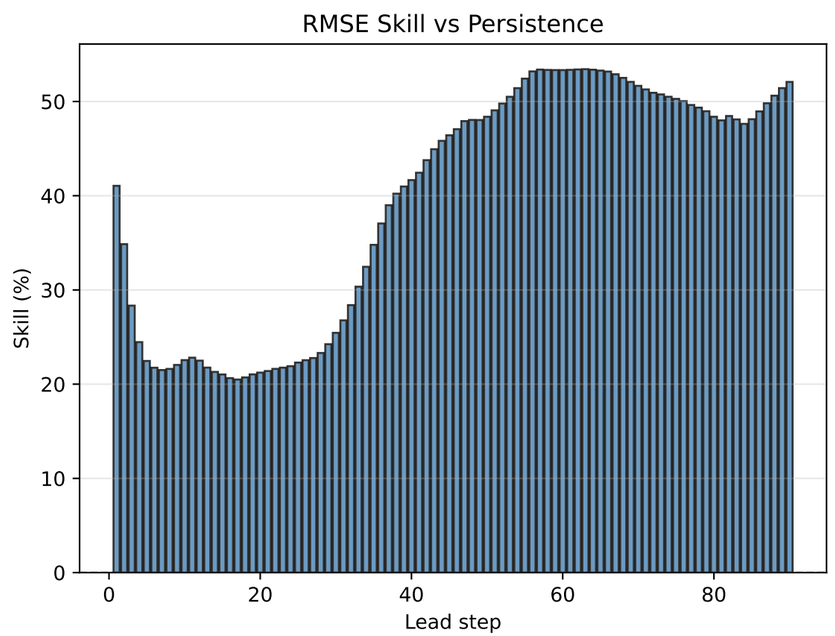


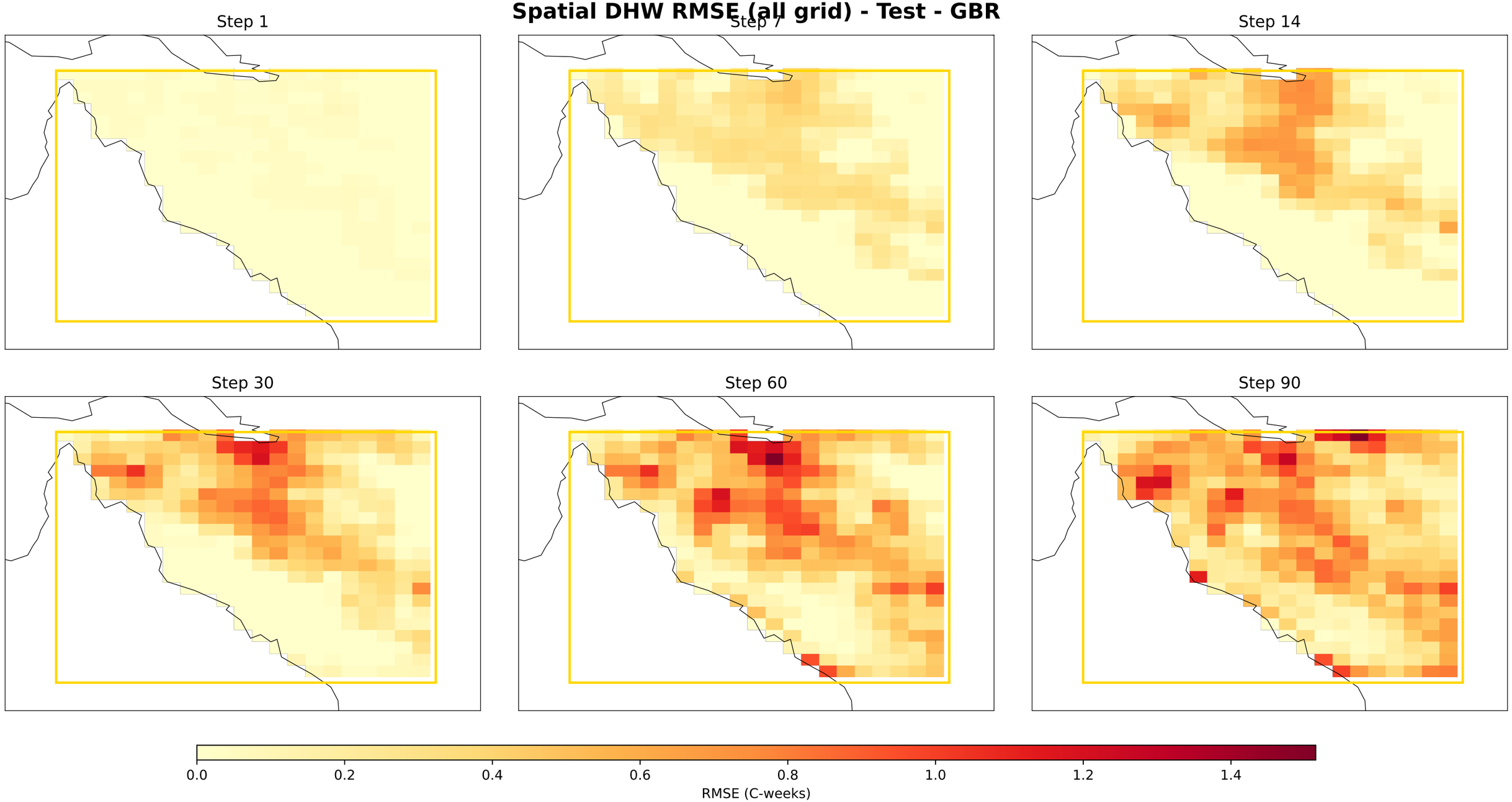


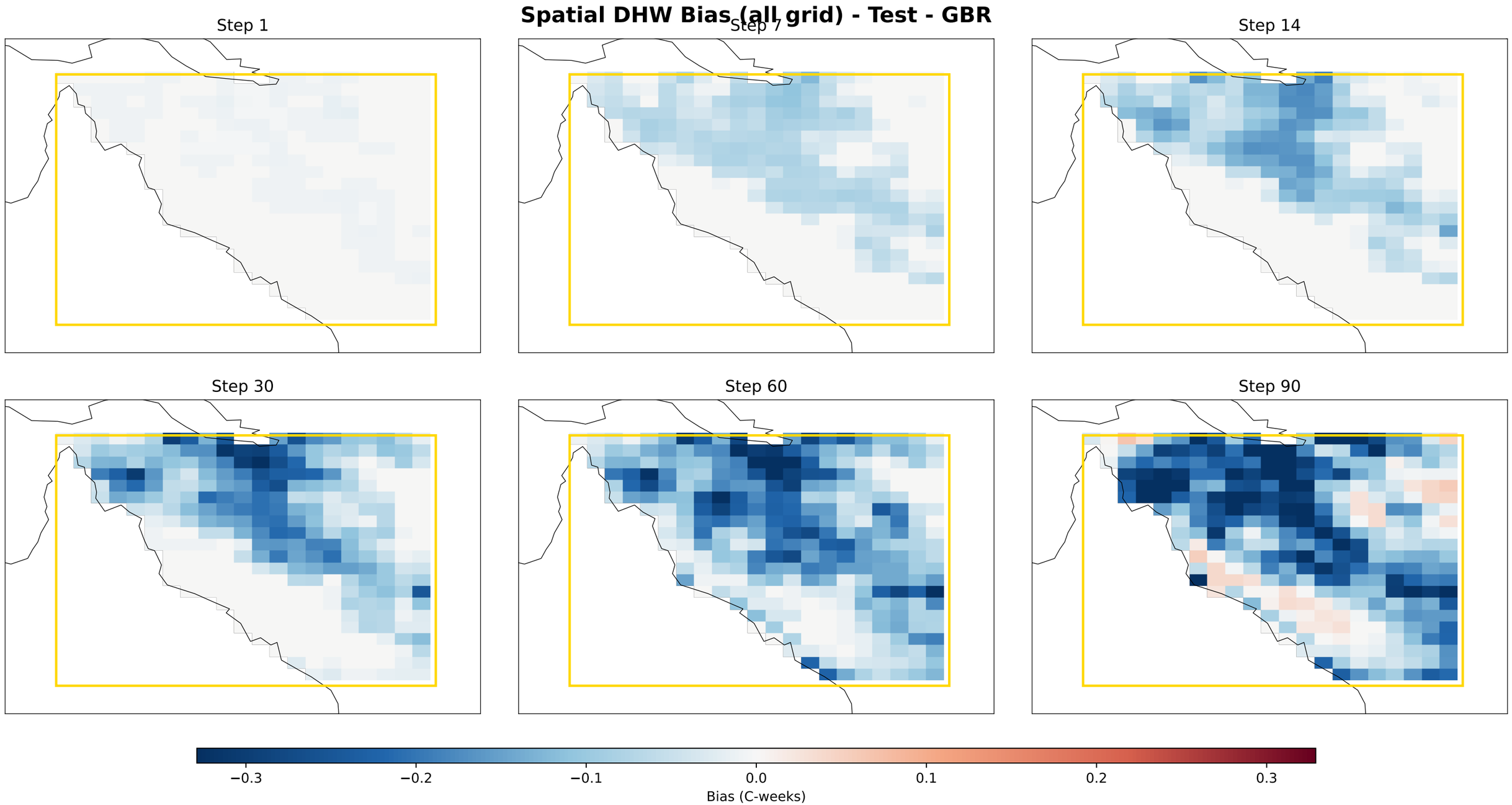


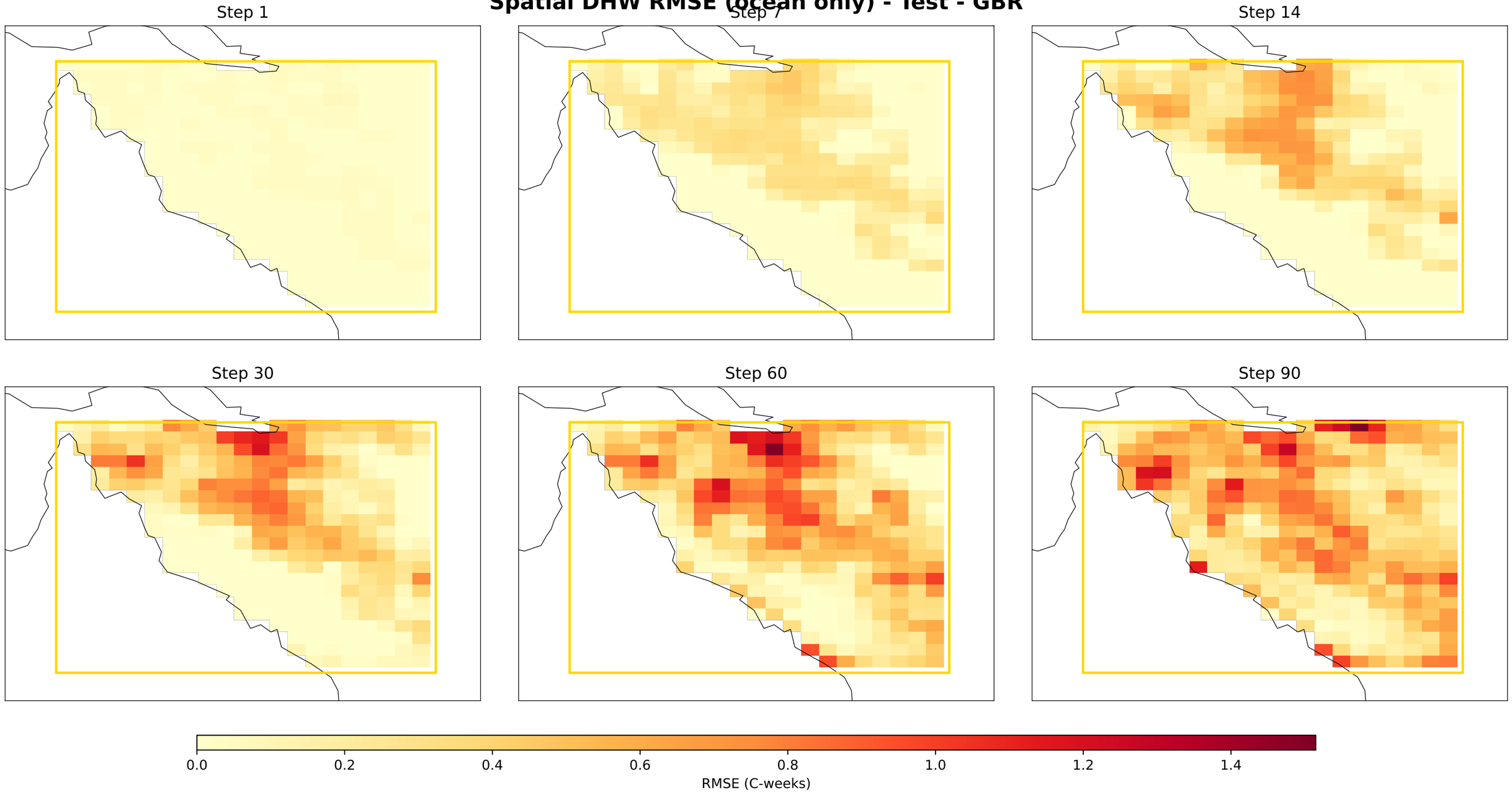


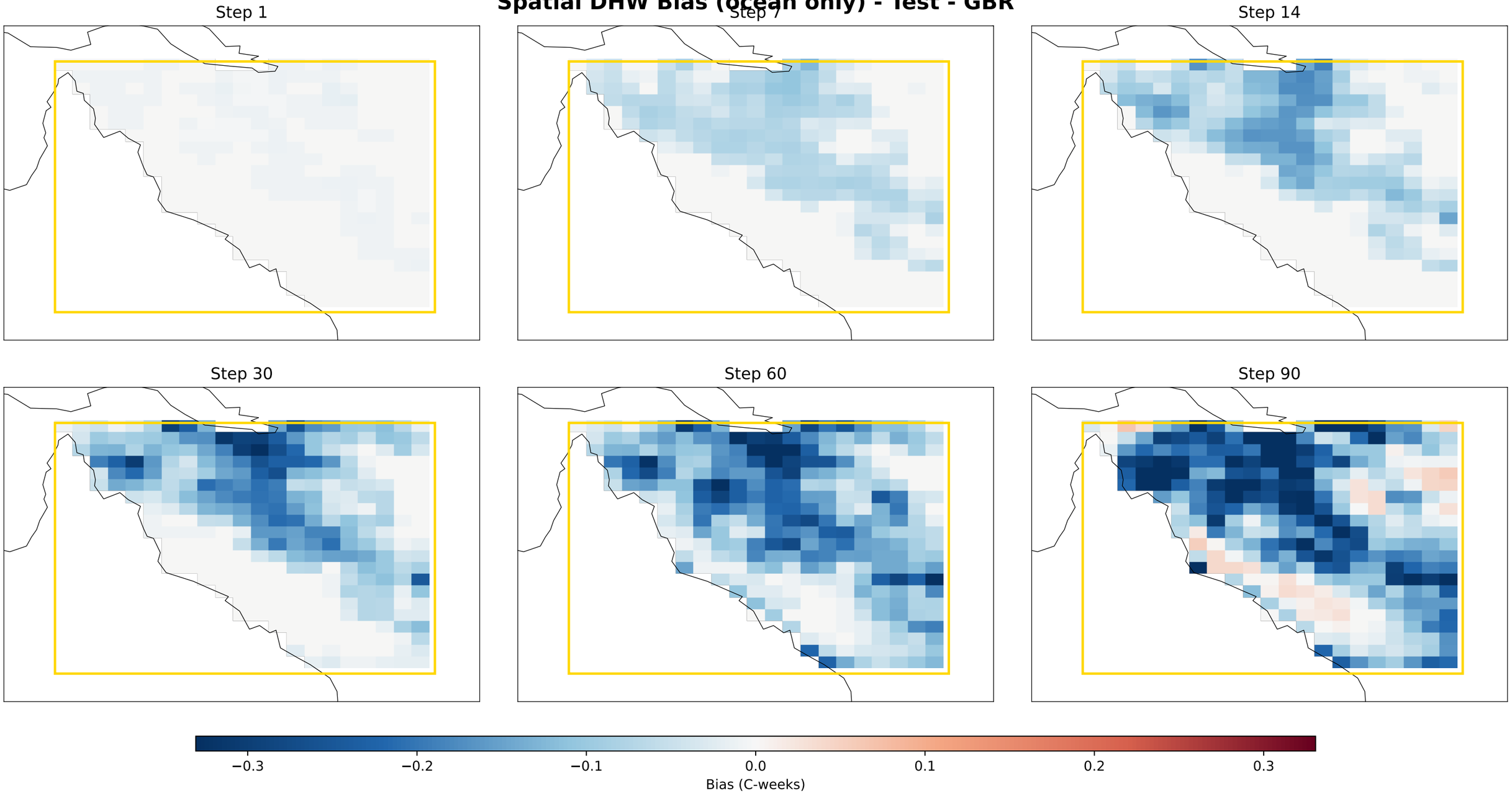





Spatial DHW RMSE (all grid) - Test - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
RMSE (C-weeks)

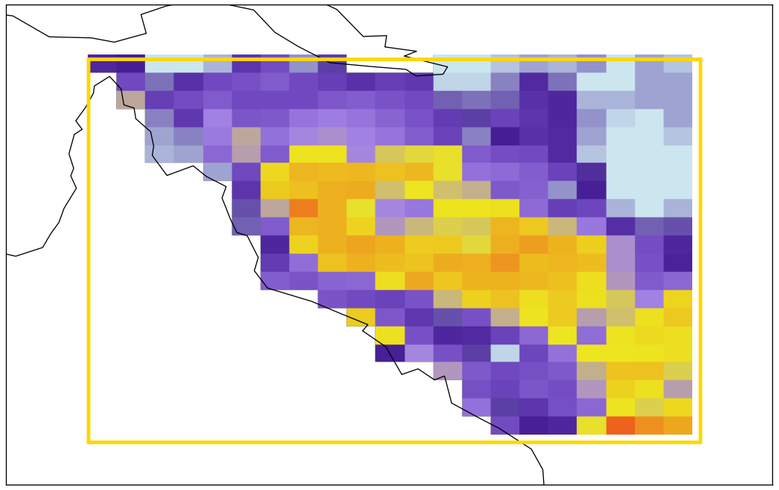

Spatial DHW Bias (all grid) - Test - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Bias (C-weeks)


Spatial DHW RMSE (ocean only) - Test - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
RMSE (C-weeks)

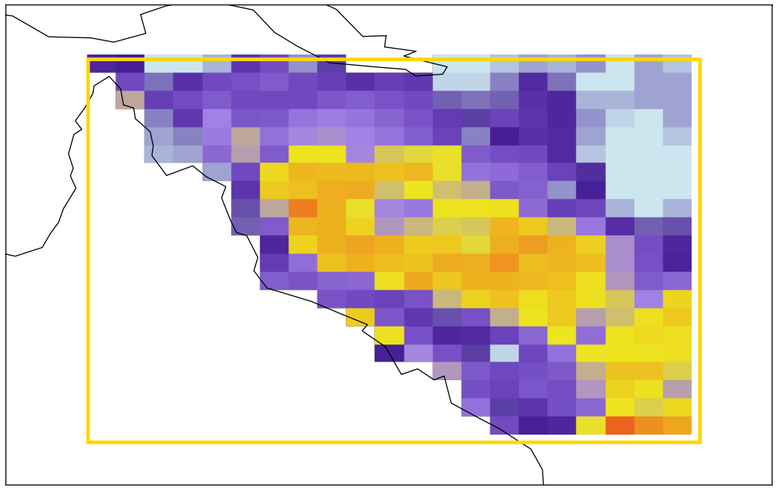

Spatial DHW Bias (ocean only) - Test - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Bias (C-weeks)

DHW Chain init: 2020-04-10 - Test - GBR (C-weeks)

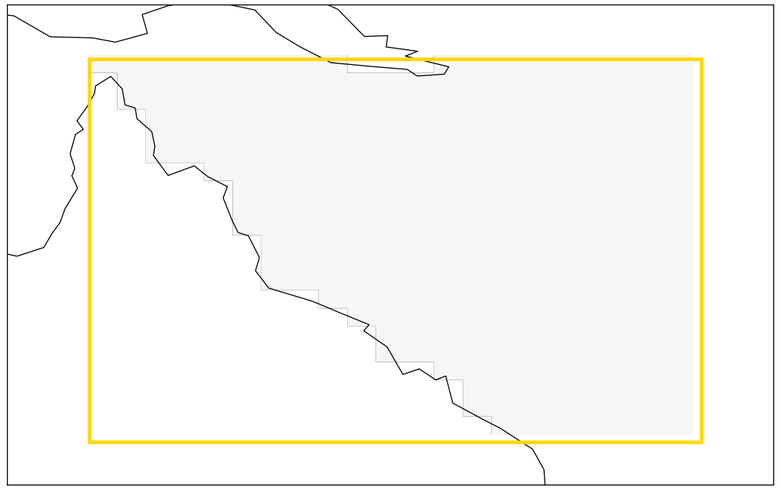
Actual DHW - Step 1



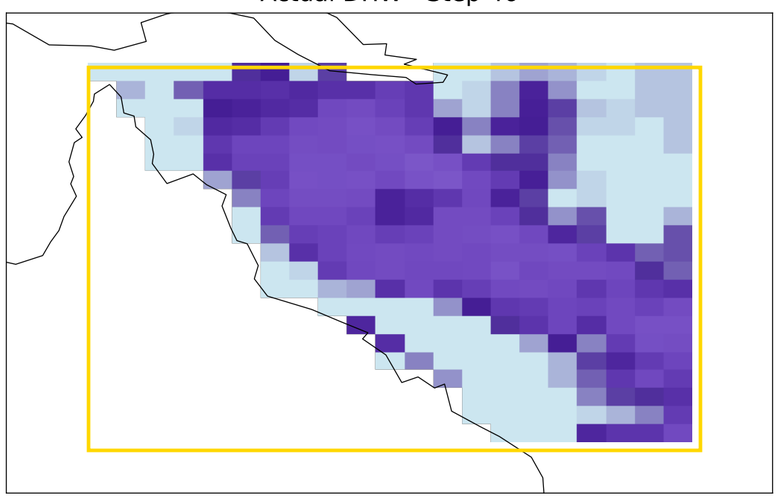
Predicted DHW - Step 1



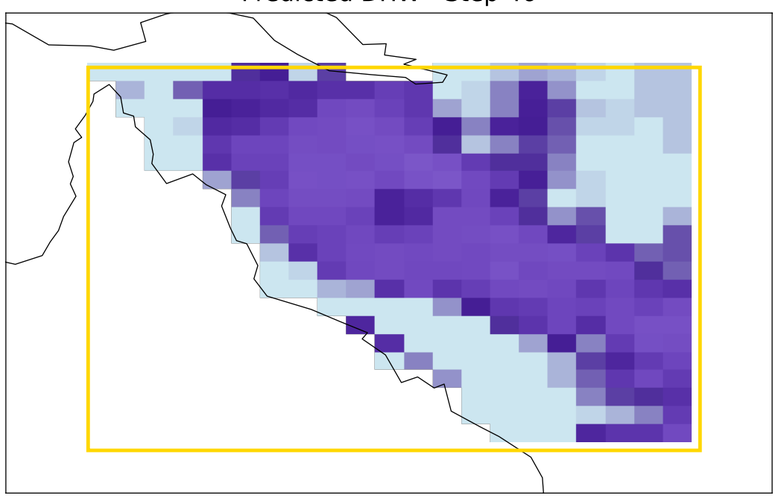
Error (Pred - Actual) - Step 1 [scale ±8]



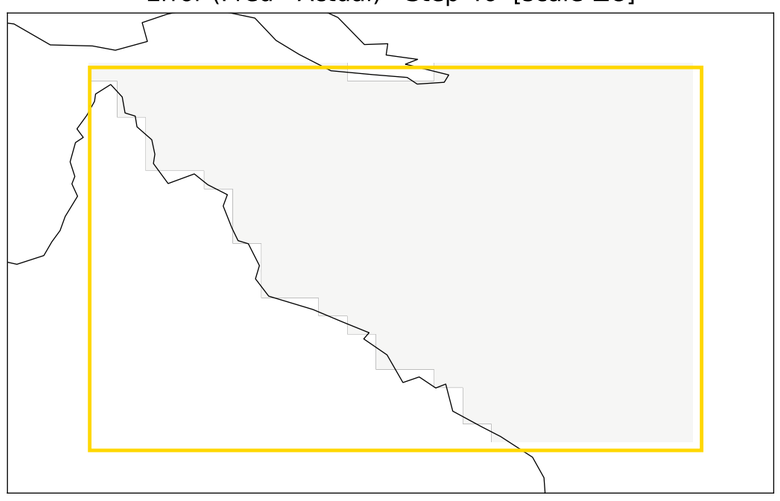
Actual DHW - Step 46



Predicted DHW - Step 46



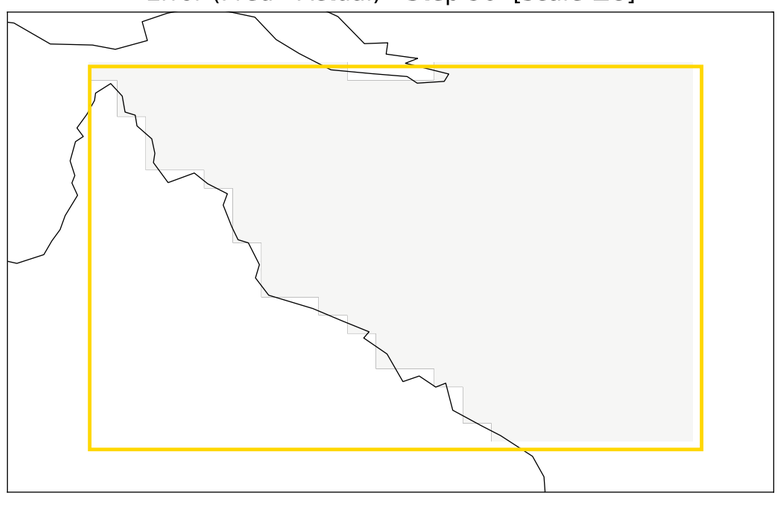
Error (Pred - Actual) - Step 46 [scale ±8]

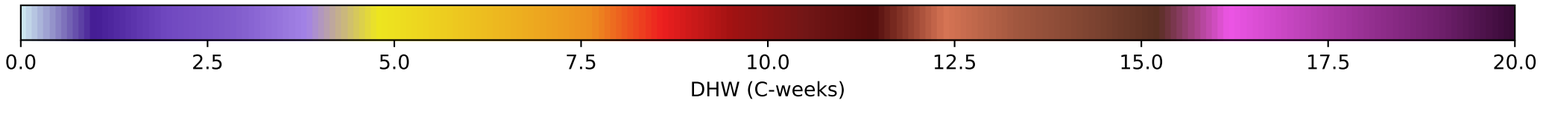


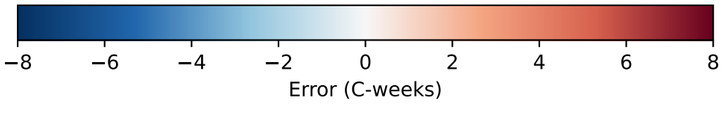
Actual DHW - Step 90

Predicted DHW - Step 90

Error (Pred - Actual) - Step 90 [scale ±8]

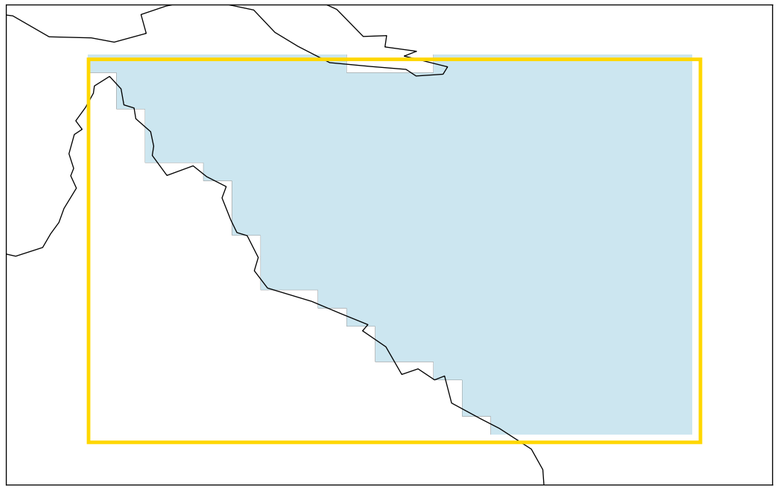




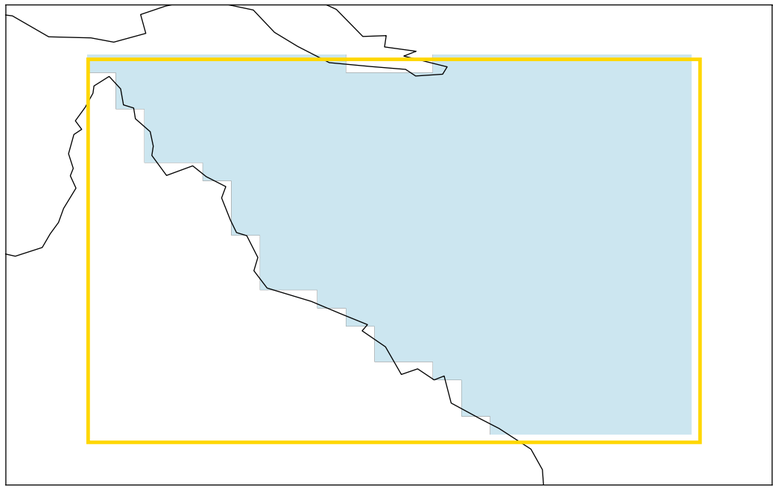


DHW Chain init: 2020-07-09 - Test - GBR (C-weeks)

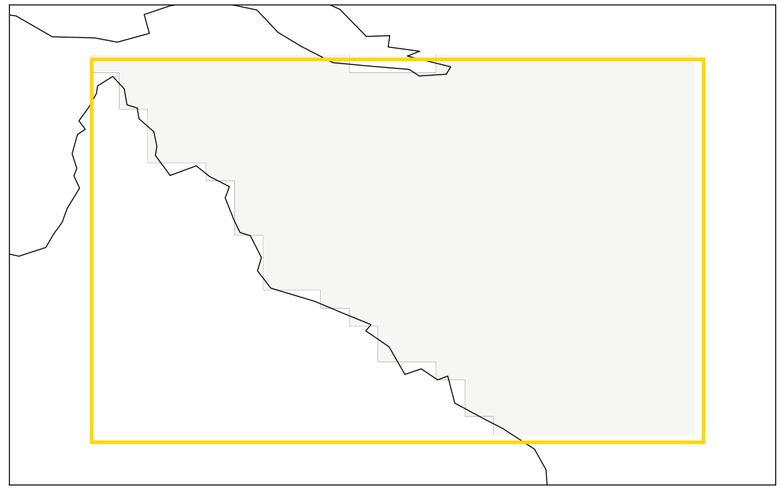
Actual DHW - Step 1



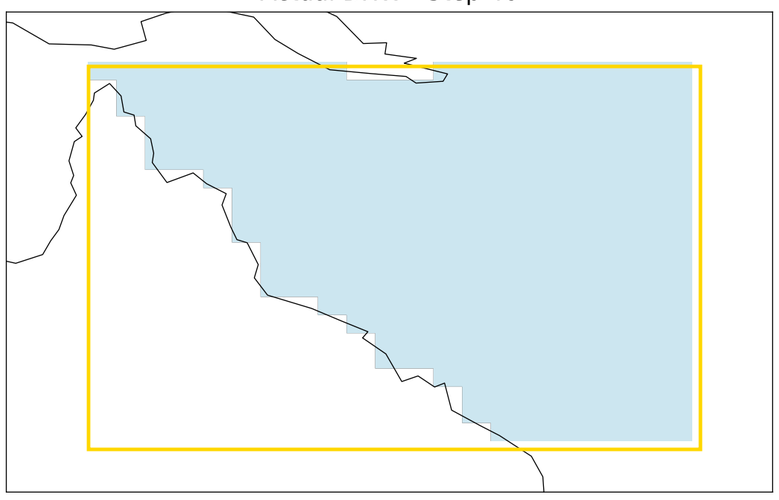
Predicted DHW - Step 1



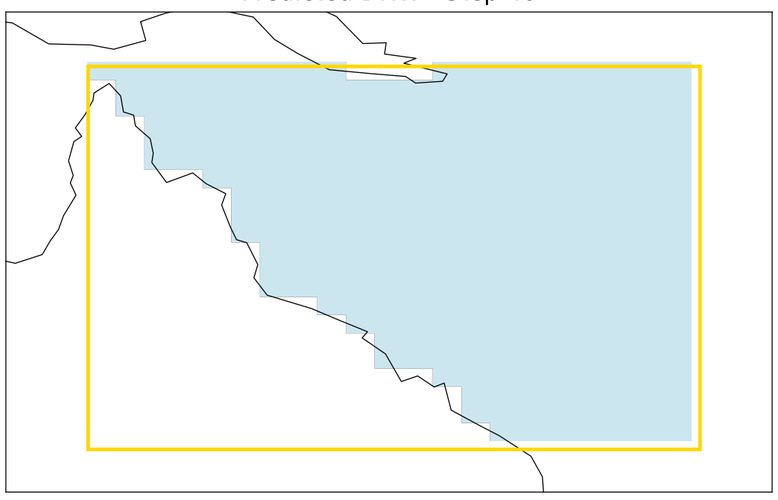
Error (Pred - Actual) - Step 1 [scale ±8]



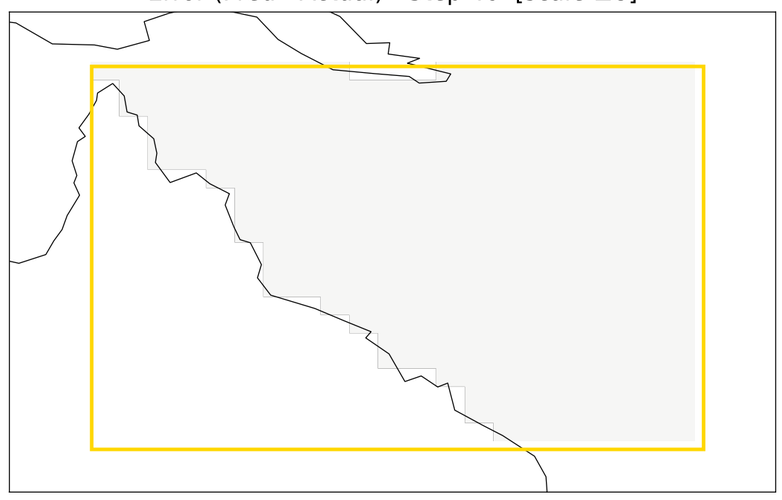
Actual DHW - Step 46



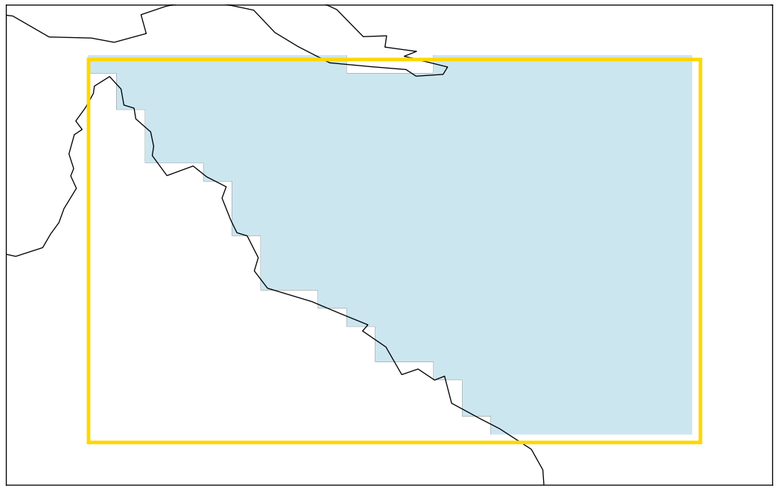
Predicted DHW - Step 46



Error (Pred - Actual) - Step 46 [scale ±8]

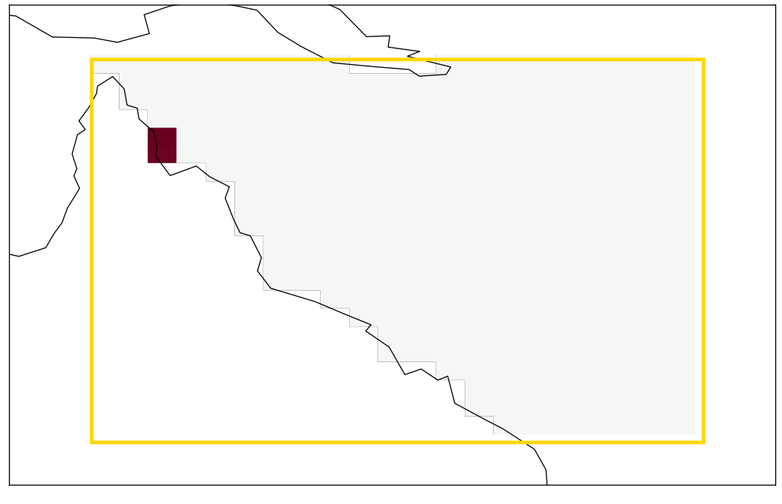


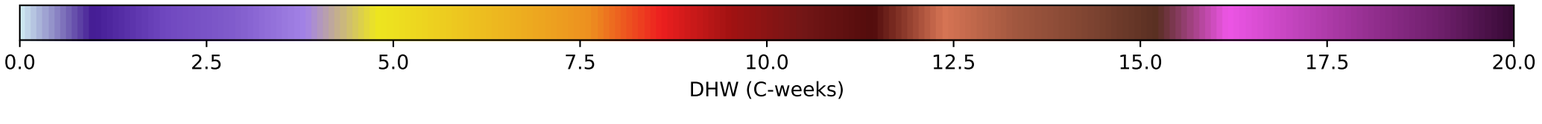
Actual DHW - Step 90

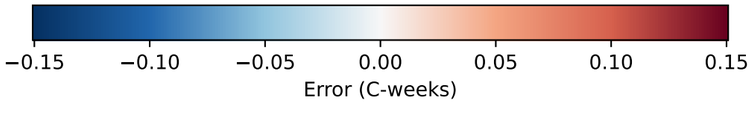


Predicted DHW - Step 90

Error (Pred - Actual) - Step 90 [scale ±0.15]

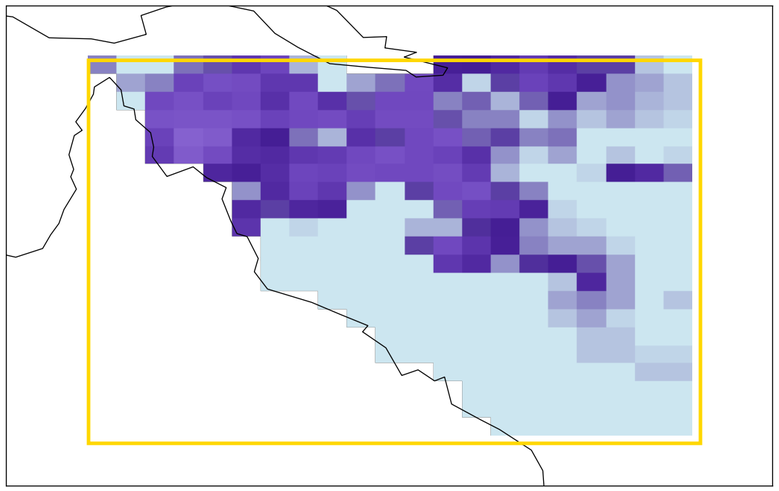




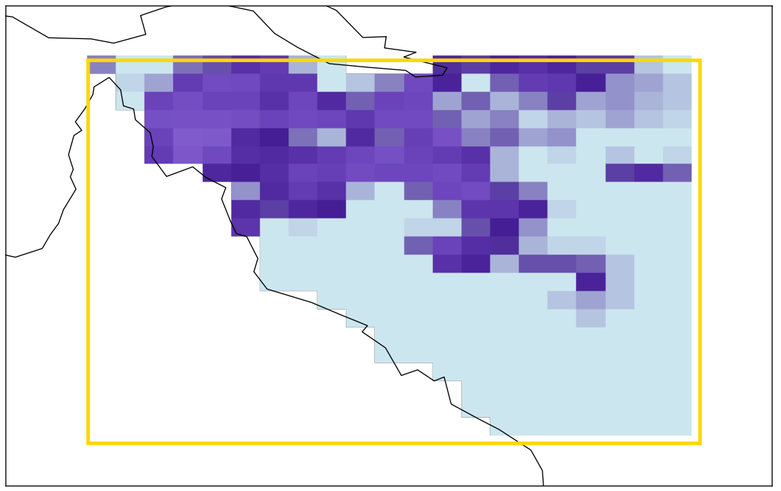


DHW Chain init: 2021-01-06 - Test - GBR (C-weeks)

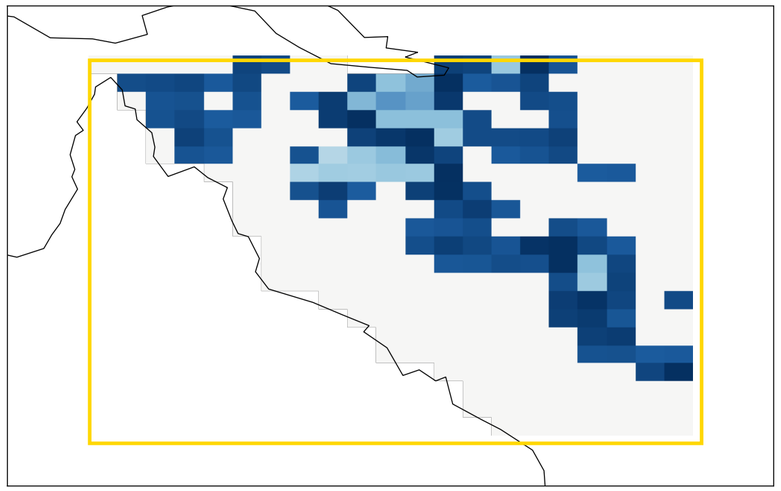
Actual DHW - Step 1



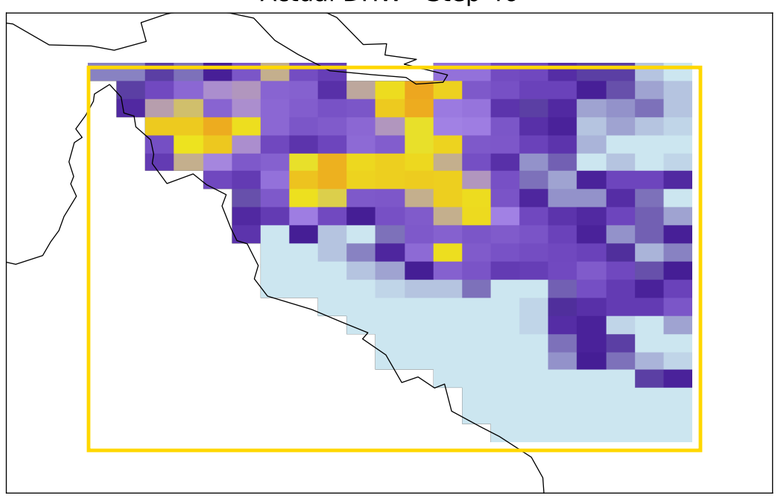
Predicted DHW - Step 1



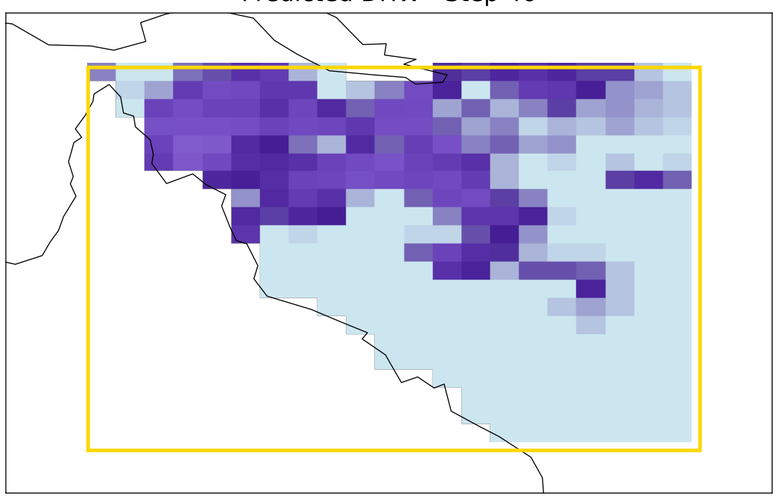
Error (Pred - Actual) - Step 1 [scale ±0.2]



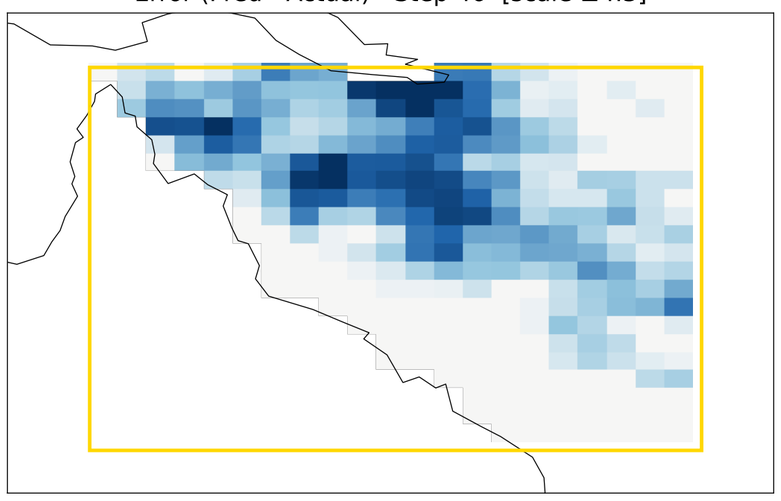
Actual DHW - Step 46



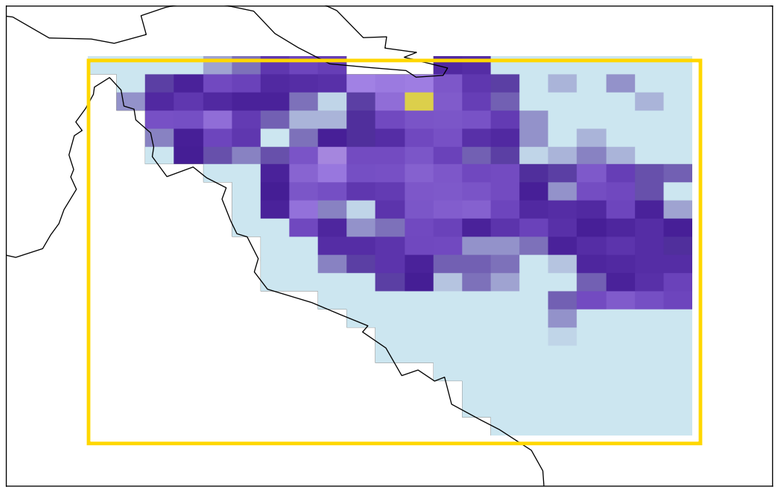
Predicted DHW - Step 46



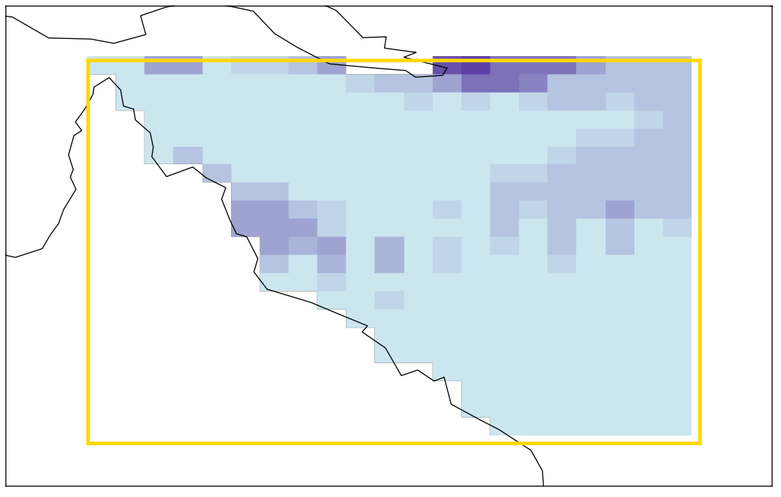
Error (Pred - Actual) - Step 46 [scale ±4.3]



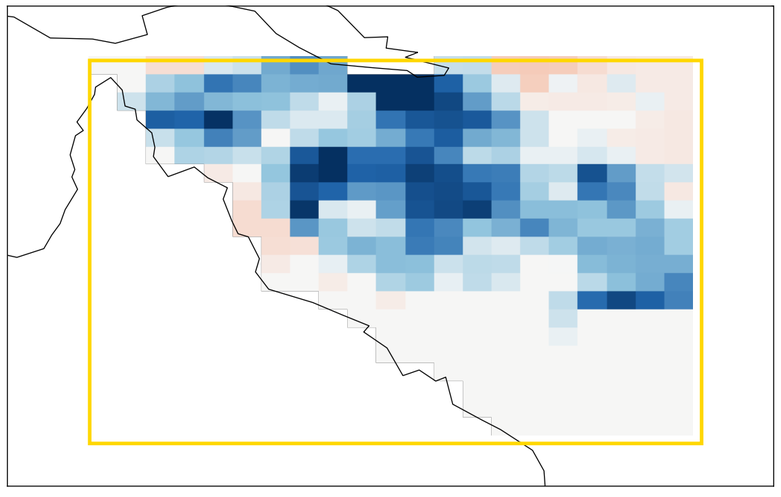
Actual DHW - Step 90

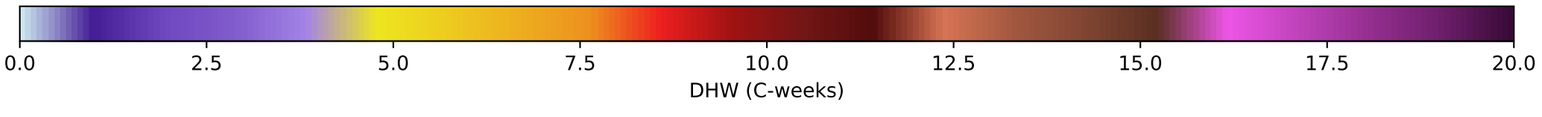


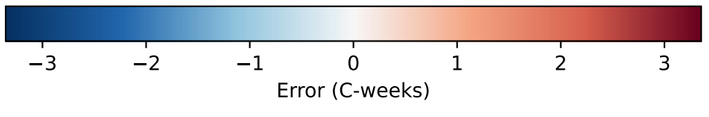
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±3.4]

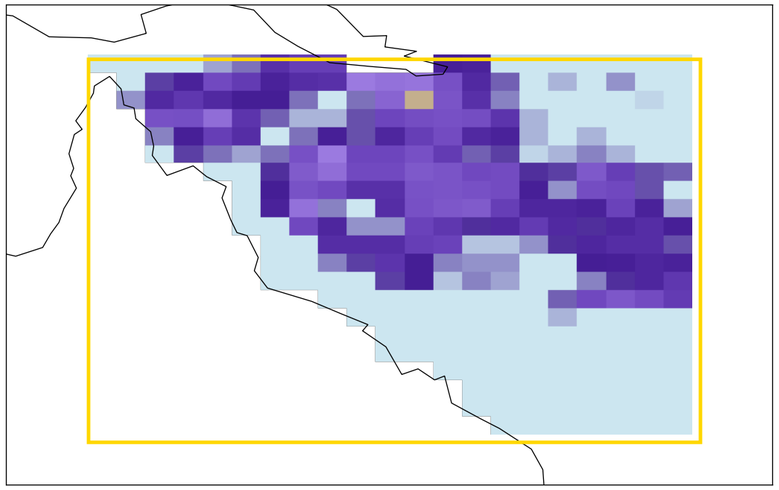




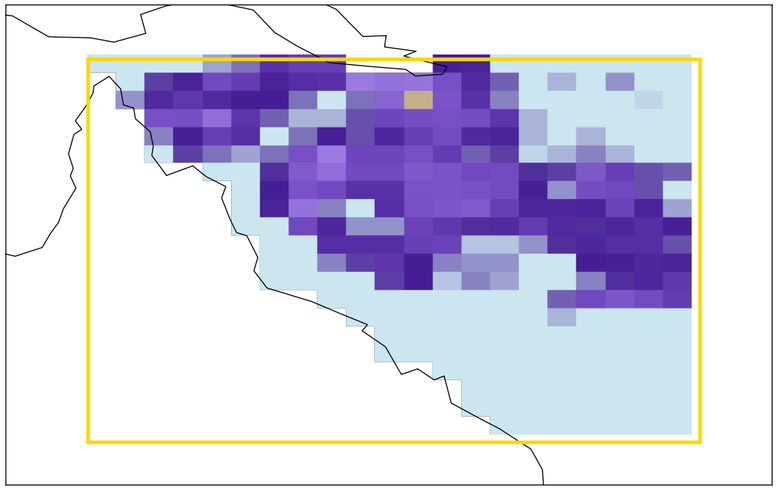


DHW Chain init: 2021-04-06 - Test - GBR (C-weeks)

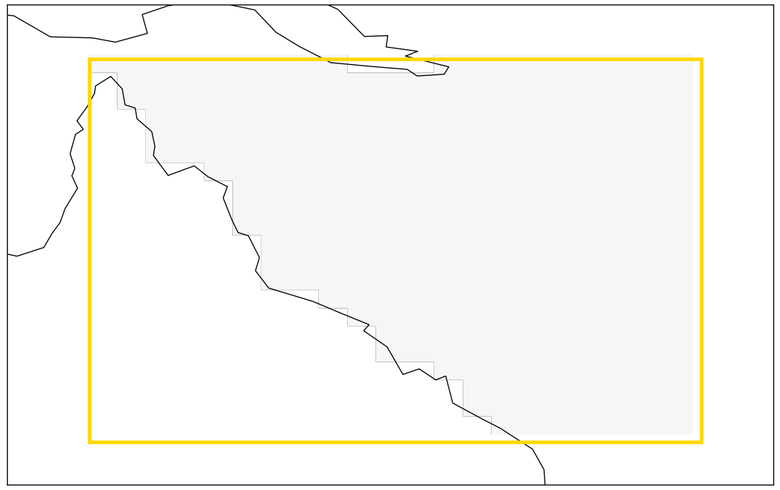
Actual DHW - Step 1



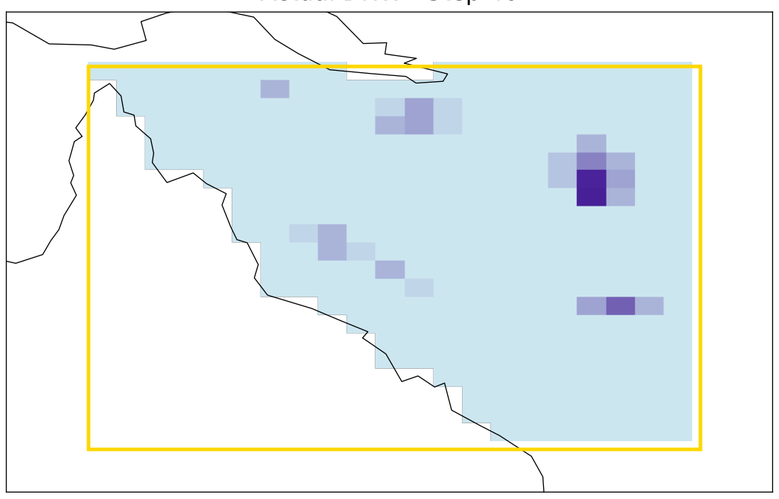
Predicted DHW - Step 1



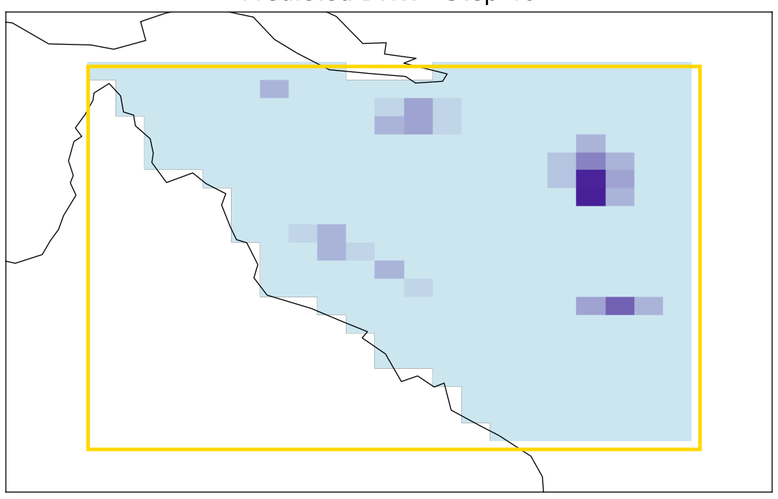
Error (Pred - Actual) - Step 1 [scale ±8]



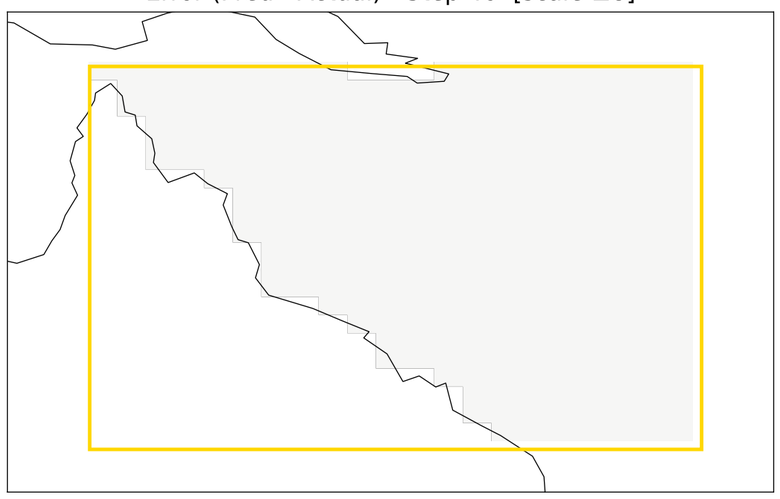
Actual DHW - Step 46



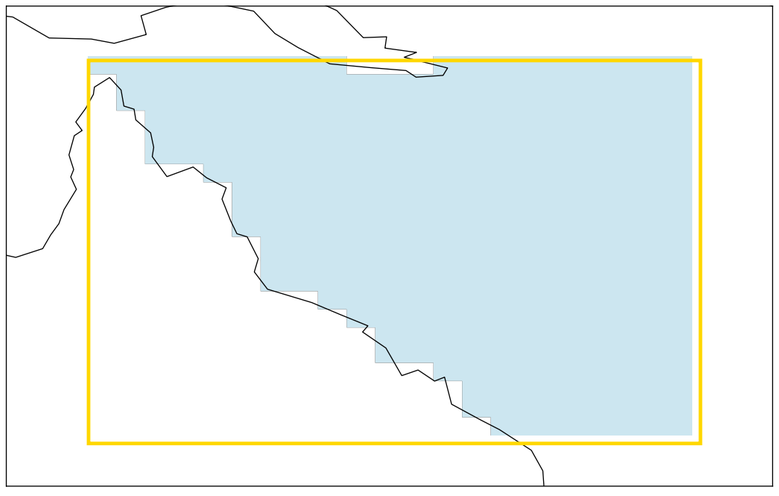
Predicted DHW - Step 46



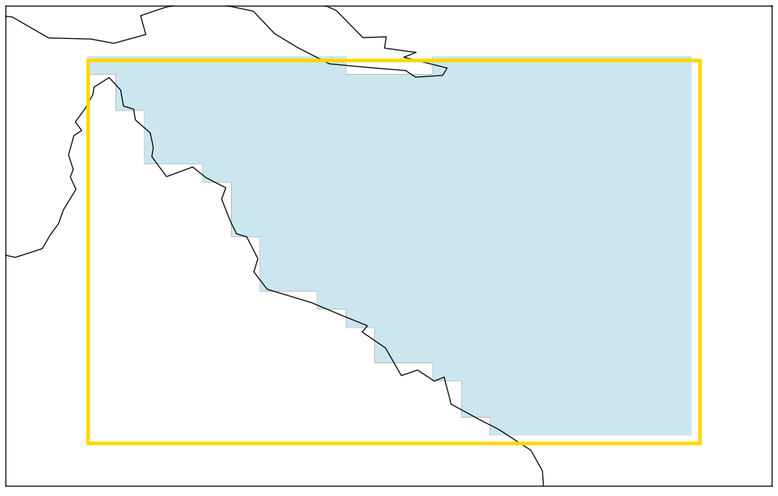
Error (Pred - Actual) - Step 46 [scale ±8]



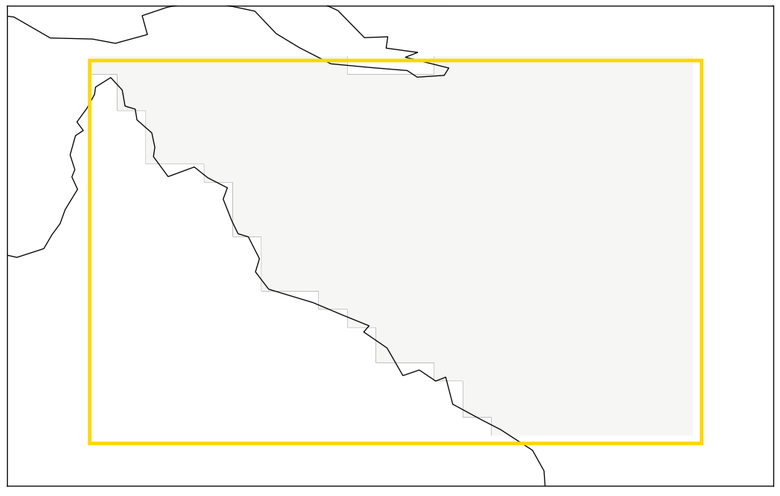
Actual DHW - Step 90

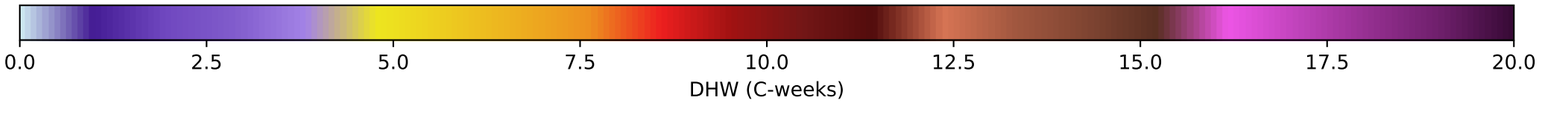


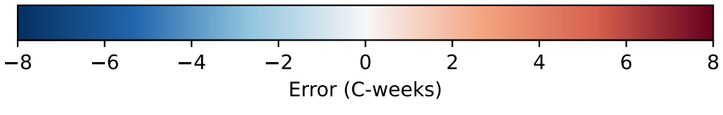
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

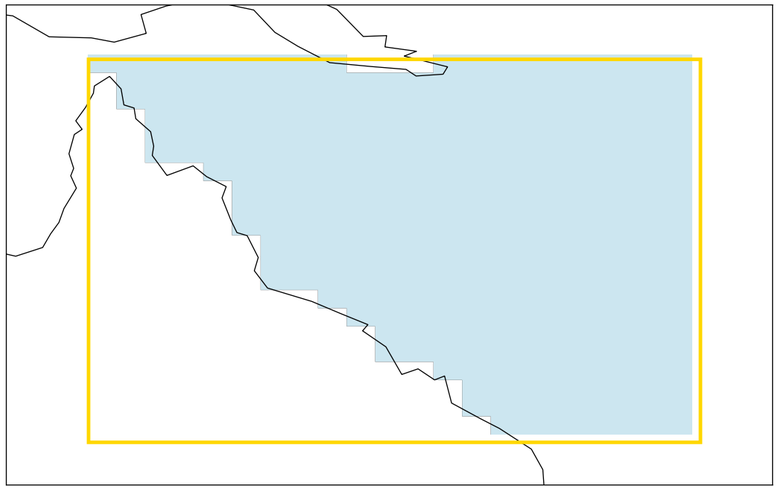




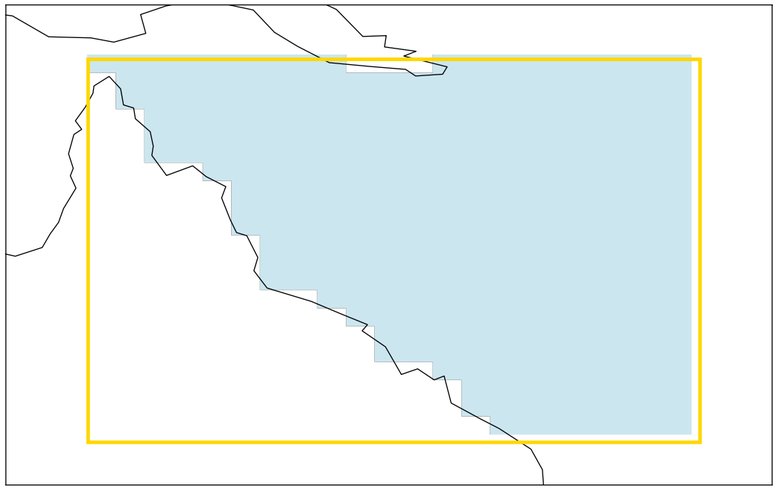


DHW Chain init: 2021-10-03 - Test - GBR (C-weeks)

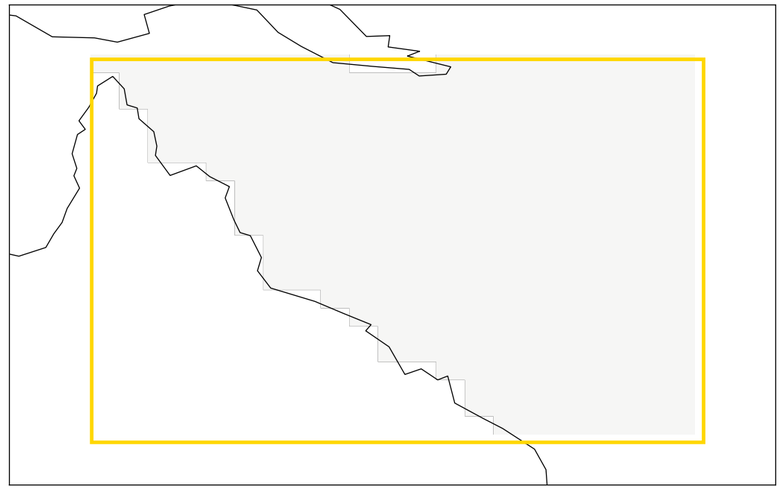
Actual DHW - Step 1



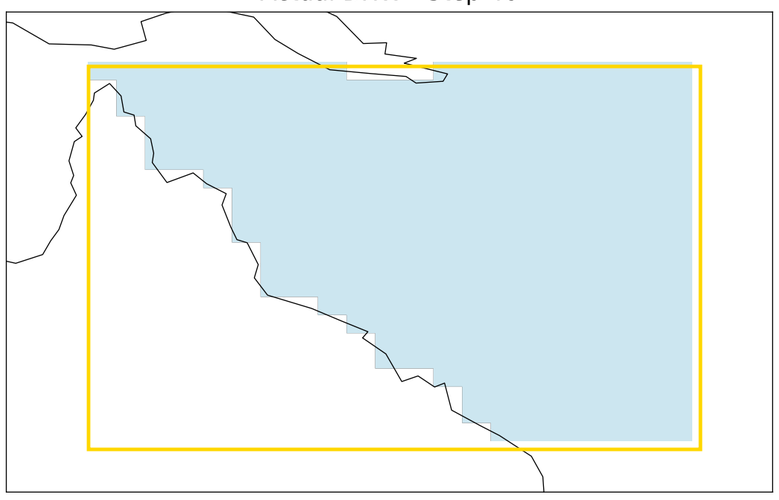
Predicted DHW - Step 1



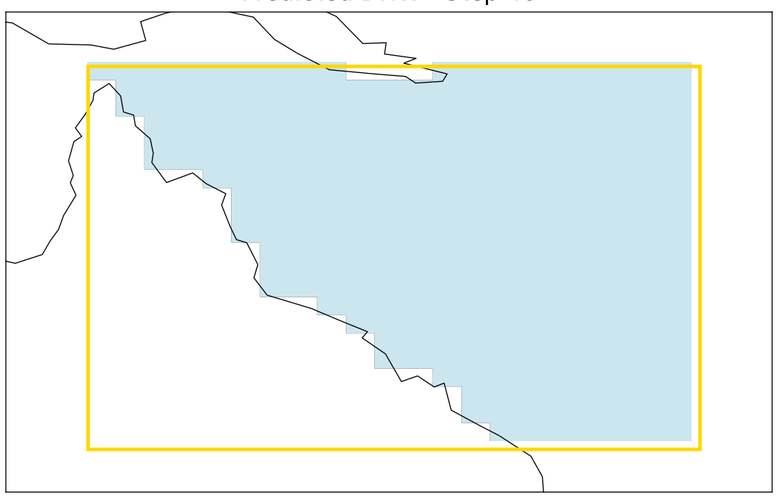
Error (Pred - Actual) - Step 1 [scale ±8]



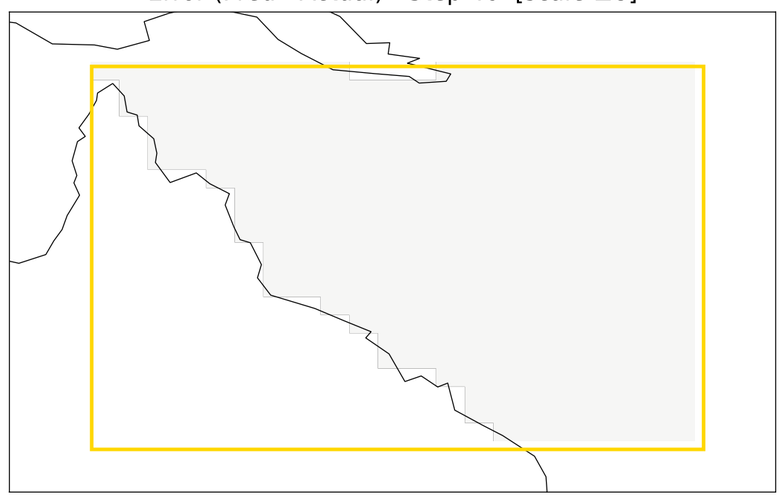
Actual DHW - Step 46



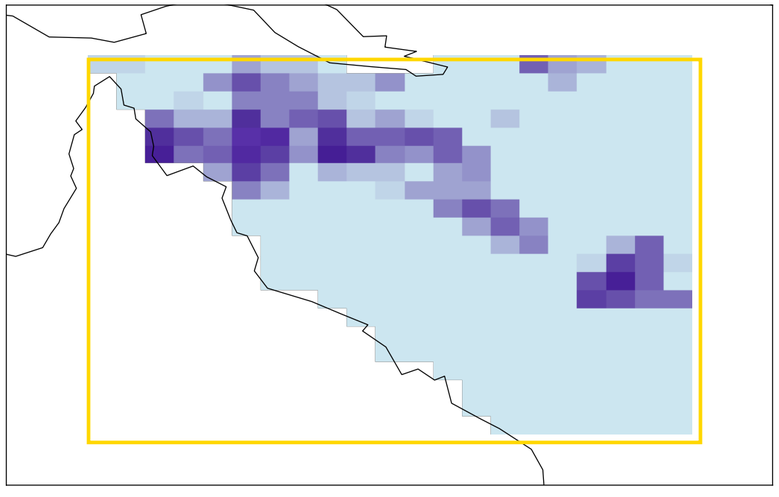
Predicted DHW - Step 46



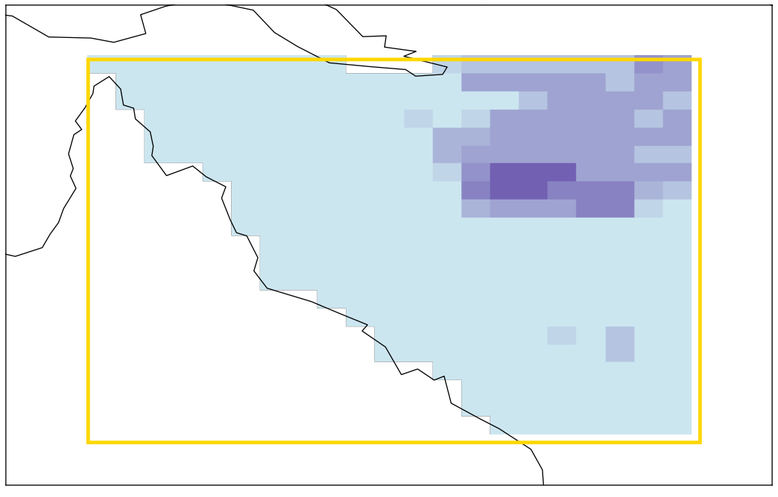
Error (Pred - Actual) - Step 46 [scale ±8]



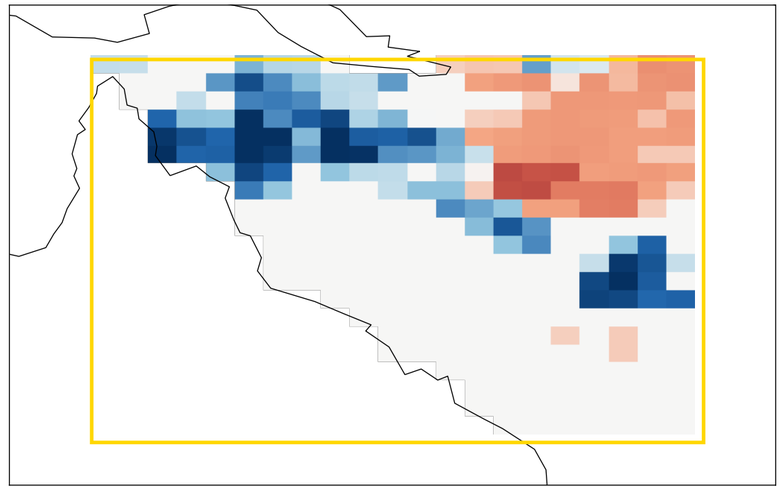
Actual DHW - Step 90

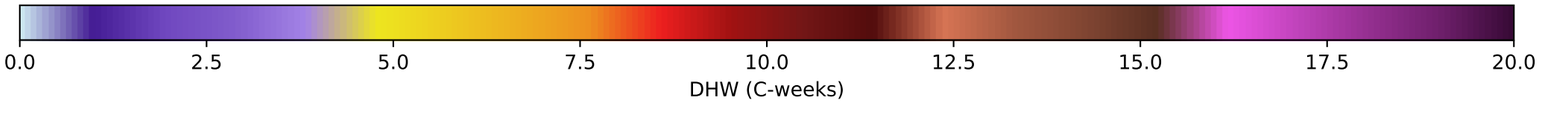


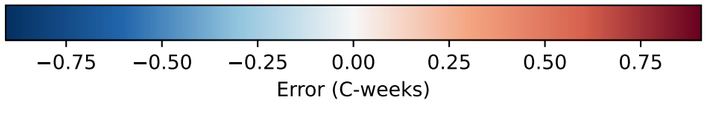
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.91]

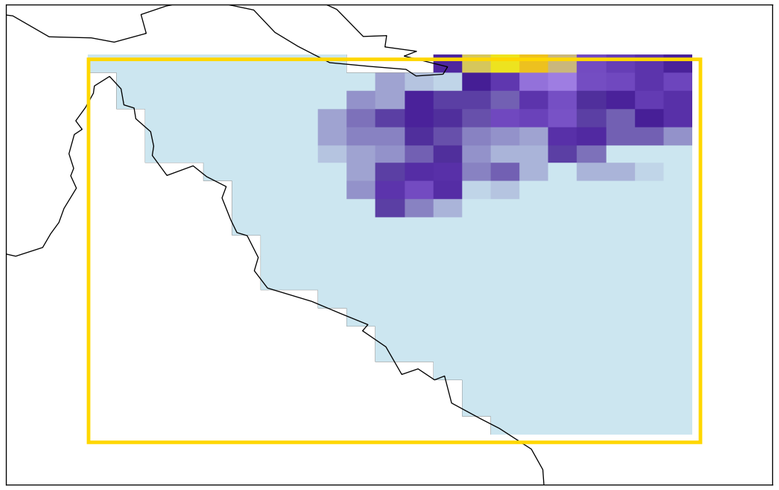




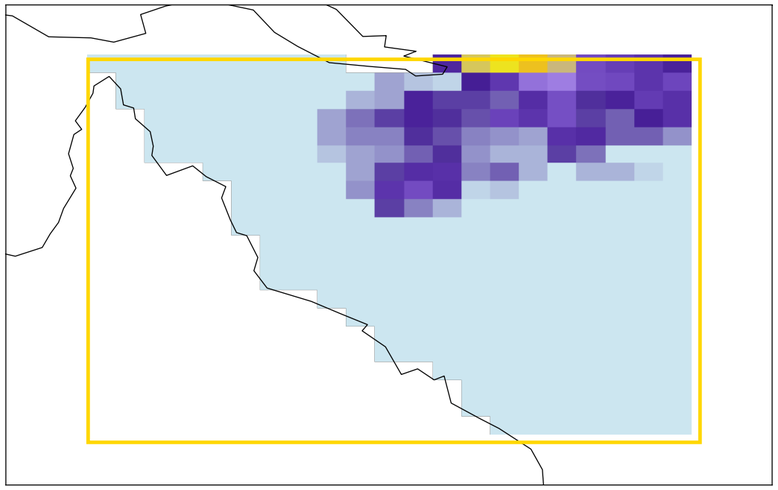


DHW Chain init: 2022-12-27 - Test - GBR (C-weeks)

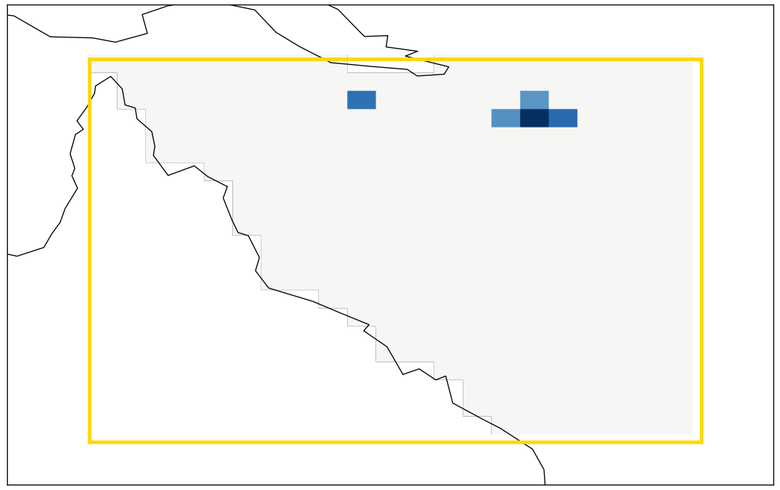
Actual DHW - Step 1



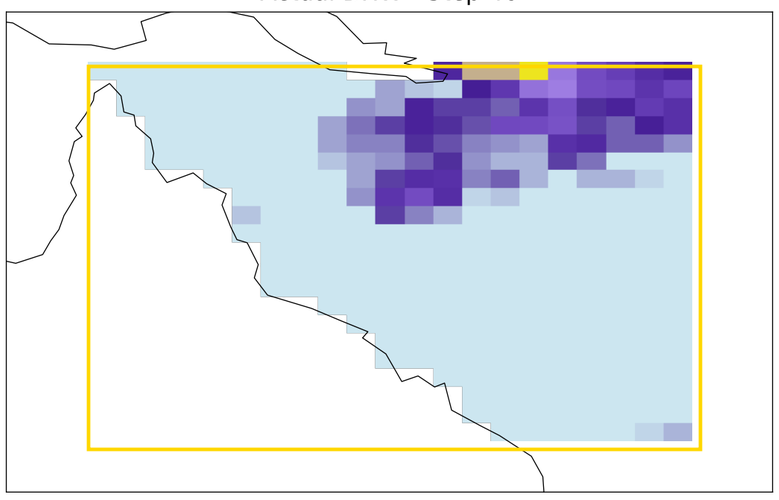
Predicted DHW - Step 1



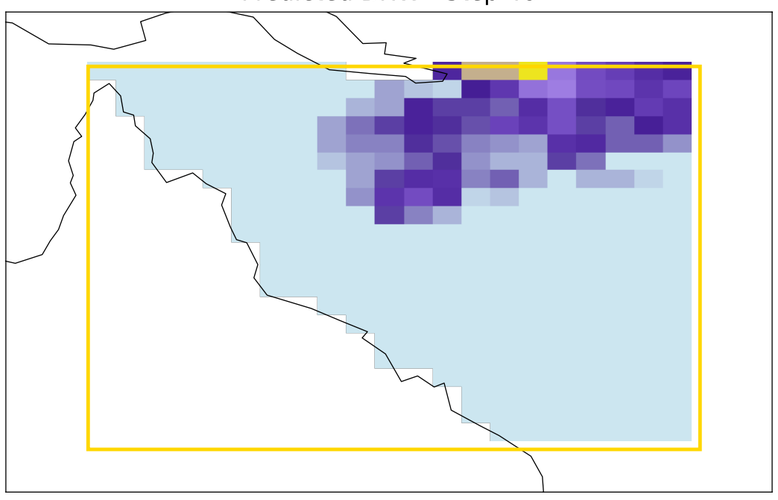
Error (Pred - Actual) - Step 1 [scale ±0.29]



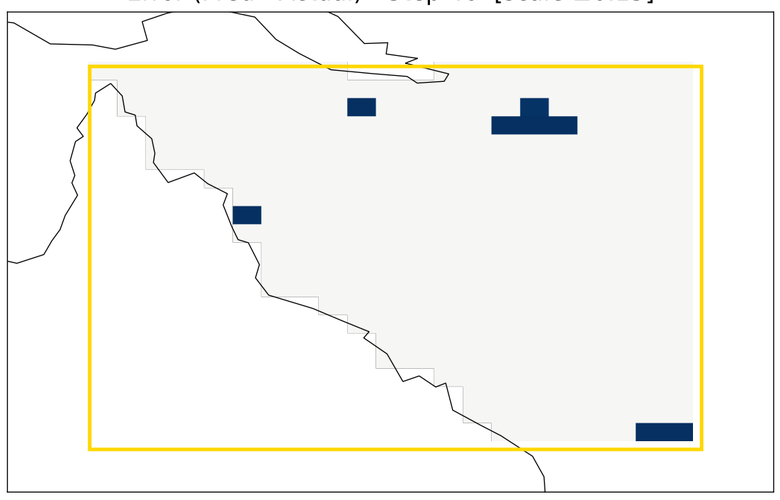
Actual DHW - Step 46



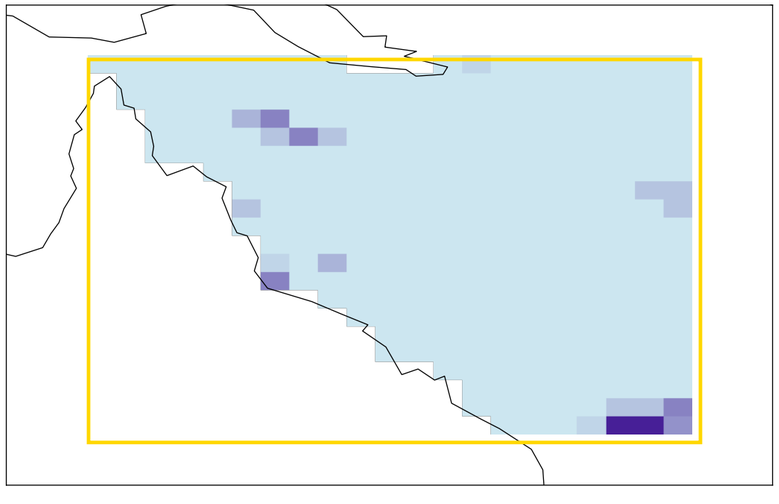
Predicted DHW - Step 46



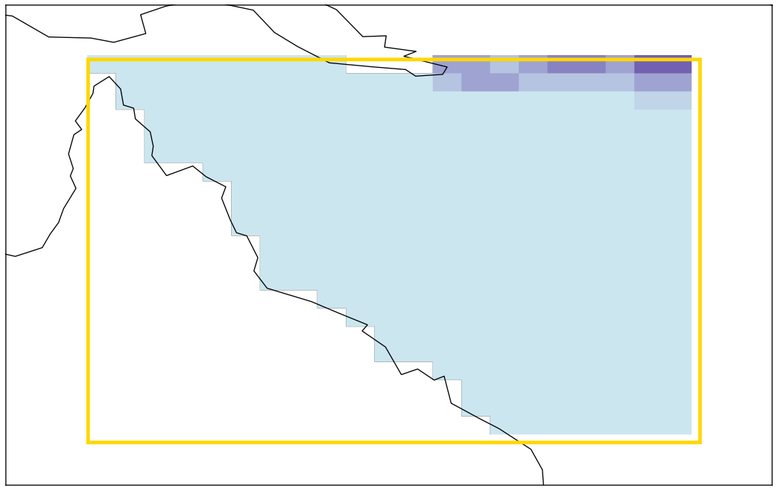
Error (Pred - Actual) - Step 46 [scale ±0.15]



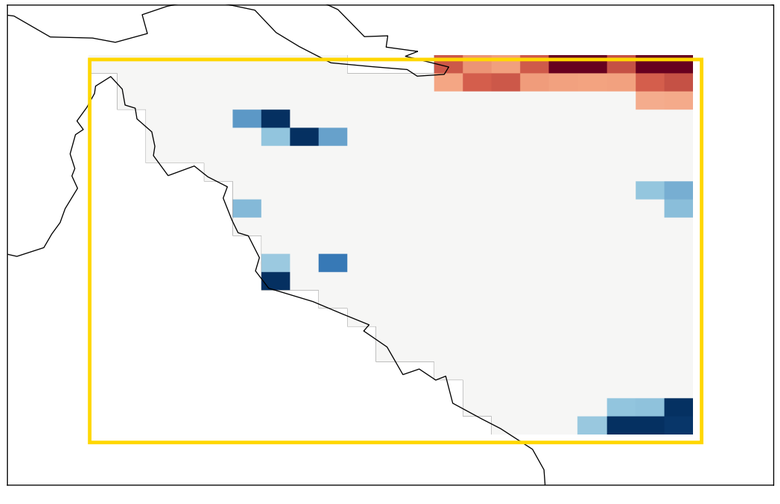
Actual DHW - Step 90

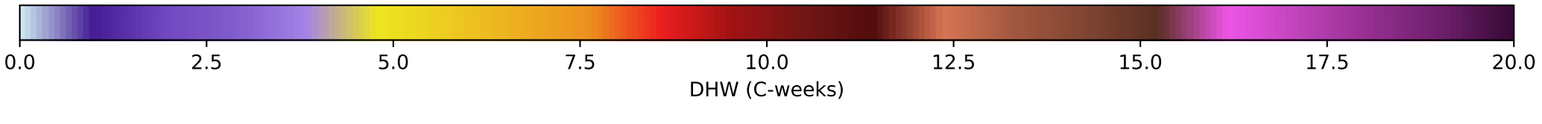


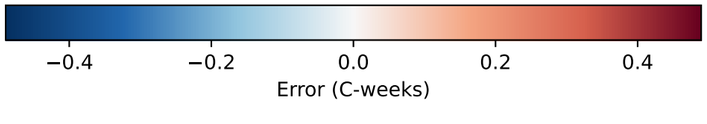
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.49]

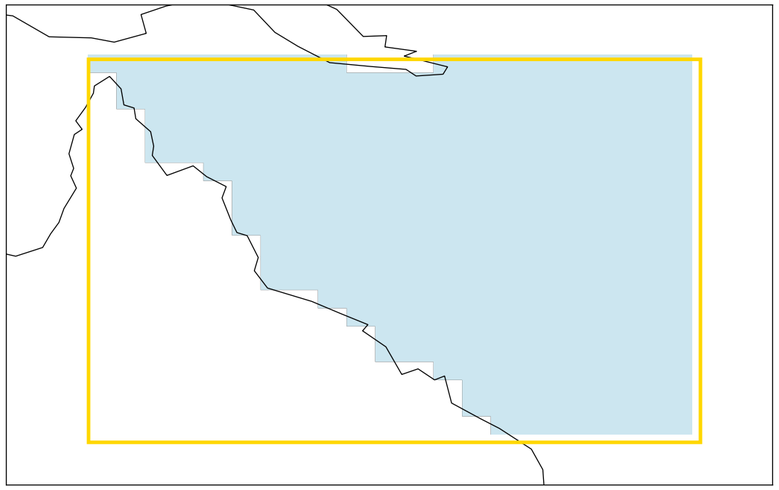




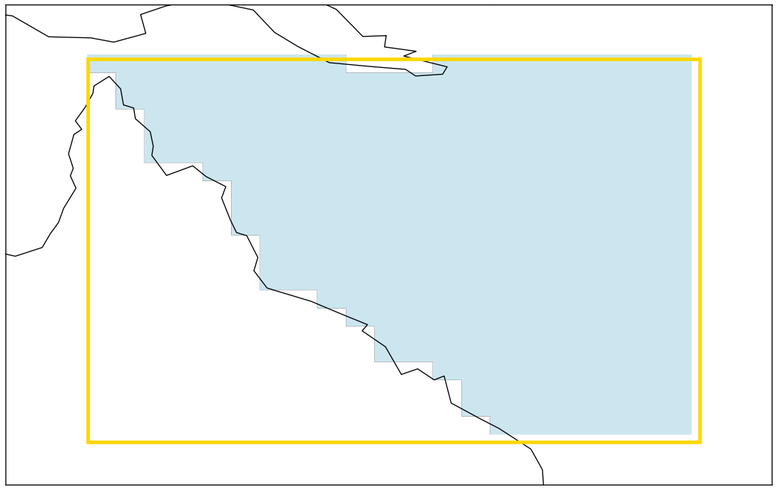


DHW Chain init: 2023-09-23 - Test - GBR (C-weeks)

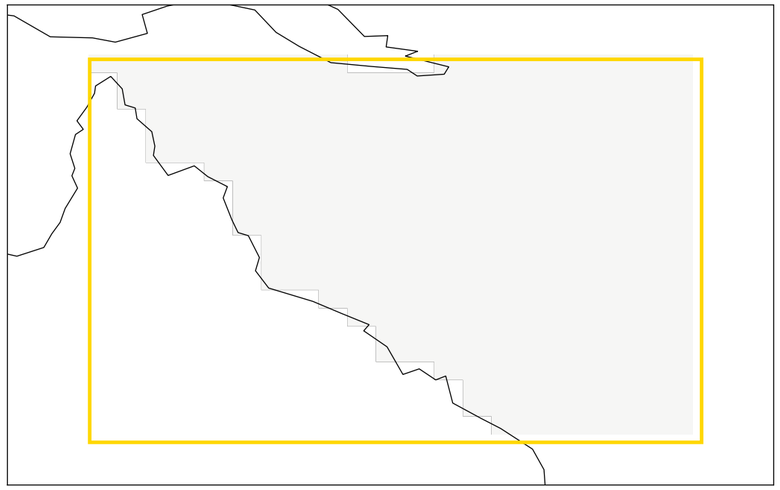
Actual DHW - Step 1



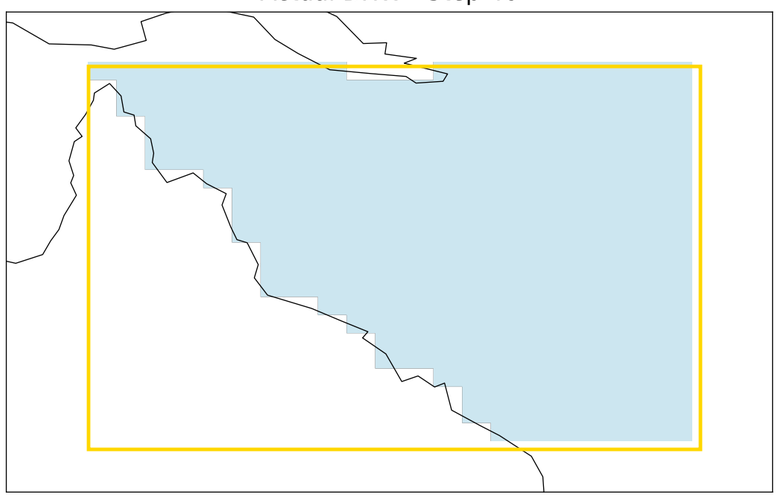
Predicted DHW - Step 1



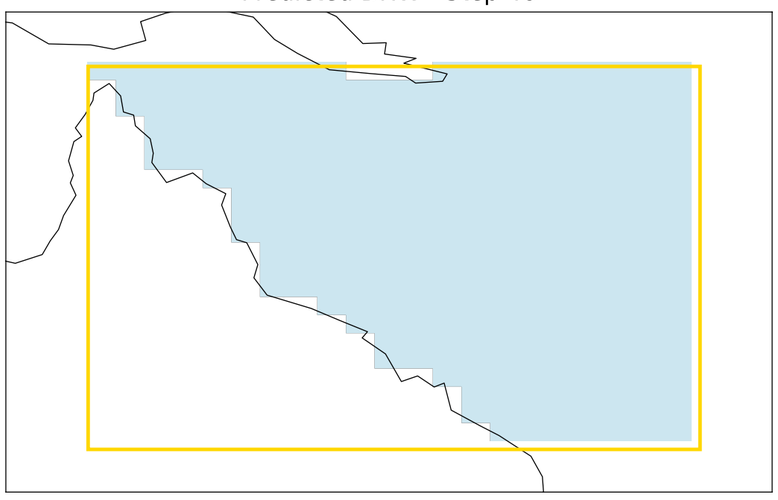
Error (Pred - Actual) - Step 1 [scale ±8]



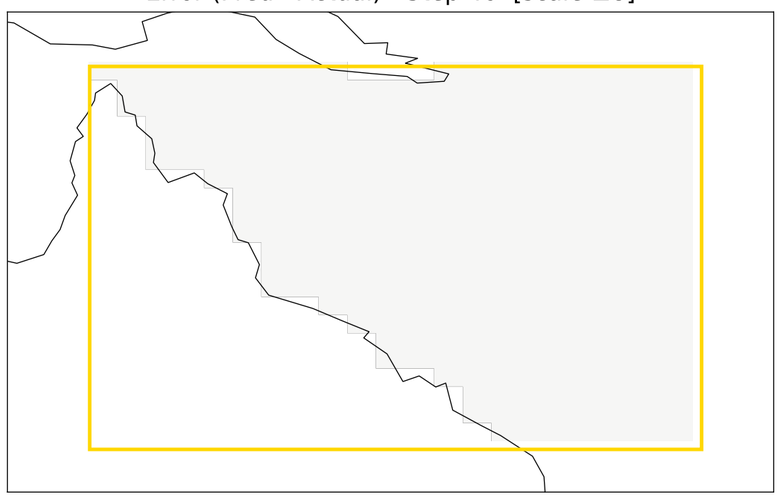
Actual DHW - Step 46



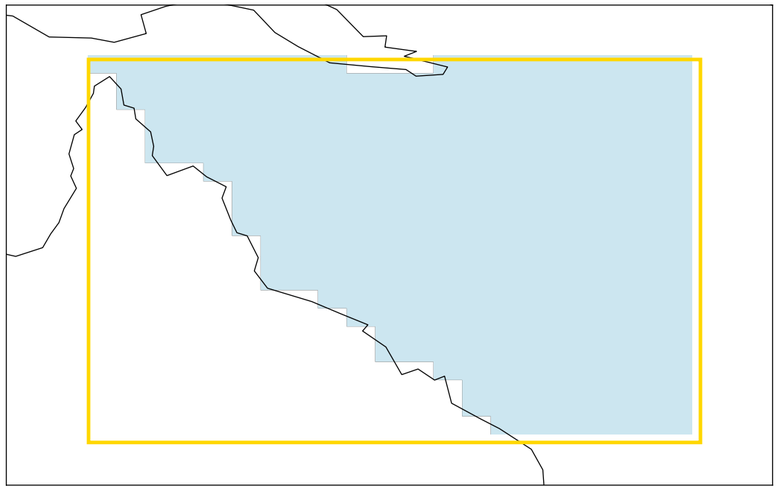
Predicted DHW - Step 46



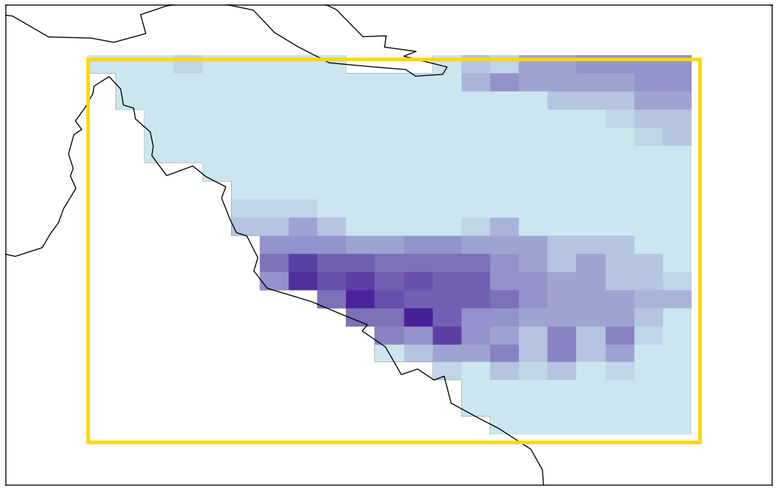
Error (Pred - Actual) - Step 46 [scale ±8]



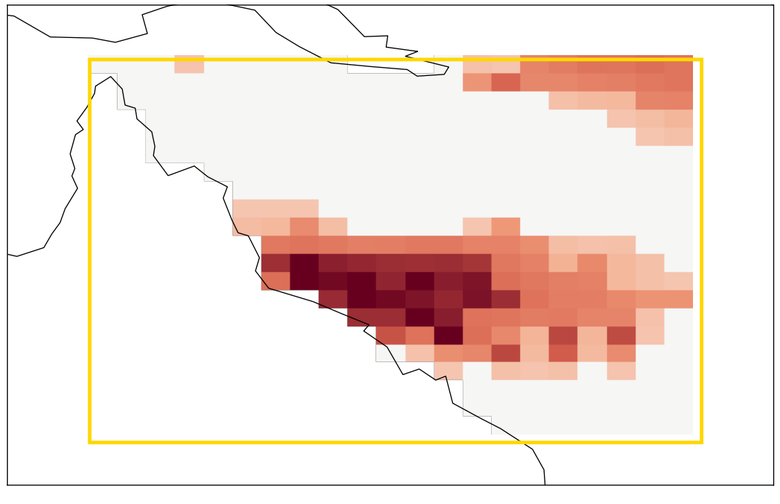
Actual DHW - Step 90

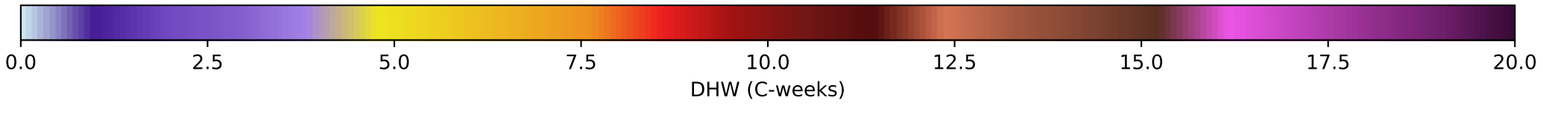


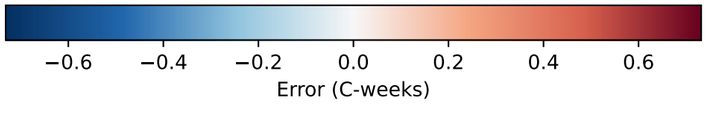
Predicted DHW - Step 90



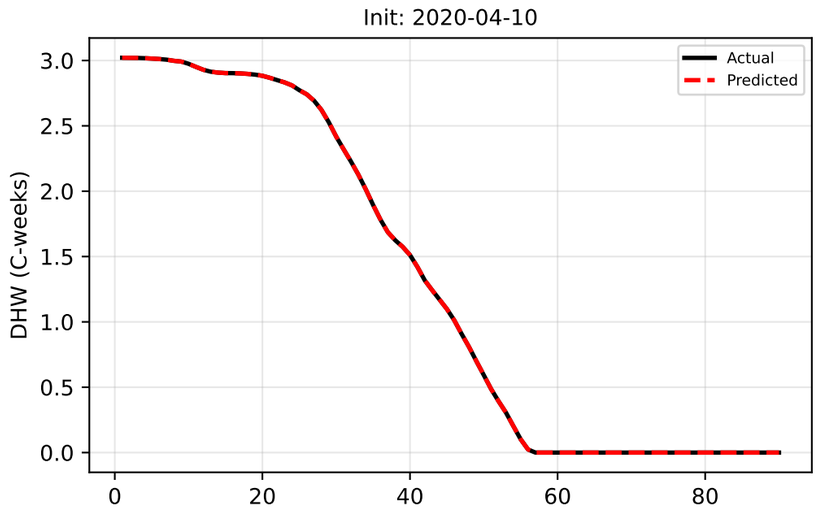
Error (Pred - Actual) - Step 90 [scale ±0.73]

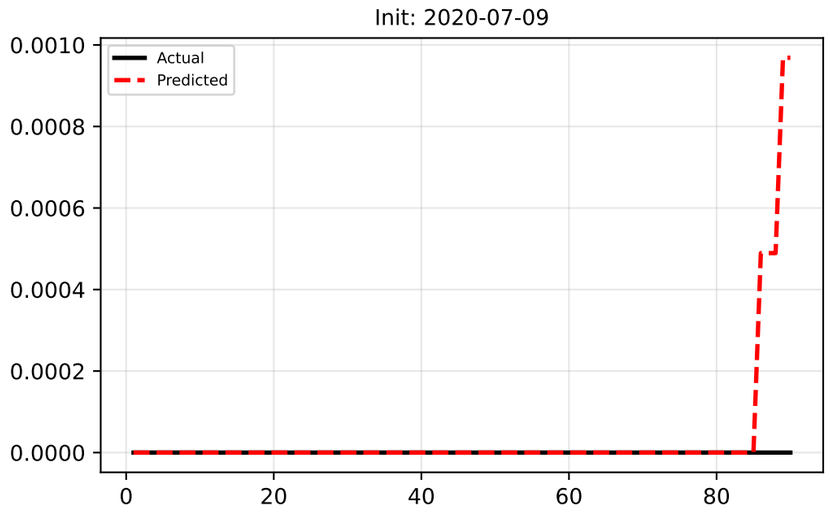


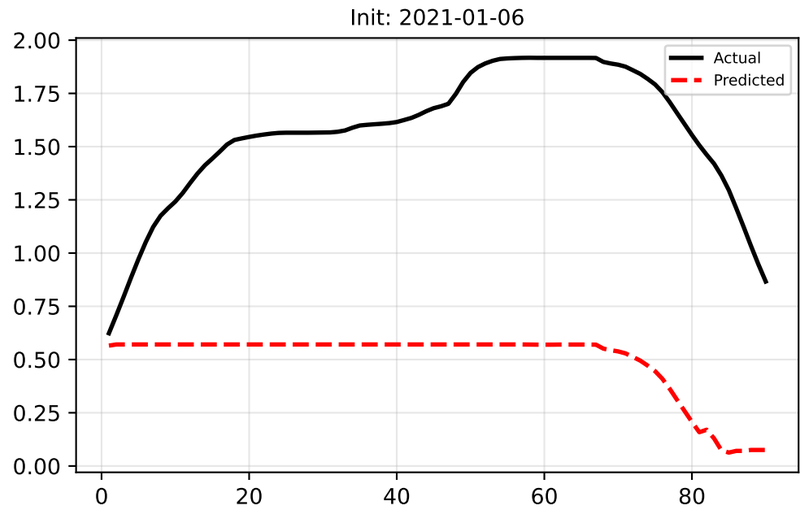


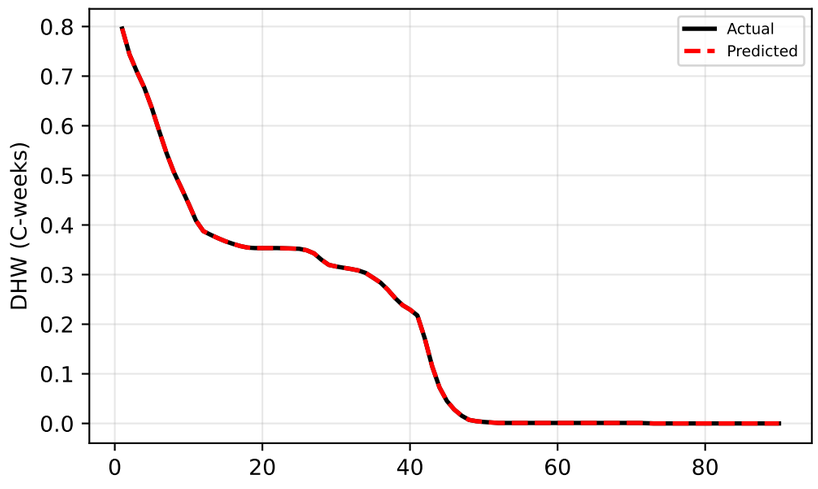


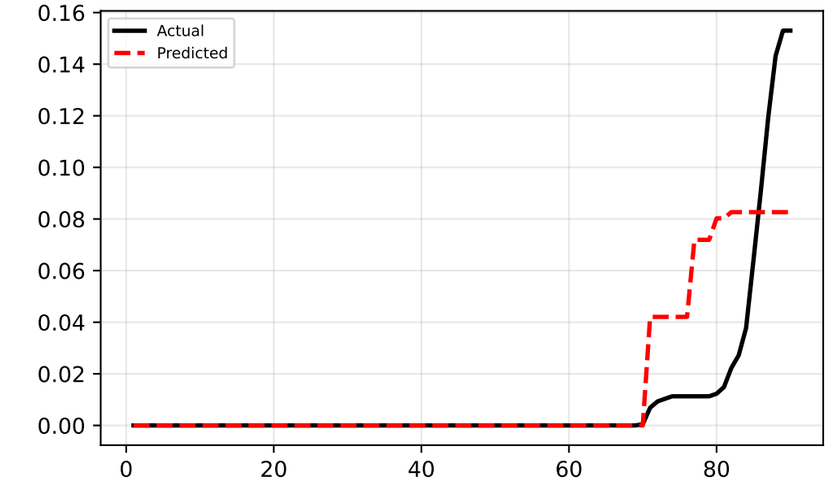
Global-Mean DHW - Test - GBR (C-weeks)

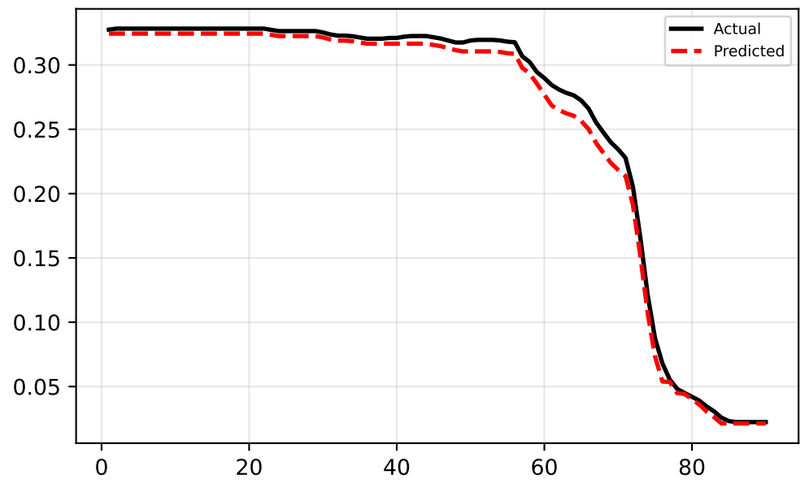


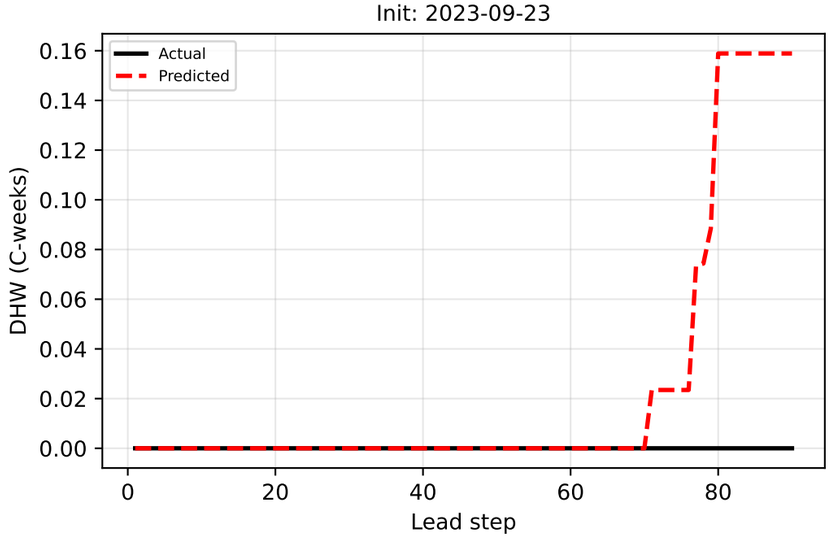


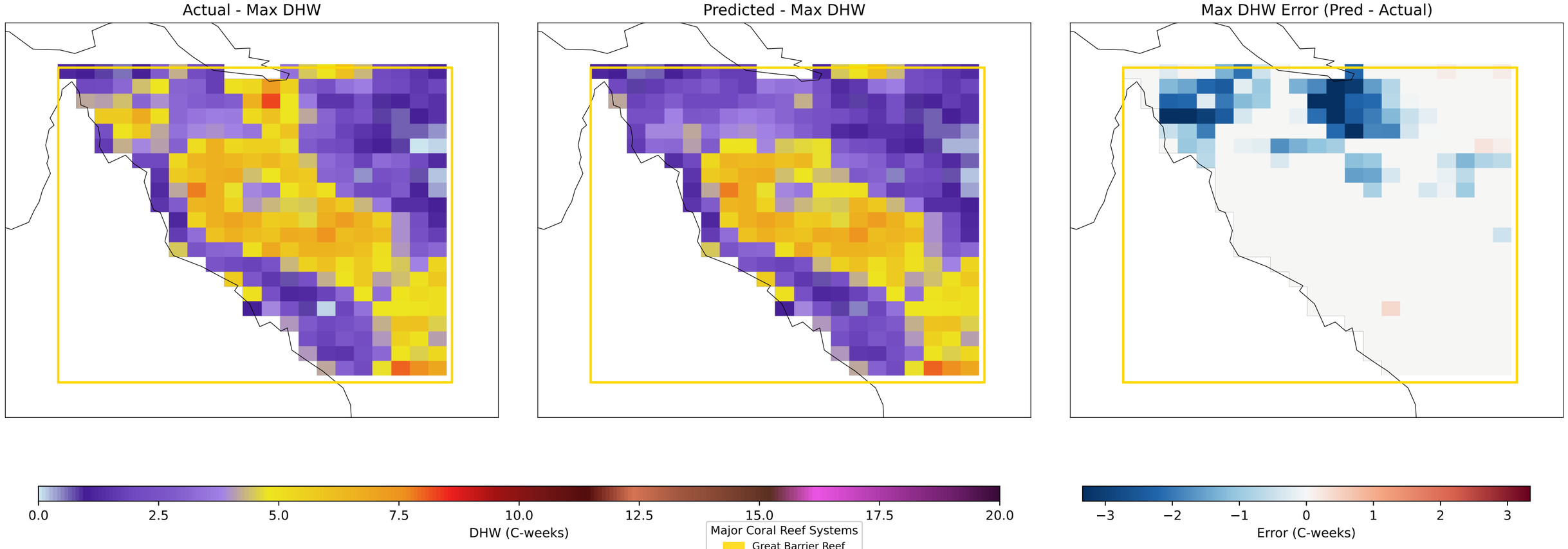


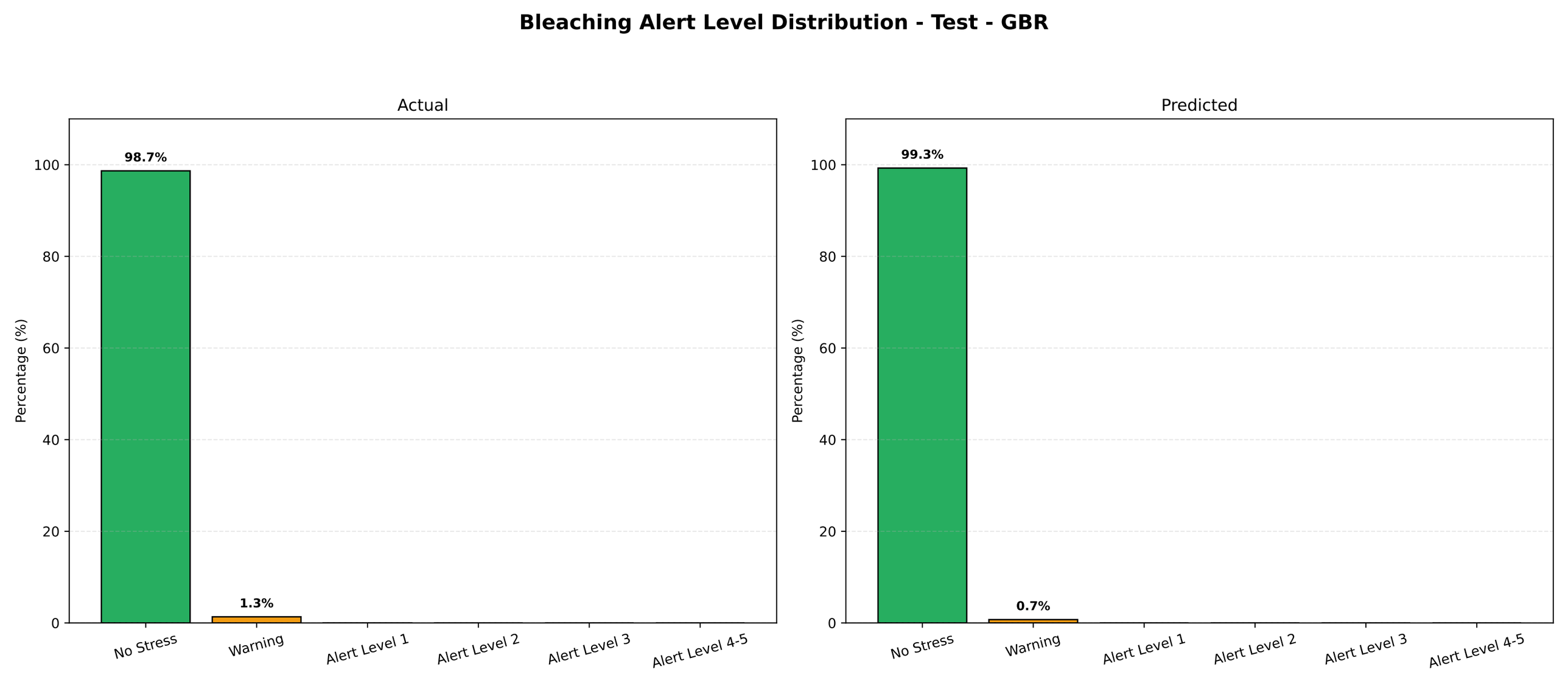






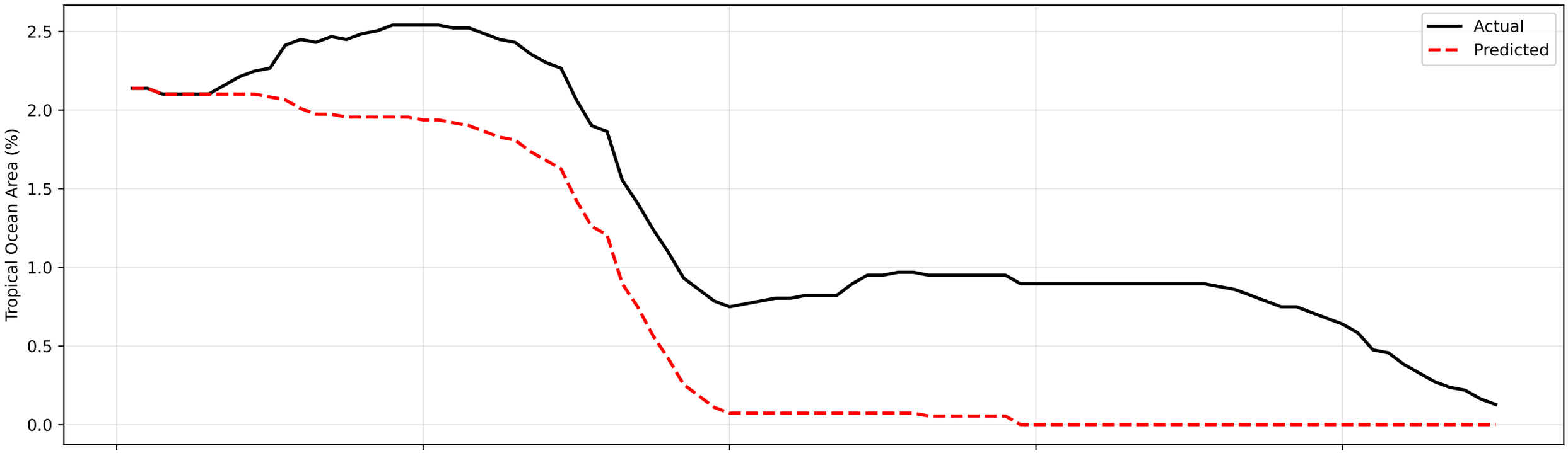



Peak DHW (max over all chains & leads) - Test - GBR (C-weeks)
Actual - Max DHW
Predicted - Max DHW
Max DHW Error (Pred - Actual)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
DHW (C-weeks)
Major Coral Reef Systems
Great Barrier Reef
−3
−2
−1
0
1
2
3
Error (C-weeks)

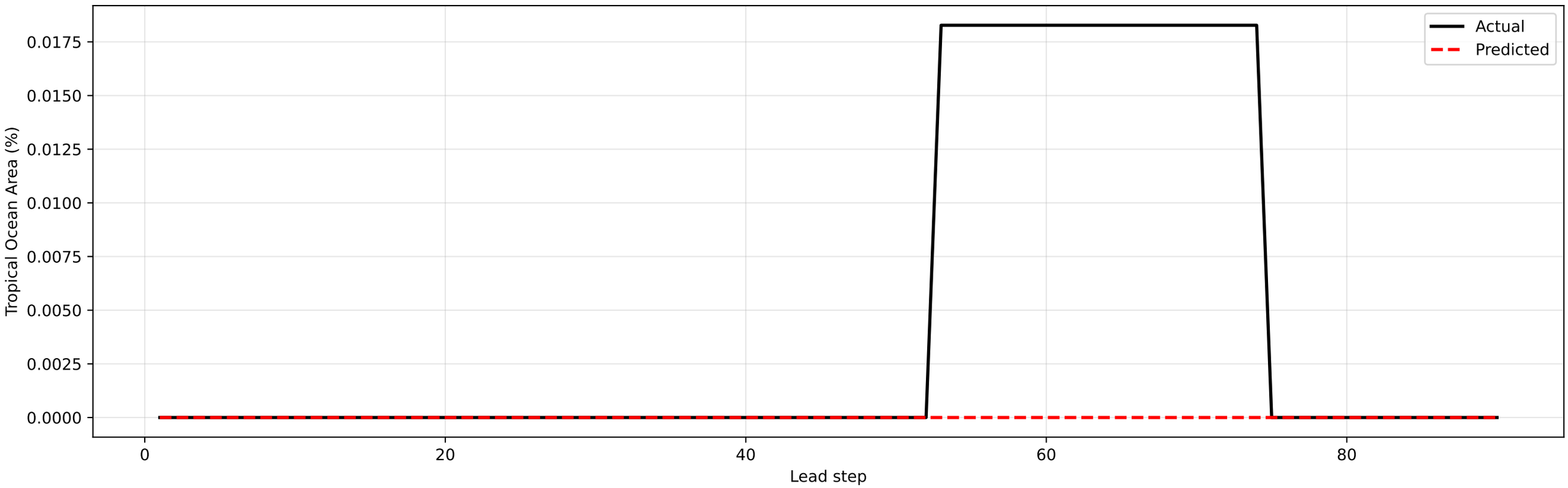

Bleaching Alert Level Distribution - Test - GBR
Actual
Predicted
Percentage (%)
0
20
40
60
80
100
98.7%
1.3%
99.3%
0.7%
No Stress
Warning
Alert Level 1
Alert Level 2
Alert Level 3
Alert Level 4-5

Tropical Ocean Area Under Bleaching Stress - Test - GBR

DHW >= 4 C-weeks (Alert Level 1 DHW threshold)



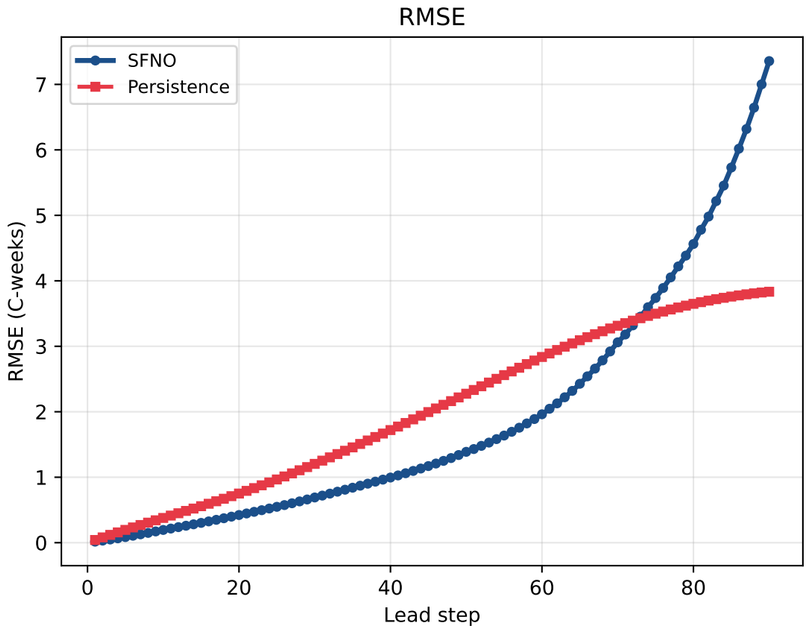
DHW >= 8 C-weeks (Alert Level 2 DHW threshold)

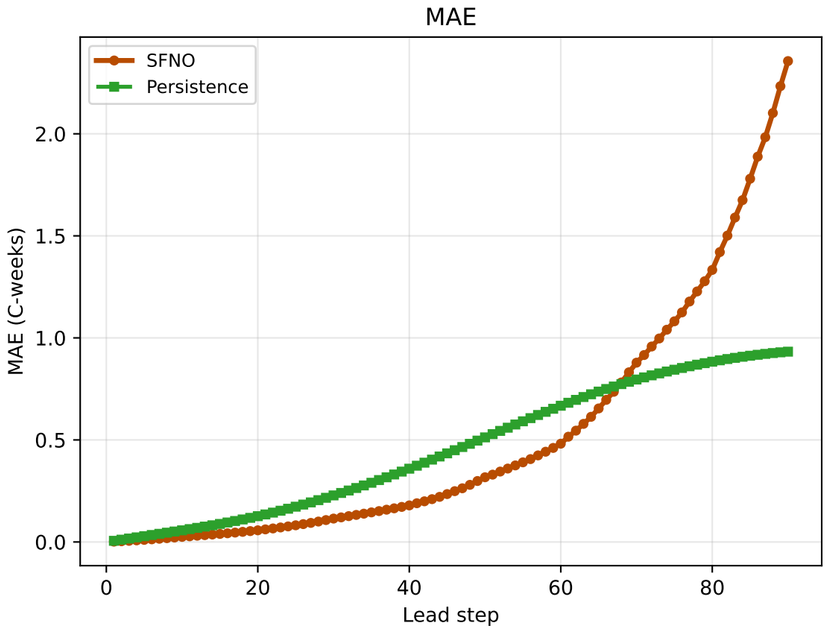


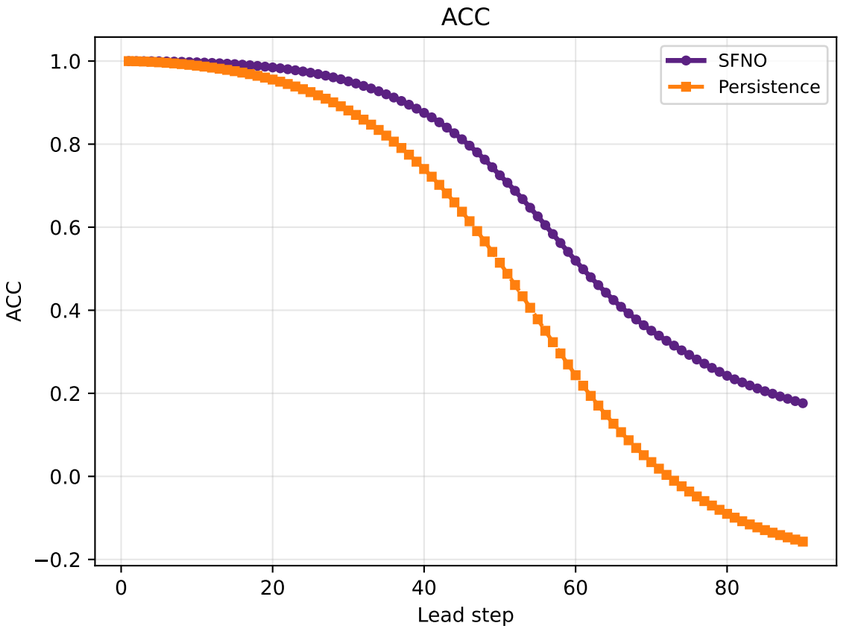
DHW Metrics by Lead Step - Val (15 chains scored at every lead)

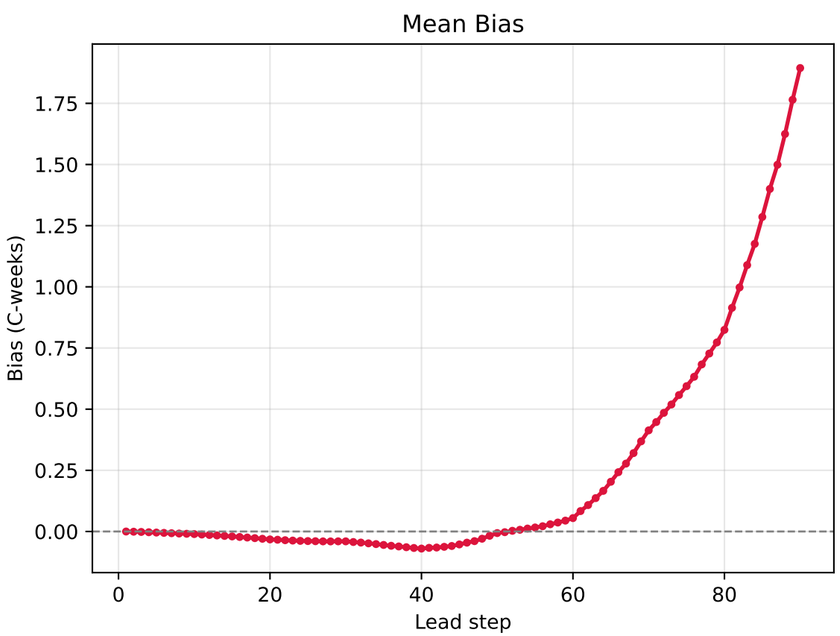
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	Persist MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0163	0.0415	60.6%	0.0017	0.0057	-0.0001	1.0000	0.9999	1.0000
2	0.0320	0.0810	60.5%	0.0035	0.0113	-0.0009	0.9999	0.9995	0.9999
3	0.0491	0.1194	58.9%	0.0056	0.0169	-0.0018	0.9998	0.9989	0.9997
4	0.0668	0.1574	57.5%	0.0078	0.0225	-0.0030	0.9997	0.9981	0.9994
5	0.0859	0.1956	56.1%	0.0103	0.0284	-0.0043	0.9994	0.9970	0.9991
6	0.1064	0.2331	54.4%	0.0130	0.0342	-0.0056	0.9991	0.9958	0.9986
7	0.1278	0.2700	52.7%	0.0158	0.0399	-0.0070	0.9987	0.9944	0.9979
8	0.1496	0.3062	51.1%	0.0187	0.0455	-0.0083	0.9983	0.9927	0.9971
9	0.1720	0.3421	49.7%	0.0219	0.0512	-0.0093	0.9977	0.9909	0.9962
10	0.1947	0.3780	48.5%	0.0250	0.0570	-0.0105	0.9970	0.9889	0.9951
11	0.2160	0.4140	47.8%	0.0278	0.0630	-0.0124	0.9963	0.9867	0.9940
12	0.2378	0.4497	47.1%	0.0307	0.0691	-0.0142	0.9955	0.9843	0.9927
13	0.2591	0.4855	46.6%	0.0336	0.0754	-0.0162	0.9947	0.9816	0.9912
14	0.2807	0.5213	46.1%	0.0365	0.0817	-0.0181	0.9937	0.9788	0.9897
15	0.3027	0.5575	45.7%	0.0394	0.0884	-0.0202	0.9926	0.9757	0.9879
16	0.3253	0.5948	45.3%	0.0426	0.0954	-0.0224	0.9914	0.9723	0.9859
17	0.3492	0.6328	44.8%	0.0461	0.1028	-0.0247	0.9899	0.9686	0.9836
18	0.3734	0.6714	44.4%	0.0496	0.1103	-0.0271	0.9884	0.9645	0.9811
19	0.3979	0.7107	44.0%	0.0533	0.1182	-0.0297	0.9866	0.9602	0.9783
20	0.4226	0.7507	43.7%	0.0571	0.1264	-0.0324	0.9847	0.9555	0.9753
21	0.4465	0.7917	43.6%	0.0617	0.1349	-0.0337	0.9826	0.9504	0.9721
22	0.4715	0.8337	43.5%	0.0663	0.1438	-0.0356	0.9804	0.9448	0.9685
23	0.4965	0.8766	43.4%	0.0711	0.1532	-0.0371	0.9779	0.9388	0.9647
24	0.5221	0.9206	43.3%	0.0762	0.1628	-0.0383	0.9751	0.9323	0.9604
25	0.5479	0.9658	43.3%	0.0817	0.1729	-0.0391	0.9720	0.9252	0.9558
26	0.5745	1.0118	43.2%	0.0875	0.1832	-0.0397	0.9687	0.9176	0.9507
27	0.6023	1.0587	43.1%	0.0935	0.1939	-0.0404	0.9650	0.9094	0.9451
28	0.6307	1.1062	43.0%	0.1001	0.2049	-0.0406	0.9608	0.9006	0.9389
29	0.6602	1.1549	42.8%	0.1073	0.2164	-0.0402	0.9563	0.8911	0.9321
30	0.6907	1.2039	42.6%	0.1146	0.2282	-0.0403	0.9512	0.8811	0.9247
31	0.7194	1.2534	42.6%	0.1205	0.2401	-0.0430	0.9460	0.8703	0.9171
32	0.7495	1.3033	42.5%	0.1266	0.2521	-0.0453	0.9402	0.8590	0.9087
33	0.7792	1.3536	42.4%	0.1325	0.2644	-0.0484	0.9340	0.8469	0.8999
34	0.8094	1.4043	42.4%	0.1387	0.2770	-0.0515	0.9273	0.8342	0.8903
35	0.8400	1.4559	42.3%	0.1450	0.2901	-0.0549	0.9200	0.8206	0.8801
36	0.8706	1.5083	42.3%	0.1514	0.3034	-0.0584	0.9122	0.8060	0.8694
37	0.9015	1.5611	42.3%	0.1581	0.3169	-0.0611	0.9039	0.7907	0.8579
38	0.9323	1.6141	42.2%	0.1648	0.3307	-0.0643	0.8950	0.7746	0.8459
39	0.9635	1.6671	42.2%	0.1718	0.3448	-0.0673	0.8855	0.7578	0.8331
40	0.9953	1.7202	42.1%	0.1792	0.3591	-0.0698	0.8753	0.7402	0.8196
41	1.0274	1.7738	42.1%	0.1898	0.3737	-0.0668	0.8645	0.7216	0.8054
42	1.0612	1.8284	42.0%	0.1992	0.3885	-0.0655	0.8526	0.7020	0.7899
43	1.0963	1.8837	41.8%	0.2098	0.4036	-0.0627	0.8398	0.6813	0.7733
44	1.1328	1.9395	41.6%	0.2210	0.4188	-0.0593	0.8261	0.6596	0.7556
45	1.1703	1.9951	41.3%	0.2347	0.4342	-0.0529	0.8116	0.6373	0.7367
46	1.2092	2.0504	41.0%	0.2490	0.4497	-0.0456	0.7963	0.6143	0.7167
47	1.2496	2.1061	40.7%	0.2631	0.4653	-0.0394	0.7801	0.5906	0.6953
48	1.2927	2.1623	40.2%	0.2799	0.4809	-0.0295	0.7628	0.5659	0.6721
49	1.3391	2.2188	39.6%	0.2987	0.4966	-0.0175	0.7443	0.5405	0.6467
50	1.3879	2.2756	39.0%	0.3171	0.5125	-0.0064	0.7250	0.5145	0.6195
51	1.4305	2.3325	38.7%	0.3299	0.5284	-0.0028	0.7072	0.4878	0.5951
52	1.4790	2.3893	38.1%	0.3450	0.5442	0.0028	0.6876	0.4608	0.5669
53	1.5290	2.4458	37.5%	0.3597	0.5599	0.0070	0.6676	0.4336	0.5377
54	1.5823	2.5028	36.8%	0.3751	0.5756	0.0120	0.6468	0.4060	0.5062
55	1.6365	2.5595	36.1%	0.3902	0.5913	0.0164	0.6261	0.3782	0.4740
56	1.6939	2.6160	35.2%	0.4061	0.6068	0.0215	0.6050	0.3506	0.4398
57	1.7563	2.6721	34.3%	0.4242	0.6223	0.0295	0.5835	0.3231	0.4024
58	1.8215	2.7277	33.2%	0.4422	0.6376	0.0366	0.5619	0.2960	0.3631
59	1.8887	2.7824	32.1%	0.4608	0.6527	0.0444	0.5406	0.2695	0.3223
60	1.9614	2.8367	30.9%	0.4815	0.6676	0.0547	0.5194	0.2436	0.2778
61	2.0458	2.8903	29.2%	0.5158	0.6821	0.0835	0.4986	0.2183	0.2245
62	2.1283	2.9429	27.7%	0.5455	0.6963	0.1080	0.4795	0.1938	0.1723
63	2.2212	2.9940	25.8%	0.5787	0.7102	0.1365	0.4605	0.1705	0.1118
64	2.3187	3.0438	23.8%	0.6133	0.7237	0.1661	0.4424	0.1481	0.0466
65	2.4274	3.0919	21.5%	0.6543	0.7369	0.2032	0.4246	0.1268	-0.0290
66	2.5409	3.1389	19.1%	0.6964	0.7496	0.2424	0.4082	0.1064	-0.1101
67	2.6588	3.1847	16.5%	0.7361	0.7617	0.2775	0.3923	0.0870	-0.1963
68	2.7844	3.2293	13.8%	0.7815	0.7736	0.3204	0.3779	0.0685	-0.2910
69	2.9211	3.2721	10.7%	0.8314	0.7850	0.3685	0.3639	0.0510	-0.3987
70	3.0606	3.3135	7.6%	0.8786	0.7959	0.4134	0.3508	0.0343	-0.5119
71	3.1800	3.3532	5.2%	0.9160	0.8064	0.4474	0.3388	0.0185	-0.6079
72	3.3194	3.3916	2.1%	0.9578	0.8165	0.4849	0.3265	0.0036	-0.7258
73	3.4524	3.4286	-0.7%	0.9971	0.8260	0.5191	0.3149	-0.0105	-0.8397
74	3.5967	3.4642	-3.8%	1.0400	0.8352	0.5577	0.3035	-0.0239	-0.9682
75	3.7377	3.4981	-6.8%	1.0813	0.8441	0.5940	0.2926	-0.0365	-1.0965
76	3.8901	3.5305	-10.2%	1.1250	0.8526	0.6324	0.2814	-0.0483	-1.2410
77	4.0528	3.5621	-13.8%	1.1782	0.8609	0.6828	0.2715	-0.0595	-1.3997
78	4.2200	3.5923	-17.5%	1.2277	0.8687	0.7274	0.2612	-0.0702	-1.5691
79	4.3847	3.6210	-21.1%	1.2776	0.8762	0.7727	0.2517	-0.0805	-1.7414
80	4.5623	3.6483	-25.1%	1.3330	0.8833	0.8241	0.2423	-0.0902	-1.9358
81	4.7800	3.6740	-30.1%	1.4204	0.8901	0.9141	0.2337	-0.0994	-2.1905
82	4.9812	3.6978	-34.7%	1.5013	0.8964	0.9976	0.2264	-0.1080	-2.4329
83	5.2169	3.7198	-40.2%	1.5897	0.9022	1.0887	0.2189	-0.1157	-2.7333
84	5.4544	3.7403	-45.8%	1.6752	0.9076	1.1756	0.2118	-0.1228	-3.0472
85	5.7302	3.7593	-52.4%	1.7799	0.9126	1.2856	0.2051	-0.1294	-3.4321
86	6.0178	3.7772	-59.3%	1.8881	0.9172	1.4002	0.1991	-0.1356	-3.8528
87	6.3183	3.7938	-66.5%	1.9835	0.9216	1.4989	0.1924	-0.1414	-4.3138
88	6.6445	3.8087	-74.5%	2.1019	0.9257	1.6247	0.1869	-0.1469	-4.8455
89	7.0011	3.8217	-83.2%	2.2335	0.9293	1.7646	0.1813	-0.1521	-5.4652
90	7.3561	3.8331	-91.9%	2.3564	0.9325	1.8943	0.1762	-0.1571	-6.1187

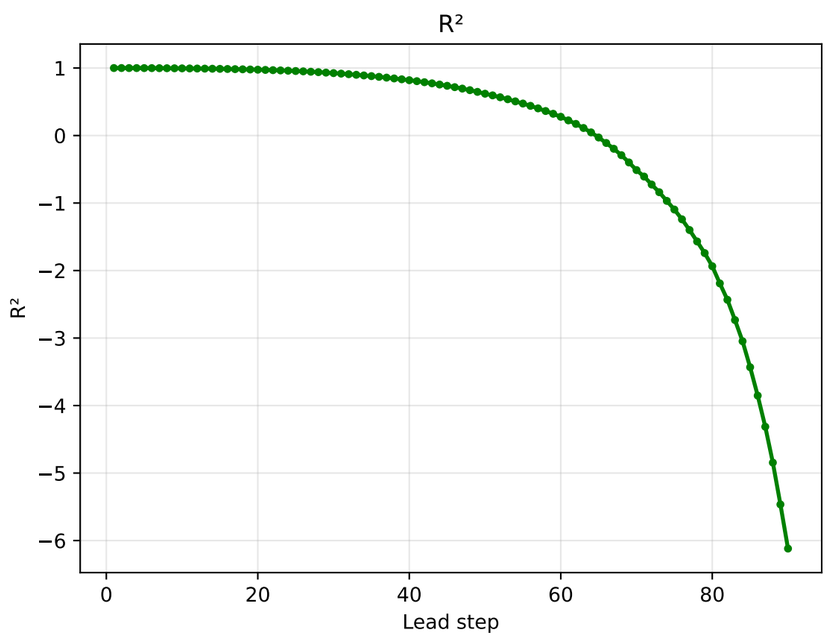
DHW Forecast Skill vs Lead Step - Val

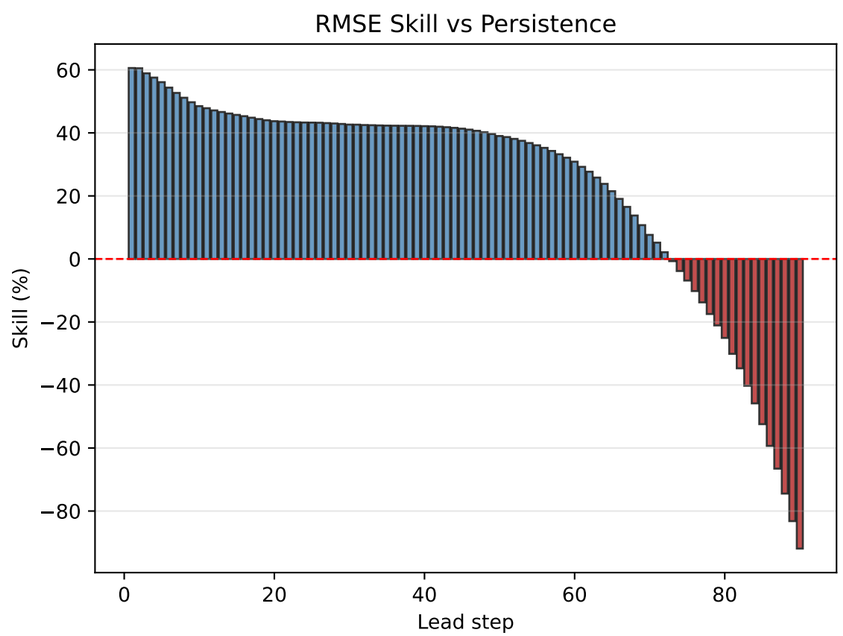


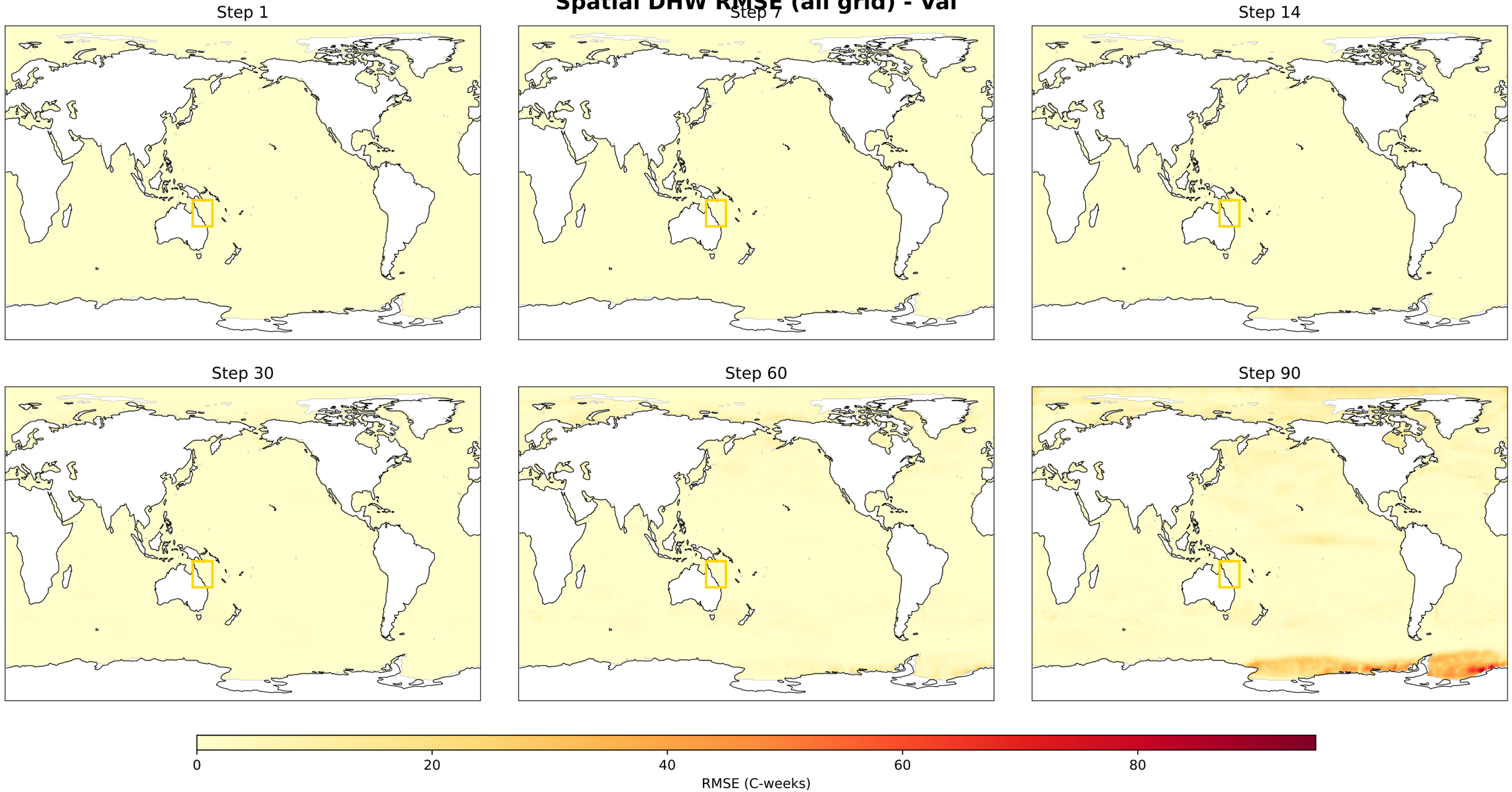


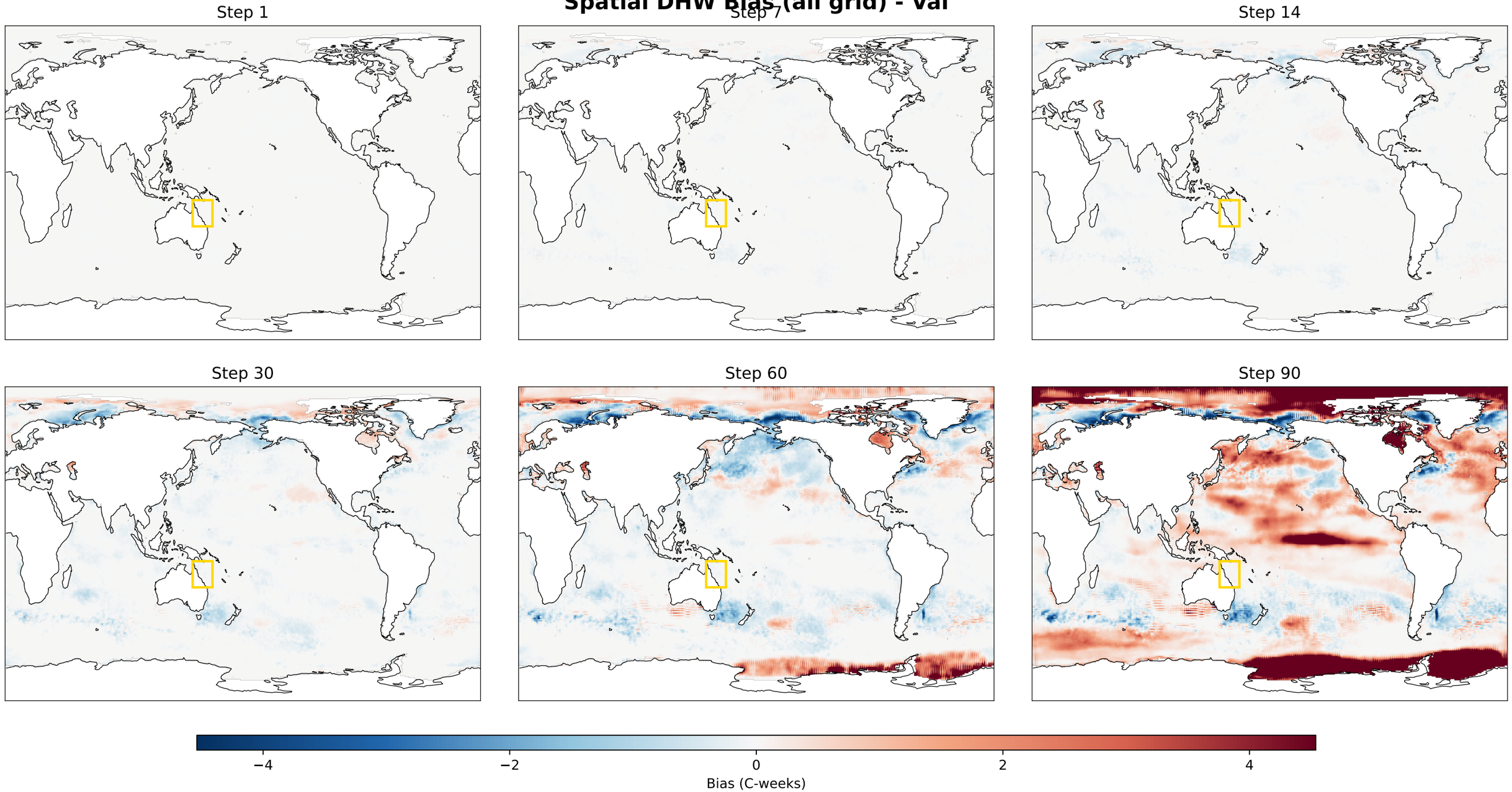


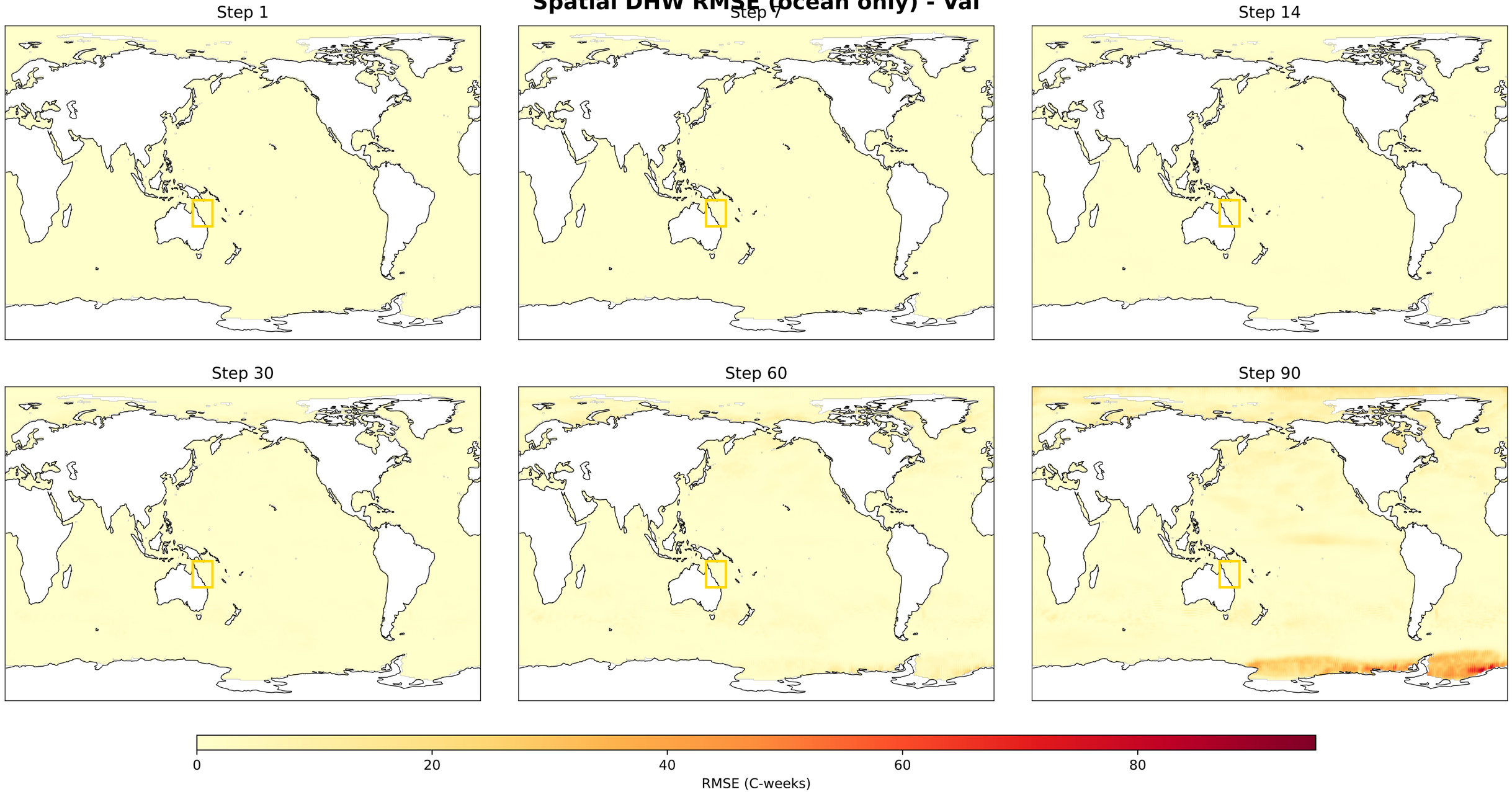


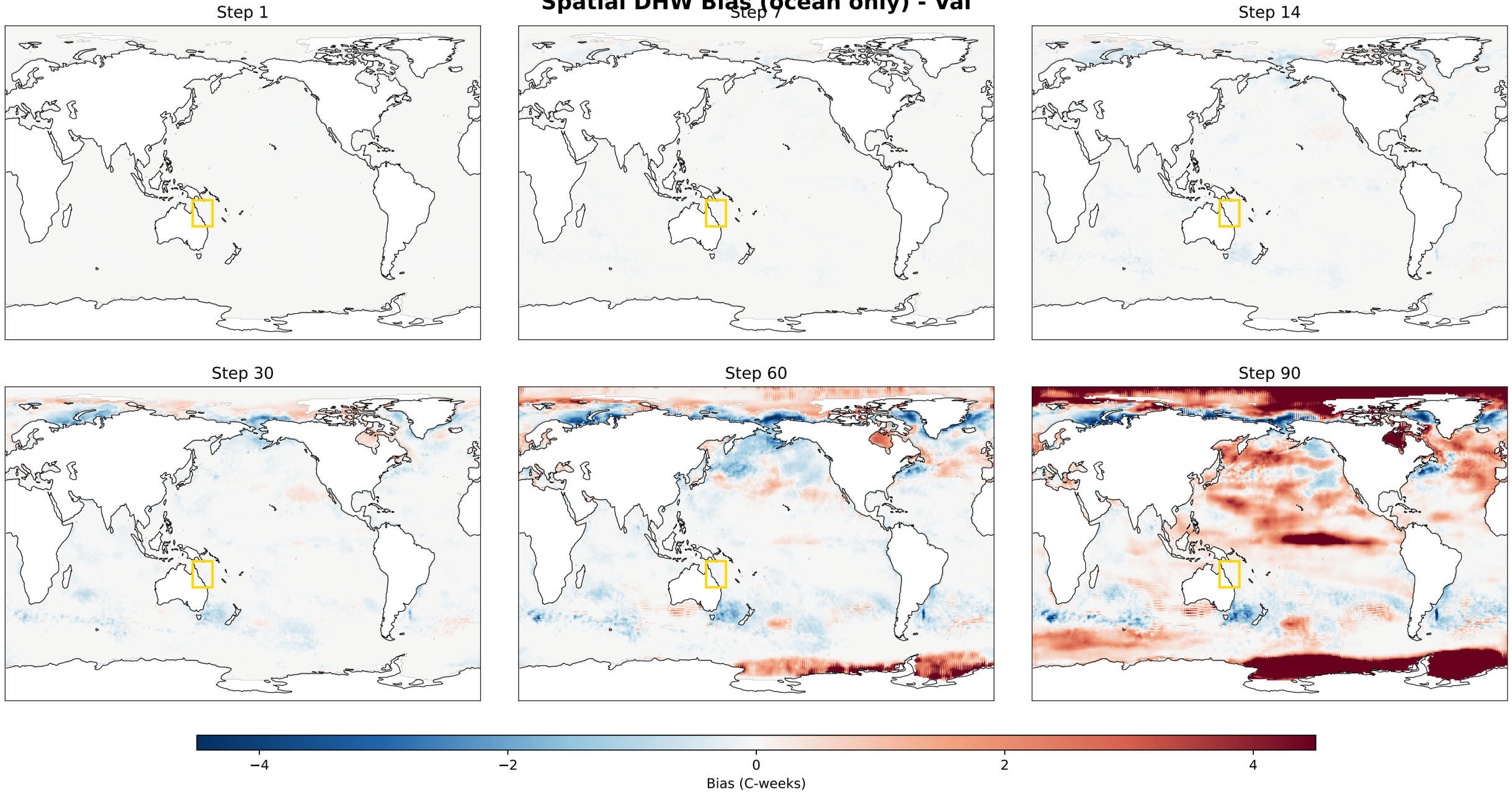





Spatial DHW RMSE (all grid) - Val
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0
20
40
60
80
RMSE (C-weeks)

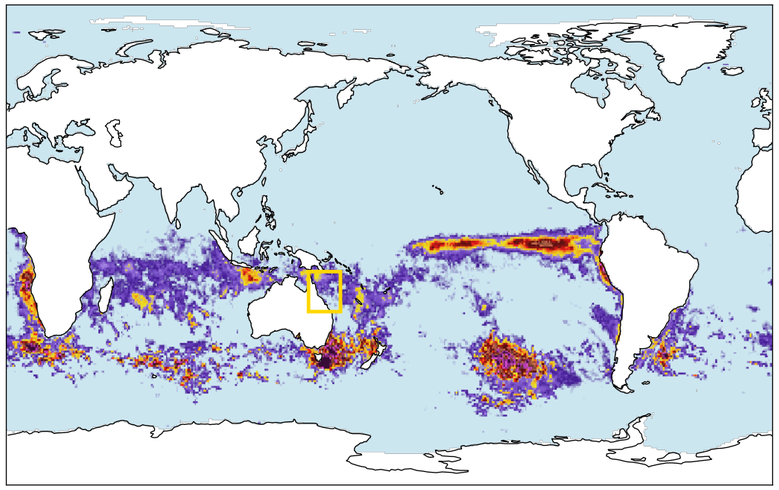

Spatial DHW Bias (all grid) - Val
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−4
−2
0
2
4
Bias (C-weeks)


Spatial DHW RMSE (ocean only) - Val
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0
20
40
60
80
RMSE (C-weeks)

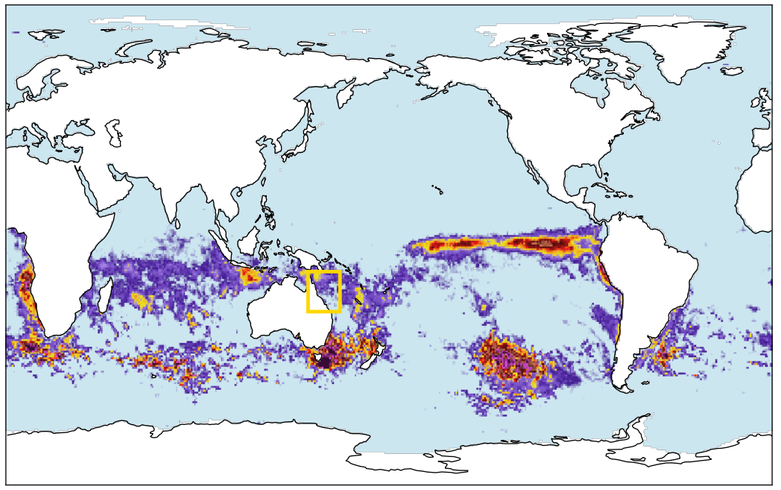

Spatial DHW Bias (ocean only) - Val
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−4
−2
0
2
4
Bias (C-weeks)

DHW Chain init: 2016-04-10 - Val (C-weeks)

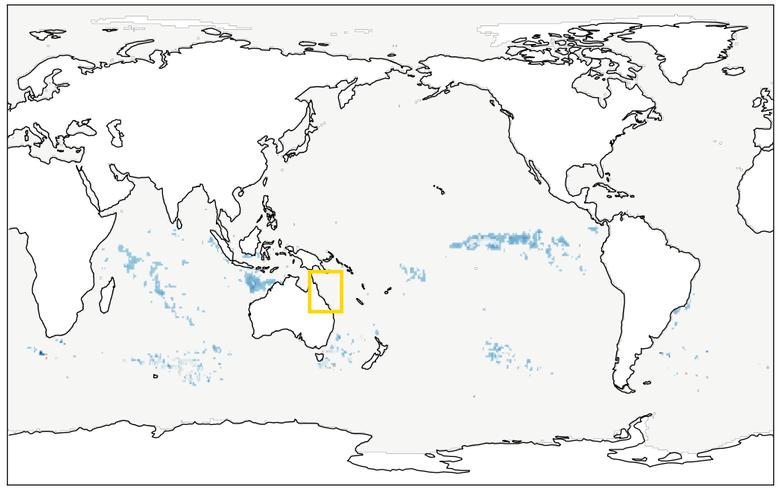
Actual DHW - Step 1



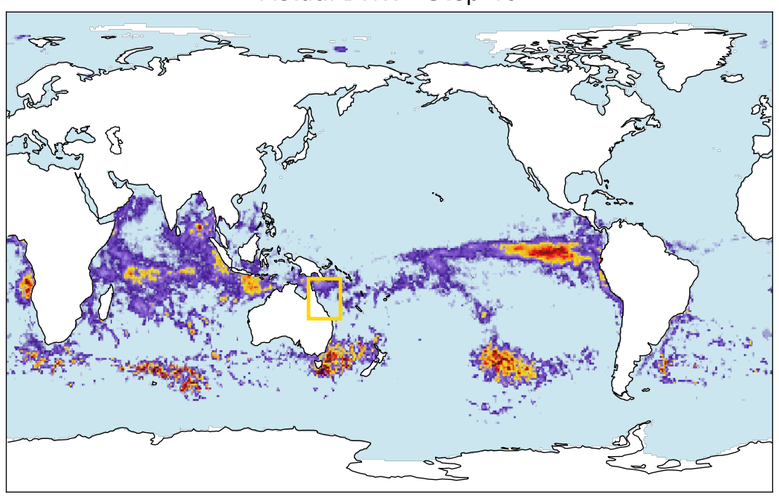
Predicted DHW - Step 1



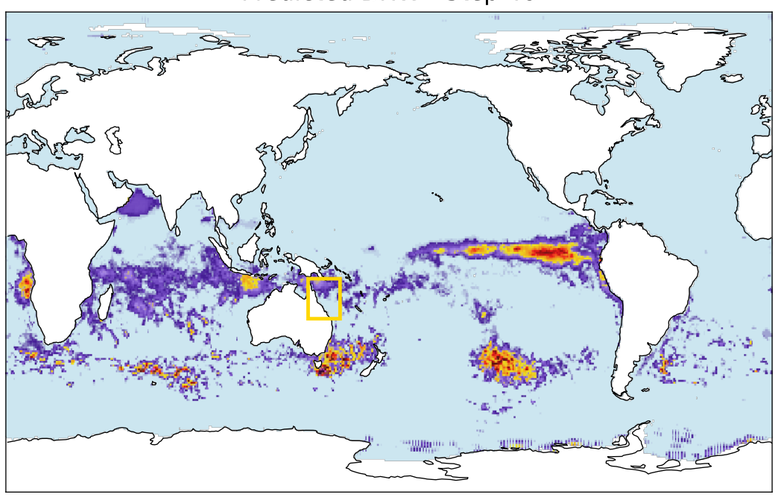
Error (Pred - Actual) - Step 1 [scale ±0.53]



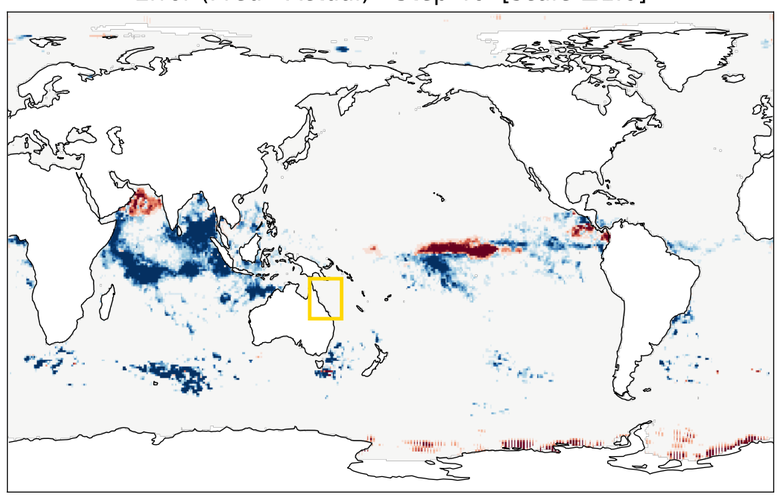
Actual DHW - Step 46



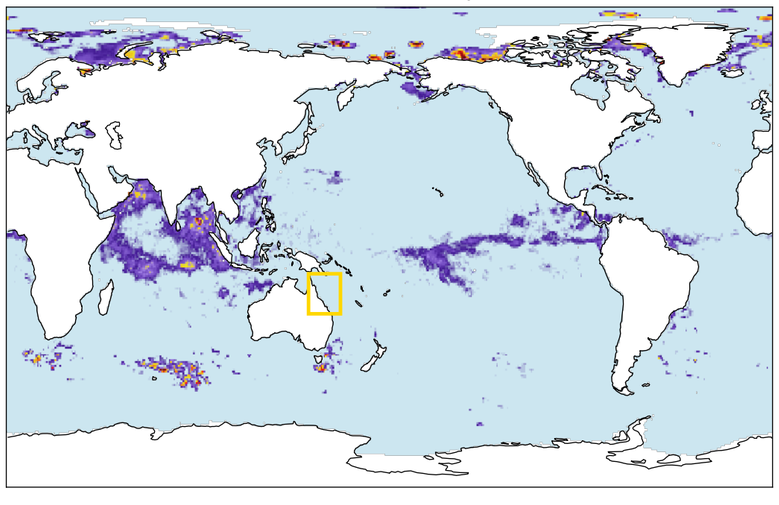
Predicted DHW - Step 46



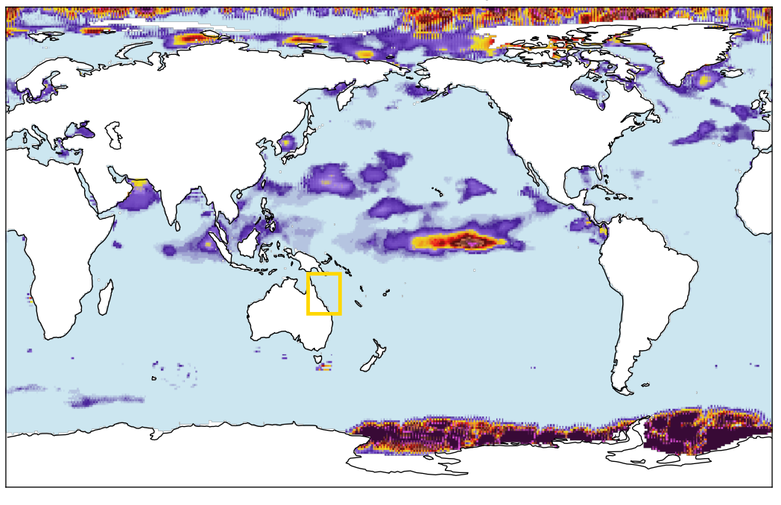
Error (Pred - Actual) - Step 46 [scale ±1.6]



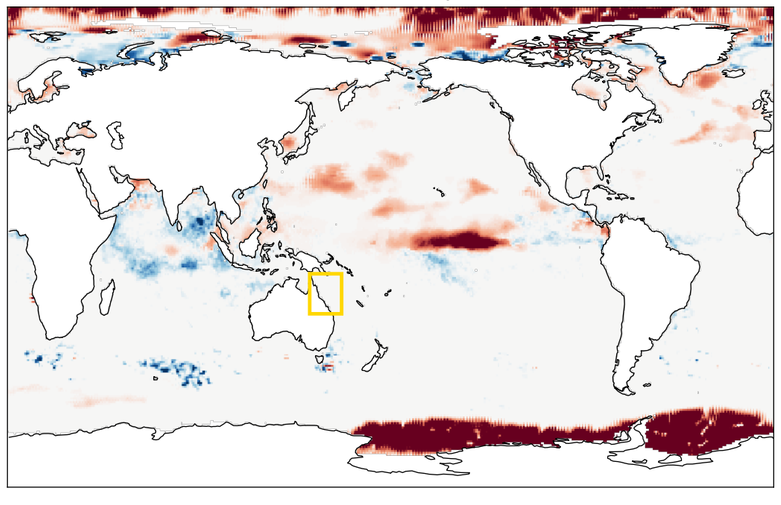
Actual DHW - Step 90

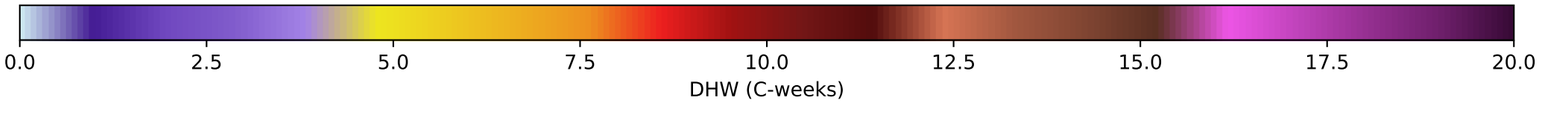


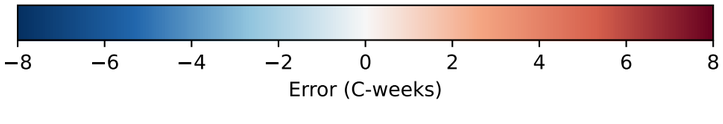
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

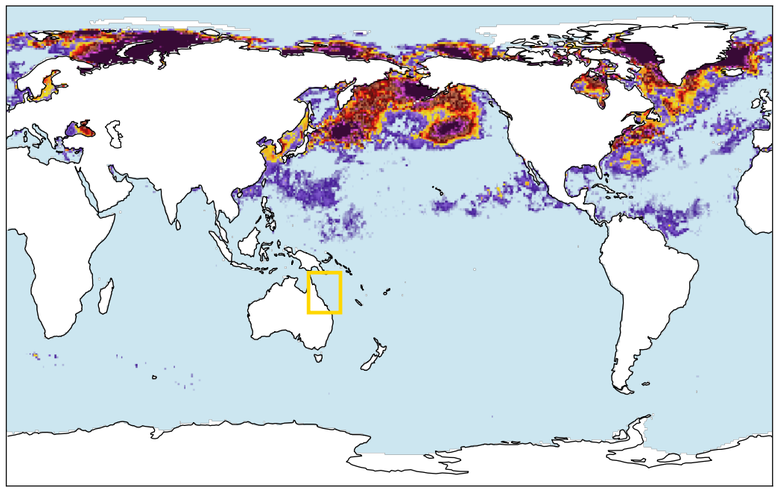




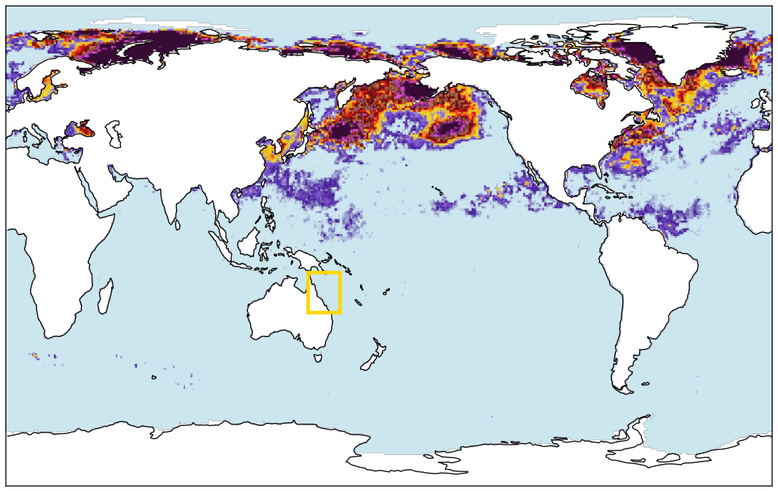


DHW Chain init: 2016-10-07 - Val (C-weeks)

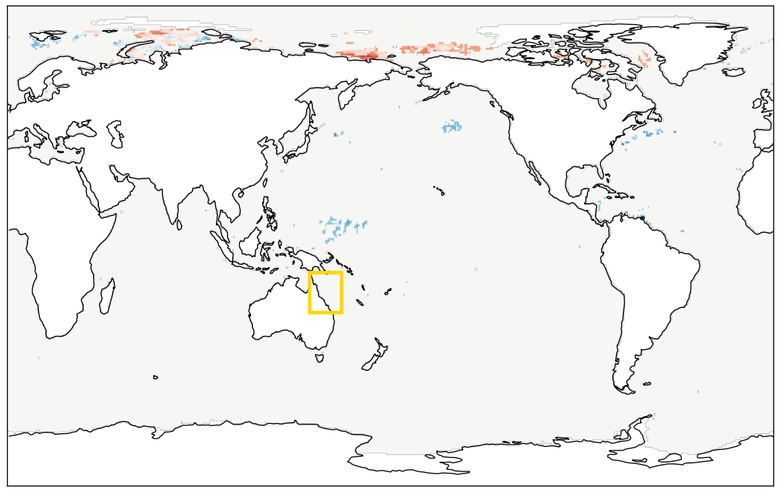
Actual DHW - Step 1



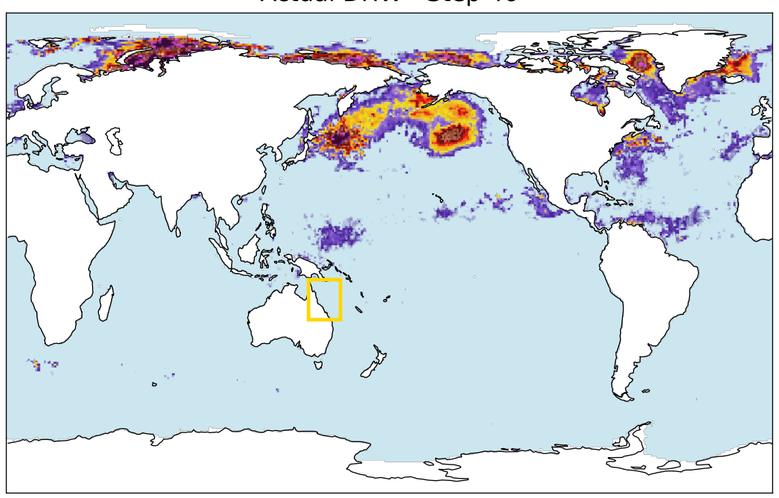
Predicted DHW - Step 1



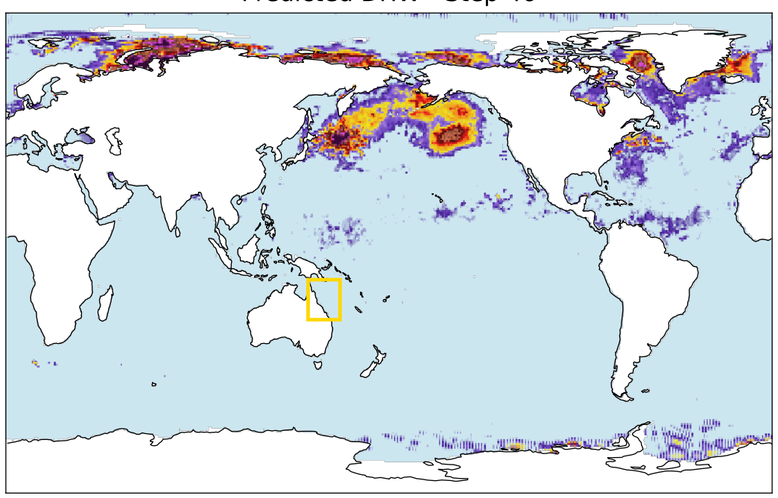
Error (Pred - Actual) - Step 1 [scale ±0.48]



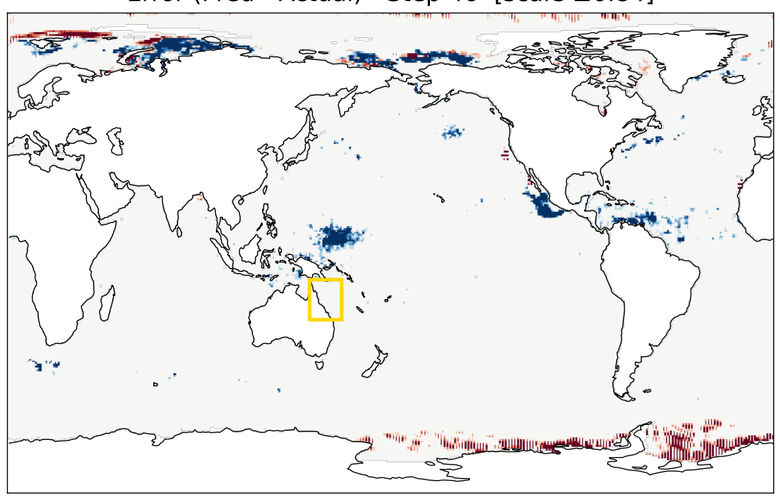
Actual DHW - Step 46



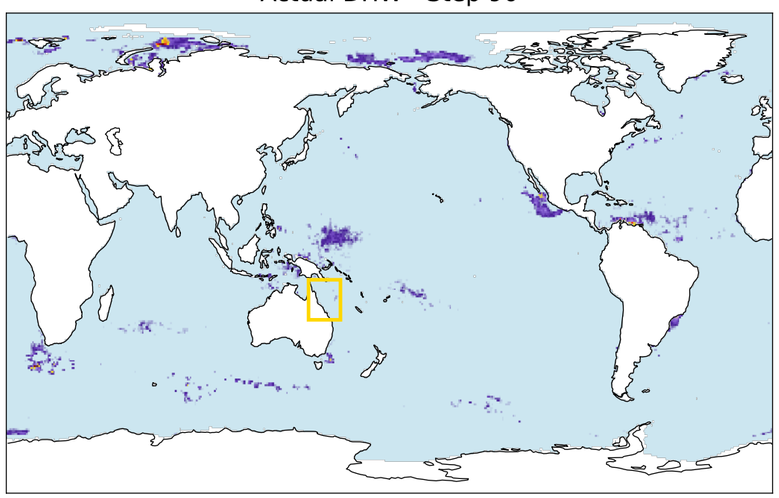
Predicted DHW - Step 46



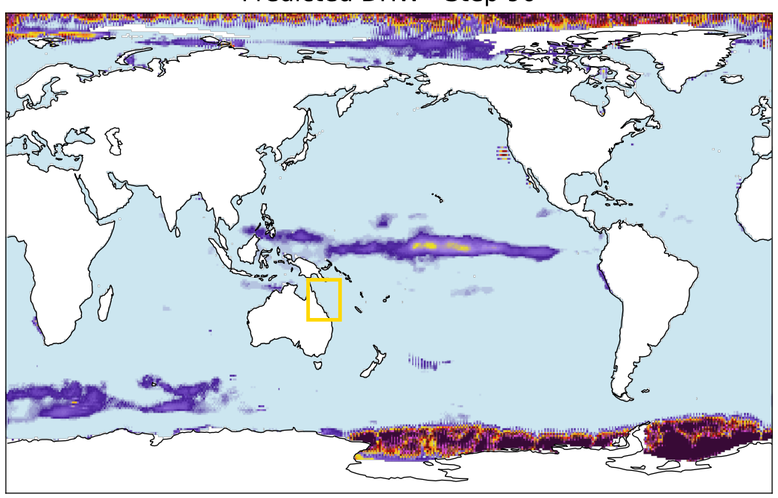
Error (Pred - Actual) - Step 46 [scale ±0.84]



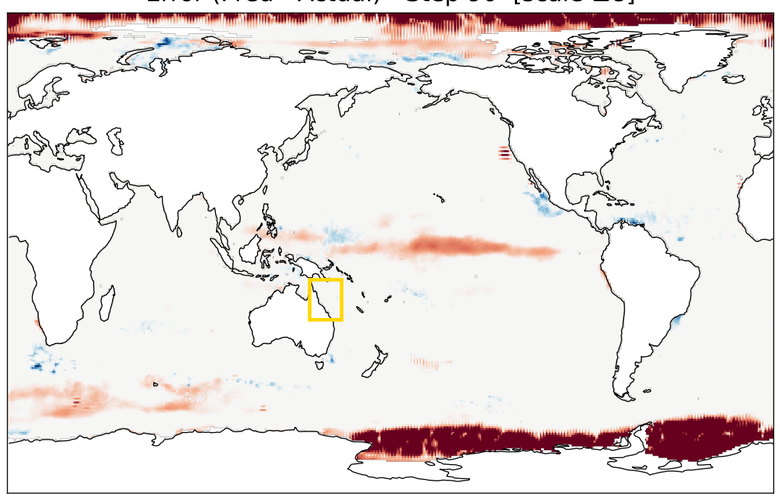
Actual DHW - Step 90

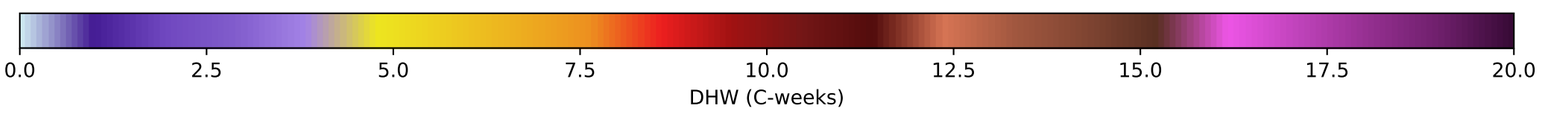


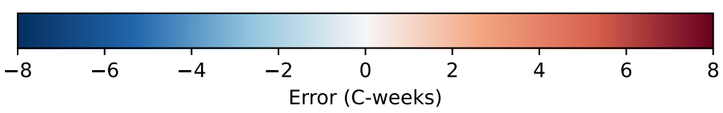
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

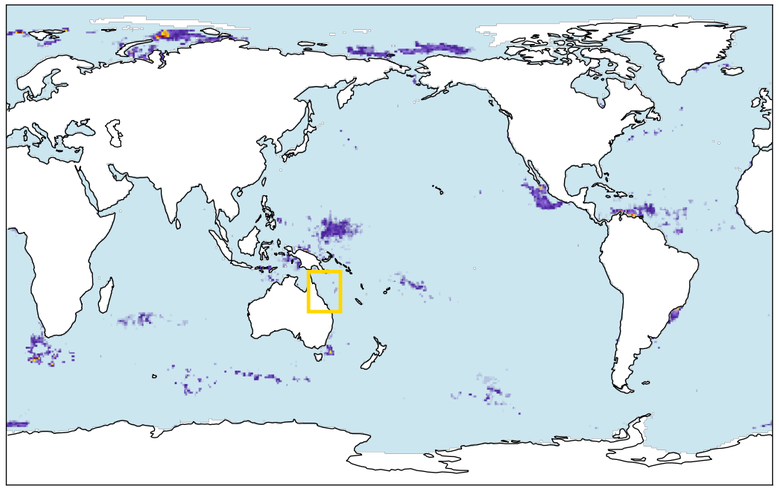




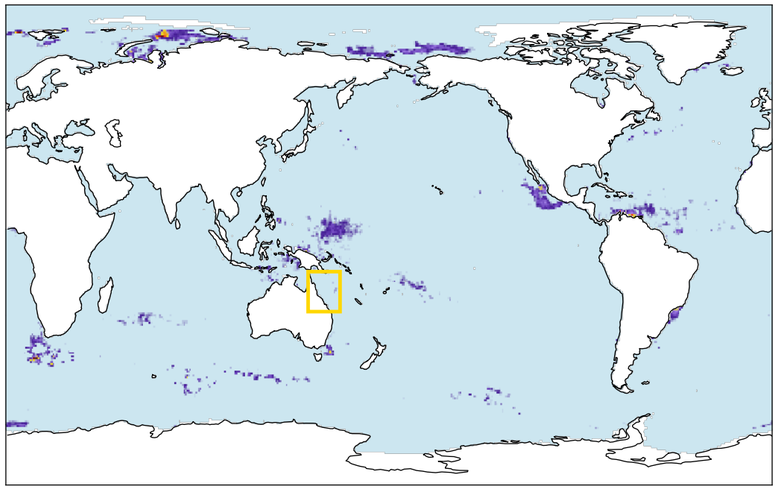


DHW Chain init: 2017-01-05 - Val (C-weeks)

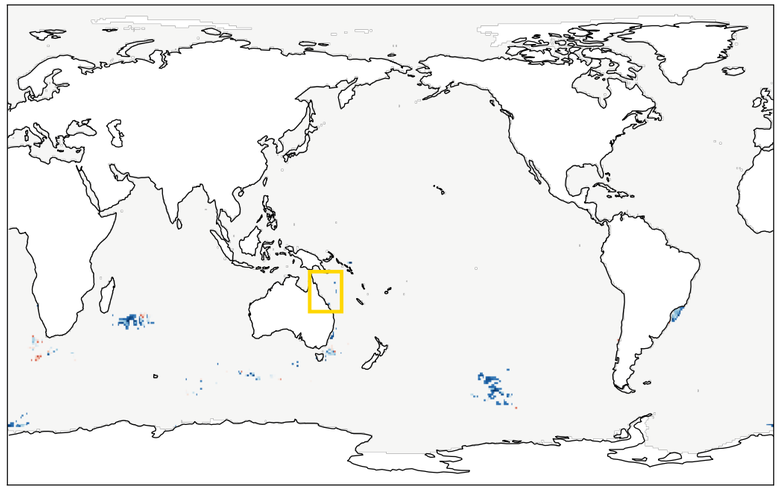
Actual DHW - Step 1



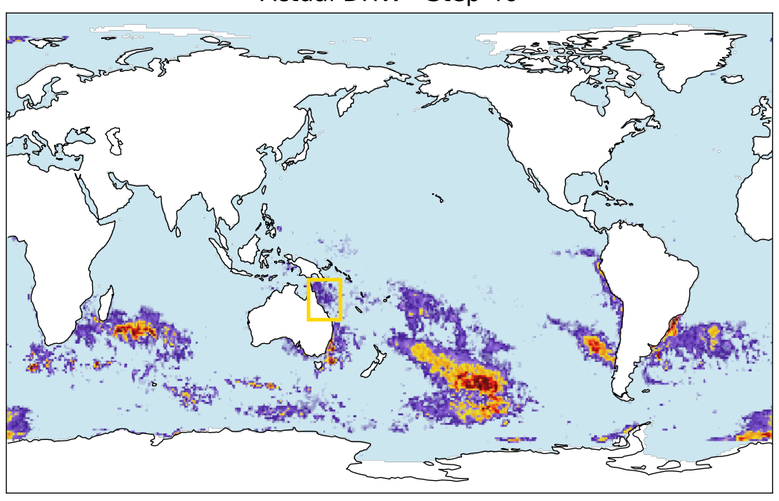
Predicted DHW - Step 1



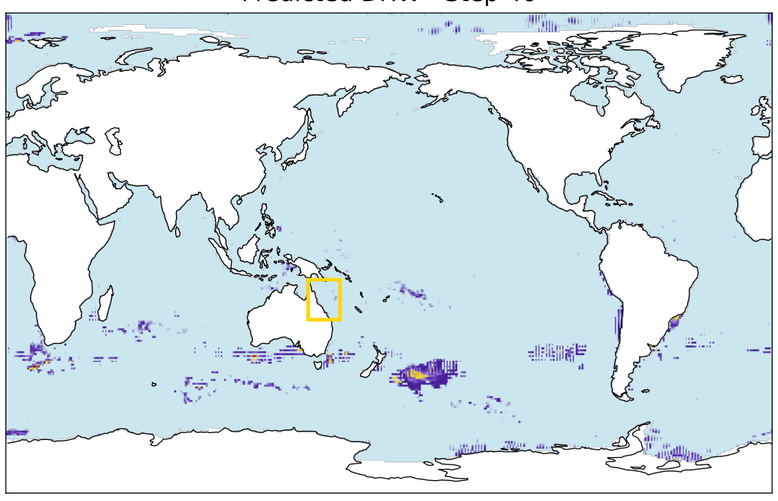
Error (Pred - Actual) - Step 1 [scale ±0.26]



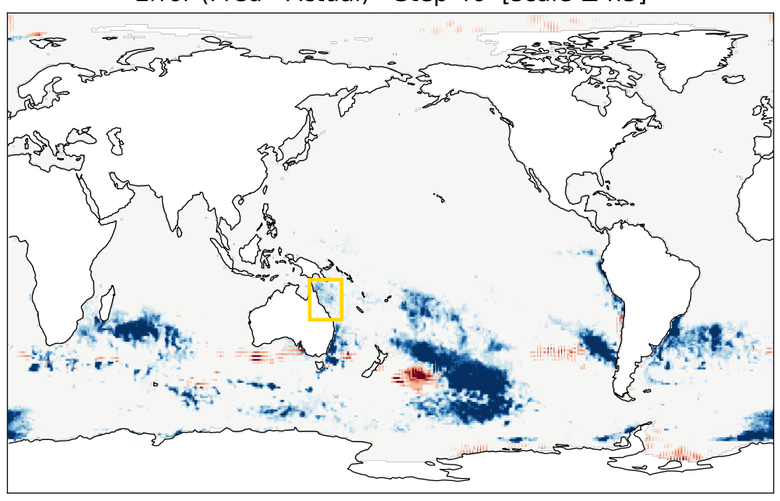
Actual DHW - Step 46



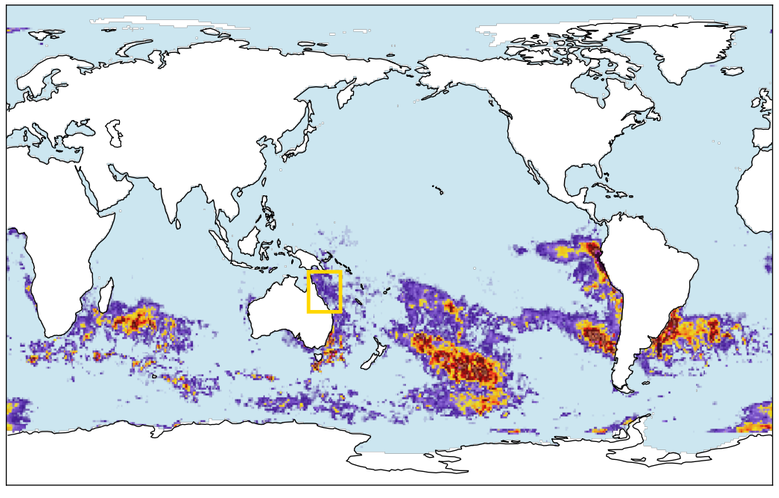
Predicted DHW - Step 46



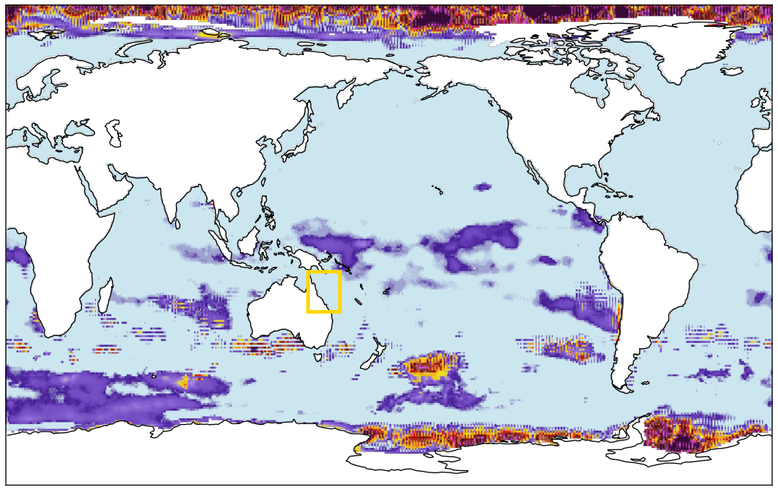
Error (Pred - Actual) - Step 46 [scale ±4.5]



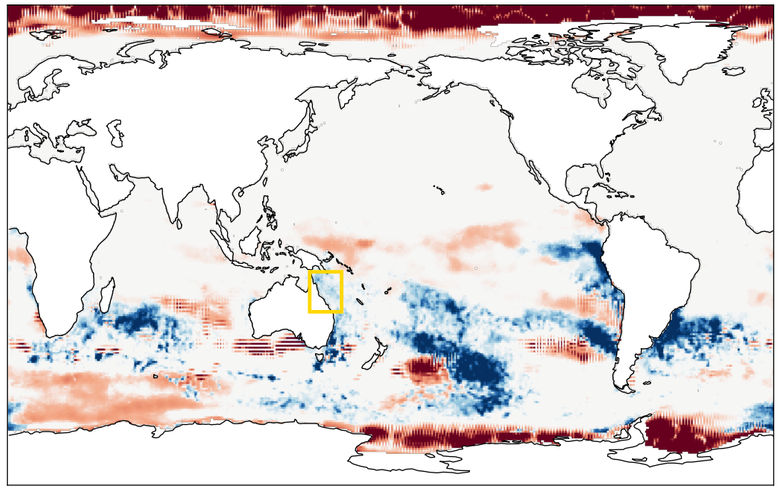
Actual DHW - Step 90

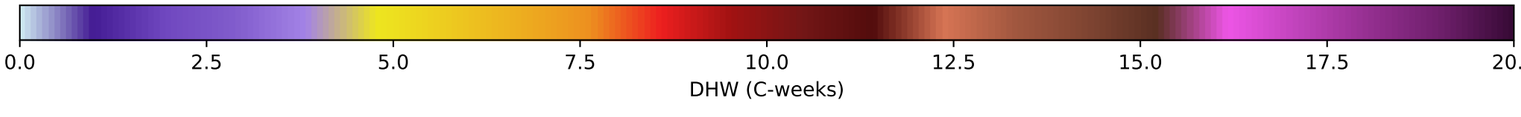


Predicted DHW - Step 90



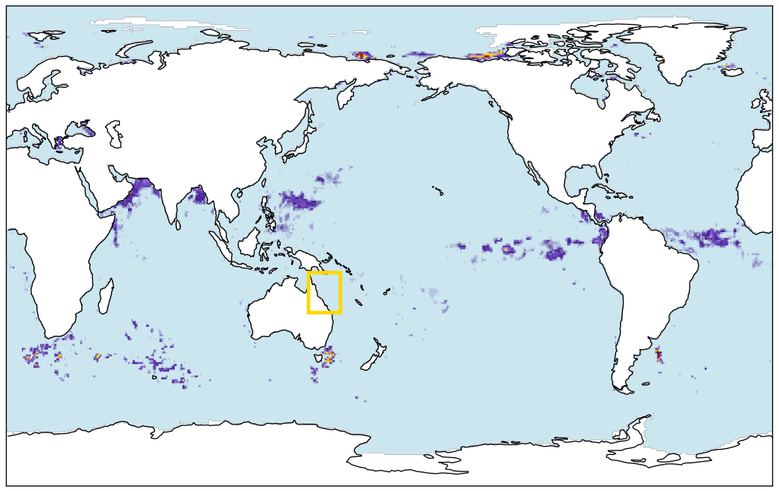
Error (Pred - Actual) - Step 90 [scale ±8]



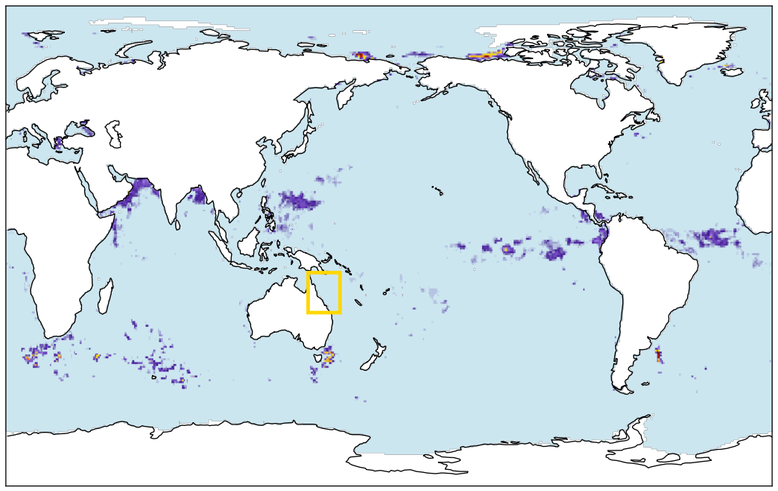


DHW Chain init: 2017-07-04 - Val (C-weeks)

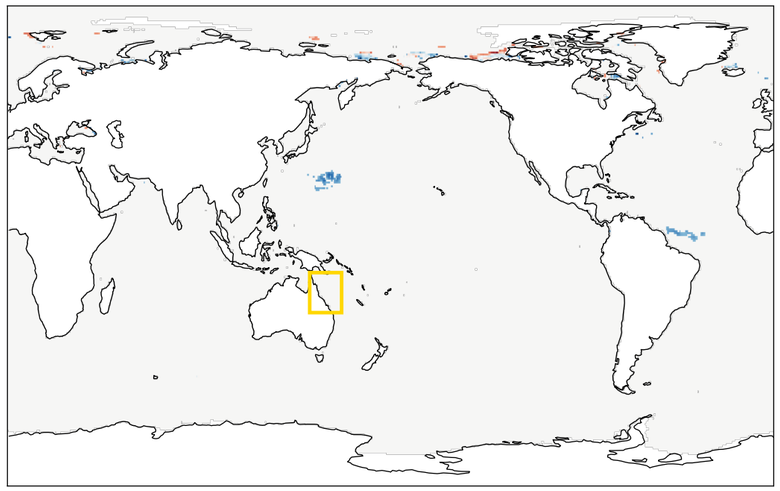
Actual DHW - Step 1



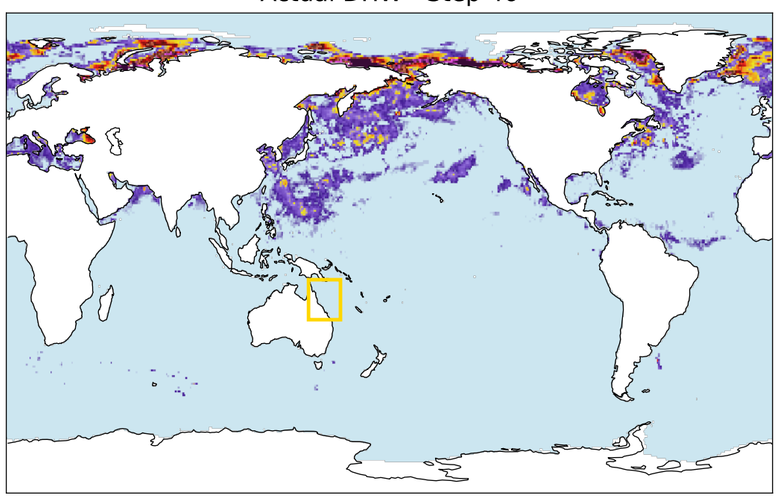
Predicted DHW - Step 1



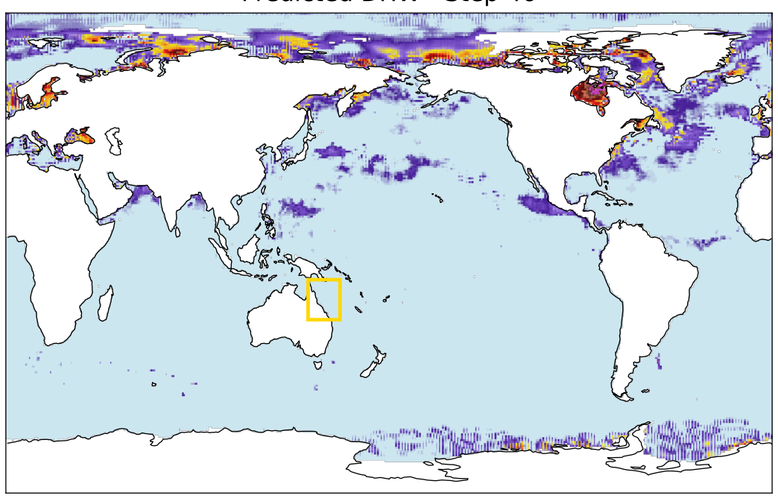
Error (Pred - Actual) - Step 1 [scale ±0.38]



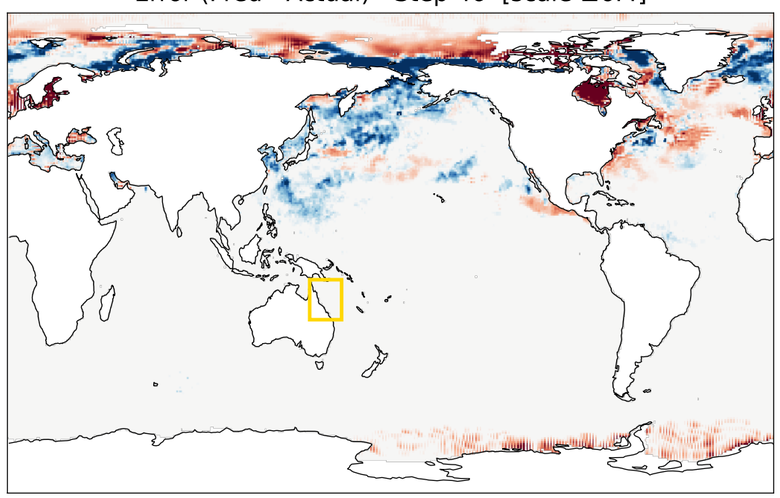
Actual DHW - Step 46



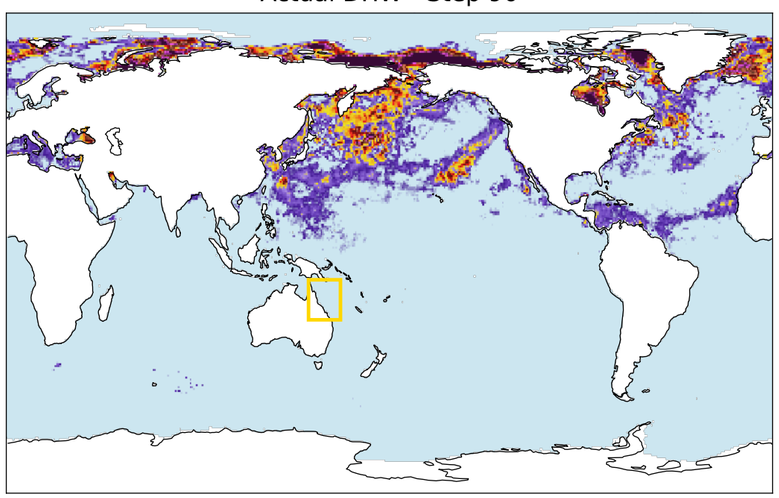
Predicted DHW - Step 46



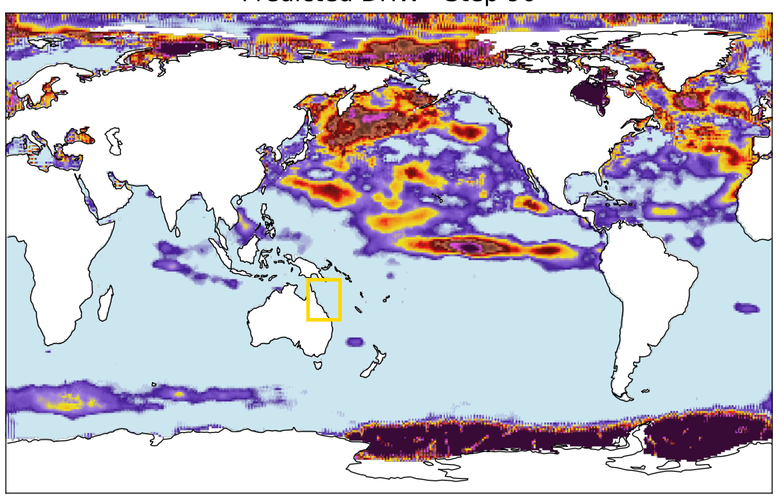
Error (Pred - Actual) - Step 46 [scale ±6.4]



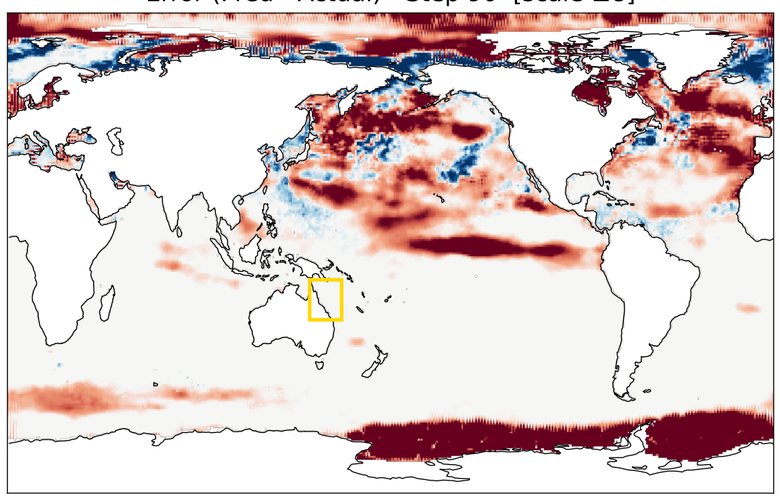
Actual DHW - Step 90

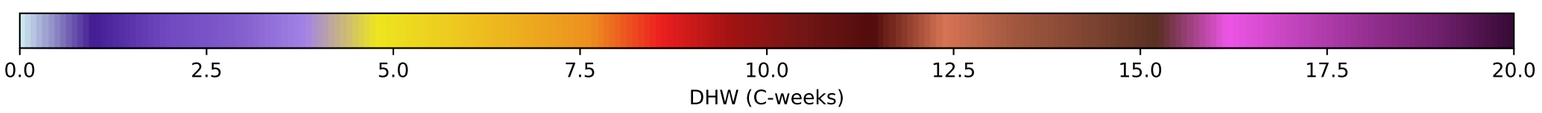


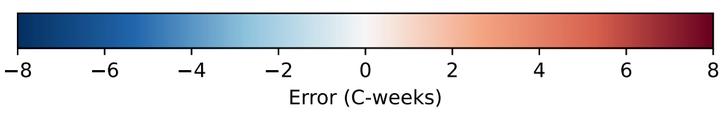
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

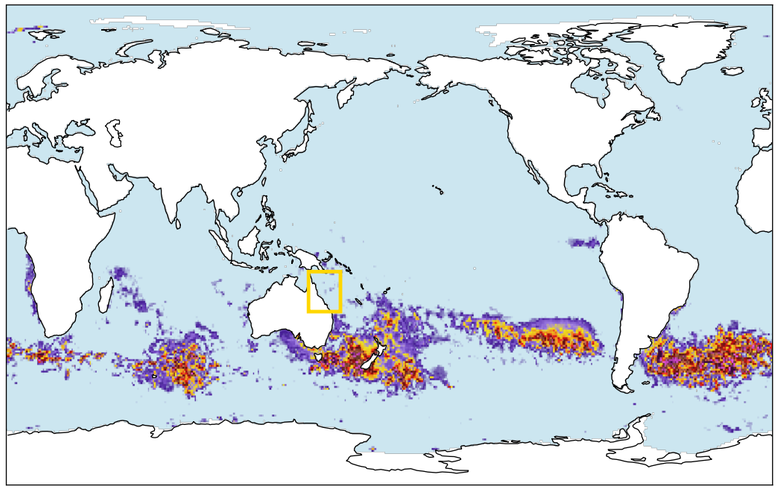




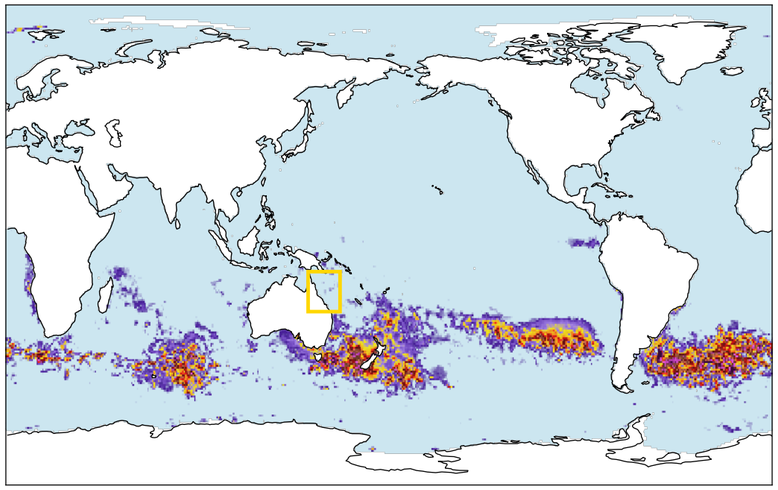


DHW Chain init: 2018-03-31 - Val (C-weeks)

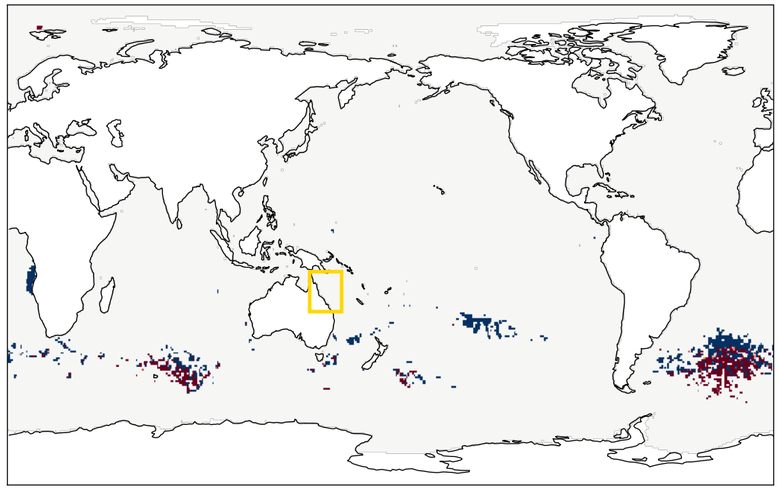
Actual DHW - Step 1



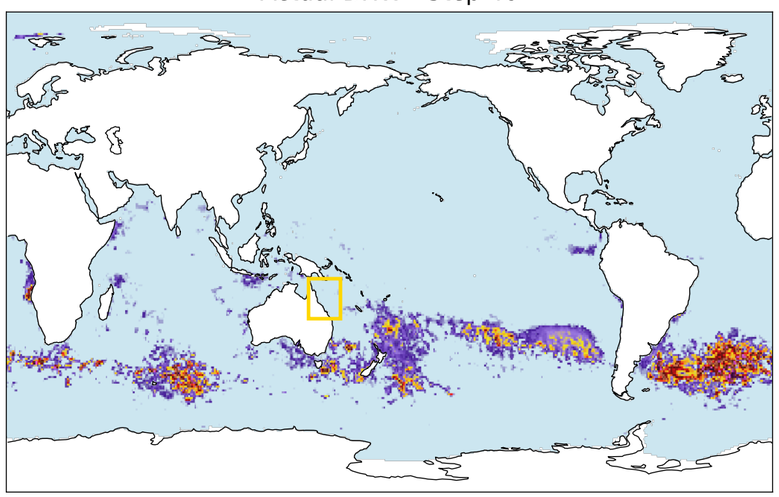
Predicted DHW - Step 1



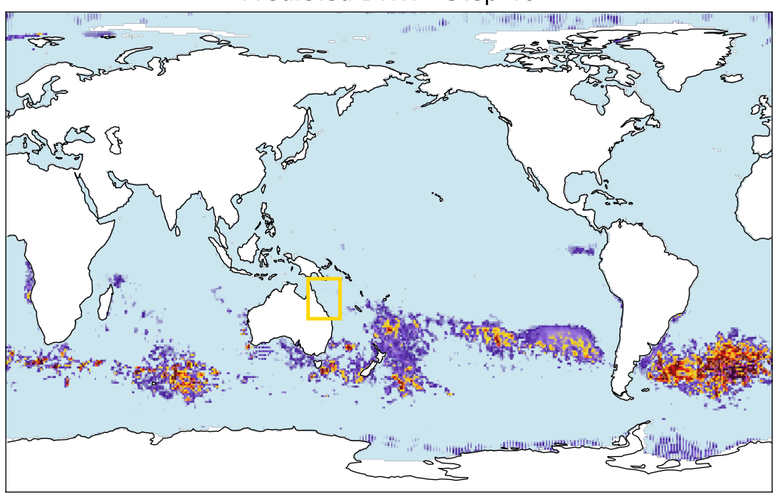
Error (Pred - Actual) - Step 1 [scale ±0.0035]



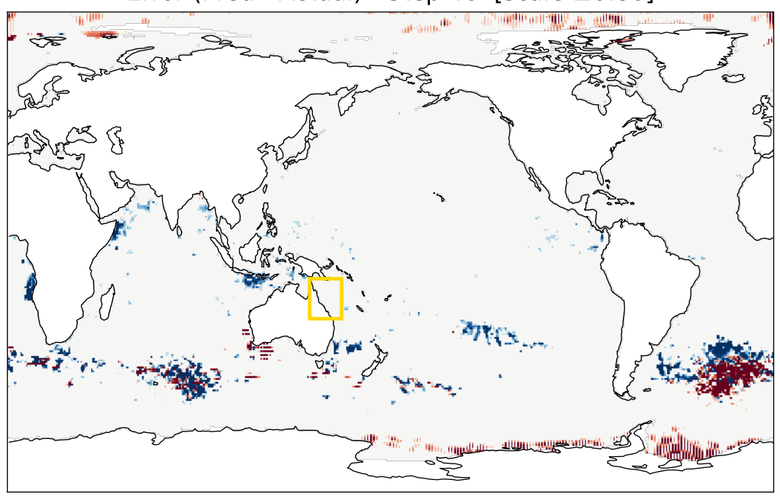
Actual DHW - Step 46



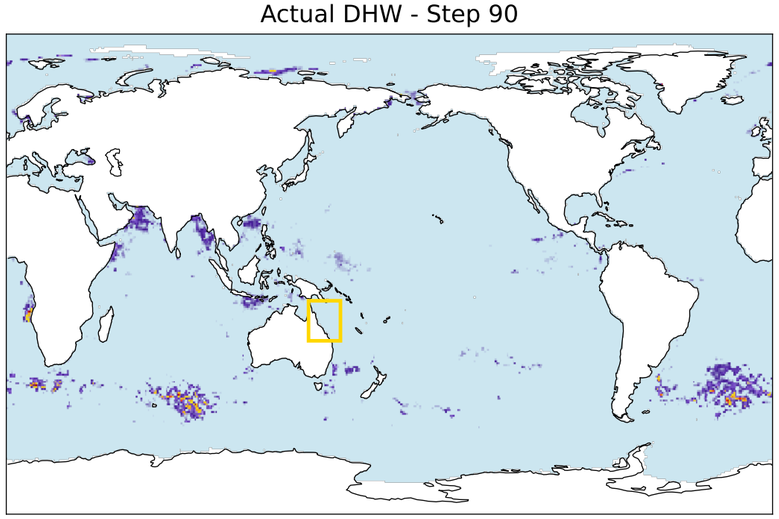
Predicted DHW - Step 46



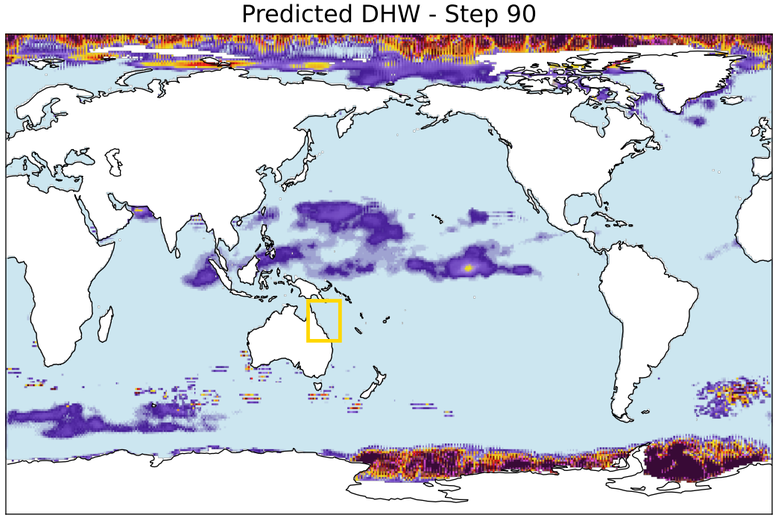
Error (Pred - Actual) - Step 46 [scale ±0.86]



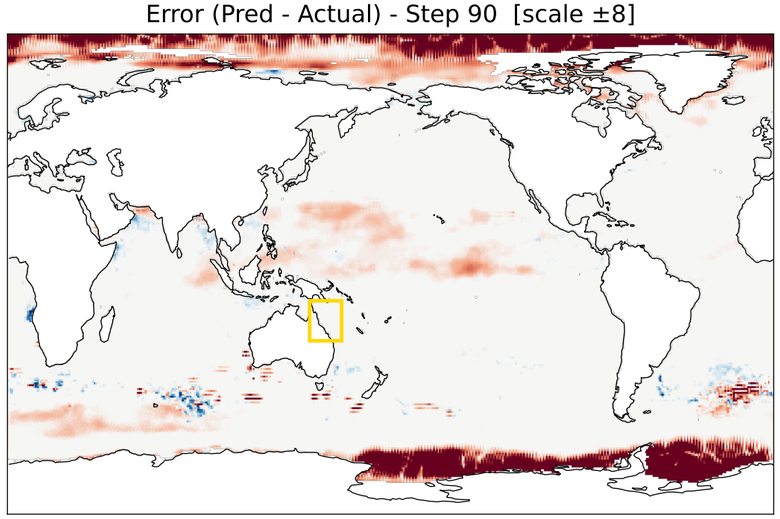
Actual DHW - Step 90

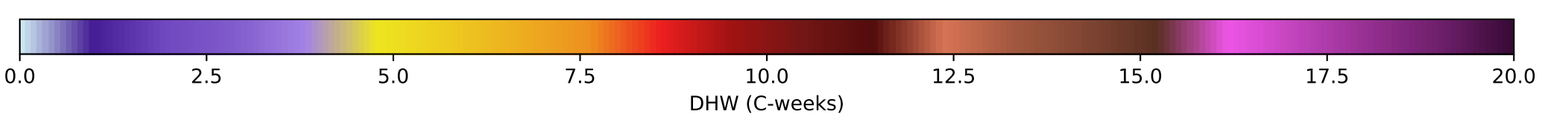


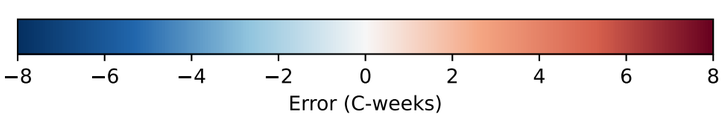
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

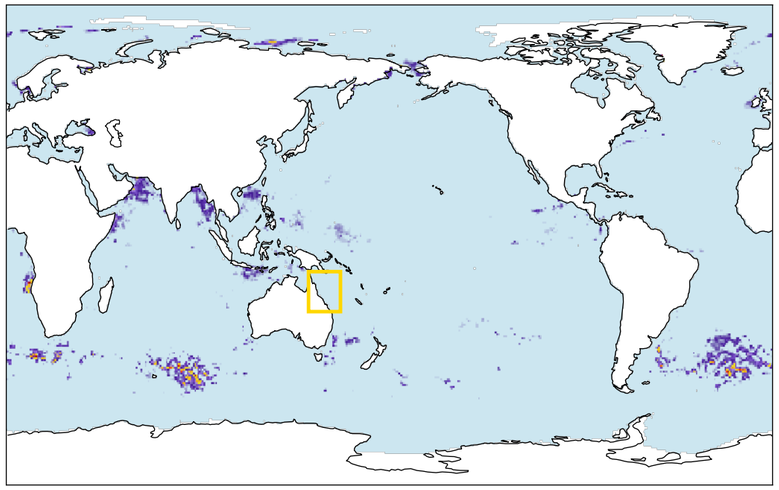




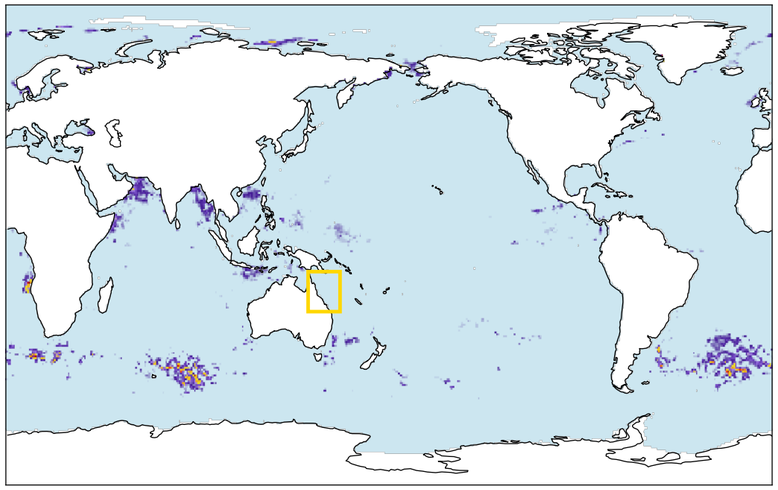


DHW Chain init: 2018-06-29 - Val (C-weeks)

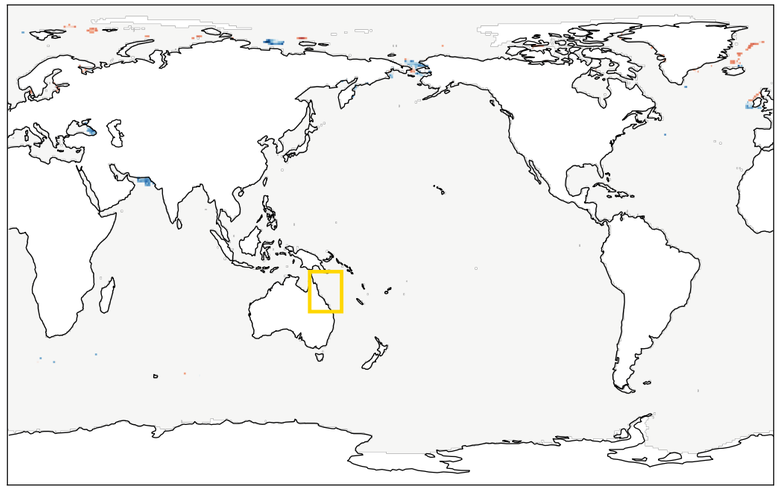
Actual DHW - Step 1



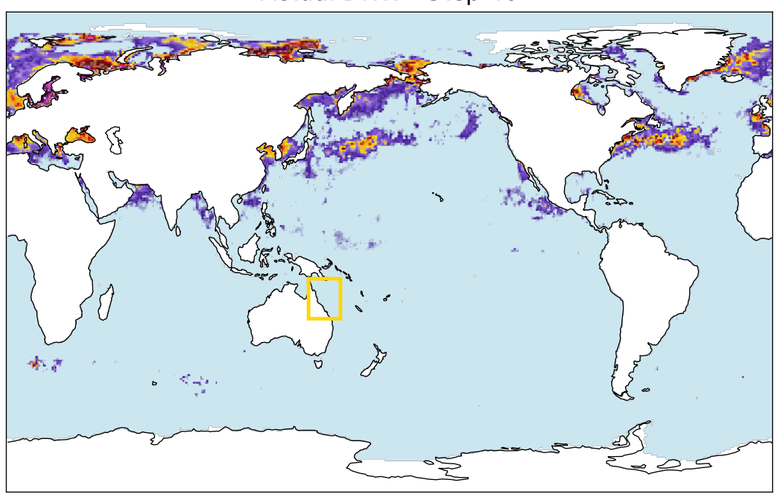
Predicted DHW - Step 1



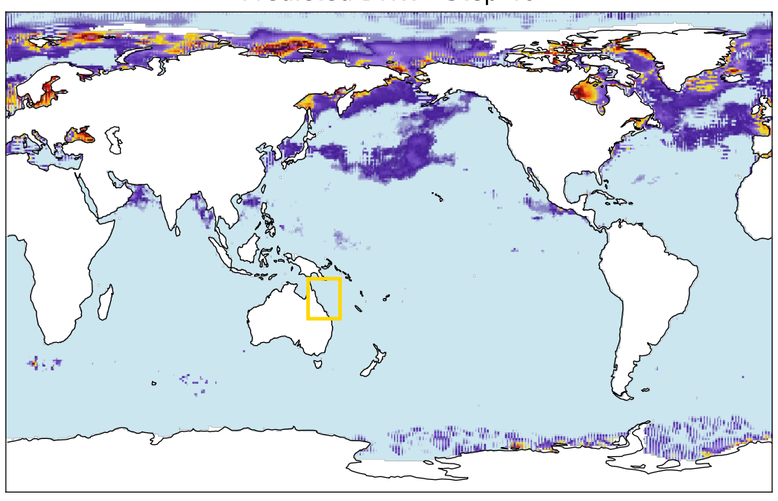
Error (Pred - Actual) - Step 1 [scale ±0.39]



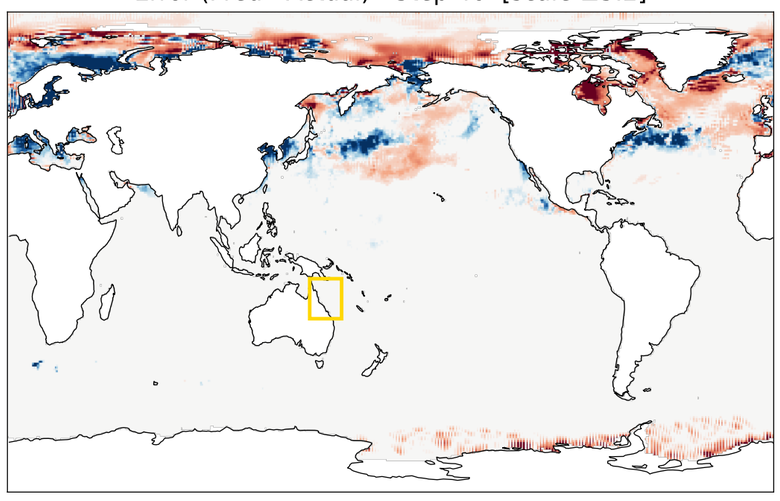
Actual DHW - Step 46



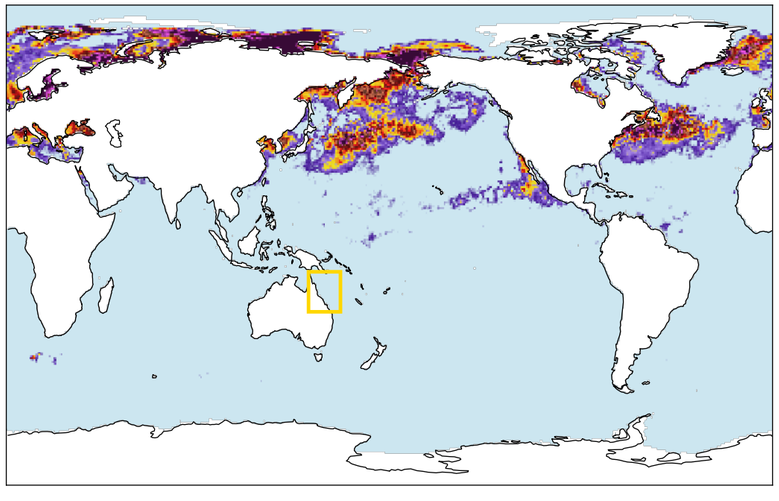
Predicted DHW - Step 46



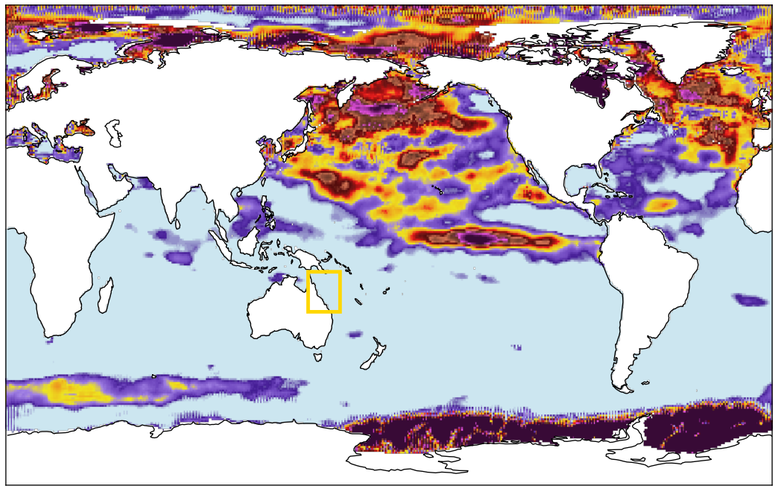
Error (Pred - Actual) - Step 46 [scale ±5.2]



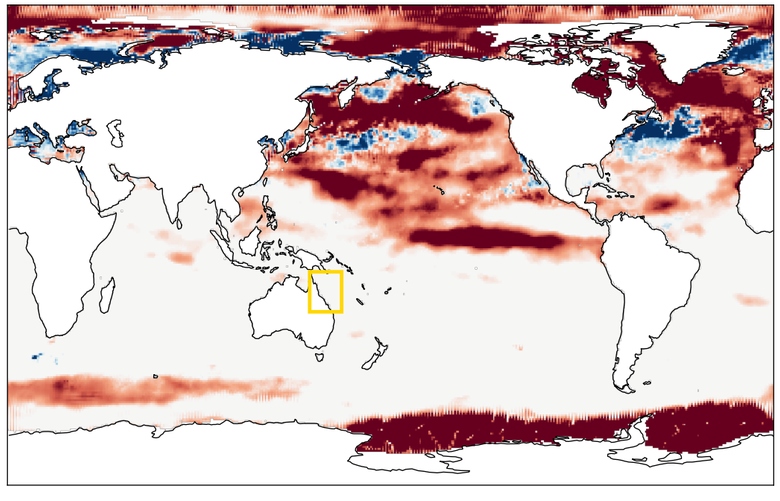
Actual DHW - Step 90

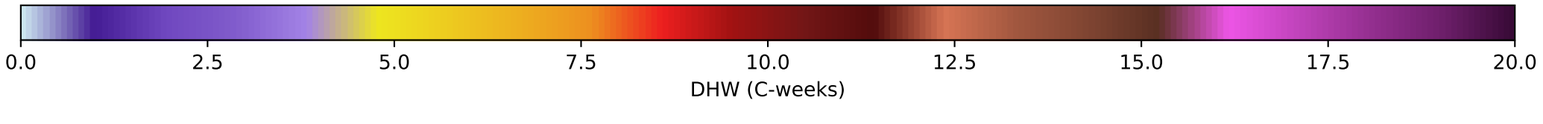


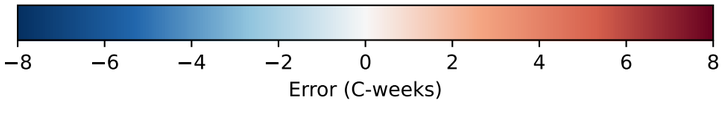
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

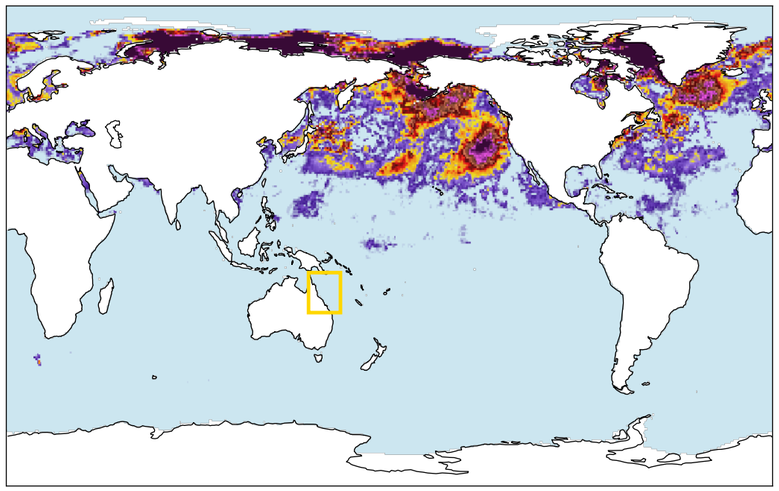




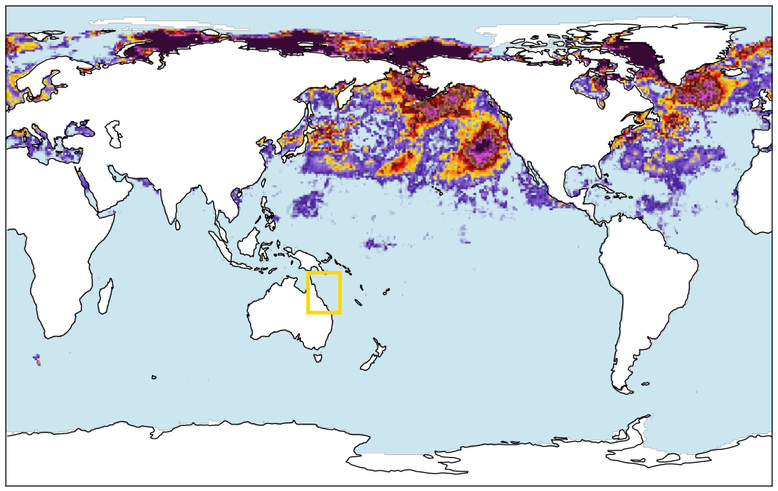


DHW Chain init: 2019-09-22 - Val (C-weeks)

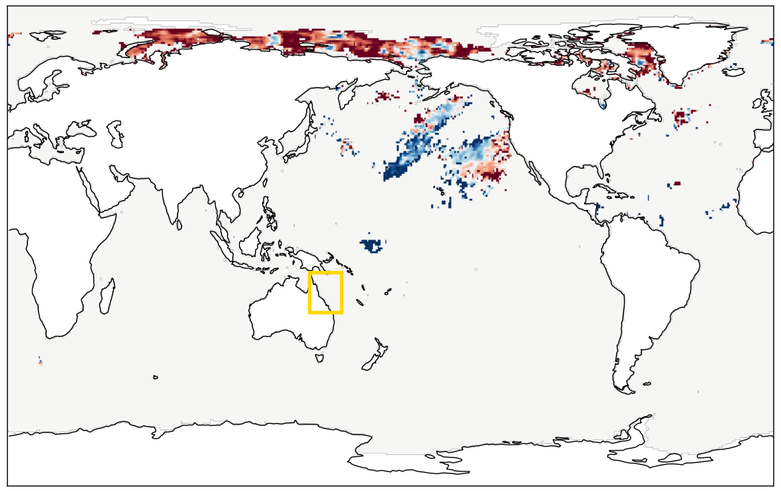
Actual DHW - Step 1



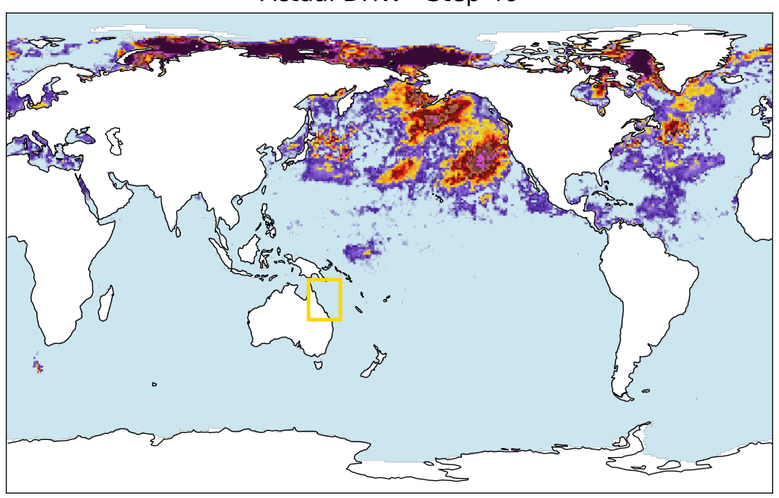
Predicted DHW - Step 1



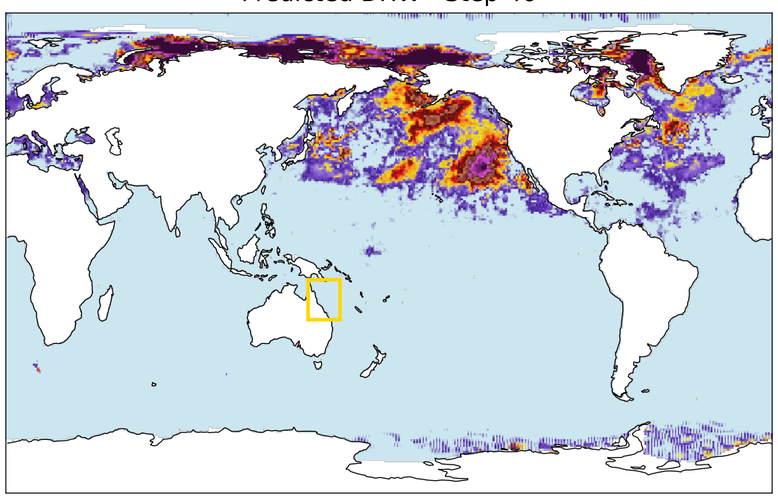
Error (Pred - Actual) - Step 1 [scale ±0.09]



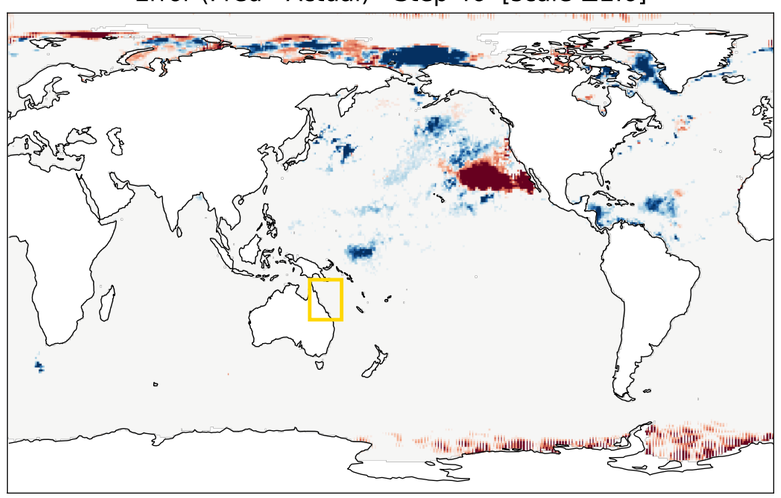
Actual DHW - Step 46



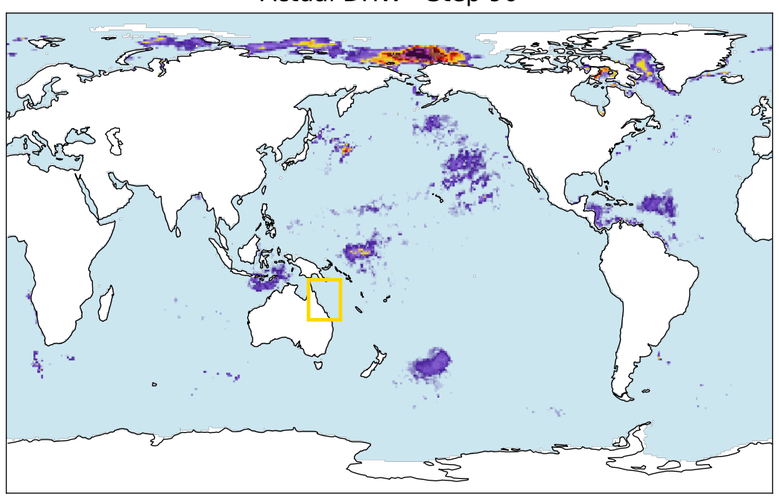
Predicted DHW - Step 46



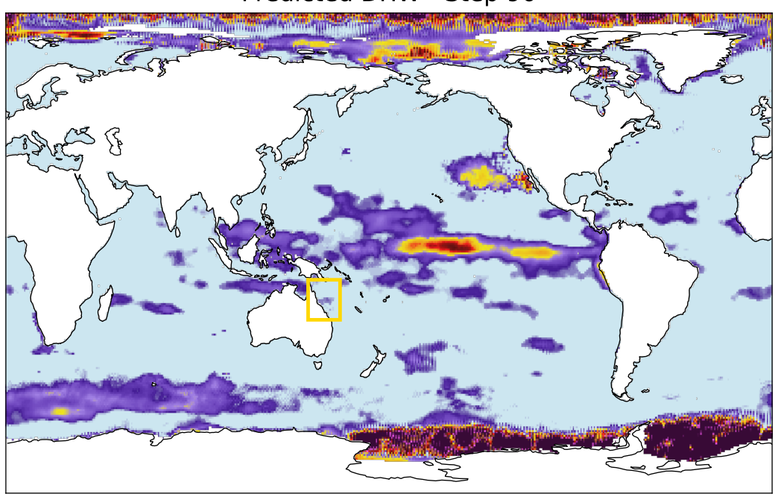
Error (Pred - Actual) - Step 46 [scale ±2.6]



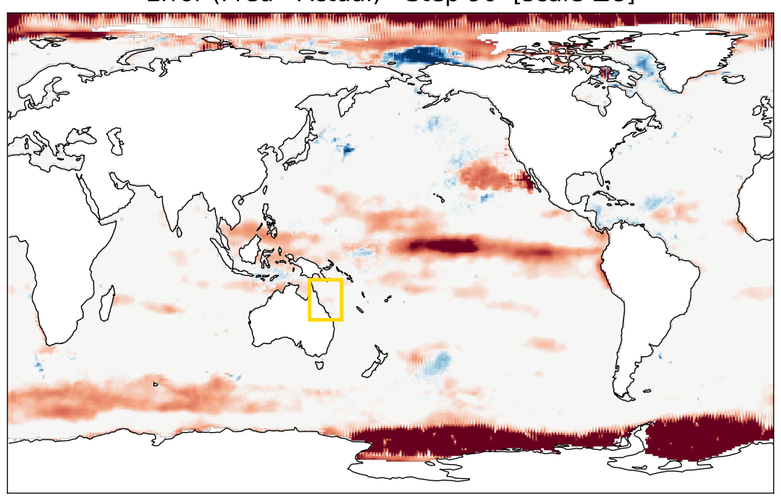
Actual DHW - Step 90

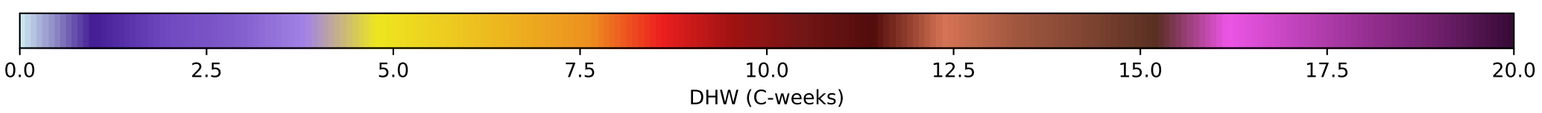


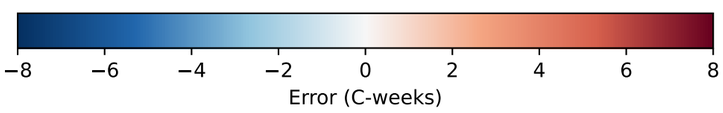
Predicted DHW - Step 90



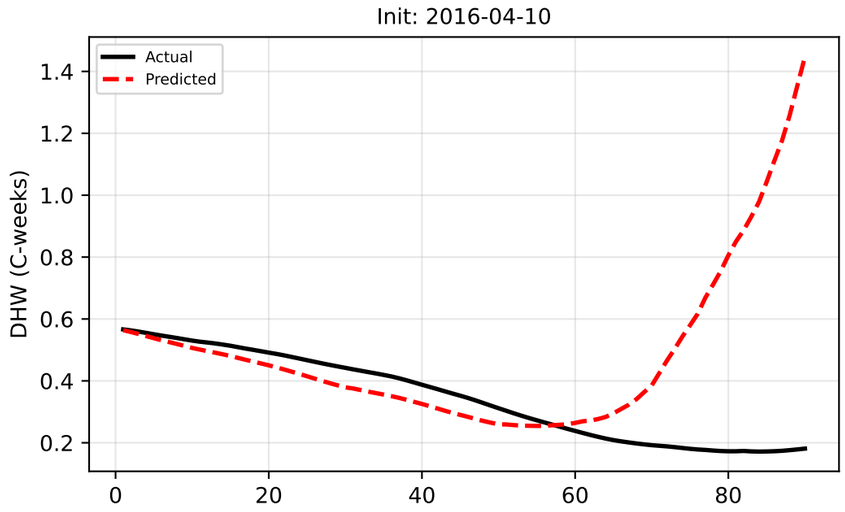
Error (Pred - Actual) - Step 90 [scale ±8]

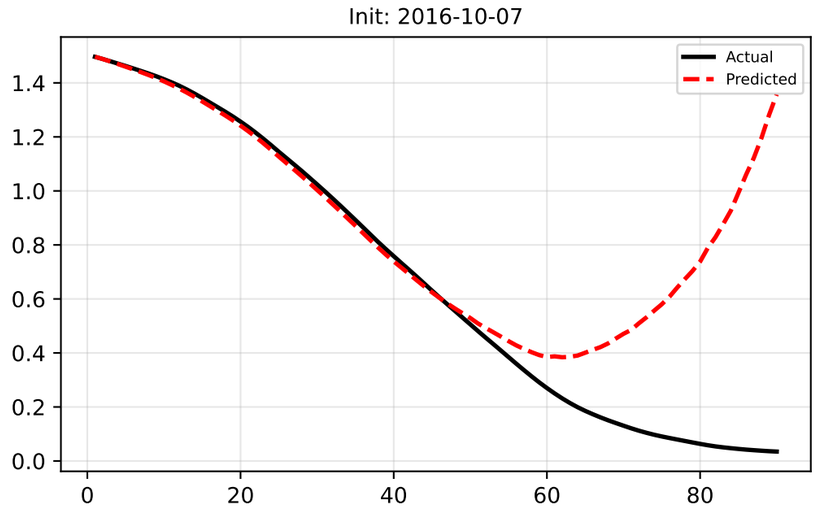


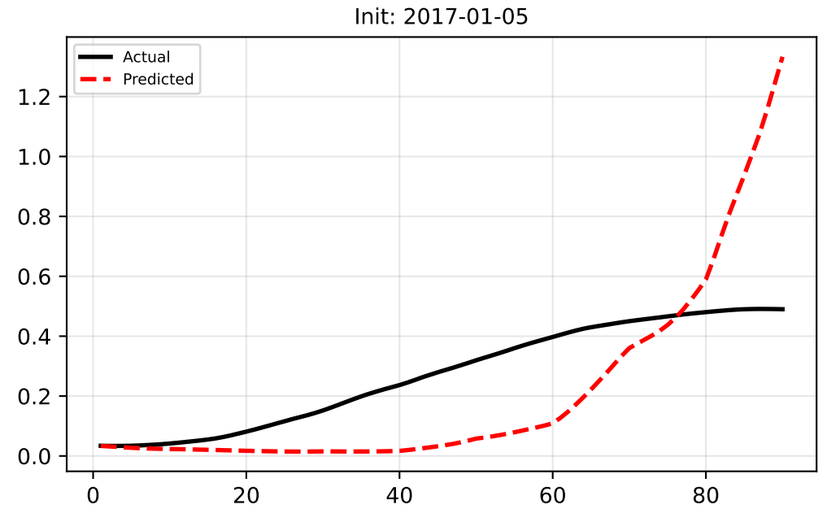


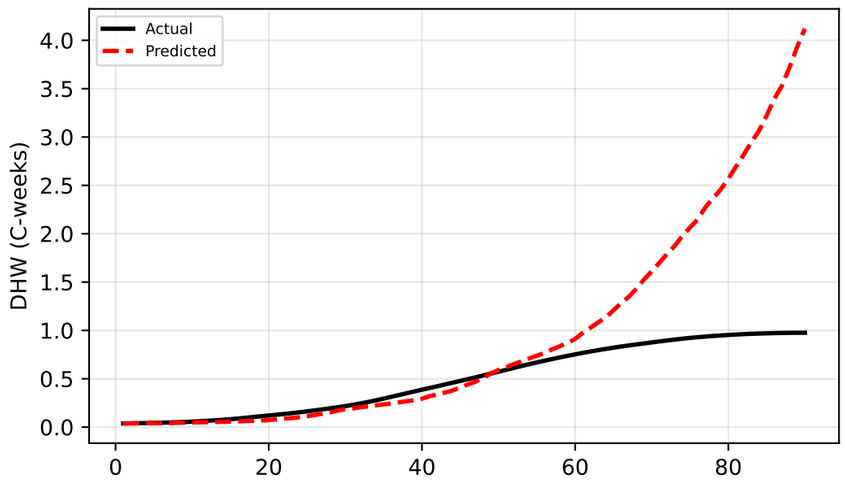


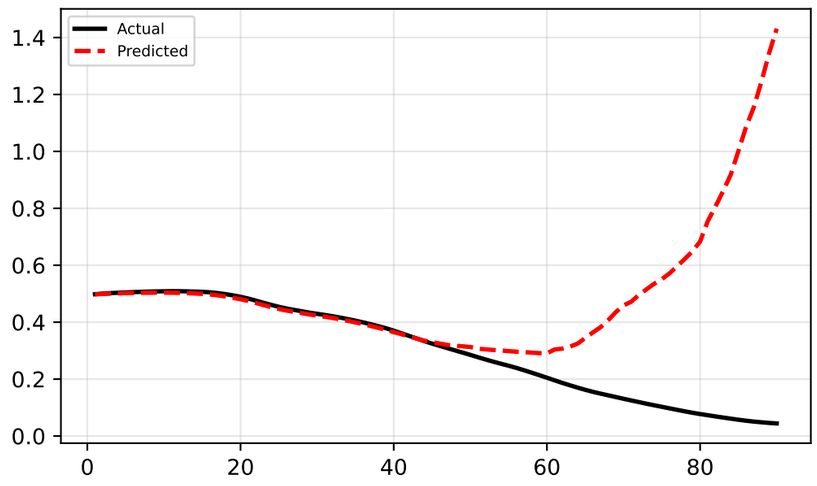
Global-Mean DHW - Val (C-weeks)

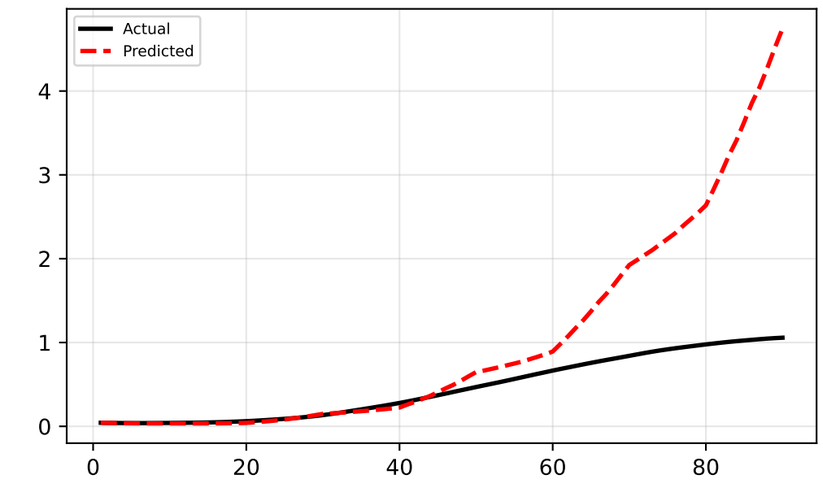


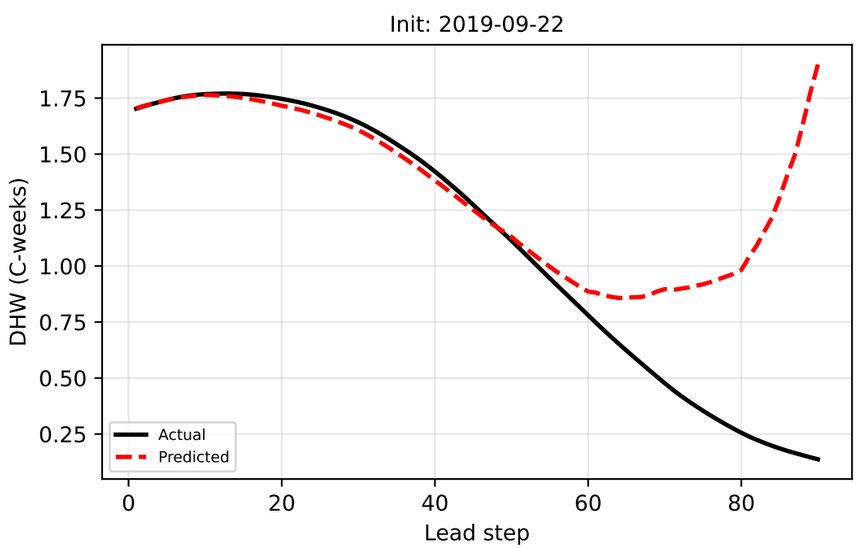




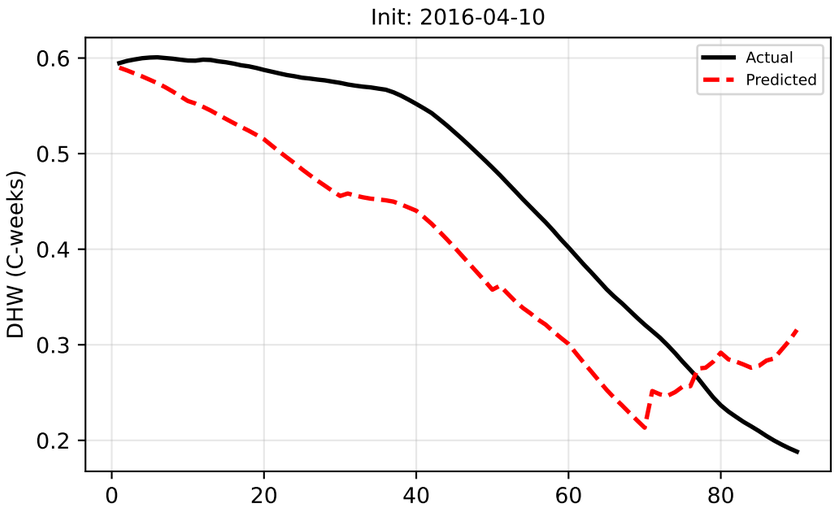


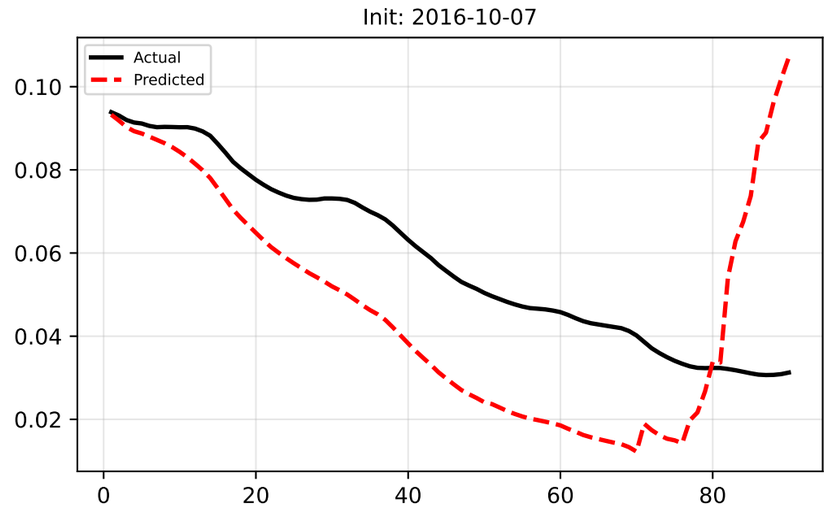


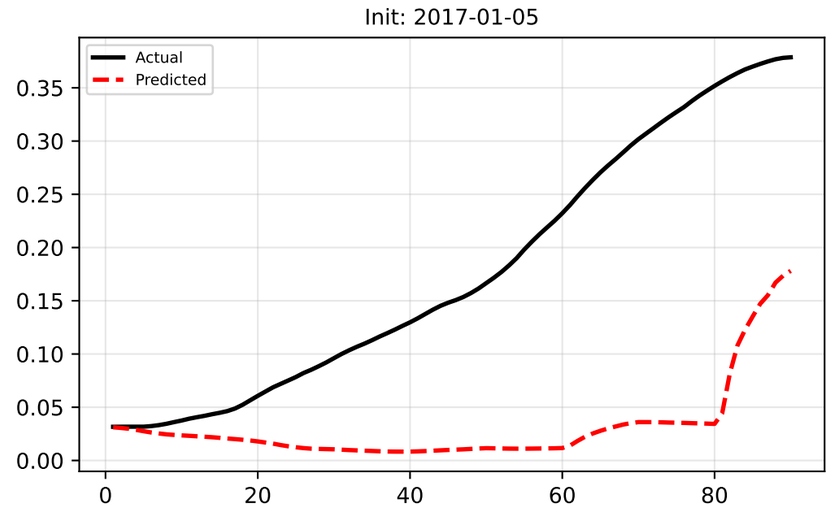


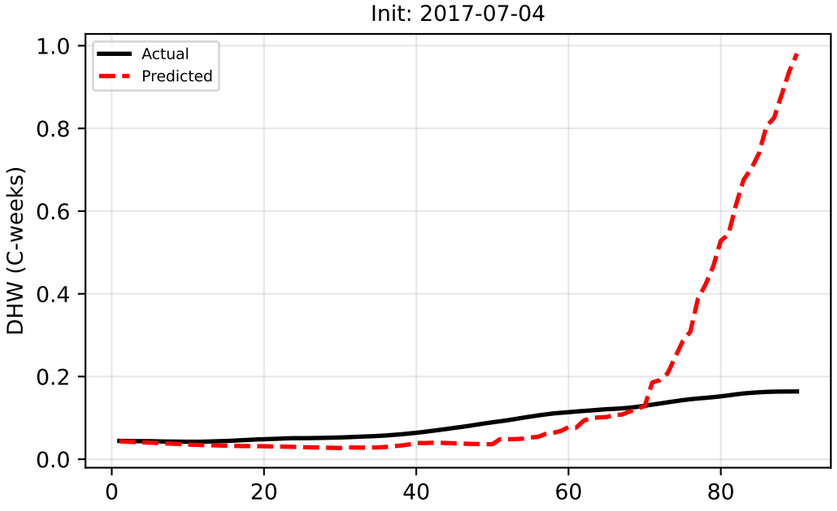


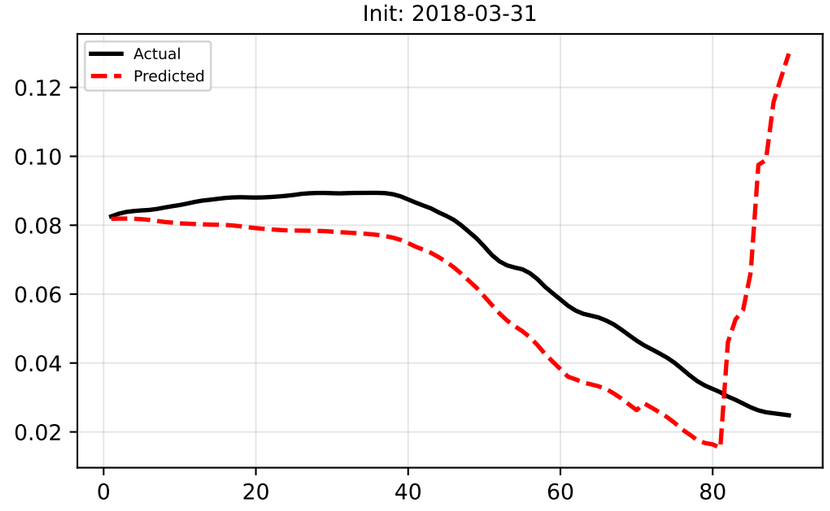
Tropical-Mean DHW (|lat|<=30) - Val (C-weeks)

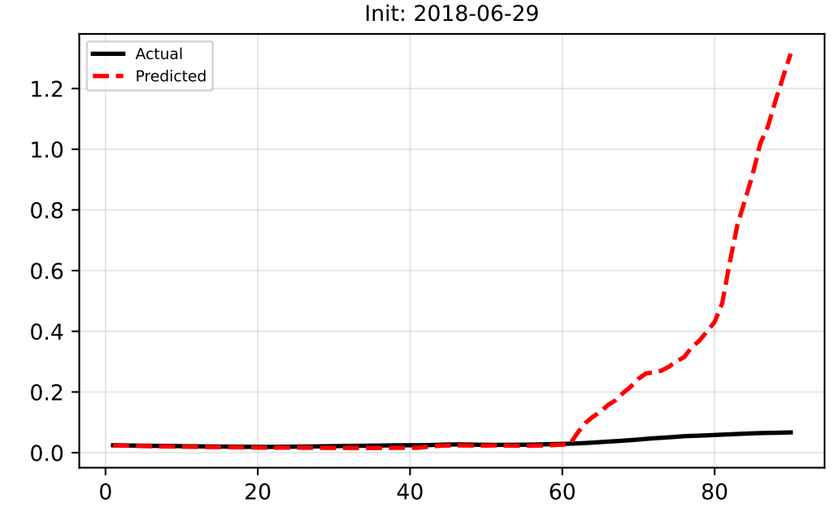


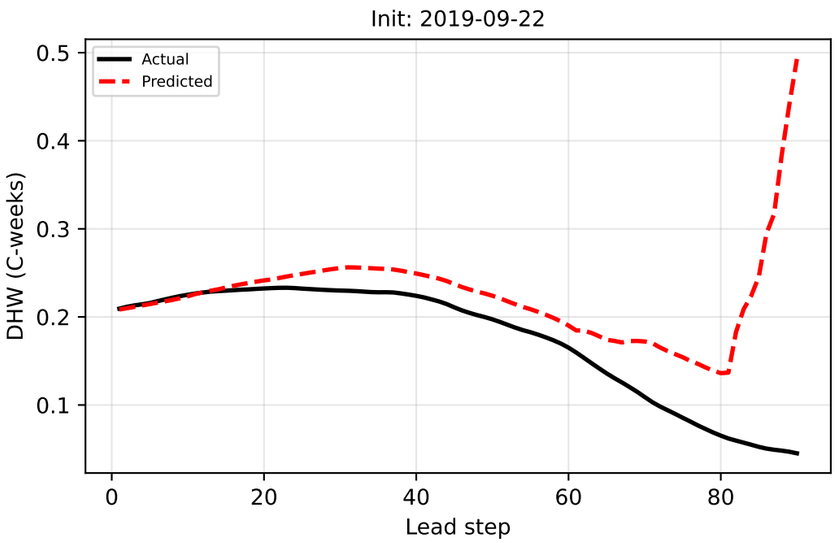




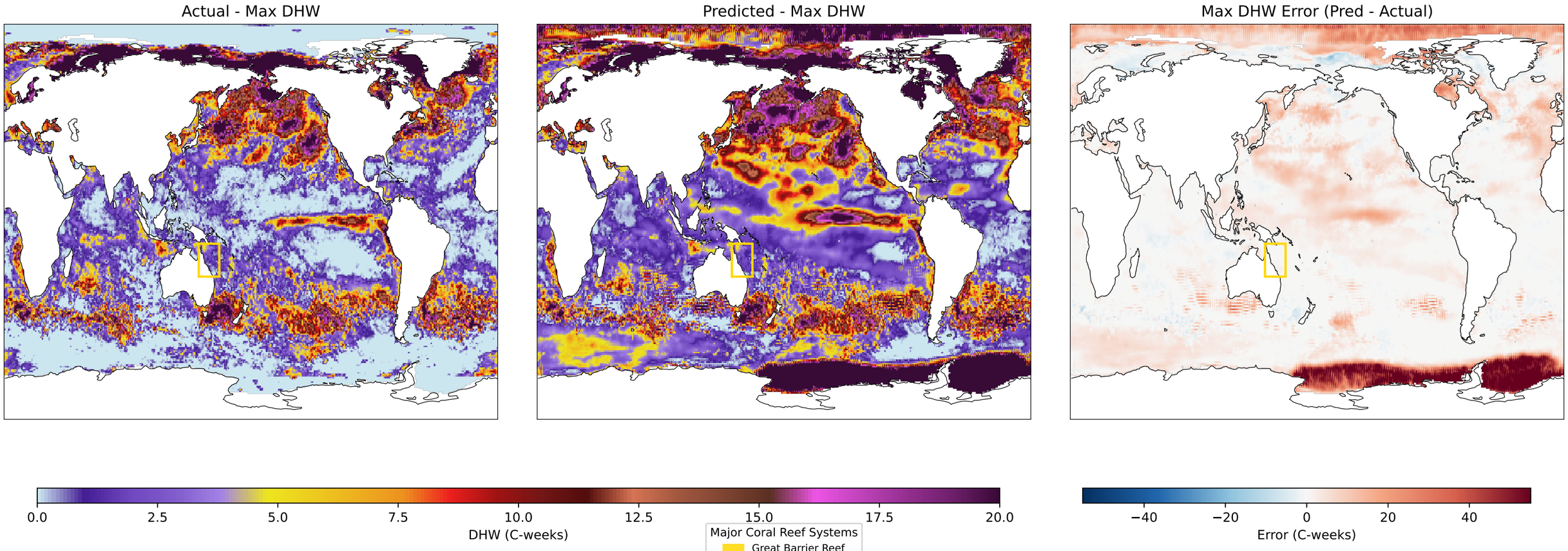


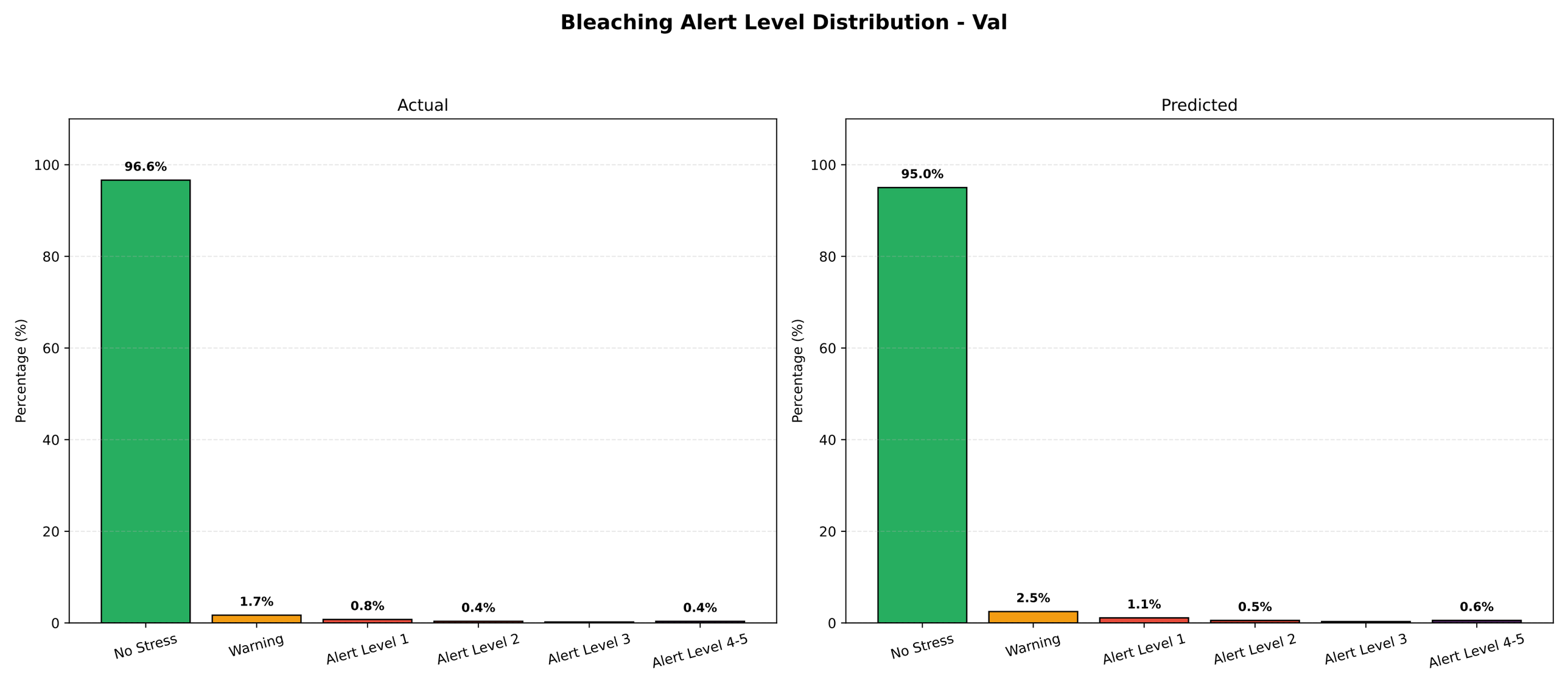






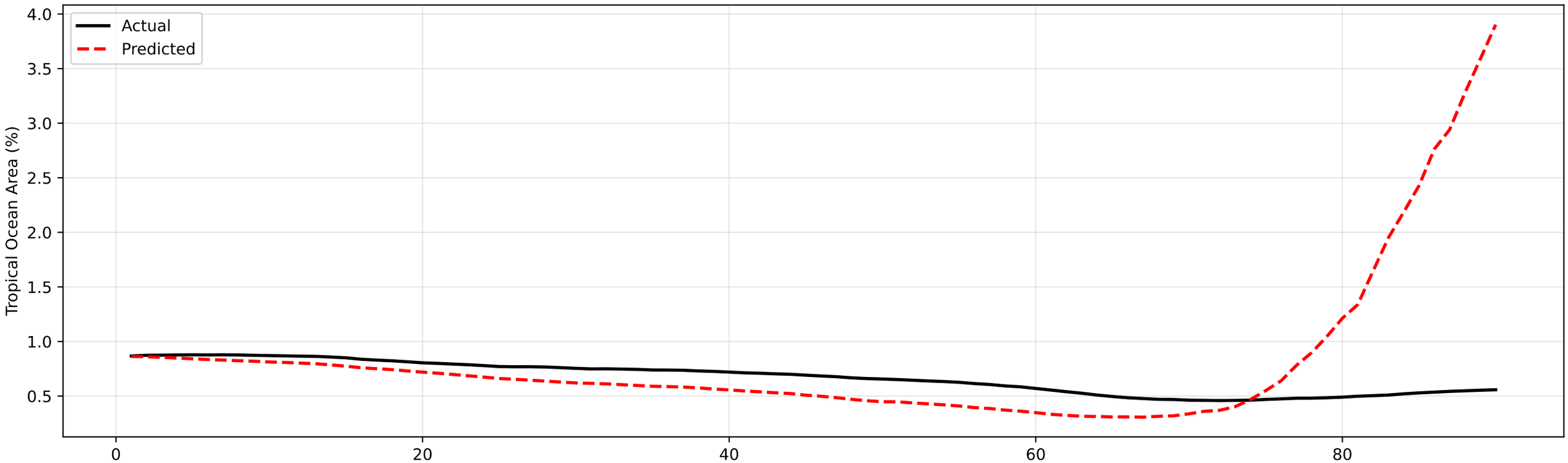
Peak DHW (max over all chains & leads) - Val (C-weeks)



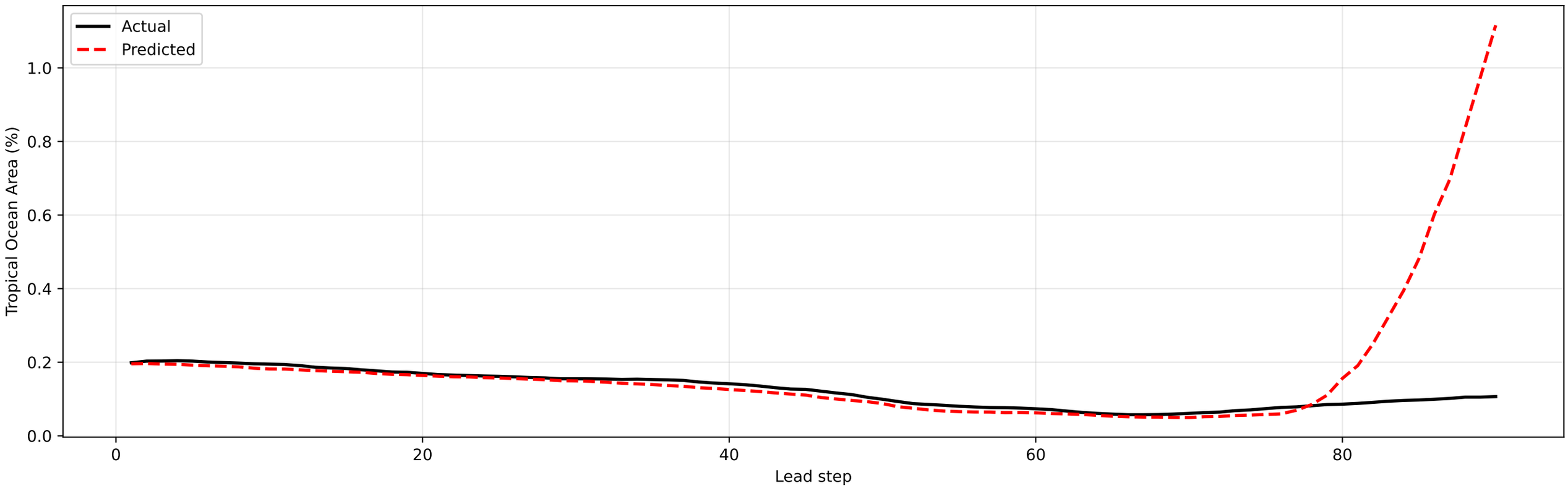

Bleaching Alert Level Distribution - Val
Actual
Predicted
Percentage (%)
0
20
40
60
80
100
96.6%
1.7%
0.8%
0.4%
0.4%
95.0%
2.5%
1.1%
0.5%
0.6%
No Stress
Warning
Alert Level 1
Alert Level 2
Alert Level 3
Alert Level 4-5

Tropical Ocean Area Under Bleaching Stress - Val

DHW >= 4 C-weeks (Alert Level 1 DHW threshold)



DHW >= 8 C-weeks (Alert Level 2 DHW threshold)



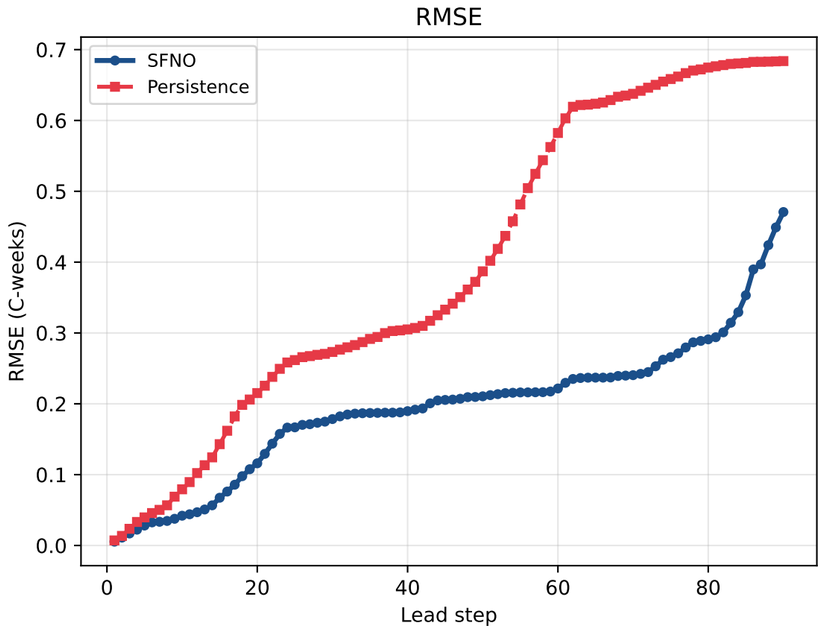
Great Barrier Reef Analysis - Val

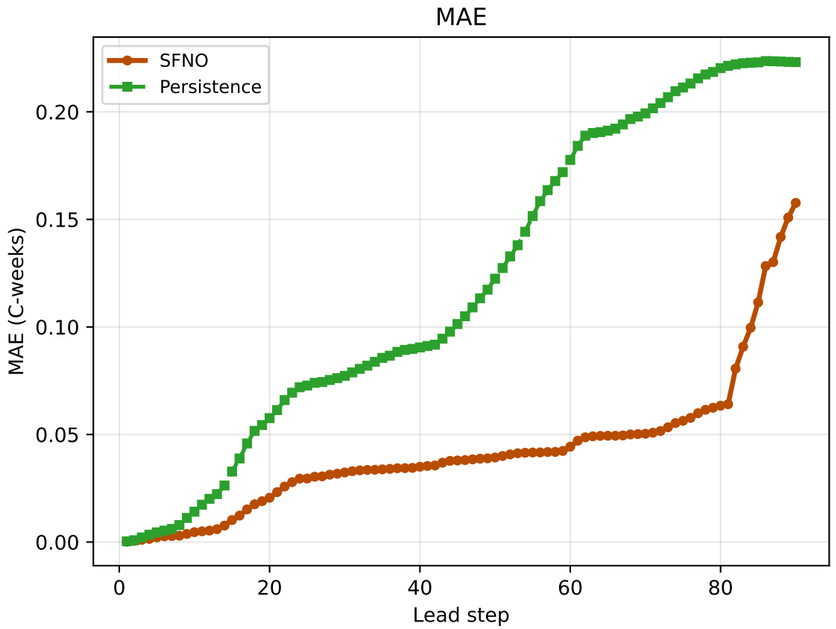
```
                    GREAT BARRIER REEF - VAL
Lat -25.0 deg to -10.0 deg  |  Lon 142.0 deg to 157.0 deg  (0-360)
                    Grid: 21 lat x 21 lon points
```

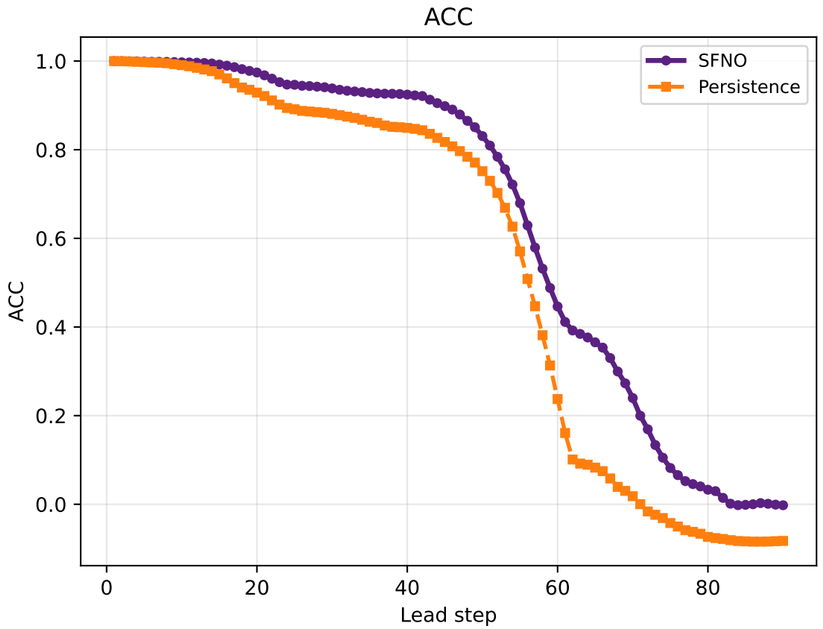
DHW Metrics by Lead Step - Val - GBR (15 chains scored at every lead)

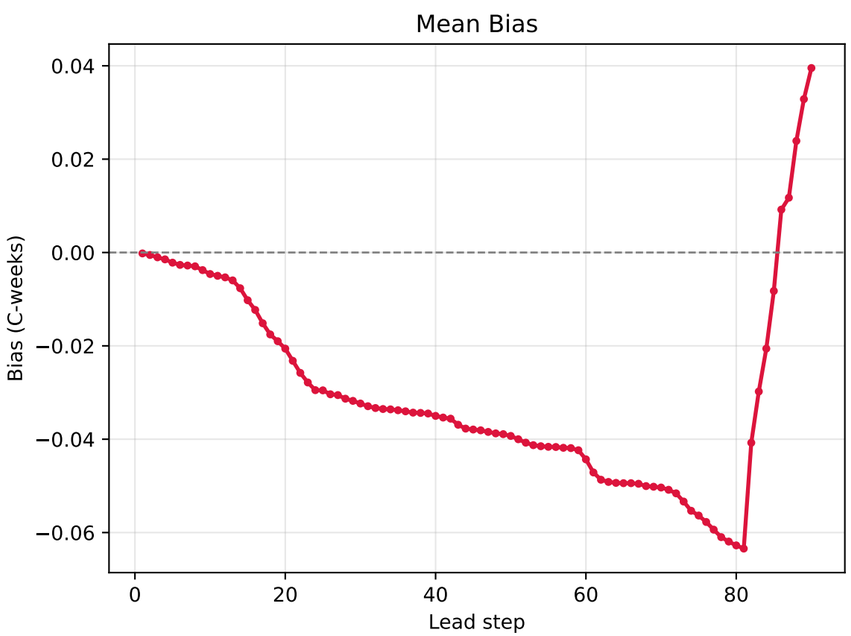
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	Persist MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0054	0.0071	23.9%	0.0002	0.0003	-0.0002	1.0000	0.9999	0.9999
2	0.0111	0.0135	17.5%	0.0005	0.0008	-0.0005	0.9998	0.9997	0.9996
3	0.0169	0.0235	28.1%	0.0010	0.0021	-0.0010	0.9996	0.9992	0.9992
4	0.0223	0.0332	32.8%	0.0015	0.0034	-0.0015	0.9992	0.9983	0.9986
5	0.0281	0.0396	29.0%	0.0022	0.0045	-0.0022	0.9988	0.9976	0.9977
6	0.0326	0.0458	28.8%	0.0027	0.0053	-0.0027	0.9984	0.9969	0.9969
7	0.0334	0.0503	33.5%	0.0028	0.0061	-0.0028	0.9983	0.9962	0.9967
8	0.0346	0.0567	39.0%	0.0030	0.0079	-0.0030	0.9981	0.9952	0.9964
9	0.0377	0.0689	45.3%	0.0038	0.0112	-0.0038	0.9978	0.9930	0.9957
10	0.0420	0.0793	47.0%	0.0046	0.0141	-0.0046	0.9972	0.9908	0.9946
11	0.0441	0.0895	50.8%	0.0050	0.0174	-0.0050	0.9968	0.9883	0.9940
12	0.0469	0.1022	54.1%	0.0053	0.0201	-0.0053	0.9963	0.9847	0.9931
13	0.0508	0.1133	55.2%	0.0060	0.0223	-0.0060	0.9957	0.9811	0.9918
14	0.0567	0.1244	54.4%	0.0077	0.0263	-0.0077	0.9945	0.9772	0.9896
15	0.0674	0.1430	52.9%	0.0102	0.0328	-0.0102	0.9920	0.9698	0.9849
16	0.0761	0.1619	53.0%	0.0123	0.0388	-0.0123	0.9895	0.9610	0.9802
17	0.0858	0.1822	52.9%	0.0152	0.0458	-0.0152	0.9863	0.9503	0.9740
18	0.0978	0.1985	50.7%	0.0176	0.0517	-0.0176	0.9818	0.9403	0.9656
19	0.1077	0.2060	47.7%	0.0190	0.0544	-0.0190	0.9779	0.9352	0.9582
20	0.1160	0.2151	46.1%	0.0206	0.0576	-0.0206	0.9742	0.9288	0.9514
21	0.1293	0.2255	42.6%	0.0232	0.0614	-0.0232	0.9679	0.9211	0.9397
22	0.1437	0.2380	39.6%	0.0258	0.0660	-0.0258	0.9603	0.9112	0.9259
23	0.1575	0.2494	36.8%	0.0278	0.0694	-0.0278	0.9524	0.9018	0.9116
24	0.1664	0.2583	35.6%	0.0295	0.0720	-0.0295	0.9469	0.8942	0.9017
25	0.1668	0.2617	36.3%	0.0295	0.0728	-0.0295	0.9465	0.8912	0.9009
26	0.1702	0.2658	36.0%	0.0304	0.0740	-0.0304	0.9442	0.8876	0.8969
27	0.1712	0.2673	36.0%	0.0305	0.0744	-0.0305	0.9437	0.8862	0.8958
28	0.1732	0.2691	35.6%	0.0313	0.0753	-0.0313	0.9423	0.8846	0.8933
29	0.1748	0.2705	35.4%	0.0318	0.0762	-0.0318	0.9410	0.8834	0.8910
30	0.1784	0.2732	34.7%	0.0324	0.0773	-0.0324	0.9383	0.8809	0.8863
31	0.1822	0.2765	34.1%	0.0329	0.0789	-0.0329	0.9352	0.8778	0.8808
32	0.1849	0.2797	33.9%	0.0333	0.0805	-0.0333	0.9329	0.8747	0.8769
33	0.1861	0.2829	34.2%	0.0335	0.0820	-0.0335	0.9314	0.8717	0.8743
34	0.1867	0.2871	35.0%	0.0336	0.0838	-0.0336	0.9300	0.8676	0.8718
35	0.1870	0.2916	35.9%	0.0338	0.0856	-0.0338	0.9281	0.8631	0.8686
36	0.1872	0.2943	36.4%	0.0340	0.0866	-0.0340	0.9271	0.8603	0.8668
37	0.1874	0.2998	37.5%	0.0343	0.0884	-0.0343	0.9264	0.8547	0.8655
38	0.1875	0.3027	38.1%	0.0344	0.0893	-0.0344	0.9262	0.8516	0.8652
39	0.1879	0.3036	38.1%	0.0345	0.0898	-0.0345	0.9256	0.8507	0.8641
40	0.1896	0.3050	37.8%	0.0350	0.0905	-0.0350	0.9243	0.8492	0.8619
41	0.1917	0.3071	37.6%	0.0354	0.0911	-0.0354	0.9226	0.8470	0.8589
42	0.1934	0.3100	37.6%	0.0356	0.0918	-0.0356	0.9211	0.8439	0.8563
43	0.2007	0.3172	36.7%	0.0369	0.0945	-0.0369	0.9129	0.8358	0.8418
44	0.2048	0.3250	37.0%	0.0377	0.0978	-0.0377	0.9051	0.8267	0.8283
45	0.2055	0.3329	38.3%	0.0379	0.1014	-0.0379	0.8983	0.8175	0.8167
46	0.2059	0.3414	39.7%	0.0381	0.1050	-0.0381	0.8905	0.8076	0.8035
47	0.2071	0.3506	40.9%	0.0384	0.1091	-0.0384	0.8794	0.7969	0.7848
48	0.2094	0.3613	42.0%	0.0388	0.1133	-0.0388	0.8652	0.7840	0.7613
49	0.2096	0.3722	43.7%	0.0389	0.1173	-0.0389	0.8507	0.7710	0.7377
50	0.2106	0.3870	45.6%	0.0393	0.1224	-0.0393	0.8307	0.7515	0.7057
51	0.2122	0.4019	47.2%	0.0400	0.1274	-0.0400	0.8094	0.7297	0.6722
52	0.2136	0.4187	49.0%	0.0407	0.1328	-0.0407	0.7841	0.7024	0.6338
53	0.2151	0.4370	50.8%	0.0413	0.1380	-0.0413	0.7558	0.6688	0.5921
54	0.2156	0.4577	52.9%	0.0415	0.1443	-0.0415	0.7214	0.6265	0.5440
55	0.2161	0.4814	55.1%	0.0416	0.1516	-0.0416	0.6794	0.5704	0.4887
56	0.2161	0.5045	57.2%	0.0417	0.1585	-0.0417	0.6292	0.5083	0.4259
57	0.2163	0.5246	58.8%	0.0418	0.1636	-0.0418	0.5791	0.4469	0.3663
58	0.2164	0.5439	60.2%	0.0419	0.1678	-0.0419	0.5316	0.3814	0.3131
59	0.2173	0.5625	61.4%	0.0424	0.1720	-0.0424	0.4881	0.3131	0.2669
60	0.2216	0.5823	61.9%	0.0443	0.1776	-0.0443	0.4463	0.2374	0.2239
61	0.2296	0.6030	61.9%	0.0471	0.1842	-0.0471	0.4114	0.1608	0.1899
62	0.2351	0.6193	62.0%	0.0487	0.1889	-0.0487	0.3921	0.1009	0.1726
63	0.2364	0.6218	62.0%	0.0492	0.1902	-0.0492	0.3844	0.0919	0.1664
64	0.2369	0.6223	61.9%	0.0494	0.1906	-0.0494	0.3763	0.0891	0.1598
65	0.2370	0.6236	62.0%	0.0494	0.1913	-0.0494	0.3654	0.0827	0.1506
66	0.2370	0.6253	62.1%	0.0494	0.1922	-0.0494	0.3534	0.0747	0.1410
67	0.2371	0.6288	62.3%	0.0495	0.1942	-0.0495	0.3299	0.0583	0.1229
68	0.2393	0.6334	62.2%	0.0500	0.1967	-0.0500	0.2996	0.0393	0.1008
69	0.2398	0.6351	62.2%	0.0502	0.1978	-0.0502	0.2729	0.0302	0.0831
70	0.2404	0.6376	62.3%	0.0504	0.1993	-0.0504	0.2396	0.0182	0.0625
71	0.2423	0.6418	62.3%	0.0508	0.2016	-0.0508	0.1998	0.0002	0.0398
72	0.2448	0.6462	62.1%	0.0516	0.2041	-0.0516	0.1692	-0.0162	0.0239
73	0.2530	0.6503	61.1%	0.0534	0.2068	-0.0534	0.1336	-0.0238	0.0076
74	0.2623	0.6549	60.0%	0.0553	0.2096	-0.0553	0.1051	-0.0311	-0.0043
75	0.2659	0.6585	59.6%	0.0564	0.2113	-0.0564	0.0818	-0.0424	-0.0138
76	0.2713	0.6620	59.0%	0.0577	0.2132	-0.0577	0.0657	-0.0502	-0.0204
77	0.2795	0.6668	58.1%	0.0599	0.2156	-0.0594	0.0521	-0.0585	-0.0264
78	0.2868	0.6704	57.2%	0.0615	0.2174	-0.0610	0.0455	-0.0622	-0.0290
79	0.2888	0.6721	57.0%	0.0624	0.2185	-0.0619	0.0403	-0.0664	-0.0317
80	0.2910	0.6747	56.9%	0.0634	0.2203	-0.0627	0.0329	-0.0737	-0.0349
81	0.2940	0.6765	56.5%	0.0641	0.2214	-0.0634	0.0297	-0.0762	-0.0358
82	0.3009	0.6781	55.6%	0.0806	0.2221	-0.0408	0.0144	-0.0782	-0.0659
83	0.3144	0.6798	53.7%	0.0908	0.2226	-0.0298	0.0013	-0.0809	-0.1460
84	0.3293	0.6805	51.6%	0.0996	0.2228	-0.0206	-0.0021	-0.0830	-0.2527
85	0.3532	0.6812	48.2%	0.1114	0.2230	-0.0082	-0.0013	-0.0836	-0.4291
86	0.3897	0.6827	42.9%	0.1283	0.2236	0.0092	0.0000	-0.0843	-0.7062
87	0.3968	0.6828	41.9%	0.1301	0.2235	0.0117	0.0027	-0.0843	-0.7664
88	0.4240	0.6830	37.9%	0.1419	0.2234	0.0239	0.0011	-0.0841	-1.0084
89	0.4490	0.6834	34.3%	0.1508	0.2232	0.0329	-0.0008	-0.0833	-1.2394
90	0.4707	0.6838	31.2%	0.1577	0.2231	0.0395	-0.0022	-0.0827	-1.4464

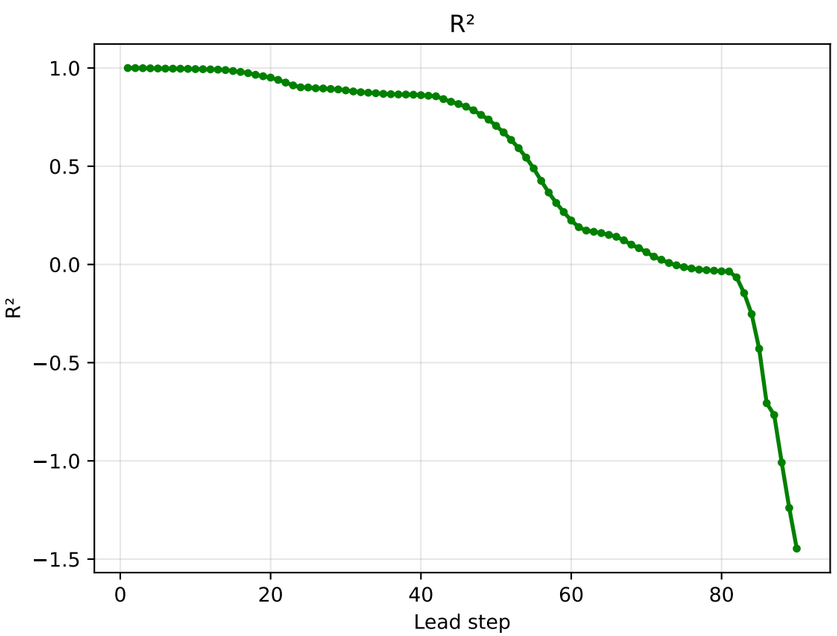
DHW Forecast Skill vs Lead Step - Val - GBR

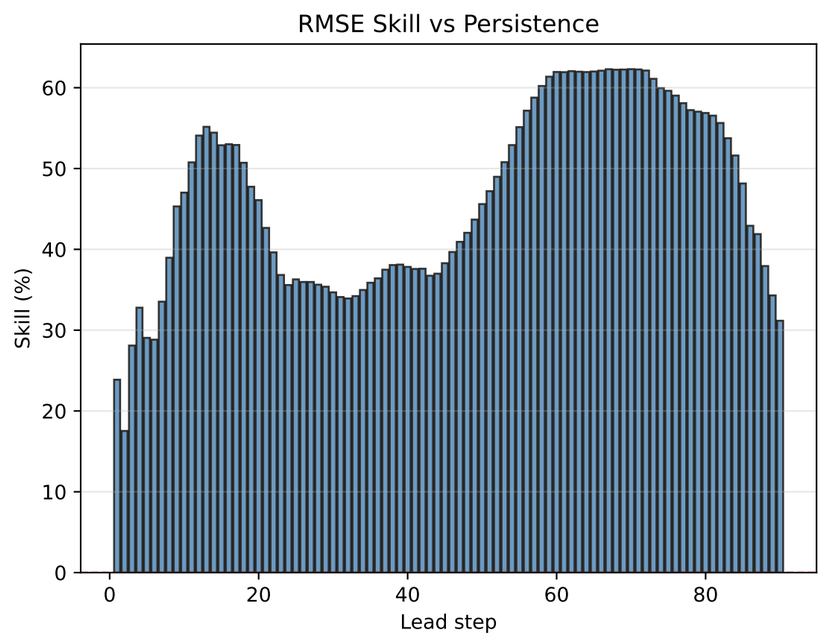


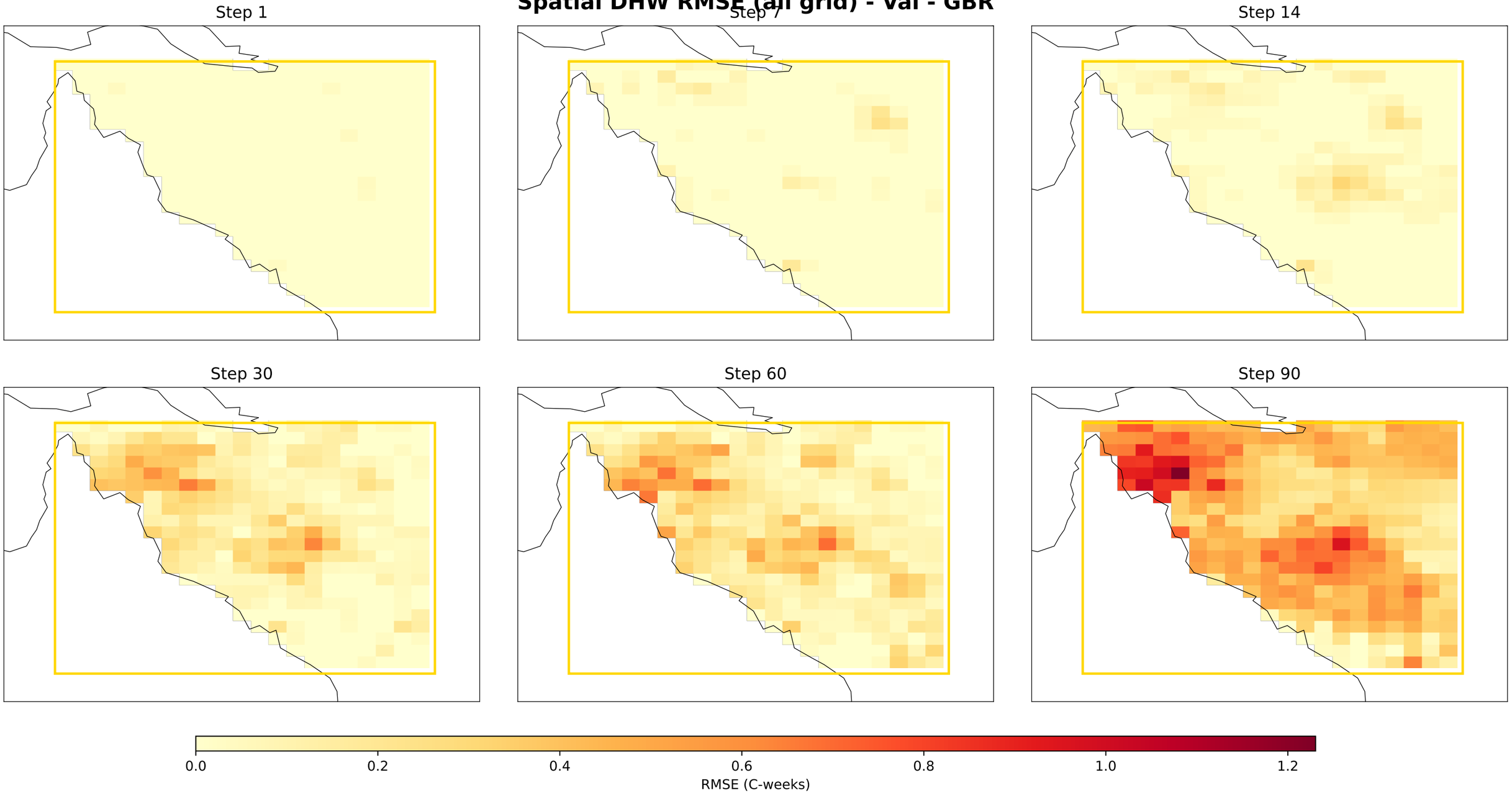


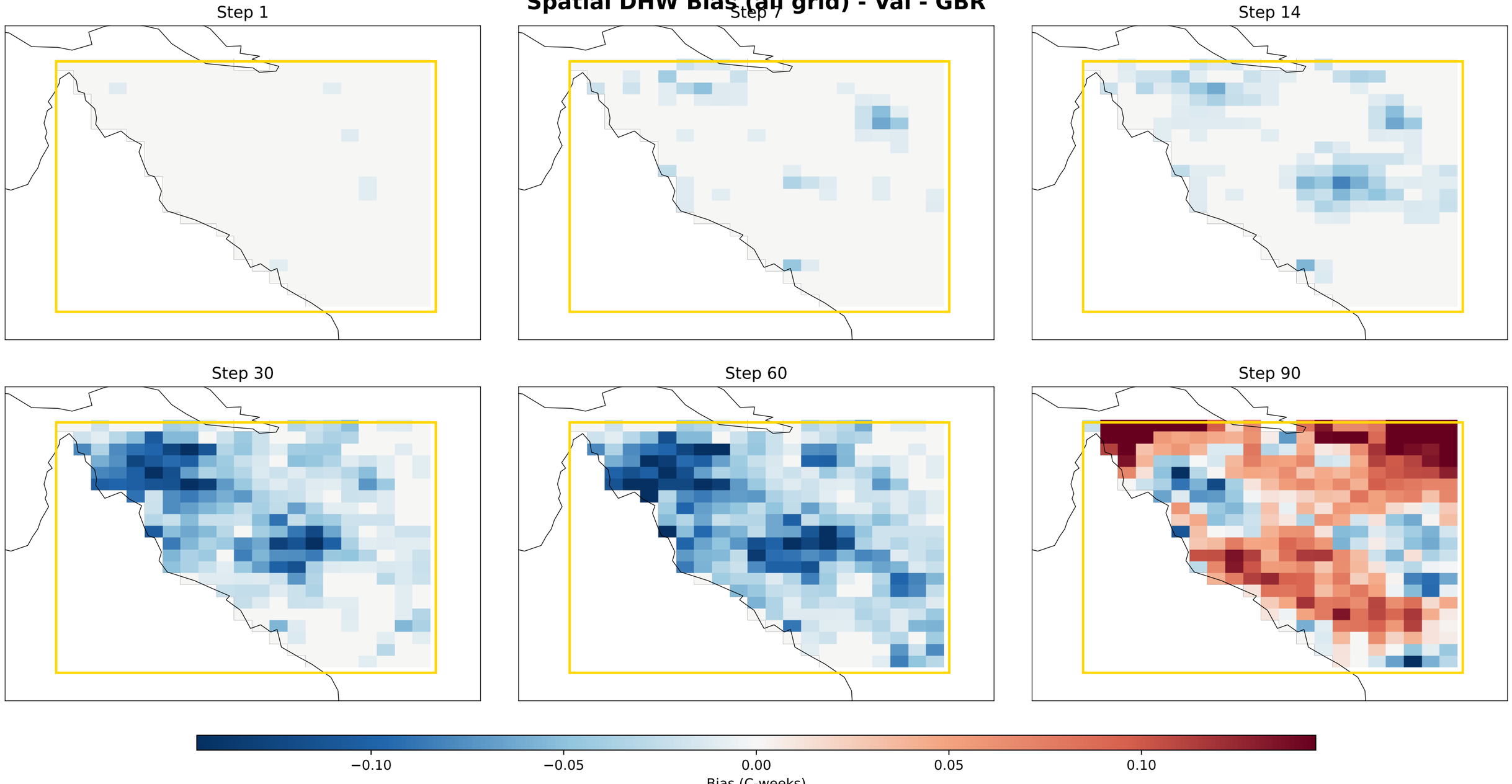


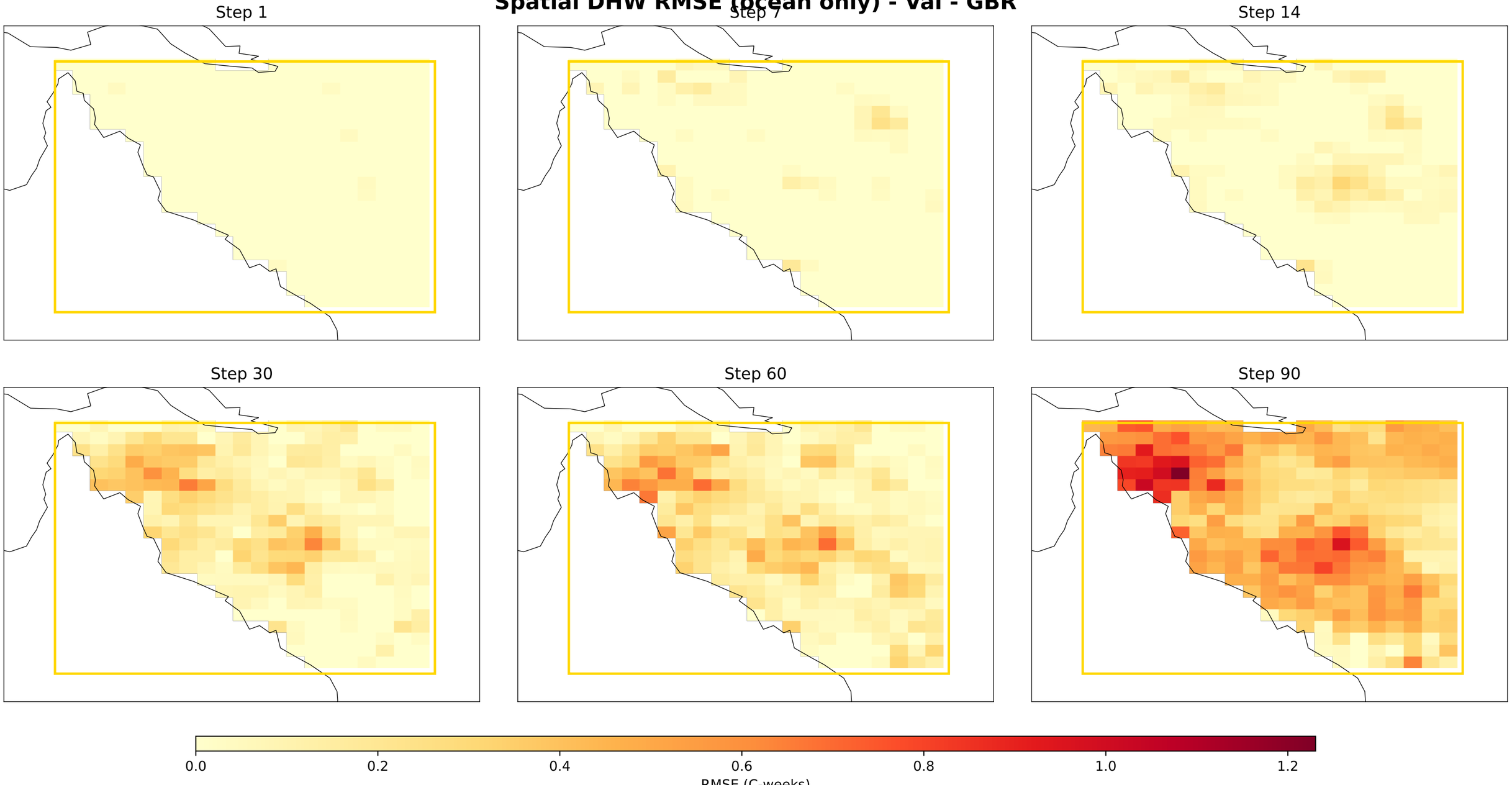


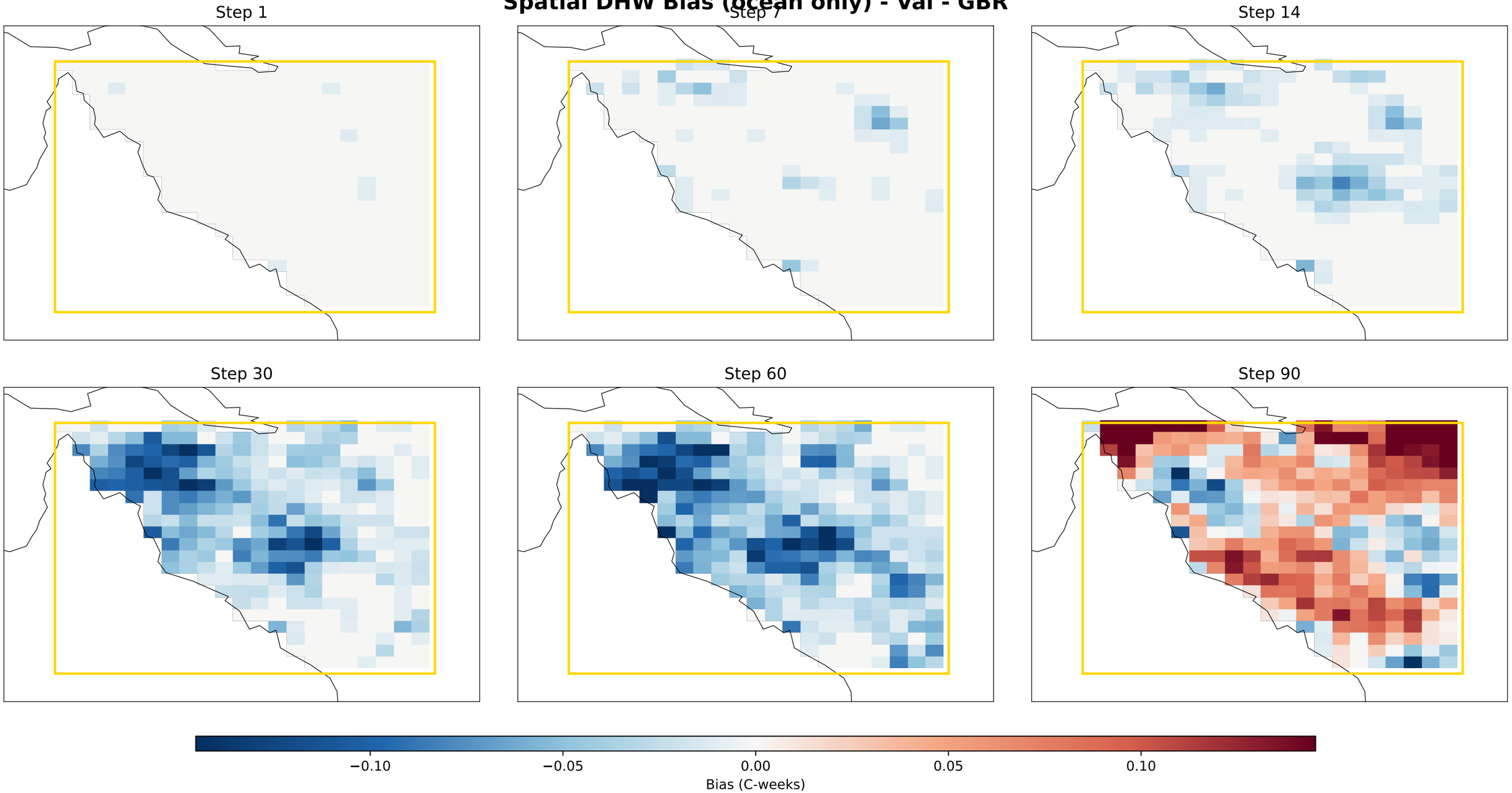





Spatial DHW RMSE (all grid) - Val - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.0
0.2
0.4
0.6
0.8
1.0
1.2
RMSE (C-weeks)

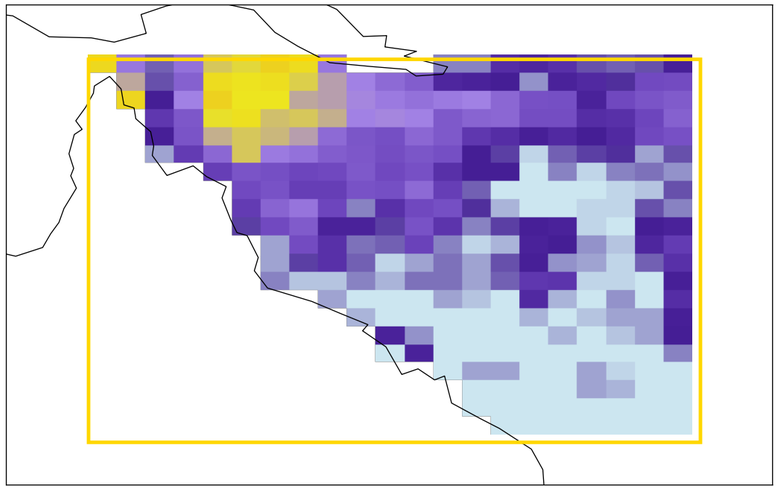

Spatial DHW Bias (all grid) - Val - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−0.10
−0.05
0.00
0.05
0.10
Bias (C-weeks)


Spatial DHW RMSE (ocean only) - Val - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.0
0.2
0.4
0.6
0.8
1.0
1.2
RMSE (C-weeks)

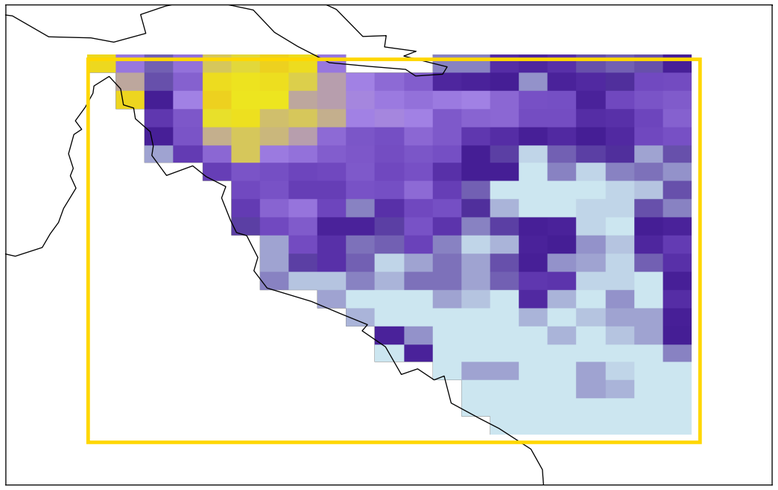

Spatial DHW Bias (ocean only) - Val - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−0.10
−0.05
0.00
0.05
0.10
Bias (C-weeks)

DHW Chain init: 2016-04-10 - Val - GBR (C-weeks)

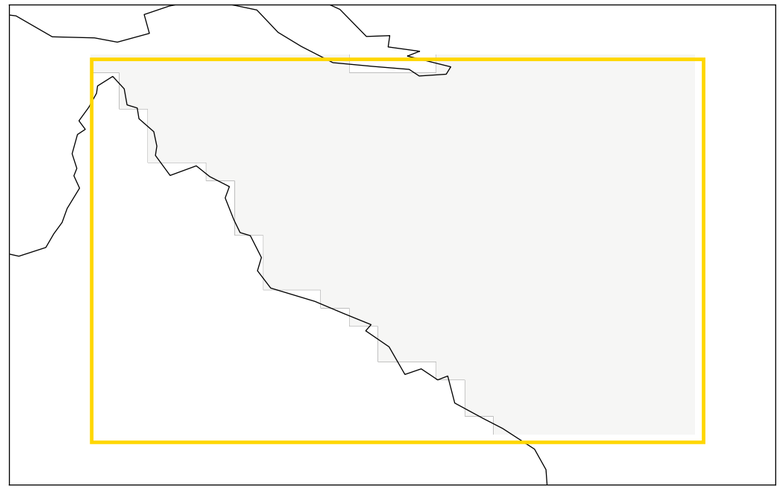
Actual DHW - Step 1



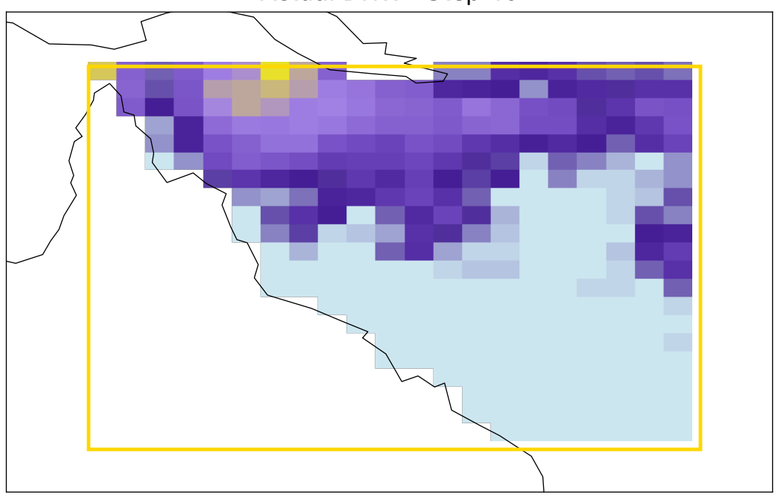
Predicted DHW - Step 1



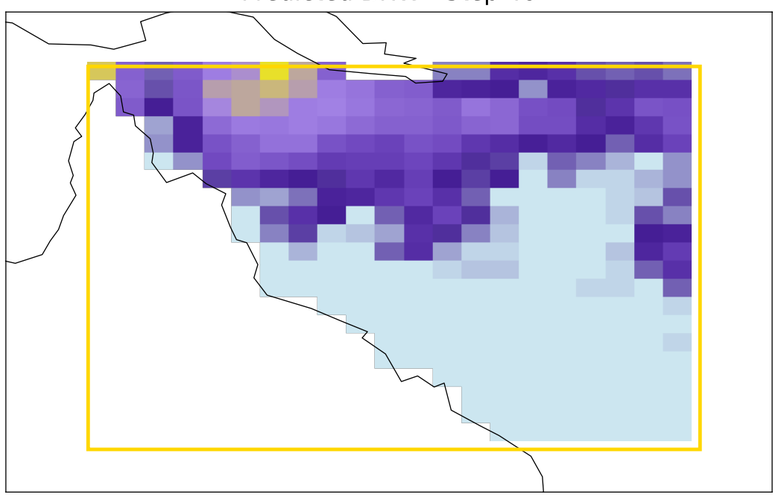
Error (Pred - Actual) - Step 1 [scale ±8]



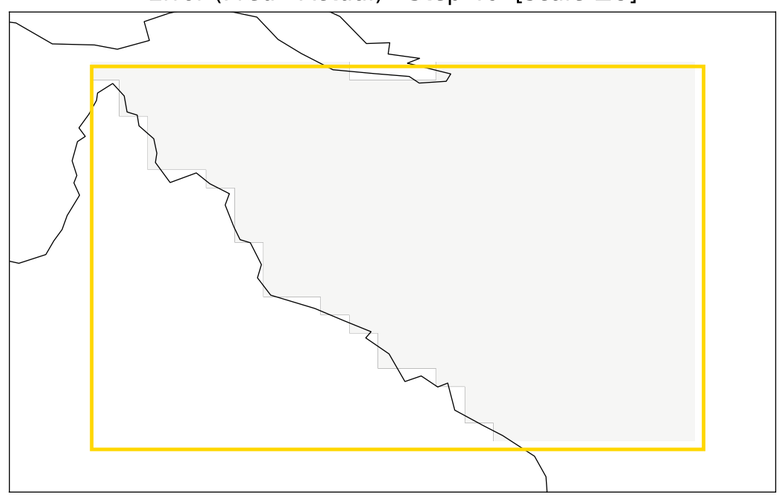
Actual DHW - Step 46



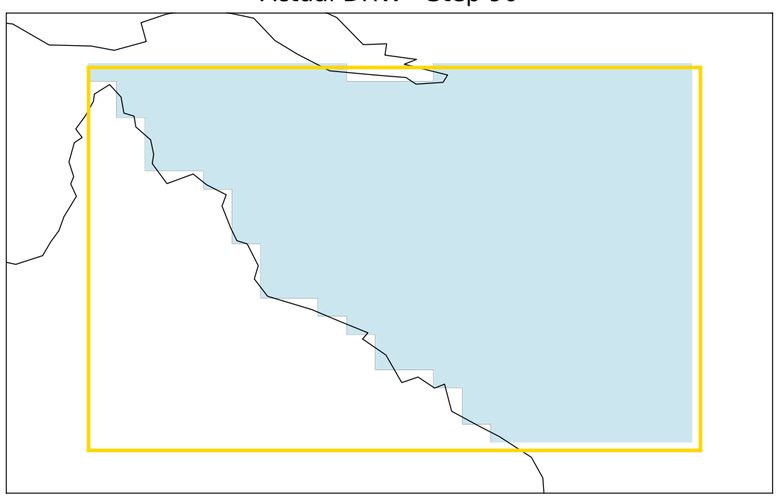
Predicted DHW - Step 46



Error (Pred - Actual) - Step 46 [scale ±8]

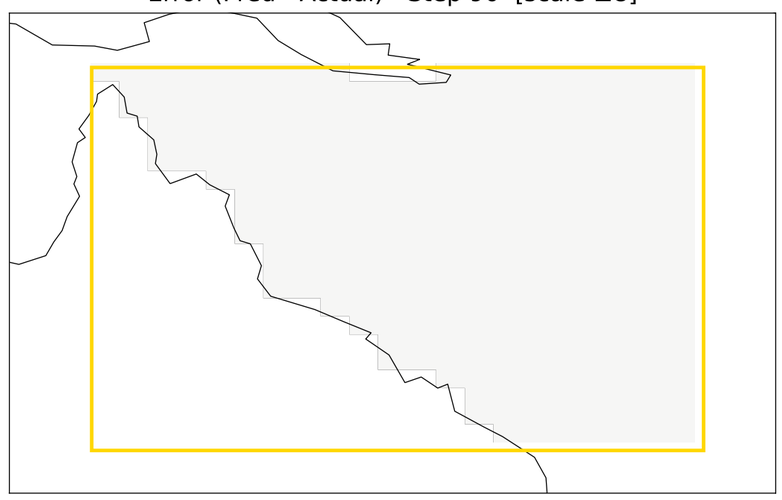


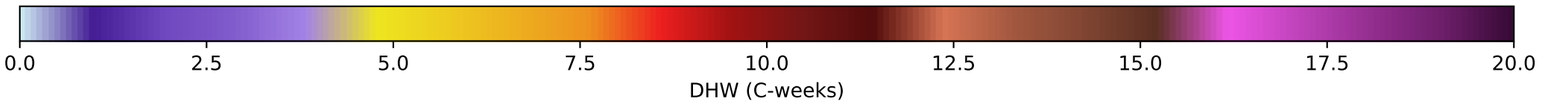
Actual DHW - Step 90



Predicted DHW - Step 90

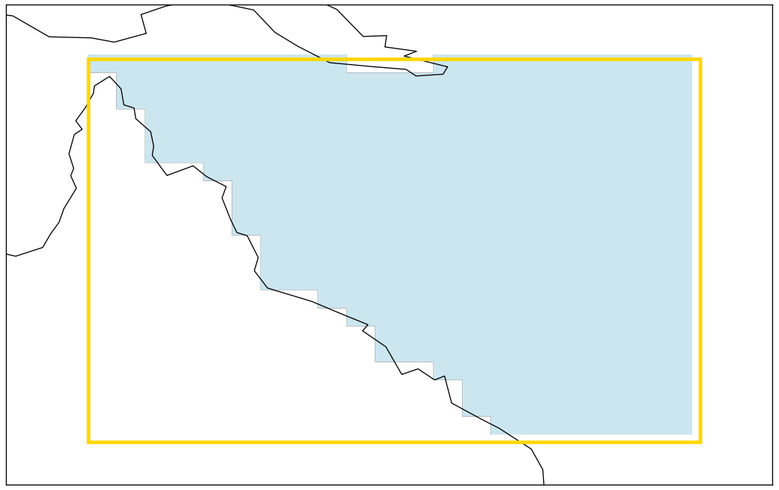
Error (Pred - Actual) - Step 90 [scale ±8]



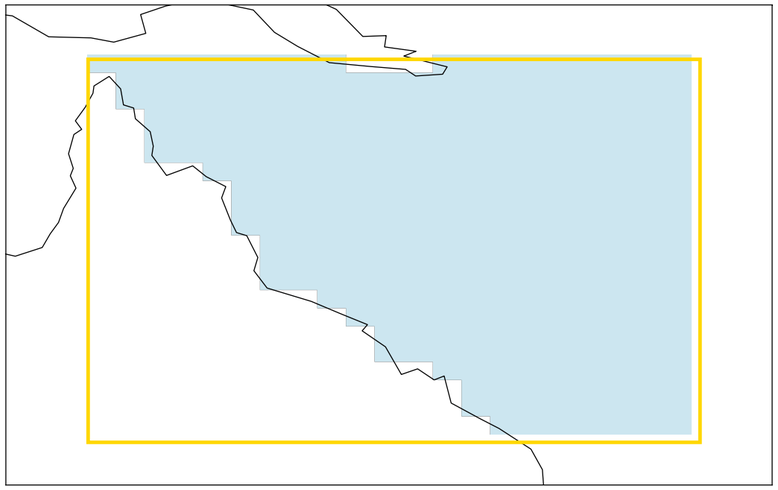


DHW Chain init: 2016-10-07 - Val - GBR (C-weeks)

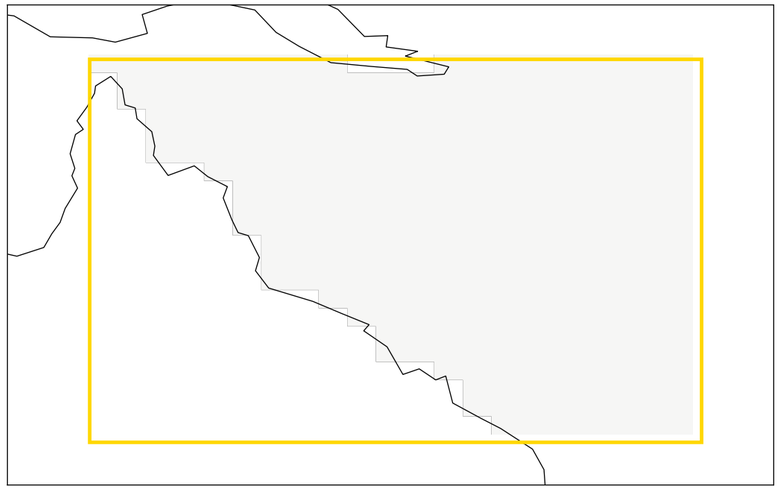
Actual DHW - Step 1



Predicted DHW - Step 1

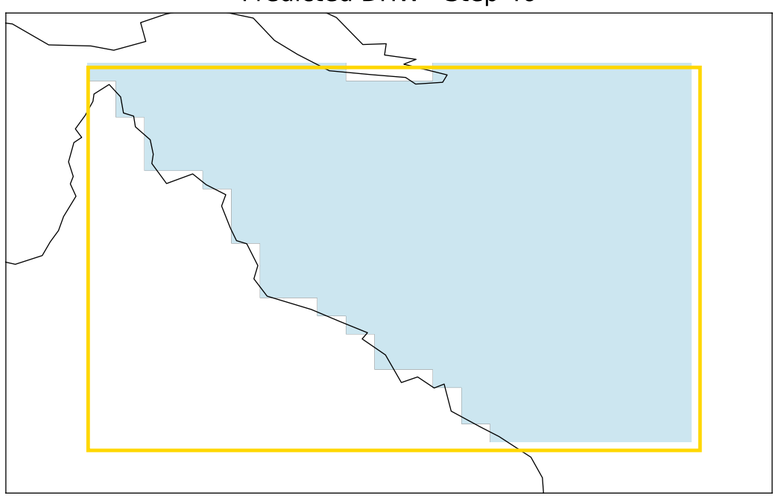


Error (Pred - Actual) - Step 1 [scale ±8]

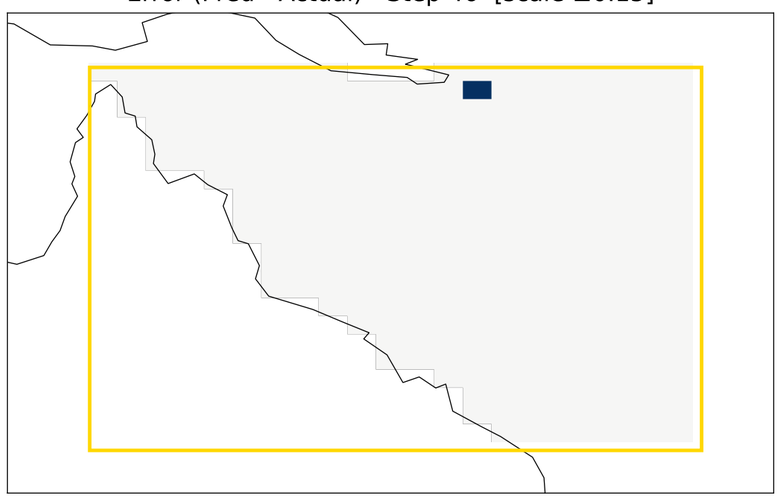


Actual DHW - Step 46

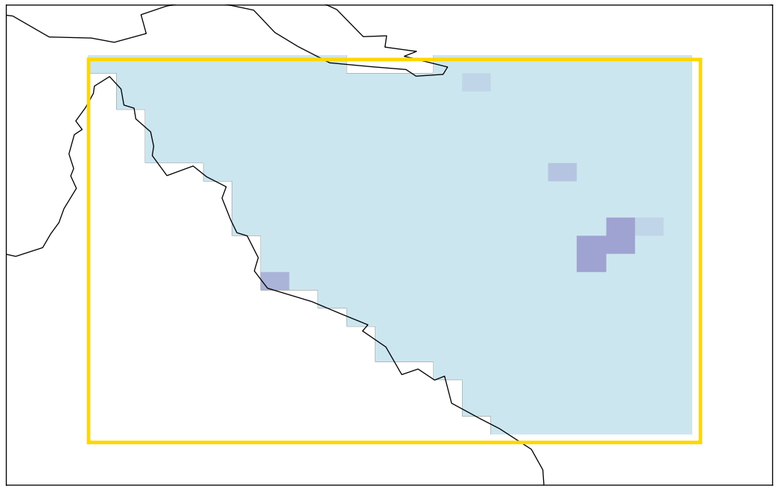
Predicted DHW - Step 46



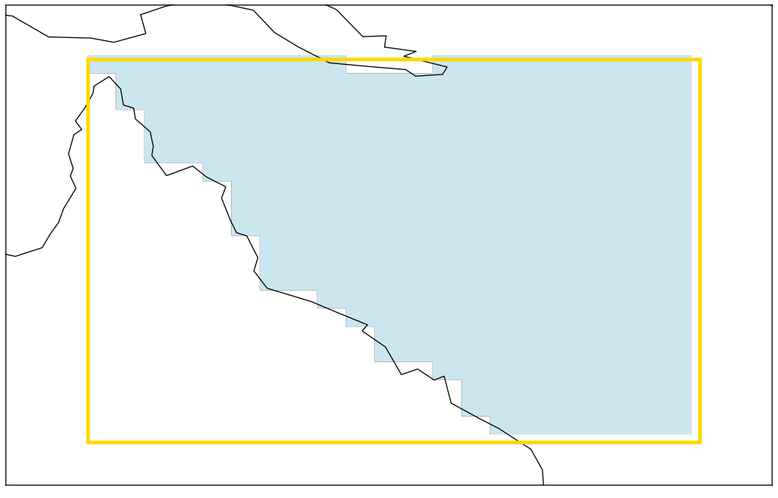
Error (Pred - Actual) - Step 46 [scale ±0.15]



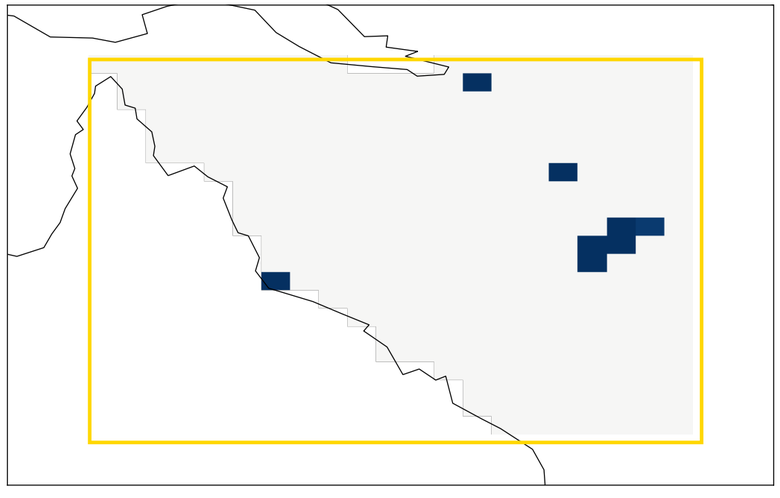
Actual DHW - Step 90

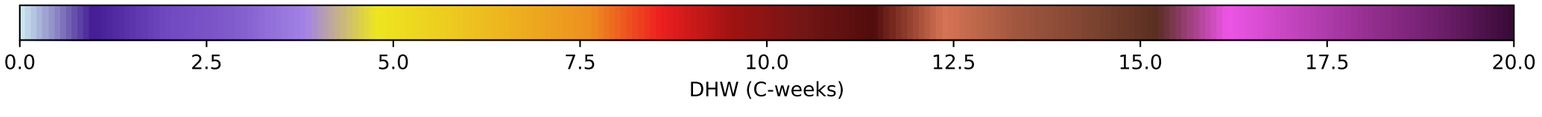


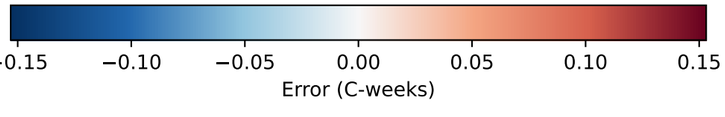
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.15]

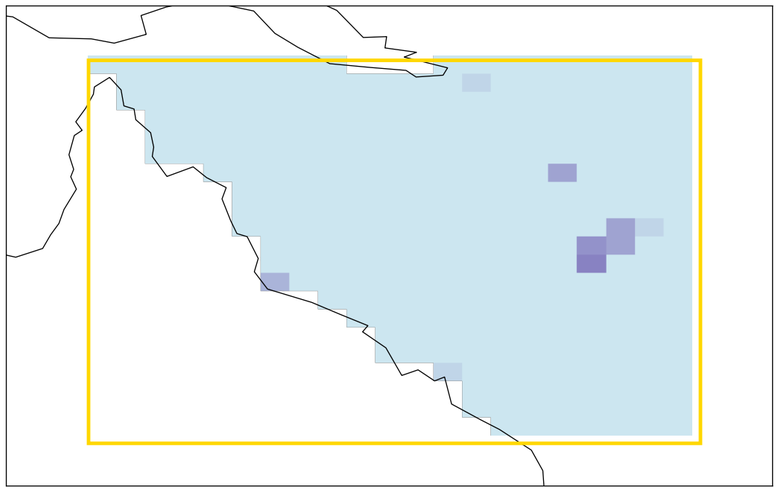




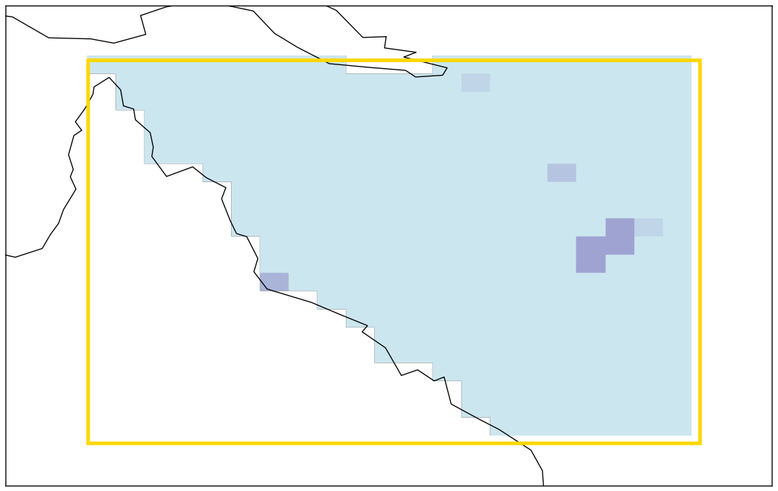


DHW Chain init: 2017-01-05 - Val - GBR (C-weeks)

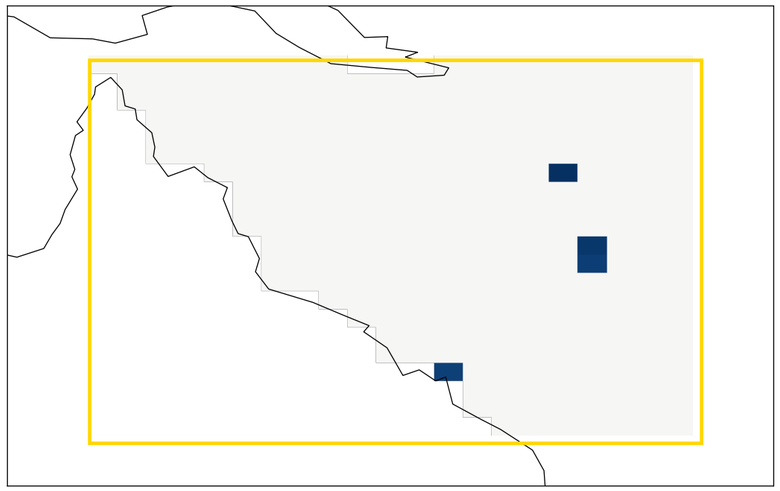
Actual DHW - Step 1



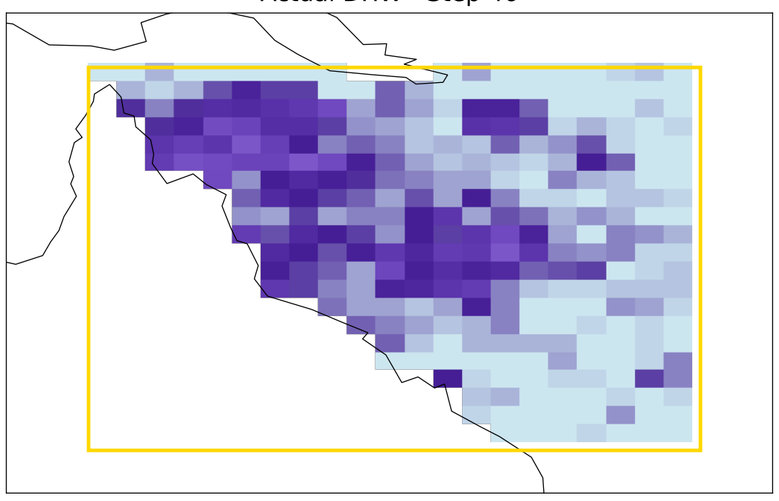
Predicted DHW - Step 1



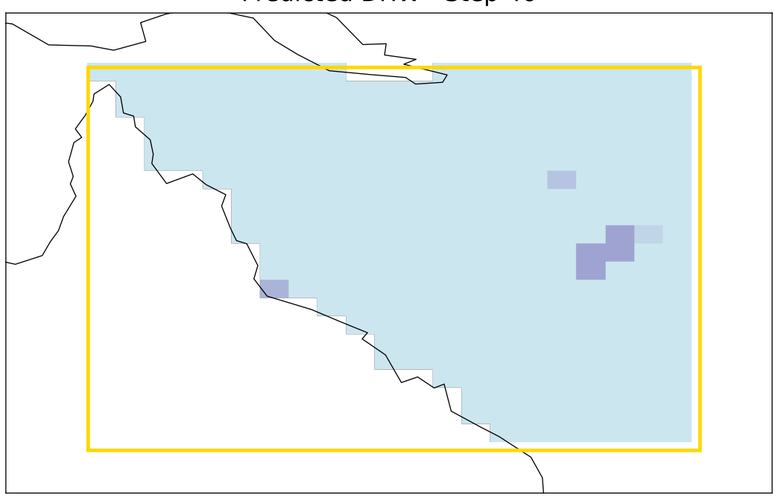
Error (Pred - Actual) - Step 1 [scale ±0.16]



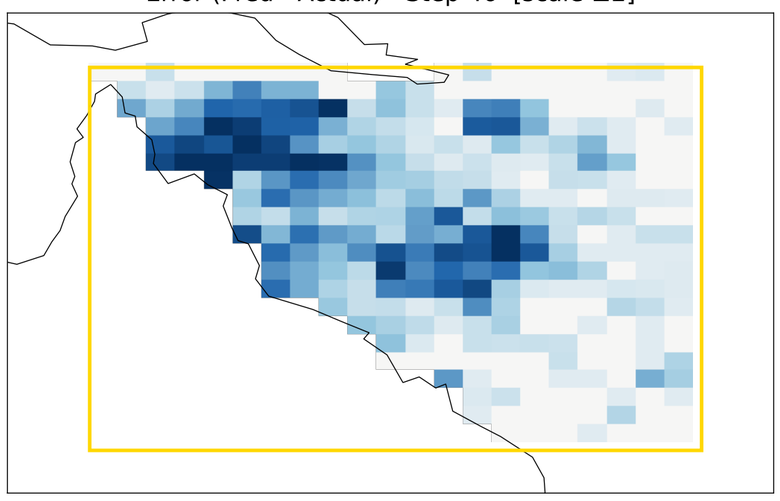
Actual DHW - Step 46



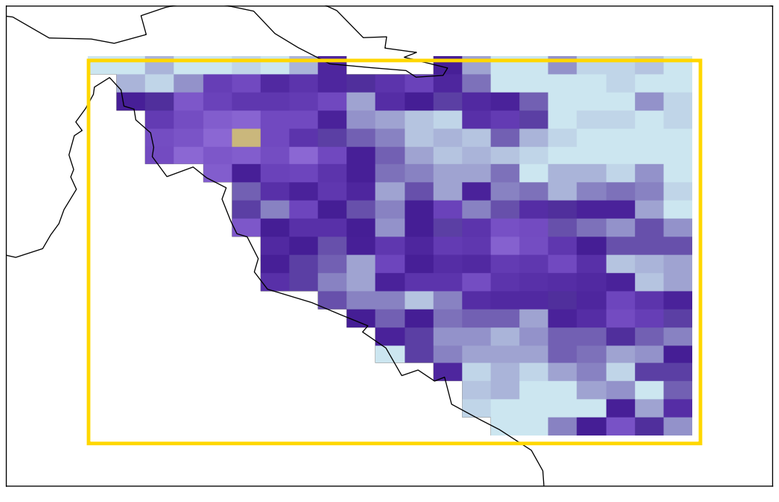
Predicted DHW - Step 46



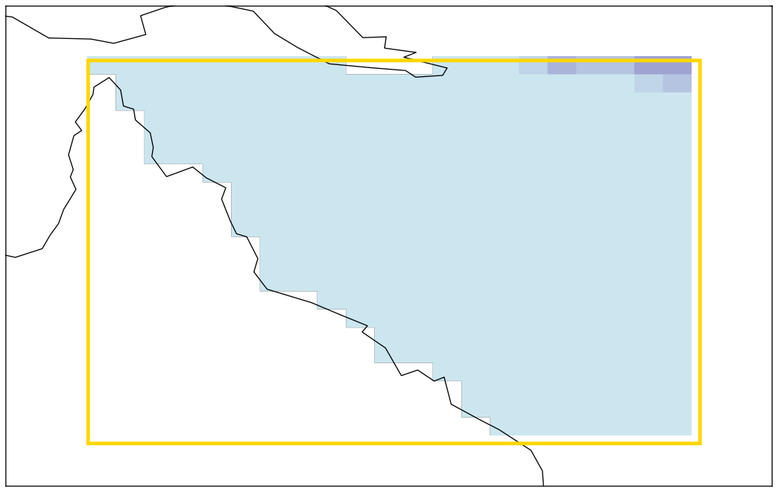
Error (Pred - Actual) - Step 46 [scale ±2]



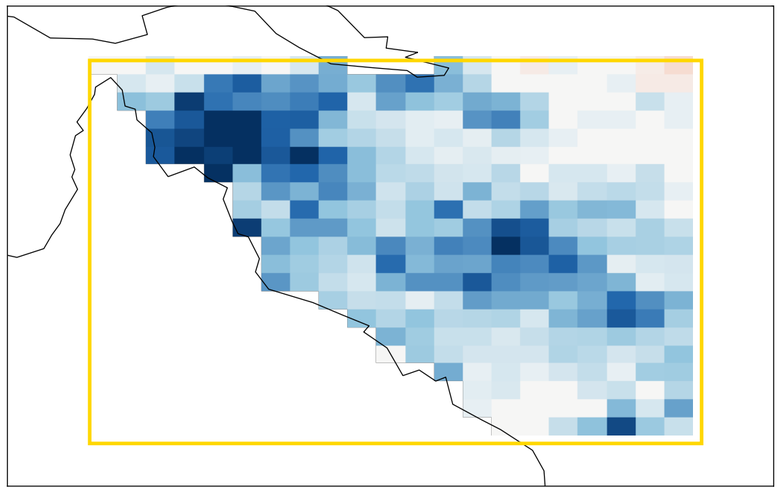
Actual DHW - Step 90

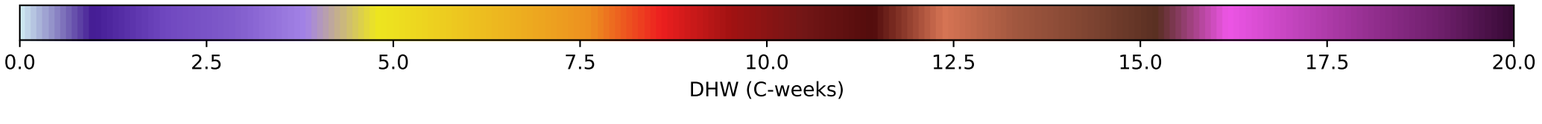


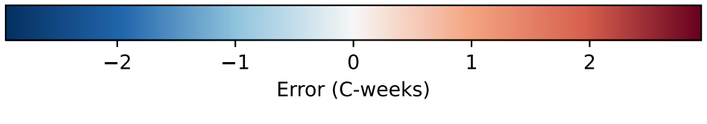
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±2.9]

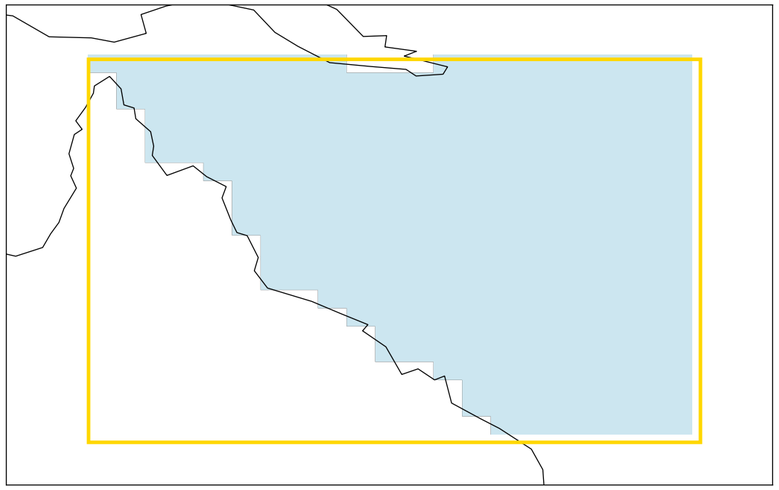




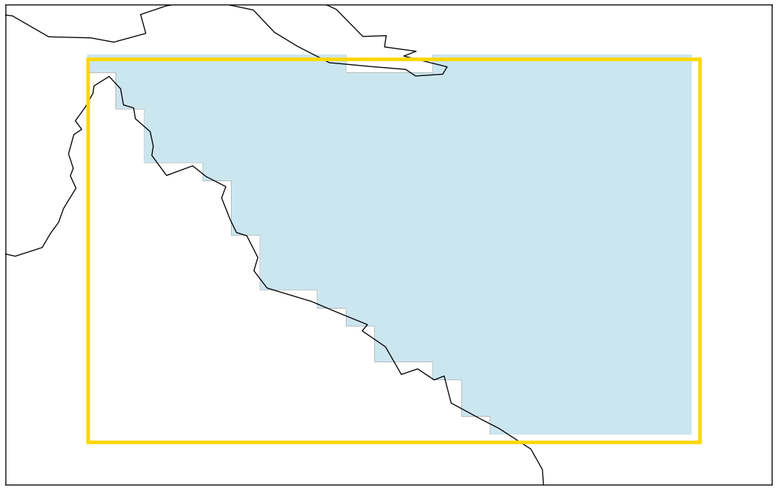


DHW Chain init: 2017-07-04 - Val - GBR (C-weeks)

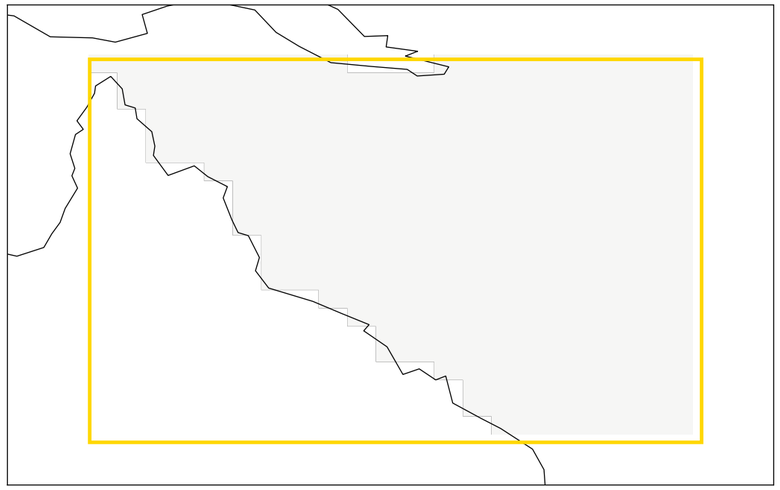
Actual DHW - Step 1



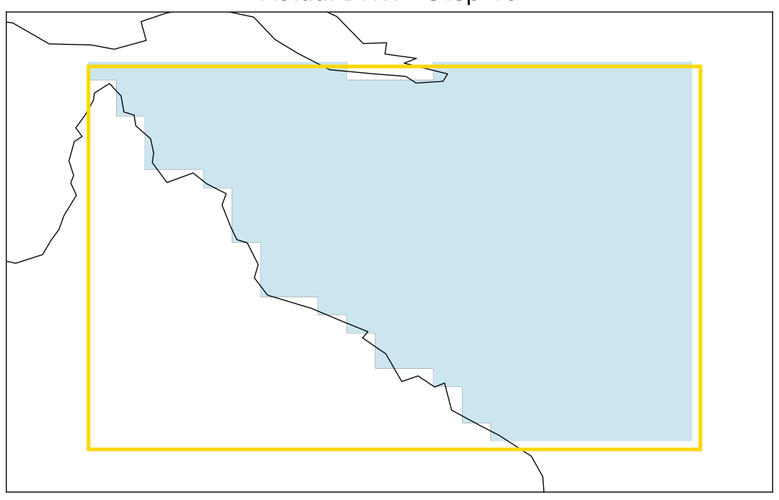
Predicted DHW - Step 1



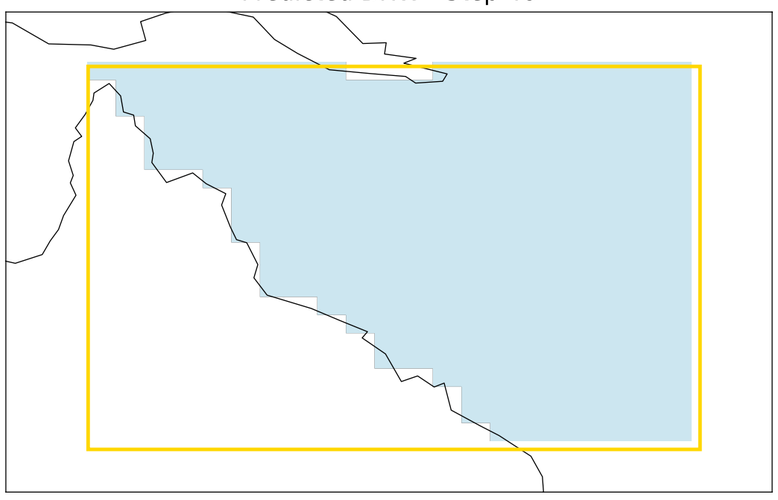
Error (Pred - Actual) - Step 1 [scale ±8]



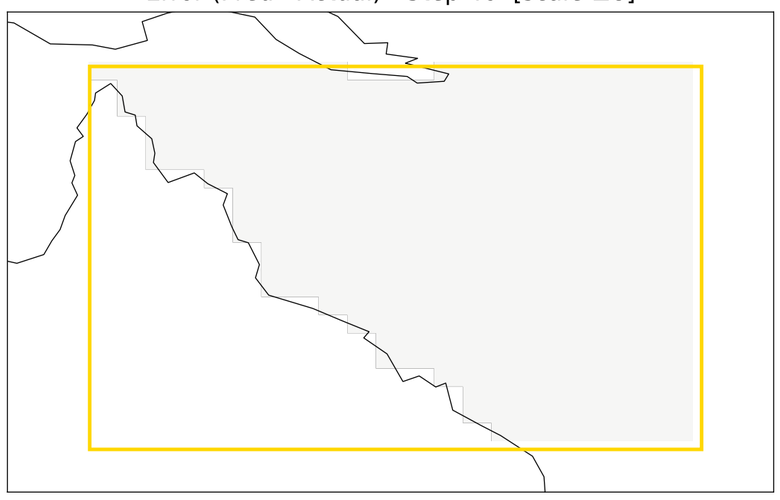
Actual DHW - Step 46



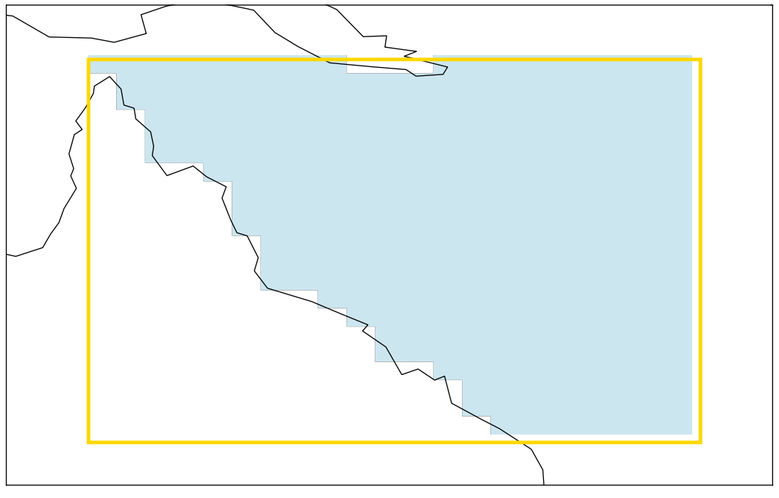
Predicted DHW - Step 46



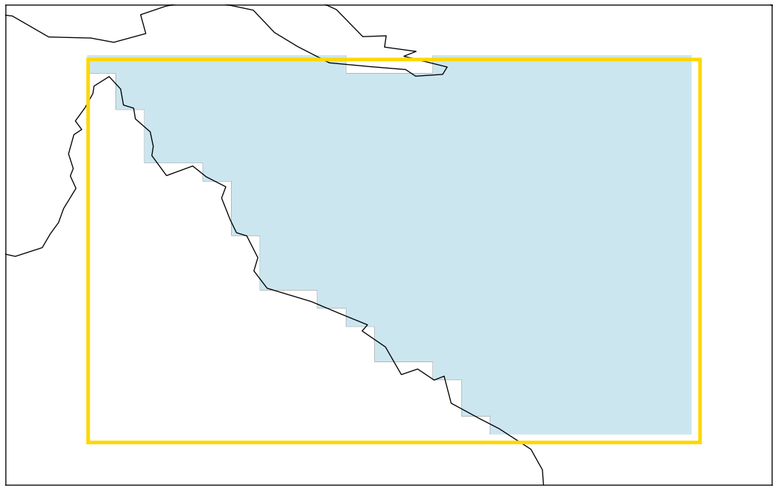
Error (Pred - Actual) - Step 46 [scale ±8]



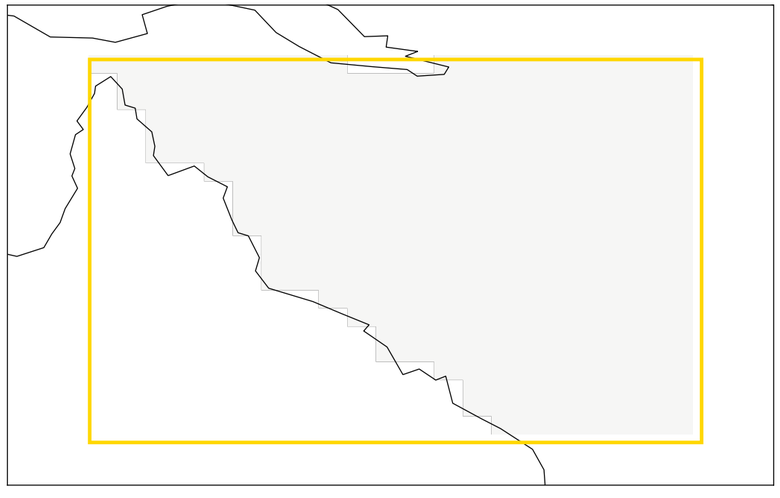
Actual DHW - Step 90

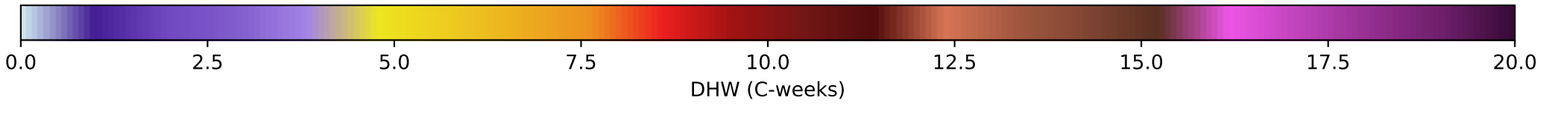


Predicted DHW - Step 90



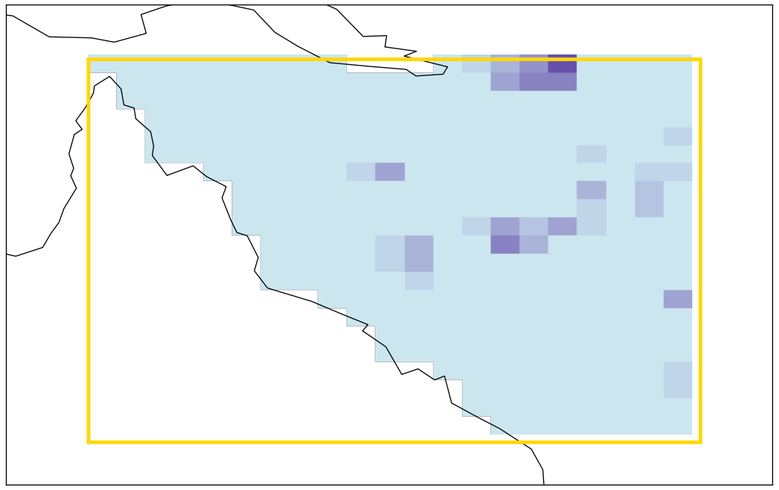
Error (Pred - Actual) - Step 90 [scale ±8]



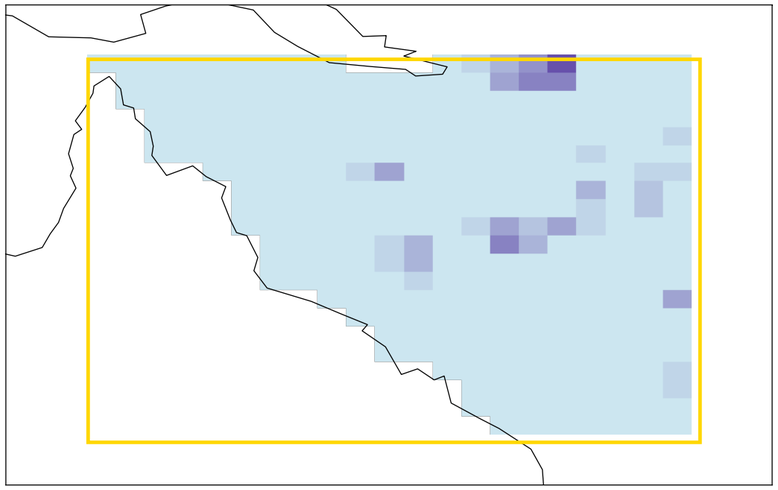


DHW Chain init: 2018-03-31 - Val - GBR (C-weeks)

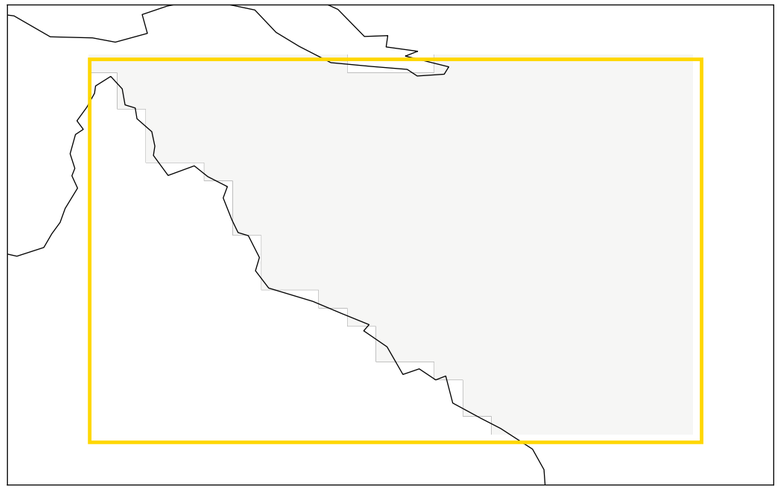
Actual DHW - Step 1



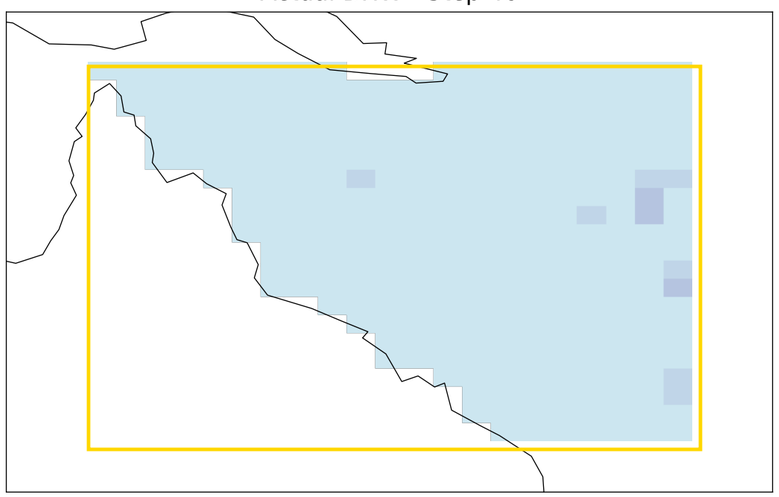
Predicted DHW - Step 1



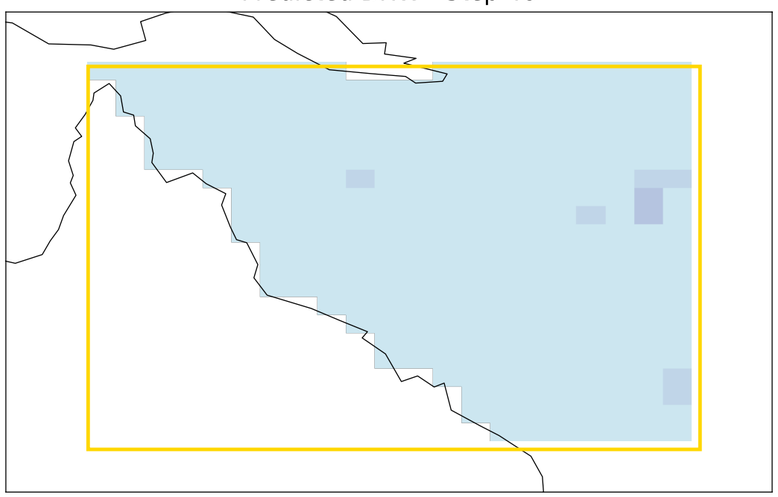
Error (Pred - Actual) - Step 1 [scale ±8]



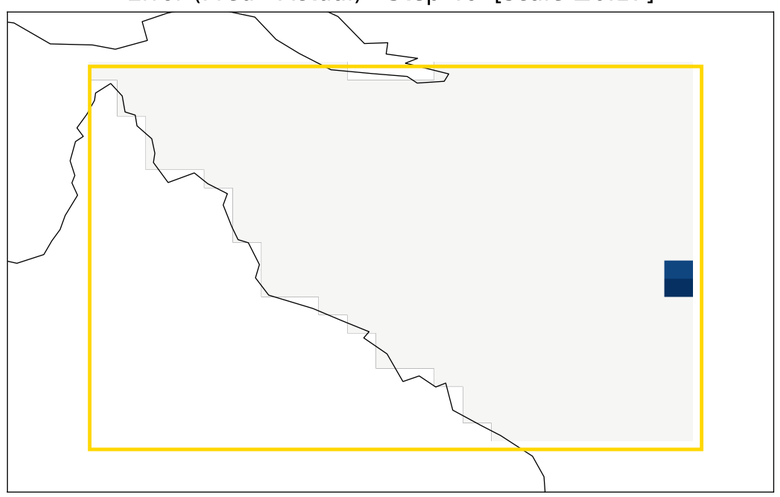
Actual DHW - Step 46



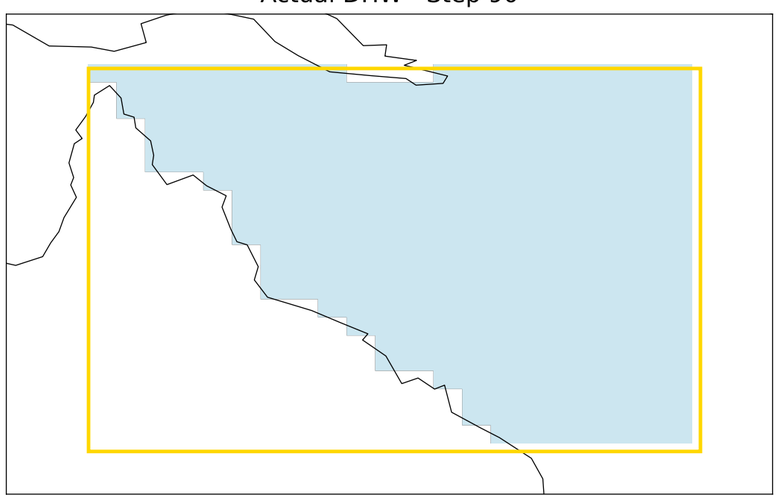
Predicted DHW - Step 46



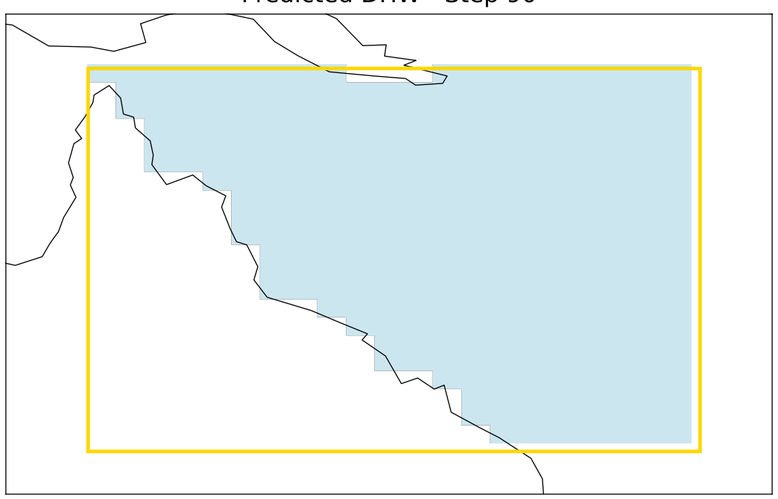
Error (Pred - Actual) - Step 46 [scale ±0.17]



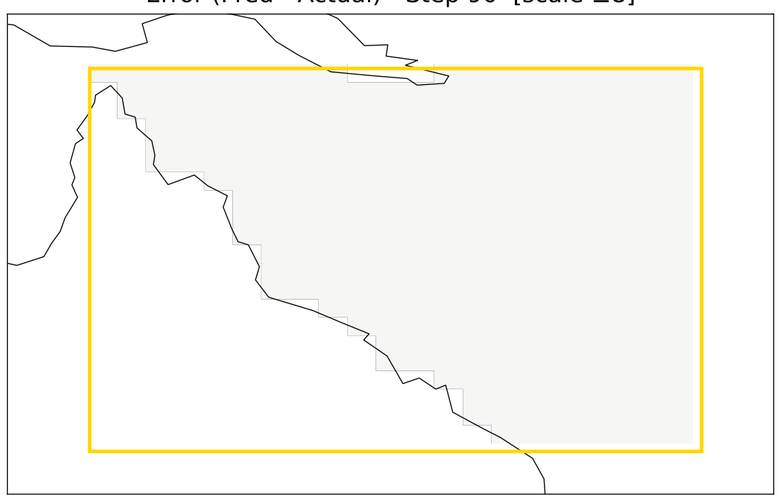
Actual DHW - Step 90

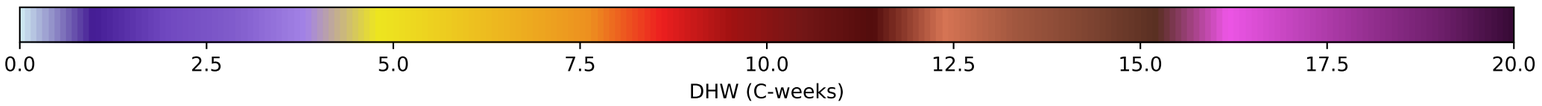


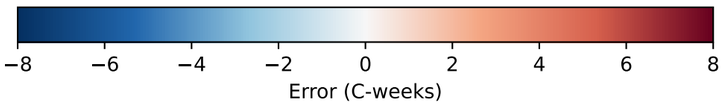
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

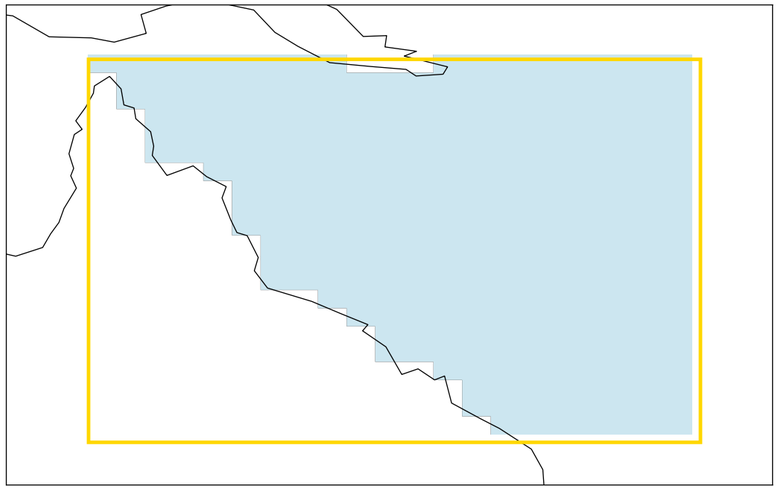




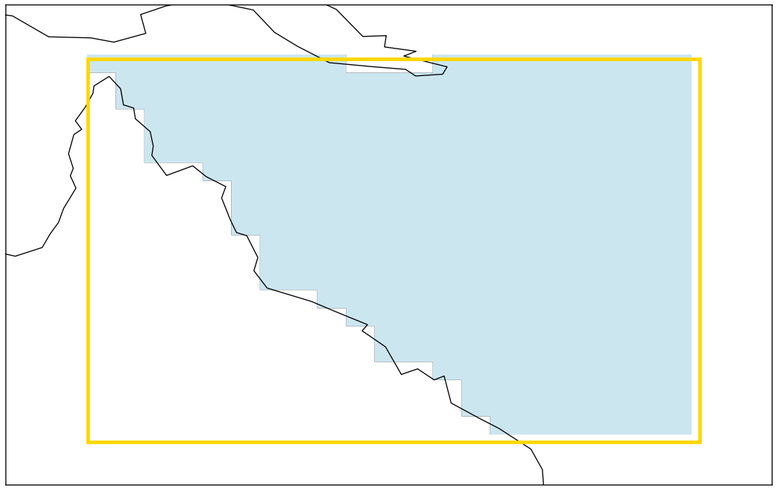


DHW Chain init: 2018-06-29 - Val - GBR (C-weeks)

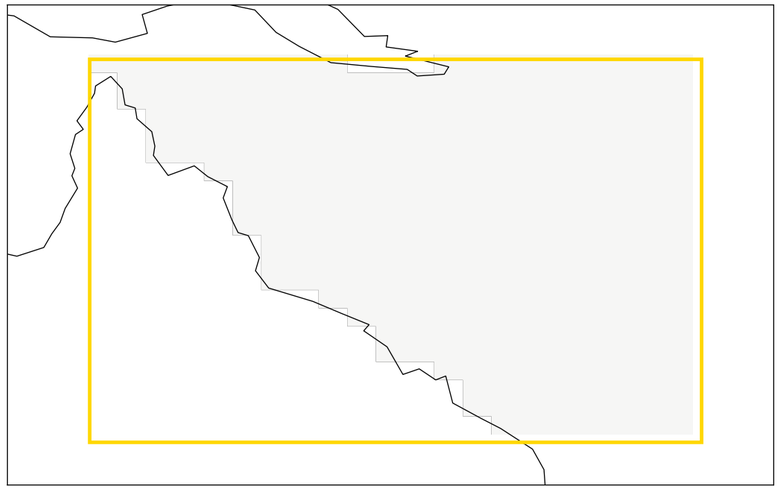
Actual DHW - Step 1



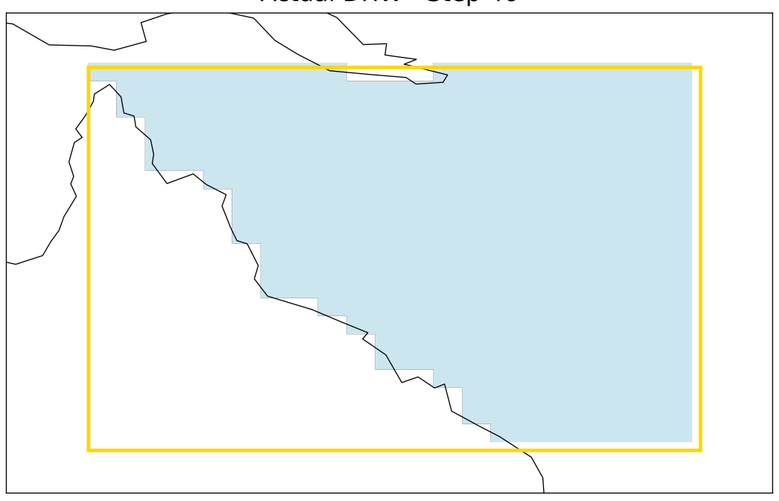
Predicted DHW - Step 1



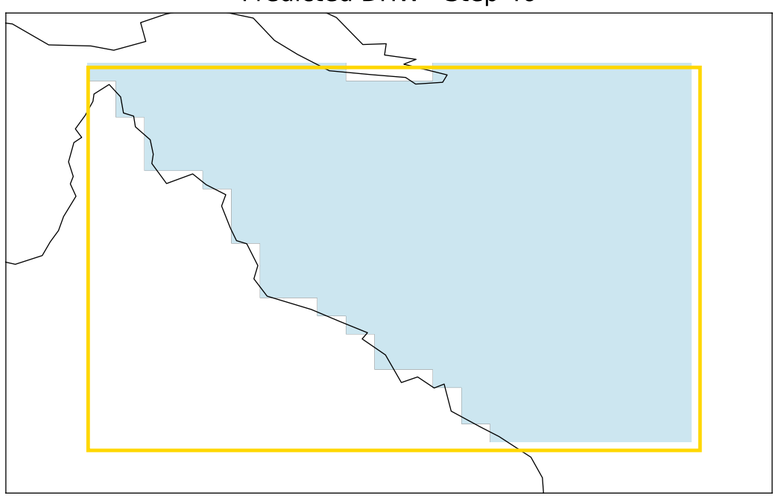
Error (Pred - Actual) - Step 1 [scale ±8]



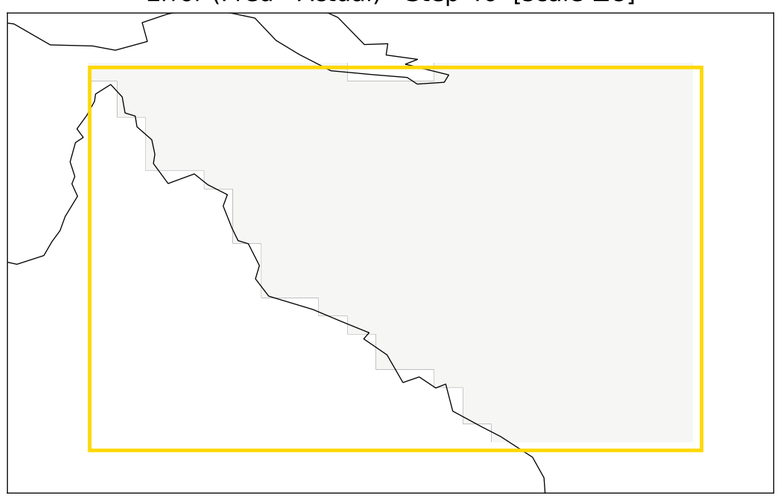
Actual DHW - Step 46



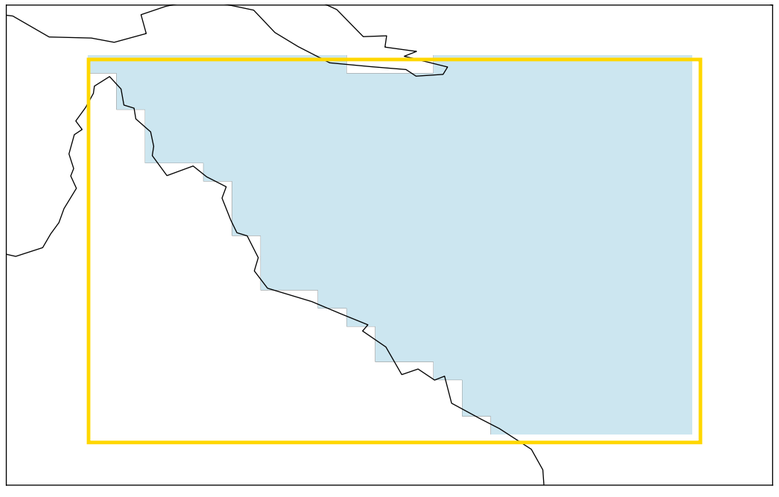
Predicted DHW - Step 46



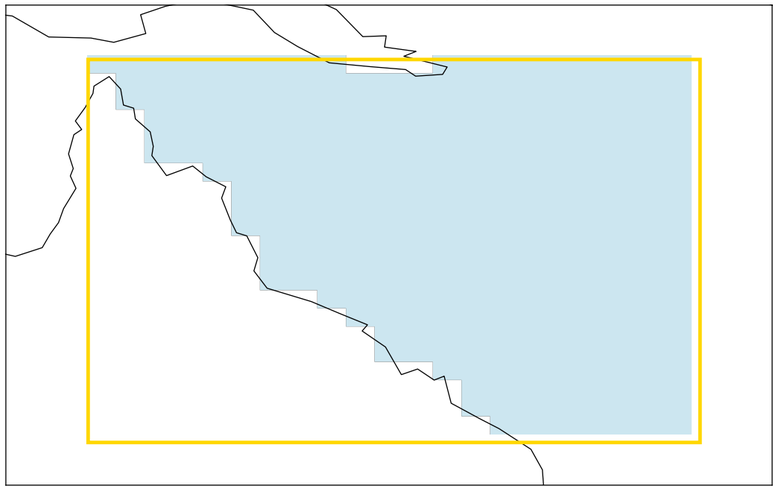
Error (Pred - Actual) - Step 46 [scale ±8]



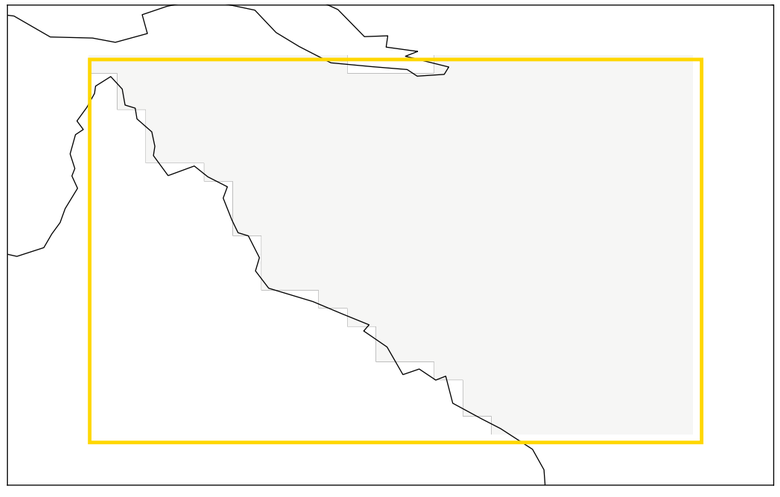
Actual DHW - Step 90

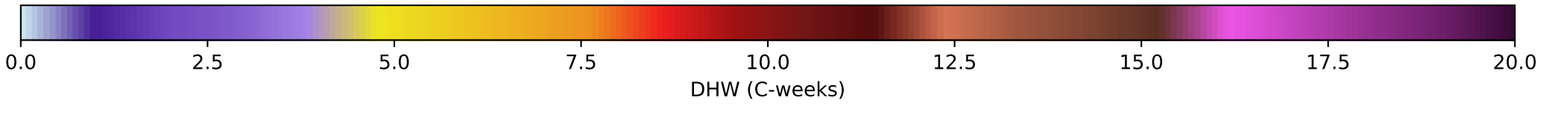


Predicted DHW - Step 90



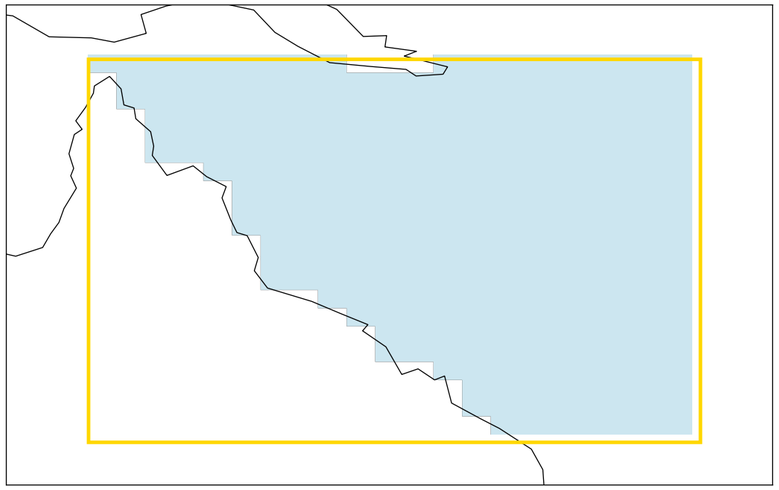
Error (Pred - Actual) - Step 90 [scale ±8]



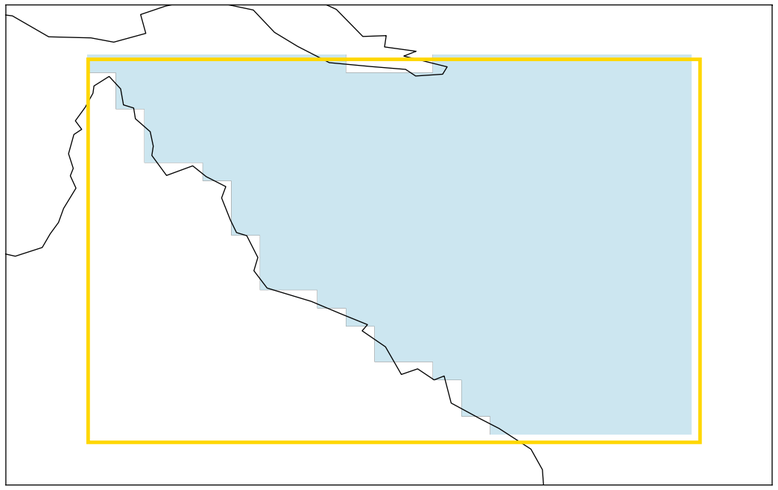


DHW Chain init: 2019-09-22 - Val - GBR (C-weeks)

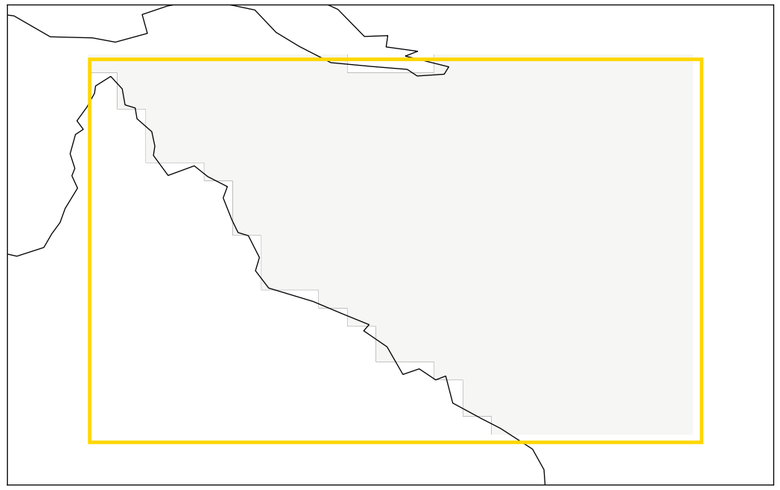
Actual DHW - Step 1



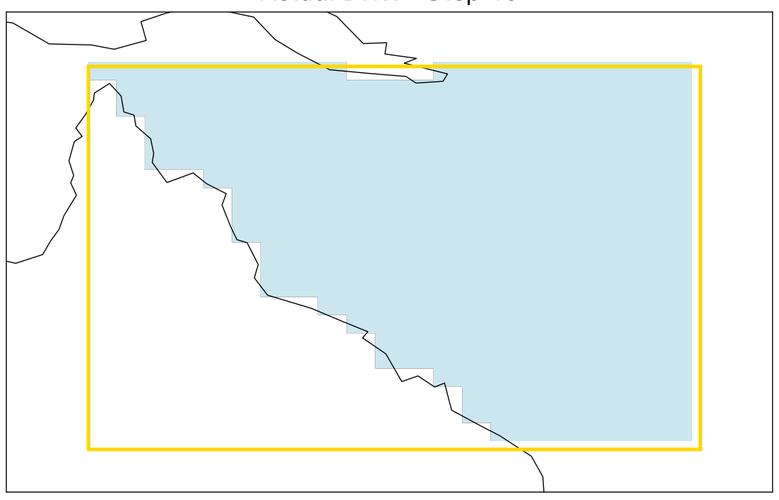
Predicted DHW - Step 1



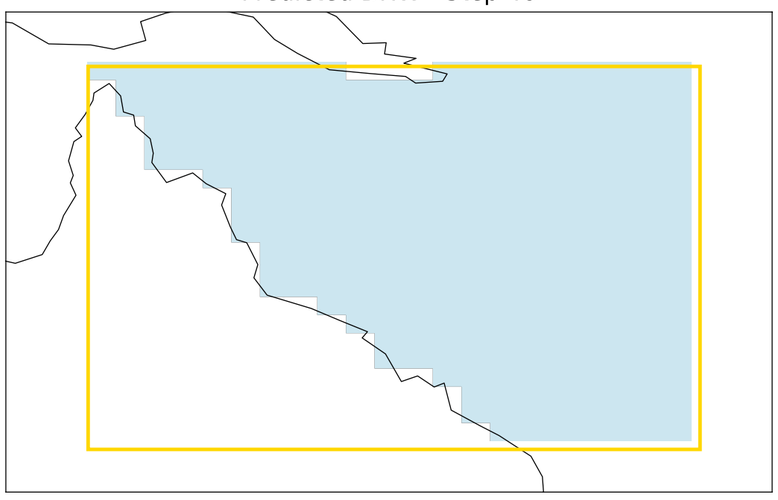
Error (Pred - Actual) - Step 1 [scale ±8]



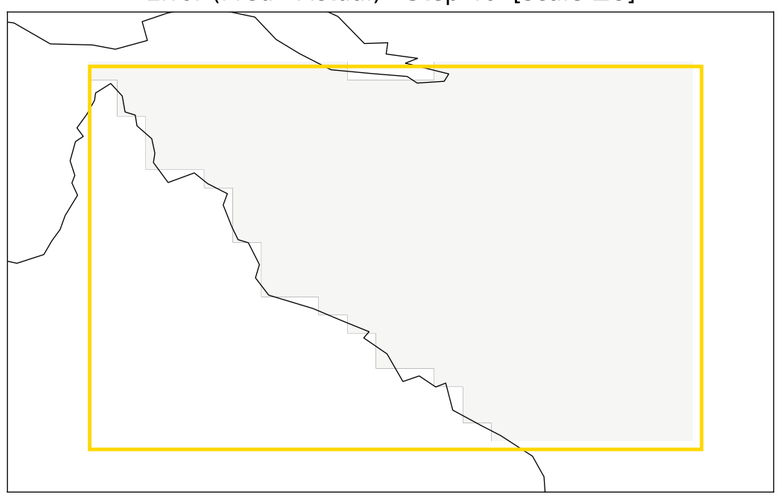
Actual DHW - Step 46



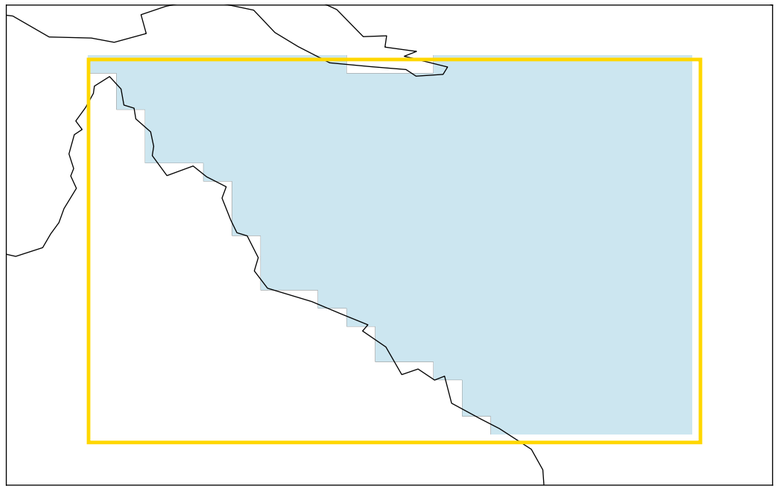
Predicted DHW - Step 46



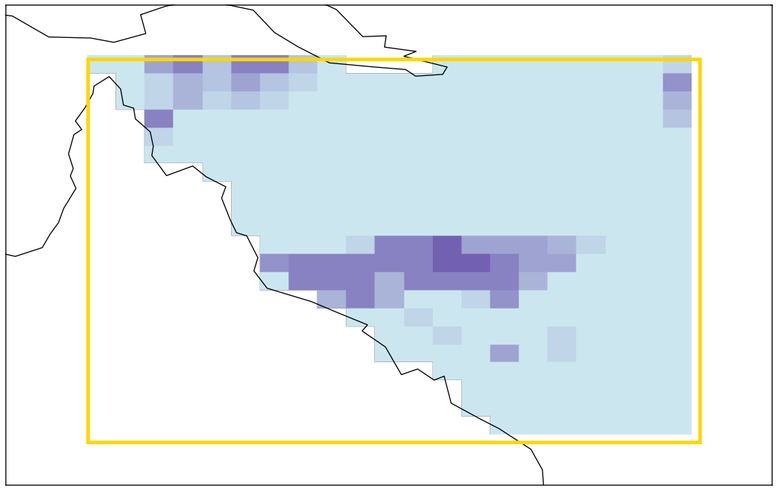
Error (Pred - Actual) - Step 46 [scale ±8]



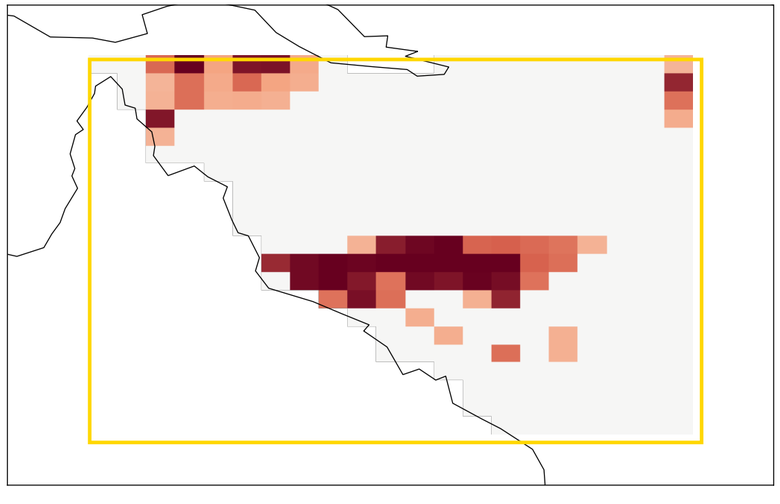
Actual DHW - Step 90

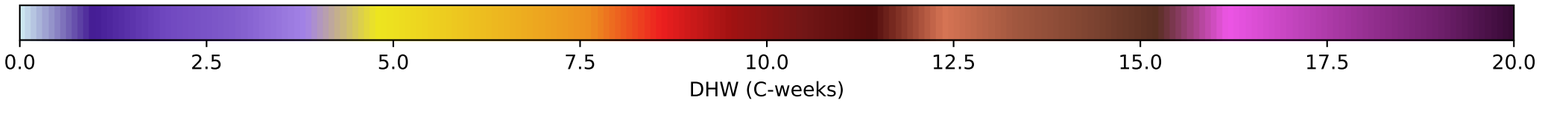


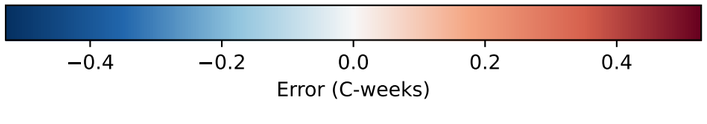
Predicted DHW - Step 90



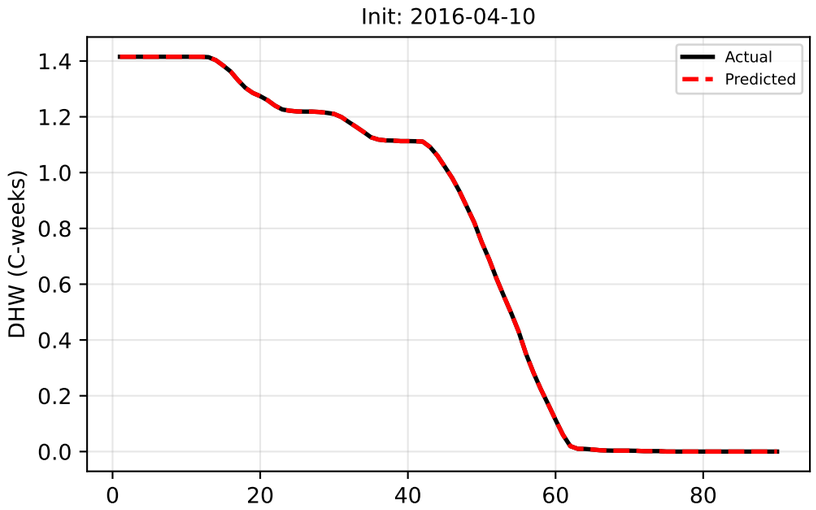
Error (Pred - Actual) - Step 90 [scale ±0.53]

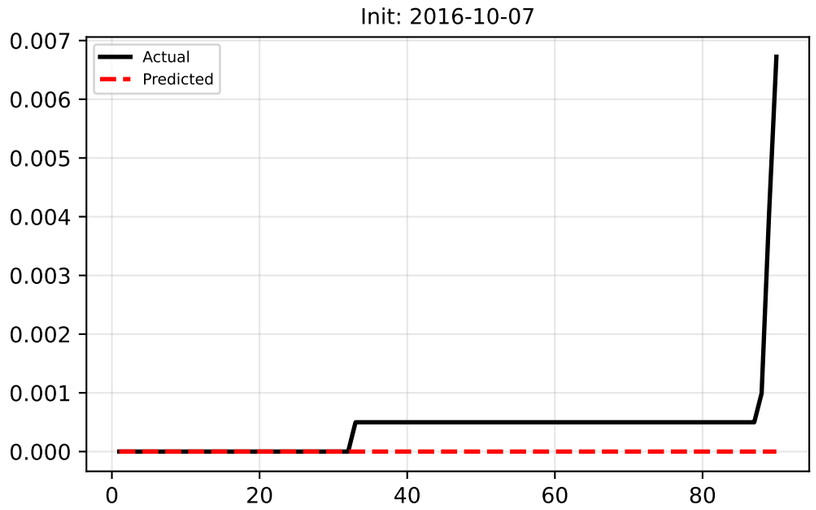


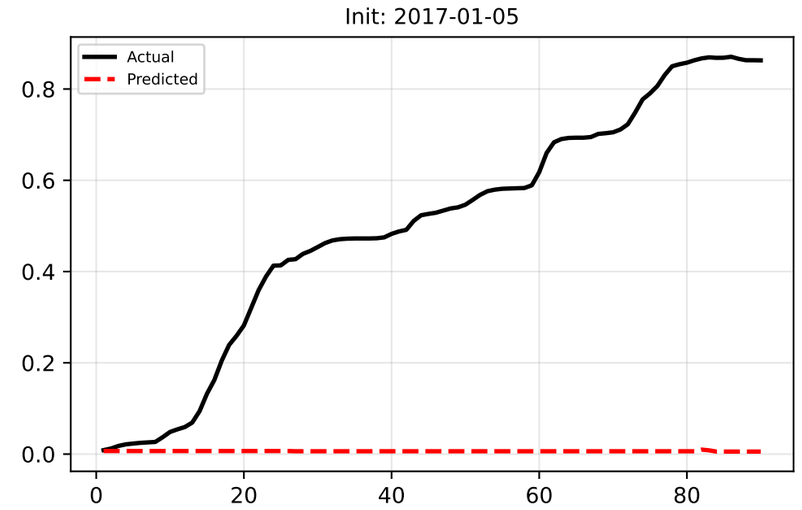


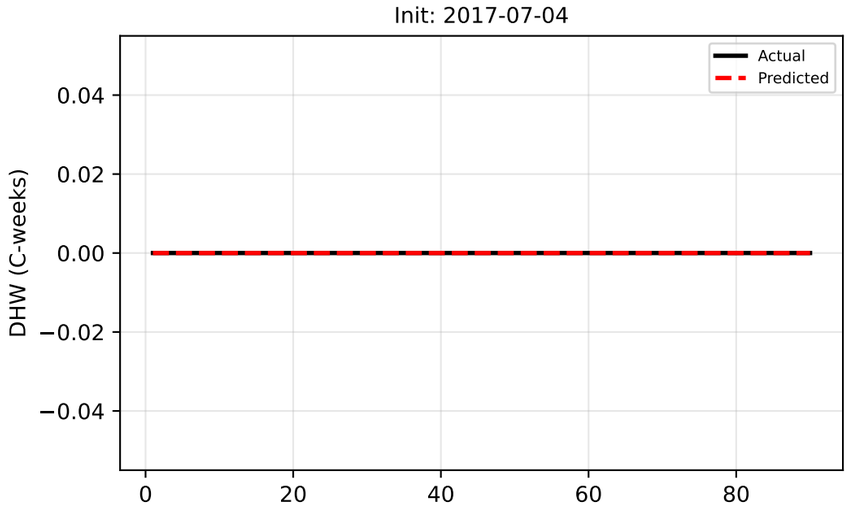


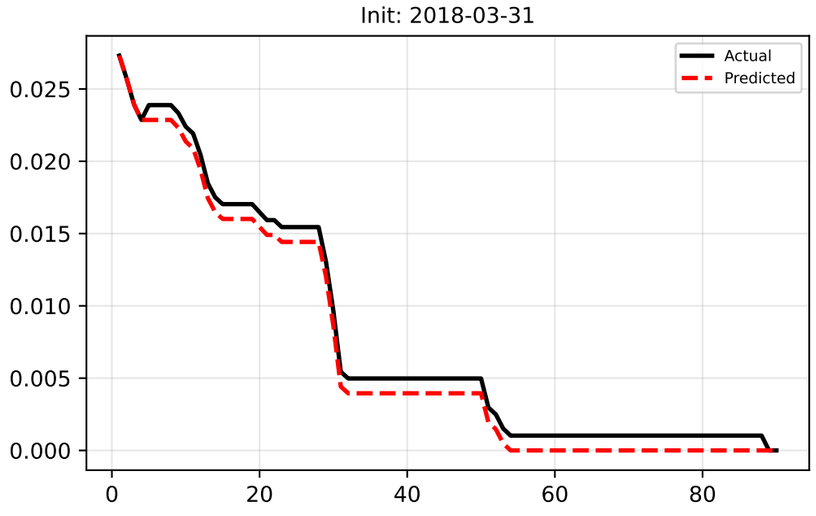
Global-Mean DHW - Val - GBR (C-weeks)

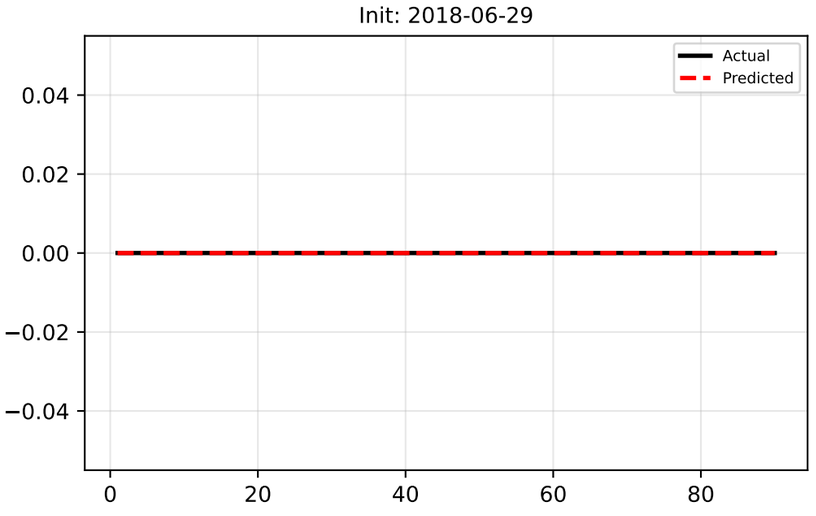


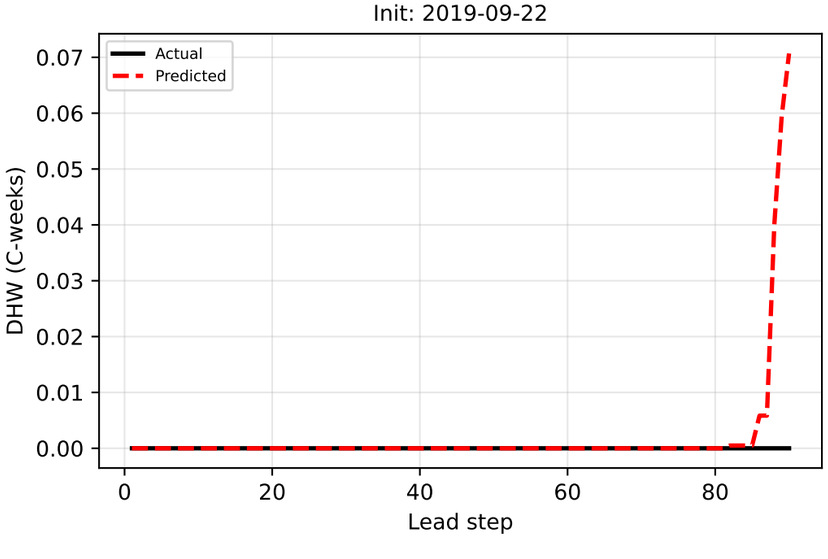


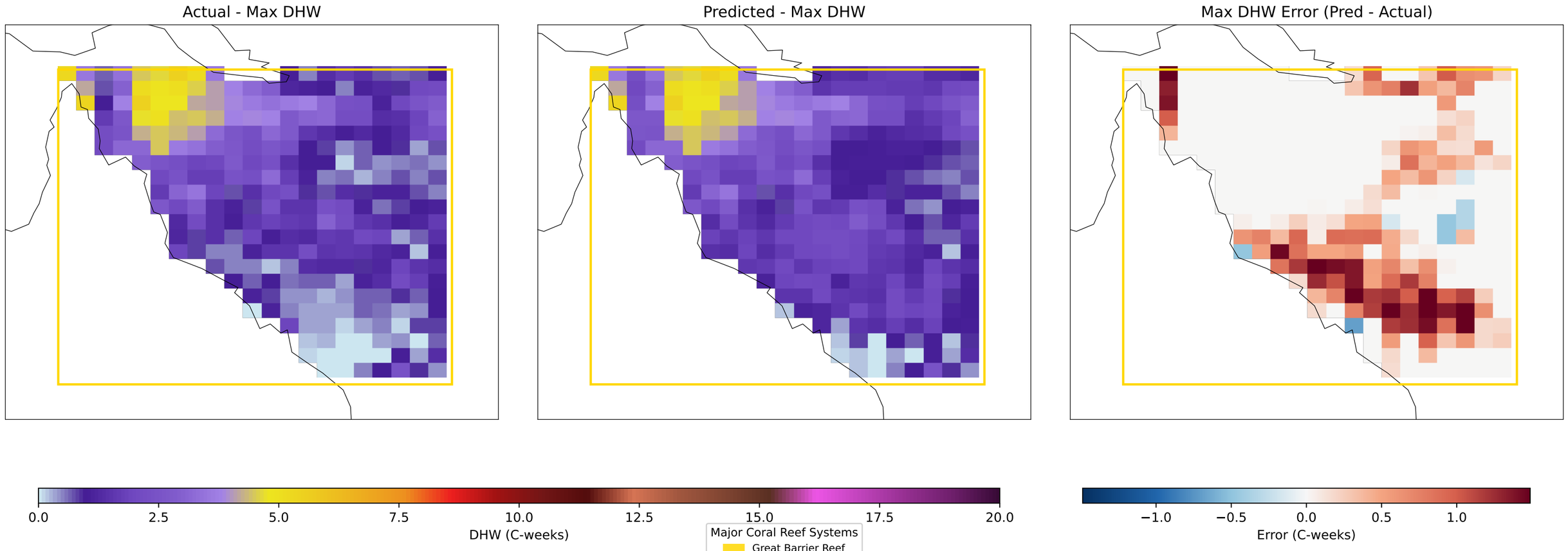


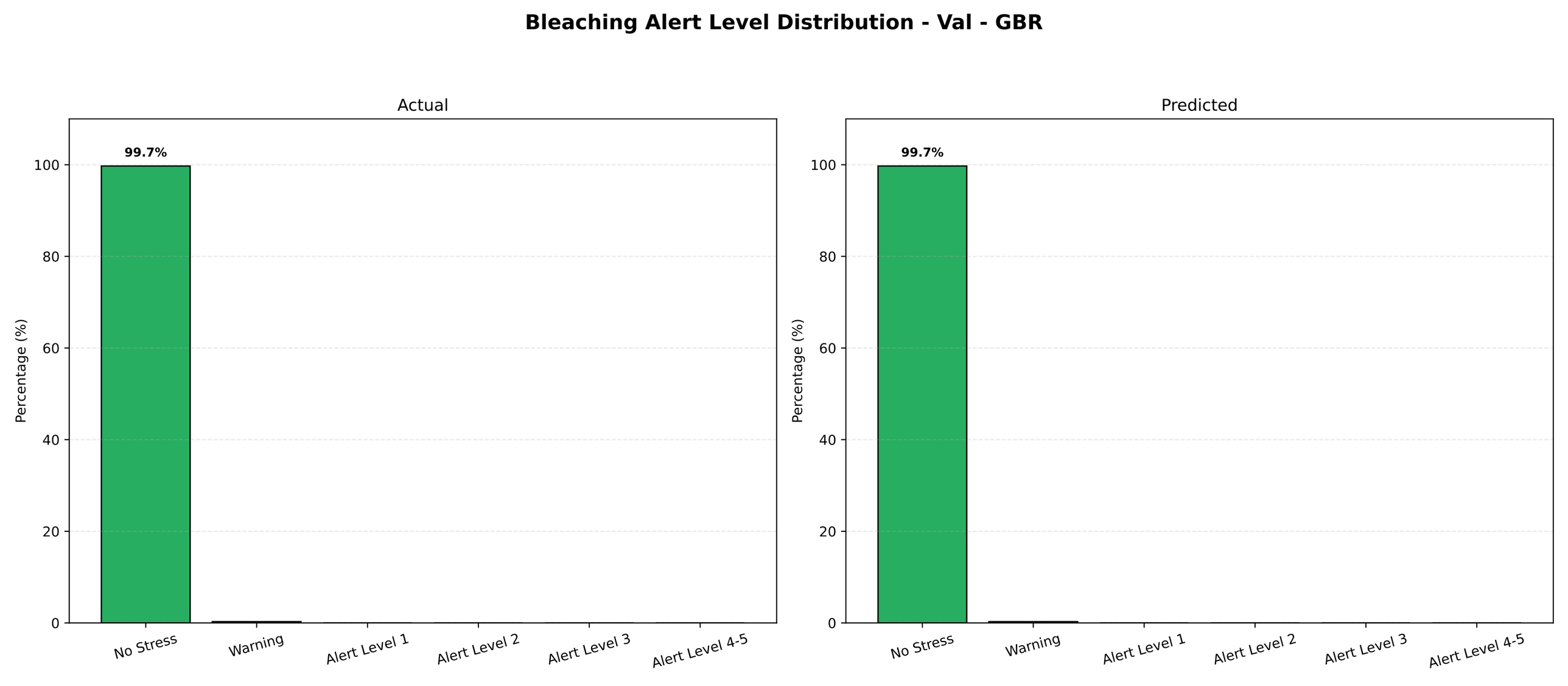






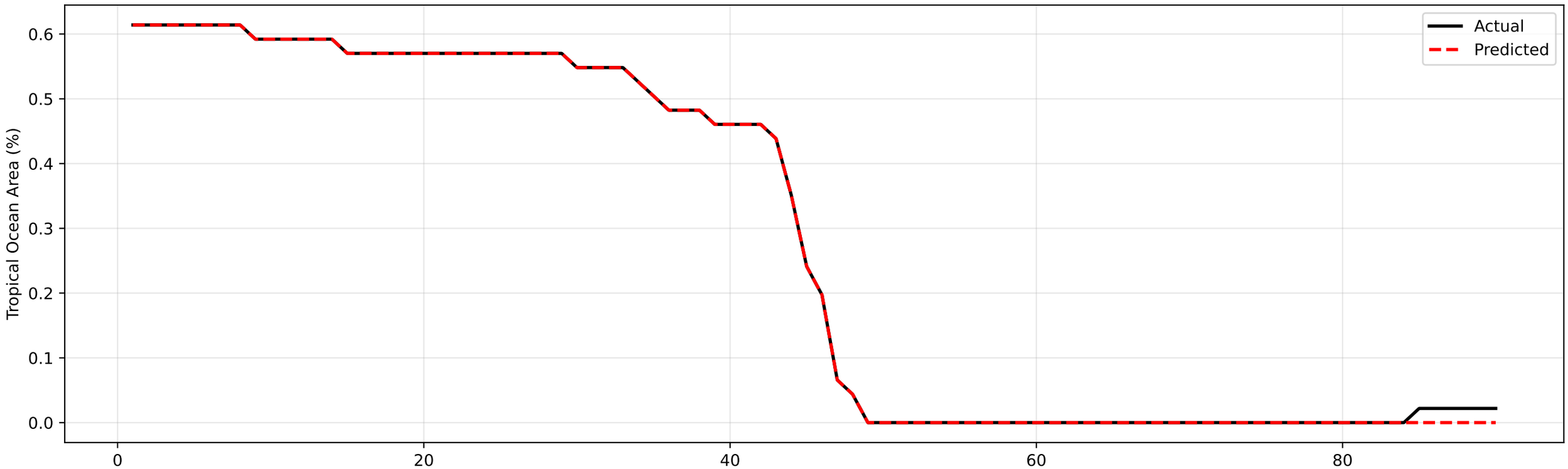



Peak DHW (max over all chains & leads) - Val - GBR (C-weeks)
Actual - Max DHW
Predicted - Max DHW
Max DHW Error (Pred - Actual)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
DHW (C-weeks)
Major Coral Reef Systems
Great Barrier Reef
−1.0
−0.5
0.0
0.5
1.0
Error (C-weeks)


Bleaching Alert Level Distribution - Val - GBR
Actual
Predicted
Percentage (%)
0
20
40
60
80
100
99.7%
No Stress
Warning
Alert Level 1
Alert Level 2
Alert Level 3
Alert Level 4-5

Tropical Ocean Area Under Bleaching Stress - Val - GBR

DHW >= 4 C-weeks (Alert Level 1 DHW threshold)



DHW >= 8 C-weeks (Alert Level 2 DHW threshold)